11.16 PROBLEMS

11.1 Consider a subsonic compressible flow in cartesian coordinates where the velocity potential is given by

$$\phi(x, y) = V_\infty x + \frac{70}{\sqrt{1 - M_\infty^2}} e^{-2\pi\sqrt{1-M_\infty^2}\,y} \sin 2\pi x$$

If the freestream properties are given by $V_\infty = 700$ ft/s, $p_\infty = 1$ atm, and $T_\infty = 519°$R, calculate the following properties at the location $(x, y) =$ (0.2 ft, 0.2 ft): M, p, and T.

11.2 Using the Prandtl-Glauert rule, calculate the lift coefficient for an NACA 2412 airfoil at 5° angle of attack in a Mach 0.6 freestream. (Refer to Figure 4.5 for the original airfoil data.)

11.3 Under low-speed incompressible flow conditions, the pressure coefficient at a given point on an airfoil is -0.54. Calculate C_p at this point when the freestream Mach number is 0.58, using

a. The Prandtl-Glauert rule

b. The Karman-Tsien rule

c. Laitone's rule

11.4 In low-speed incompressible flow, the peak pressure coefficient (at the minimum pressure point) on an airfoil is -0.41. Estimate the critical Mach number for this airfoil, using the Prandtl-Glauert rule.

11.5 For a given airfoil, the critical Mach number is 0.8. Calculate the value of p/p_∞ at the minimum pressure point when $M_\infty = 0.8$.

11.6 Consider an airfoil in a Mach 0.5 freestream. At a given point on the airfoil, the local Mach number is 0.86. Using the compressible flow tables at the back of this book, calculate the pressure coefficient at that point. Check your answer using the appropriate analytical equation from this chapter. [Note: This problem is analogous to an incompressible problem where the freestream velocity and the velocity at a point are given, and the pressure coefficient is calculated from Equation (3.38). In an incompressible flow, the pressure coefficient at any point in the flow is a unique function of the local velocity at that point and the freestream velocity. In the present problem, we see that Mach number is the relevant property for a compressible flow—not velocity. The pressure coefficient for an inviscid compressible flow is a unique function of the local Mach number and the freestream Mach number.]

11.7 Figure 11.5 shows four cases for the flow over the same airfoil wherein M_∞ is progressively increased from 0.3 to $M_{cr} = 0.61$. Have you wondered where the numbers on Figure 11.5 came from? Here is your chance to find out. Point A on the airfoil is the point of minimum pressure (hence maximum M) on the airfoil. Assume that the minimum pressure (maximum Mach number) continues to occur at this same point as M_∞ is increased. In part (a) of Figure 11.5, for $M_\infty = 0.3$, the local Mach

number at point A was arbitrarily chosen as $M_A = 0.435$, this arbitrariness is legitimate because we have not specified the airfoil shape, but rather are stating that, whatever the shape is, a maximum Mach number of 0.435 occurs at point A on the airfoil surface. However, once the numbers are given for part (a), then the numbers for parts (b), (c), and (d) are *not* arbitrary. Rather, M_A is a unique function of M_∞ for the remaining pictures. With all this as background information, starting with the data shown in Figure 11.5*a*, *calculate* M_A when $M_\infty = 0.61$. Obviously, from Figure 11.5*d*, your result should turn out to be $M_A = 1.0$ because $M_\infty = 0.61$ is said to be the critical Mach number. Said in another way, you are being asked to prove that the critical Mach number for this airfoil is 0.61. *Hint:* For simplicity, assume that the Prandtl-Glauert rule holds for the conditions of this problem.

11.8 Consider the flow over a circular cylinder; the incompressible flow over such a cylinder is discussed in Section 3.13. Consider also the flow over a sphere; the incompressible flow over a sphere is described in Section 6.4. The subsonic compressible flow over both the cylinder and the sphere is qualitatively similar but quantitatively different from their incompressible counterparts. Indeed, because of the "bluntness" of these bodies, their critical Mach numbers are relatively low. In particular:

For a cylinder: $M_{\text{cr}} = 0.404$

For a sphere: $M_{\text{cr}} = 0.57$

Explain on a physical basis why the sphere has a higher M_{cr} than the cylinder.

11.9 In Problem 11.8, the critical Mach number for a circular cylinder is given as $M_{cr} = 0.404$. This value is based on experimental measurements, and therefore is considered reasonably accurate. Calculate M_{cr} for a circular cylinder using the incompressible result for C_p and the Prandtl-Glauert compressibility correction, and compare your result with the experimental value. *Note*: The Prandtl-Glauert rule is based on linear theory assuming small perturbations, and therefore we would not expect that it would be valid for the case of flow over a circular cylinder. Nevertheless, when you use it to make this calculation of M_{cr}, you will find your calculated result to be within 3.5 percent of the experimental value. Interesting.

CHAPTER 12

Linearized Supersonic Flow

With the stabilizer setting at 2° the speed was allowed to increase to approximately 0.98 to 0.99 Mach number where elevator and rudder effectiveness were regained and the airplane seemed to smooth out to normal flying characteristics. This development lent added confidence and the airplane was allowed to continue until an indication of 1.02 on the cockpit Mach meter was obtained. At this indication the meter momentarily stopped and then jumped at 1.06, and this hesitation was assumed to be caused by the effect of shock waves on the static source. At this time the power units were cut and the airplane allowed to decelerate back to the subsonic flight condition.

Captain Charles Yeager, describing his flight on October 14, 1947—the first manned flight to exceed the speed of sound.

PREVIEW BOX

The calculation of lift and drag for an airfoil at supersonic speeds is about as different from that for lower-speed airfoils as night is from day. The physics of a supersonic flow is completely different from that of a subsonic flow, and therefore virtually nothing discussed in Chapter 4 or Chapter 11 can be used to calculate the properties of an airfoil at supersonic speeds. So what can you do? The answer is that you can read this chapter. It is short and sweet, and it provides some simple results that you can use to estimate the aerodynamic properties of supersonic airfoils.

12.1 INTRODUCTION

The linearized perturbation velocity potential equation derived in Chapter 11, Equation (11.18), is

$$\left(1 - M_\infty^2\right)\frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0 \tag{11.18}$$

and holds for both subsonic and supersonic flow. In Chapter 11, we treated the case of subsonic flow, where $1 - M_\infty^2 > 0$ in Equation (11.18). However, for supersonic flow, $1 - M_\infty^2 < 0$. This seemingly innocent change in sign on the first term of Equation (11.18) is, in reality, a very dramatic change. Mathematically, when $1 - M_\infty^2 > 0$ for subsonic flow, Equation (11.18) is an *elliptic* partial differential equation, whereas when $1 - M_\infty^2 < 0$ for supersonic flow, Equation (11.18) becomes a *hyperbolic* differential equation. The details of this mathematical difference are beyond the scope of this book; however, the important point is that there *is* a difference. Moreover, this portends a fundamental difference in the physical aspects of subsonic and supersonic flow—something we have already demonstrated in previous chapters.

The purpose of this chapter is to obtain a solution of Equation (11.18) for supersonic flow and to apply this solution to the calculation of supersonic airfoil properties. Since our purpose is straightforward, and since this chapter is relatively short, there is no need for a chapter road map to provide guidance on the flow of our ideas.

12.2 DERIVATION OF THE LINEARIZED SUPERSONIC PRESSURE COEFFICIENT FORMULA

For the case of supersonic flow, let us write Equation (11.18) as

$$\lambda^2 \frac{\partial^2 \hat{\phi}}{\partial x^2} - \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0 \tag{12.1}$$

where $\lambda = \sqrt{M_\infty^2 - 1}$. A solution to this equation is the functional relation

$$\hat{\phi} = f(x - \lambda y) \tag{12.2}$$

We can demonstrate this by substituting Equation (12.2) into Equation (12.1) as follows. The partial derivative of Equation (12.2) with respect to x can be written as

$$\frac{\partial \hat{\phi}}{\partial x} = f'(x - \lambda y)\frac{\partial (x - \lambda y)}{\partial x}$$

or

$$\frac{\partial \hat{\phi}}{\partial x} = f' \tag{12.3}$$

In Equation (12.3), the prime denotes differentiation of f with respect to its argument, $x - \lambda y$. Differentiating Equation (12.3) again with respect to x, we obtain

$$\frac{\partial^2 \hat{\phi}}{\partial x^2} = f'' \tag{12.4}$$

Similarly,
$$\frac{\partial \hat{\phi}}{\partial y} = f'(x - \lambda y)\frac{\partial (x - \lambda y)}{\partial y}$$

or
$$\frac{\partial \hat{\phi}}{\partial y} = f'(-\lambda) \tag{12.5}$$

Differentiating Equation (12.5) again with respect to y, we have
$$\frac{\partial^2 \hat{\phi}}{\partial y^2} = \lambda^2 f'' \tag{12.6}$$

Substituting Equations (12.4) and (12.6) into (12.1), we obtain the identity
$$\lambda^2 f'' - \lambda^2 f'' = 0$$

Hence, Equation (12.2) is indeed a solution of Equation (12.1).

Examine Equation (12.2) closely. This solution is not very specific, because f can be *any* function of $x - \lambda y$. However, Equation (12.2) tells us something specific about the flow, namely, that $\hat{\phi}$ is *constant* along lines of $x - \lambda y =$ constant. The slope of these lines is obtained from
$$x - \lambda y = \text{const}$$

Hence,
$$\frac{dy}{dx} = \frac{1}{\lambda} = \frac{1}{\sqrt{M_\infty^2 - 1}} \tag{12.7}$$

From Equation (9.31) and the accompanying Figure 9.25, we know that
$$\tan \mu = \frac{1}{\sqrt{M_\infty^2 - 1}} \tag{12.8}$$

where μ is the Mach angle. Therefore, comparing Equations (12.7) and (12.8), we see that a line along which $\hat{\phi}$ is constant is a *Mach line*. This result is sketched in Figure 12.1, which shows supersonic flow over a surface with a small hump in the middle, where θ is the angle of the surface relative to the horizontal. According to Equations (12.1) to (12.8), all disturbances created at the wall (represented by the perturbation potential $\hat{\phi}$) propagate unchanged away from the wall along Mach waves. All the Mach waves have the same slope, namely, $dy/dx = (M_\infty^2 - 1)^{-1/2}$. Note that the Mach waves slope *downstream* above the wall. Hence, *any disturbance at the wall cannot propagate upstream;* its effect

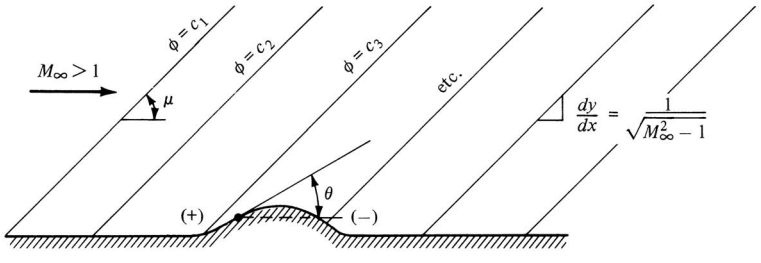


Figure 12.1 Linearized supersonic flow.

is limited to the region of the flow downstream of the Mach wave emanating from the point of the disturbance. This is a further substantiation of the major difference between subsonic and supersonic flows mentioned in previous chapters, namely, disturbances propagate *everywhere* throughout a subsonic flow, whereas they cannot propagate upstream in a steady supersonic flow.

Keep in mind that the above results, as well as the picture in Figure 12.1, pertain to *linearized* supersonic flow [because Equation (12.1) is a linear equation]. Hence, these results assume *small perturbations;* that is, the hump in Figure 12.1 is small, and thus θ is small. Of course, we know from Chapter 9 that in reality a shock wave will be induced by the forward part of the hump, and an expansion wave will emanate from the remainder of the hump. These are waves of finite strength and are not a part of linearized theory. Linearized theory is approximate; one of the consequences of this approximation is that waves of finite strength (shock and expansion waves) are not admitted.

The above results allow us to obtain a simple expression for the pressure coefficient in supersonic flow, as follows. From Equation (12.3),

$$\hat{u} = \frac{\partial \hat{\phi}}{\partial x} = f' \tag{12.9}$$

and from Equation (12.5),

$$\hat{v} = \frac{\partial \hat{\phi}}{\partial y} = -\lambda f' \tag{12.10}$$

Eliminating f' from Equations (12.9) and (12.10), we obtain

$$\hat{u} = -\frac{\hat{v}}{\lambda} \tag{12.11}$$

Recall the linearized boundary condition given by Equation (11.34), repeated below:

$$\hat{v} = \frac{\partial \hat{\phi}}{\partial y} = V_\infty \tan\theta \tag{12.12}$$

We can further reduce Equation (12.12) by noting that, for small perturbations, θ is small. Hence, $\tan\theta \approx \theta$, and Equation (12.12) becomes

$$\hat{v} = V_\infty \theta \tag{12.13}$$

Substituting Equation (12.13) into (12.11), we obtain

$$\hat{u} = -\frac{V_\infty \theta}{\lambda} \tag{12.14}$$

Recall the linearized pressure coefficient given by Equation (11.32):

$$C_p = -\frac{2\hat{u}}{V_\infty} \tag{11.32}$$

Substituting Equation (12.14) into (11.32), and recalling that $\lambda \equiv \sqrt{M_\infty^2 - 1}$, we have

$$\boxed{C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}}} \tag{12.15}$$

Equation (12.15) is important. It is the linearized supersonic pressure coefficient, and it states that C_p is directly proportional to the local surface inclination with respect to the freestream. It holds for any slender two-dimensional body where θ is small.

Return again to Figure 12.1. Note that θ is positive when measured above the horizontal, and negative when measured below the horizontal. Hence, from Equation (12.15), C_p is positive on the forward portion of the hump, and negative on the rear portion. This is denoted by the (+) and (−) signs in front of and behind the hump shown in Figure 12.1. This is also somewhat consistent with our discussions in Chapter 9; in the real flow over the hump, a shock wave forms above the front portion where the flow is being turned into itself, and hence $p > p_\infty$, whereas an expansion wave occurs over the remainder of the hump, and the pressure decreases. Think about the picture shown in Figure 12.1; the pressure is higher on the front section of the hump, and lower on the rear section. As a result, a drag force exists on the hump. This drag is called *wave drag* and is a characteristic of supersonic flows. Wave drag was discussed in Section 9.7 in conjunction with shock-expansion theory applied to supersonic airfoils. It is interesting that linearized supersonic theory also predicts a finite wave drag, although shock waves themselves are not treated in such linearized theory.

Examining Equation (12.15), we note that $C_p \propto (M_\infty^2 - 1)^{-1/2}$; hence, for supersonic flow, C_p decreases as M_∞ increases. This is in direct contrast with subsonic flow, where Equation (11.51) shows that $C_p \propto (1 - M_\infty^2)^{-1/2}$; hence, for subsonic flow, C_p increases as M_∞ increases. These trends are illustrated in Figure 12.2. Note that both results predict $C_p \to \infty$ as $M_\infty \to 1$ from either side. However, keep in mind that neither Equation (12.15) nor (11.51) is valid in the transonic range around Mach 1.

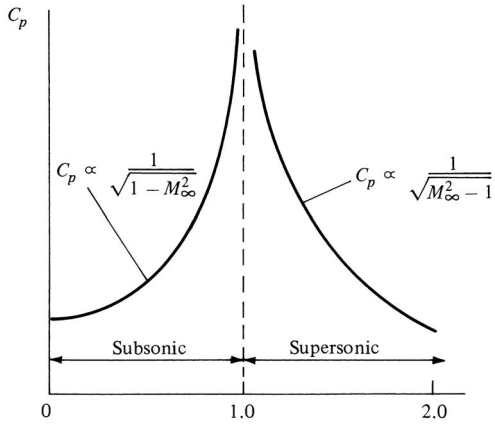


Figure 12.2 Variation of the linearized pressure coefficient with Mach number (schematic).

12.3 APPLICATION TO SUPERSONIC AIRFOILS

Equation (12.15) is very handy for estimating the lift and wave drag for thin supersonic airfoils, such as sketched in Figure 12.3. When applying Equation (12.15) to any surface, one can follow a formal sign convention for θ, which is different for regions of left-running waves (such as above the airfoil in Figure 12.3) than for regions of right-running waves (such as below the airfoil in Figure 12.3). This sign convention is developed in detail in Reference 21. However, for our purpose here, there is no need to be concerned about the sign associated with θ in Equation (12.15). Rather, keep in mind that when the surface is inclined *into* the freestream direction, linearized theory predicts a positive C_p. For example, points A and B in Figure 12.3 are on surfaces inclined into the freestream, and hence $C_{p,A}$ and $C_{p,B}$ are positive values given by

$$C_{p,A} = \frac{2\theta_A}{\sqrt{M_\infty^2 - 1}} \quad \text{and} \quad C_{p,B} = \frac{2\theta_B}{\sqrt{M_\infty^2 - 1}}$$

In contrast, when the surface is inclined *away from* the freestream direction, linearized theory predicts a negative C_p. For example, points C and D in Figure 12.3 are on surfaces inclined away from the freestream, and hence $C_{p,C}$ and $C_{p,D}$ are negative values, given by

$$C_{p,C} = -\frac{2\theta_C}{\sqrt{M_\infty^2 - 1}} \quad \text{and} \quad C_{p,D} = -\frac{2\theta_D}{\sqrt{M_\infty^2 - 1}}$$

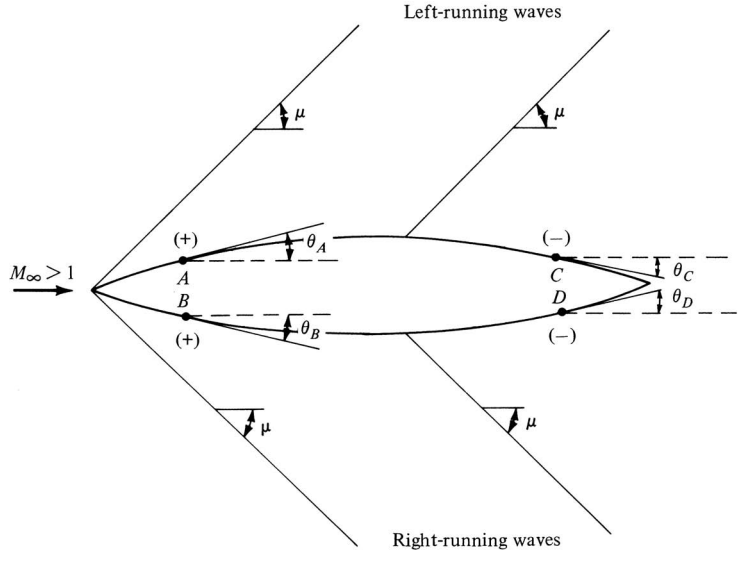


Figure 12.3 Linearized supersonic flow over an airfoil.

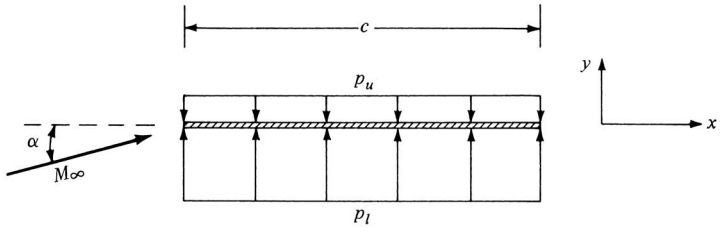


Figure 12.4 A flat plate at angle of attack in a supersonic flow.

In the above expressions, θ is always treated as a positive quantity, and the sign of C_p is determined simply by looking at the body and noting whether the surface is inclined into or away from the freestream.

With the distribution of C_p over the airfoil surface given by Equation (12.15), the lift and drag coefficients, c_l and c_d, respectively, can be obtained from the integrals given by Equations (1.15) to (1.19).

Let us consider the simplest possible airfoil, namely, a flat plate at a small angle of attack α as shown in Figure 12.4. Looking at this picture, the lower surface of the plate is a compression surface inclined at the angle α into the freestream, and from Equation (12.15),

$$C_{p,l} = \frac{2\alpha}{\sqrt{M_\infty^2 - 1}} \tag{12.16}$$

Since the surface inclination angle is constant along the entire lower surface, $C_{p,l}$ is a constant value over the lower surface. Similarly, the top surface is an expansion surface inclined at the angle α away from the freestream, and from Equation (12.15),

$$C_{p,u} = -\frac{2\alpha}{\sqrt{M_\infty^2 - 1}} \tag{12.17}$$

$C_{p,u}$ is constant over the upper surface. The normal force coefficient for the flat plate can be obtained from Equation (1.15):

$$c_n = \frac{1}{c}\int_0^c (C_{p,l} - C_{p,u})\, dx \tag{12.18}$$

Substituting Equations (12.16) and (12.17) into (12.18), we obtain

$$c_n = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}\frac{1}{c}\int_0^c dx = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \tag{12.19}$$

The axial force coefficient is given by Equation (1.16):

$$c_a = \frac{1}{c}\int_{\text{LE}}^{\text{TE}} (C_{p,u} - C_{p,l})\, dy \tag{12.20}$$

However, the flat plate has (theoretically) zero thickness. Hence, in Equation (12.20), $dy = 0$, and as a result, $c_a = 0$. This is also clearly seen in

Figure 12.4; the pressures act normal to the surface, and hence there is no component of the pressure force in the x direction. From Equations (1.18) and (1.19),

$$c_l = c_n \cos\alpha - c_a \sin\alpha \tag{1.18}$$

$$c_d = c_n \sin\alpha + c_a \cos\alpha \tag{1.19}$$

and, along with the assumption that α is small and hence $\cos\alpha \approx 1$ and $\sin\alpha \approx \alpha$, we have

$$c_l = c_n - c_a\alpha \tag{12.21}$$

$$c_d = c_n\alpha + c_a \tag{12.22}$$

Substituting Equation (12.19) and the fact that $c_a = 0$ into Equations (12.21) and (12.22), we obtain

$$\boxed{c_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}} \tag{12.23}$$

$$\boxed{c_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}}} \tag{12.24}$$

Equations (12.23) and (12.24) give the lift and wave-drag coefficients, respectively, for the supersonic flow over a flat plate. Keep in mind that they are results from linearized theory and therefore are valid only for small α.

For a thin airfoil of arbitrary shape at small angle of attack, linearized theory gives an expression for c_l identical to Equation (12.23); that is,

$$c_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

Within the approximation of linearized theory, c_l depends only on α and is independent of the airfoil shape and thickness. However, the same linearized theory gives a wave-drag coefficient in the form of

$$c_d = \frac{4}{\sqrt{M_\infty^2 - 1}}(\alpha^2 + g_c^2 + g_t^2)$$

where g_c and g_t are functions of the airfoil camber and thickness, respectively. For more details, see References 25 and 26.

EXAMPLE 12.1

Using linearized theory, calculate the lift and drag coefficients for a flat plate at a 5° angle of attack in a Mach 3 flow. Compare with the exact results obtained in Example 9.11.

■ Solution

$$\alpha = 5^\circ = 0.087 \text{ rad}$$

From Equation (12.23),

$$c_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} = \frac{(4)(0.087)}{\sqrt{(3)^2 - 1}} = \boxed{0.123}$$

From Equation (12.24),

$$c_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} = \frac{4(0.087)}{\sqrt{(3)^2 - 1}} = \boxed{0.011}$$

The results calculated in Example 9.11 for the same problem are *exact results,* utilizing the exact oblique shock theory and the exact Prandtl-Meyer expansion-wave analysis. These results were

$$\left.\begin{aligned} c_l &= 0.125 \\ c_d &= 0.011 \end{aligned}\right\} \quad \text{exact results from Example 9.11}$$

Note that, for the relatively small angle of attack of 5°, the linearized theory results are quite accurate—to within 1.6 percent.

EXAMPLE 12.2

The Lockheed F-104 supersonic fighter, shown in Figure 12.5, was the first fighter designed for sustained flight at Mach 2. The F-104 embodies good supersonic aircraft design—long slender fuselage, sharp pointed nose, and a wing with an extremely thin airfoil of 3.4 percent thickness and a razor sharp leading edge (so sharp that a protective covering is placed on the leading edge for ground handling). All these features have one purpose—to reduce supersonic wave drag. The planform area of the wing is 18.21 m^2. Consider the case of the F-104 in steady, level flight at Mach 2 at 11 km altitude. The weight of the airplane is at its combat weight of 9400 kg_f. Assume that all the lift of the airplane comes from the lift on the wings (i.e., ignore the lift of the fuselage and tail). Calculate the angle of attack of the wing relative to the freestream.

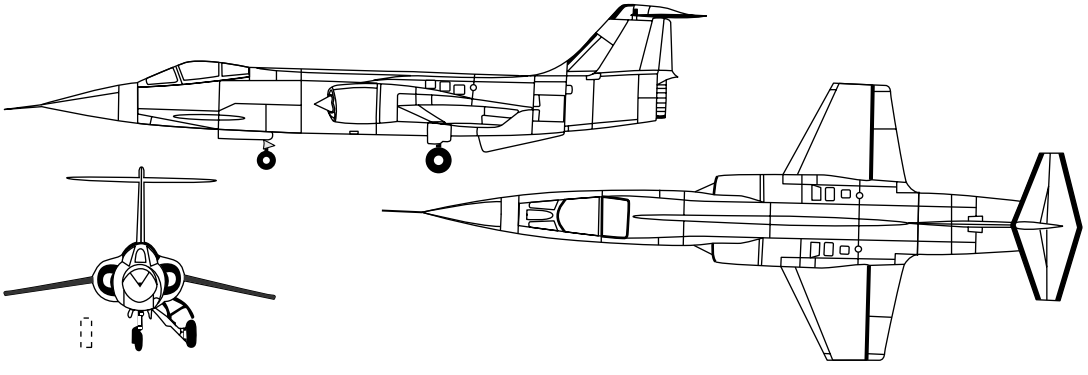


Figure 12.5 Lockheed F-104.

■ Solution

We first calculate the lift coefficient associated with the given flight conditions, recognizing that for level flight, the lift of the airplane must equal its weight:

$$C_L = \frac{L}{q_\infty S} = \frac{W}{q_\infty S}$$

At an altitude of 11 km, from Appendix D, $\rho_\infty = 0.3648$ kg/m^3 and $T_\infty = 216.78$ K. Hence,

$$a_\infty = \sqrt{\gamma R T_\infty} = \sqrt{(1.4)(287)(216.78)} = 295 \text{ m/s}$$

$$V_\infty = M_\infty a_\infty = (2)(295) = 590 \text{ m/s}$$

$$q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2 = \tfrac{1}{2}(0.3648)(590)^2 = 6.35 \times 10^4 \text{ m/s}$$

Also, the weight is given as 9400 kg_f. This is an inconsistent unit of force; a newton is the consistent unit of force in the SI system. Since 1 $\text{kg}_\text{f} = 9.8$ N, we have

$$W = (9400)(9.8) = 9.212 \times 10^4 \text{ N}$$

$$C_L = \frac{W}{q_\infty S} = \frac{9.212 \times 10^4}{(6.35 \times 10^4)(18.21)} = 0.08$$

We make the assumption that the lift coefficient of the wing, C_L, is the same as the lift coefficient for the airfoil section making up the wing, c_l. For subsonic flight, we know from our discussion in Chapter 5 that such is not the case. For supersonic flight, such is also not the case. For a straight wing in supersonic inviscid flight, however, the wing tip effects are effectively limited to the region inside the Mach cone with its vertex at the tip leading edge. At Mach 2, the semivertex angle of the Mach cone is $\mu = \sin^{-1}\frac{1}{2} = 30^\circ$. Therefore, much of the wing is unaffected by the tip effects, and experiences the two-dimensional flow discussed in this chapter. The airfoil section of the F-104 is thin, and as we will soon see, is at a small angle of attack. We conclude, therefore, that Equation (12.23) should be a good approximation for the lift coefficient of the wing of the F-104. Thus, in Equation (12.23), we use $c_l = C_L = 0.08$. From Equation (12.23),

$$c_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

or,

$$\alpha = \frac{c_\ell}{4}\sqrt{M_\infty^2 - 1} = \frac{0.08}{4}\sqrt{(2)^2 - 1} = 0.035 \text{ rad}$$

In degrees, the wing angle of attack is

$$\alpha = (0.035)(57.3) = \boxed{1.98^\circ}$$

This is a small angle of attack, and clearly satisfies the small-angle approximation embodied in Equation (12.23).

This result, which reflects actual flight conditions of supersonic airplanes, indicates that such airplanes fly at small angles of attack, for which Equations (12.23) and (12.24) are valid. Thus, the linear supersonic theory discussed in this chapter clearly has a practical application.

DESIGN BOX

The area rule discussed in Section 11.8 can be labeled the *transonic* area rule, because its proper application in an airplane design will reduce the peak drag in the transonic flight regime, allowing the airplane to more readily achieve supersonic flight. The transonic area rule calls for a smooth variation of the cross-sectional area of the airplane measured *normal* to the freestream direction. For example, the transonic area ruling of the F-16 fighter is shown in Figure 12.6; here, the cross-sectional area normal to the freestream is plotted versus distance along the fuselage.

The area rule also applies at supersonic speeds; here, however, the relevant cross-sectional area is not that perpendicular to the freestream relative wind, but rather the area cut by an oblique plane at the freestream Mach angle. For reduced supersonic wave drag, the *supersonic* area rule calls for a smooth distribution of this oblique section area. For example, consider the F-16 flying at $M_\infty = 1.6$, for which the Mach angle is $\mu = \sin^{-1}(1/M_\infty) = \sin^{-1}(1/1.6) = 38.7$ deg. In Figure 12.7*a*, a side view of the F-16 is shown, with the oblique cross-sections drawn at the angle $\mu = 38.7$ deg. The variation of this oblique cross-sectional area with distance along the fuselage is shown at the top of Figure 12.7*b*. Note that this area distribution is quite smooth; the F-16 design satisfies the supersonic area rule at $M_\infty = 1.6$. The same holds true for $M_\infty = 1.2$. The oblique cross-sectional area for $\mu = \sin^{-1}(1/1.2) = 56.4$ deg as a function of distance along the fuselage is plotted at the bottom of Figure 12.7*b*; a smooth distribution holds for this case as well. Figures 12.6 and 12.7 show that the designers of the F-16 incorporated both the transonic and supersonic area rules.

The solid curves in Figure 12.7 show the actual area distributions for the F-16. The adjacent dashed curves illustrate the area distributions proposed in the early stage of the F-16 design process; based on wind tunnel tests the final configuration was obtained by filling and shaving areas from parts of the airplane (see Figure 12.7*a*) in order to more faithfully obey the supersonic area rule.

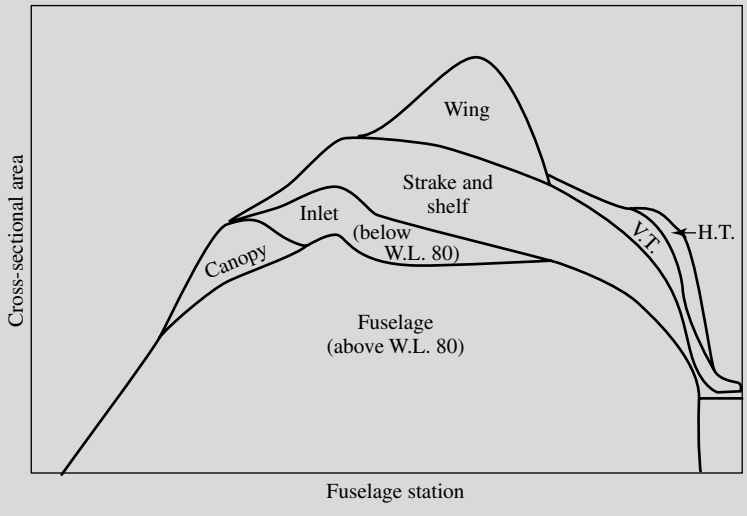
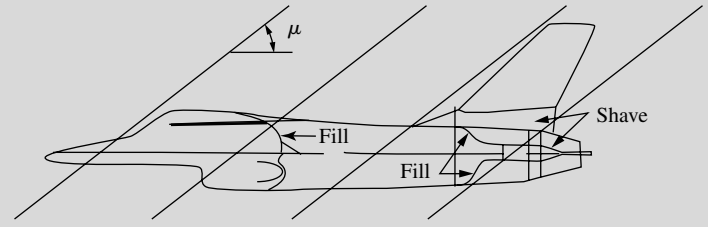
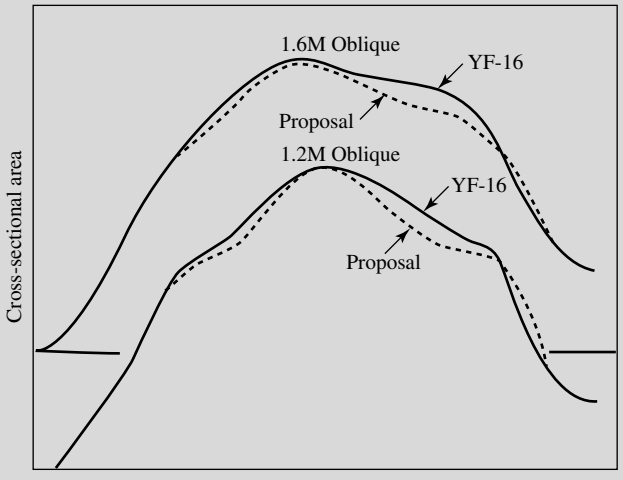


Figure 12.6 Transonic area ruling for the F-16. Variation of normal cross-sectional area as a function of location along the fuselage axis.



(*a*)



(*b*)

Figure 12.7 Supersonic area ruling for the F-16. Variation for *oblique* cross-sectional area as a function of location along the fuselage axis. Comparison between the actual area distribution and that proposed in an early design study.

The supersonic area rule was first conceived by the famous NACA and NASA aerodynamicist, R. T. Jones, and his work on this subject was first published in "Theory of Wing-Body Drag at Supersonic Speeds," NACA TR 1284, July 8, 1953 (printed in the NACA Annual Report for 1956). This report should be consulted for more details on the application of the supersonic area rule.

12.4 VISCOUS FLOW: SUPERSONIC AIRFOIL DRAG

Drag for an airfoil in low-speed flow was discussed in Section 4.12. Amazingly enough, the influence of compressibility, even supersonic flow, does not change the fundamental approach to the estimation of skin-friction drag as outlined in Section 4.12—only the numbers are different. The details behind the change in numbers are given in Chapters 18 and 19. Basically, the story goes as follows. For

laminar incompressible flow the flat plate skin-friction drag coefficient is given by Equation (4.86) as

$$C_f = \frac{1.328}{\sqrt{\text{Re}_c}} \qquad \text{(laminar incompressible)}$$

And for turbulent incompressible flow approximately by Equation (4.88) as

$$C_f = \frac{0.074}{\text{Re}_c^{1/5}} \qquad \text{(turbulent incompressible)}$$

For compressible flow, the numerators of these equations are no longer constants, but rather can be viewed as functions of Mach number, the ratio of wall temperature to the temperature at the outer edge of the boundary layer T_w/T_e, and Prandtl number Pr. The Prandtl number is defined as $\text{Pr} = \mu c_p/k$ where μ, c_p, and k are the viscosity coefficient, specific heat at constant pressure, and thermal conductivity, respectively. The importance of Prandtl number is discussed at length in Chapter 15. In essence, for compressible laminar flow we have

$$C_f = \frac{F(M_e, \text{Pr}, T_w/T_e)}{\sqrt{\text{Re}_c}} \tag{12.25}$$

and for compressible turbulent flow we have approximately

$$C_f = \frac{G(M_e, \text{Pr}, T_w/T_e)}{(\text{Re}_c)^{1/5}} \tag{12.26}$$

For a given set of values of M_e, Pr, and T_w/T_e, the numerical value of the numerators in Equations (12.25) and (12.26) are obtained from numerical solutions of the boundary layer equations as discussed in Chapters 18 and 19. Some classic results are given in Figure 19.1; leap ahead and examine this figure. The results in Figure 19.1 are presented for $\text{Pr} = 0.75$ and for the ratio T_w/T_e corresponding to that for no heat transfer at the wall (an adiabatic wall). *The most important phenomena to observe in this figure is that C_f decreases as M_∞ increases,* and that the decrease is more dramatic for a turbulent boundary layer than for a laminar boundary layer.

Keep in mind that, in addition to the skin-friction drag, a supersonic airfoil also experiences supersonic wave drag as discussed in Sections 9.7 and 12.3. The source of wave drag is the pressure distribution exerted over the airfoil surface and is a result of the shock-wave and expansion-wave pattern in the flow over the airfoil. The source of skin-friction drag is, of course, the shear stress exerted over the airfoil surface and is the result of friction in the flow. The physical mechanisms of wave drag and skin-friction drag clearly are quite different. How do these two types of drag compare in a practical flow situation? The following example addresses this question.

EXAMPLE 12.3

Consider the same Lockheed F-104 supersonic fighter described in Example 12.2, with the same flight conditions of Mach 2 at an altitude of 11 km. As calculated in Example 12.2,

for these conditions the wing angle of attack is $\alpha = 0.035$ rad $= 1.98^\circ$. Assume the chord length of the airfoil is 2.2 m, which is approximately the mean chord length for the wing. Also, assume fully turbulent flow over the airfoil. Calculate: (*a*) the airfoil skin friction drag coefficient, and (*b*) the airfoil wave drag coefficient. Compare the two values of drag.

■ Solution

(*a*) To calculate the skin-friction drag coefficient, C_f, we need the Reynolds number. From Appendix D at 11 km, $\rho_\infty = 0.3648$ kg/m^3 and $T_\infty = 216.78$ K. The speed of sound at 11 km is therefore

$$a_\infty = \sqrt{\gamma R T_\infty} = \sqrt{(1.4)(287)(216.78)} = 295 \text{ m/s}$$

Thus, $$V_\infty = M_\infty a_\infty = (2)(295) = 590 \text{ m/s}$$

As discussed in Sections 1.11 and 15.3, the viscosity coefficient is a function of temperature. We borrow Equation (15.3), which is Sutherland's law for the temperature variation of viscosity coefficient, given by

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + 110}{T + 110}$$

where T is in Kelvin, and μ_0 is a reference viscosity at a reference temperature, T_0. We choose reference conditions to be the standard sea level values of $\mu_0 = 1.7894 \times 10^{-5}$ kg/(m)(sec) and $T_0 = 288.16$ K. Thus, from Sutherland's law, the value of μ at $T = 216.78$ K is

$$\mu = (1.7894 \times 10^{-5}) \left(\frac{216.78}{288.16}\right)^{3/2} \left(\frac{288.16 + 110}{216.78 + 110}\right) = 1.4226 \times 10^{-5} \text{ kg/(m)(s)}$$

The Reynolds number is

$$\text{Re} = \frac{\rho_\infty V_\infty c}{\mu_\infty} = \frac{(0.3648)(590)(2.2)}{1.4226 \times 10^{-5}} = 3.33 \times 10^7$$

Reading Figure 19.1 very carefully, for a turbulent boundary at Mach 2 with $\text{Re} = 3.33 \times 10^7$, we have $C_f = 2.15 \times 10^{-3}$. This is for one side of a flat plate. As we have done before, we assume the skin-friction drag of the thin airfoil (and recall that the airfoil on the F-104 is really thin) to be represented by that on a flat plate. Since the skin-friction drag acts on both the upper and lower surface of the airfoil, we have, for the net skin-friction drag coefficient for the F-104 airfoil,

$$\text{Net } C_f = 2(2.15 \times 10^{-3}) = \boxed{4.3 \times 10^{-3}}$$

(*b*) The wave drag coefficient is given by Equation (12.24) as

$$c_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}}$$

Here, $\alpha = 0.035$ rad, and $M_\infty = 2$. Hence

$$c_d = \frac{4(0.035)^2}{\sqrt{(2)^2 - 1}} = \boxed{2.83 \times 10^{-3}}$$

From Example 12.3, we see that the total drag coefficient for the supersonic airfoil is $c_d + C_f = 2.83 \times 10^{-3} + 4.3 \times 10^{-3} = 7.13 \times 10^{-3}$. The skin friction is 60 percent of the total drag. At the low angle of attack for the airfoil in Example 12.3, the supersonic wave drag is reasonably small, in this case smaller than the skin friction drag. The wave drag, however, varies as the square of the angle of attack. For the conditions of Example 12.3, the wave drag would equal the skin-friction drag at an angle of attack of 2.47°, and as the angle of attack increases above this value, the wave drag would rapidly become the dominant type of drag.

Let us take a look at the lift-to-drag ratio, L/D, of the airfoil in Example 12.3. First of all, for an inviscid flow (i.e., no friction drag), the results of Examples 12.2 and 12.3 show that

$$\frac{L}{D} = \frac{c_\ell}{c_d} = \frac{0.08}{2.83 \times 10^{-3}} = 28.3 \quad \text{(inviscid)}$$

By including the skin friction drag, we have

$$\frac{L}{D} = \frac{c_\ell}{(c_d)_{\text{total}}} = \frac{0.08}{7.13 \times 10^{-3}} = 11.2$$

Clearly, the skin-friction drag greatly diminishes the lift-to-drag ratio of the airfoil. Since the value of L/D is an important measure of aerodynamic efficiency, we see the importance of trying to reduce the skin-friction drag, say by encouraging laminar rather than turbulent boundary layers on the airfoil. Indeed, this is essentially a generic statement that holds true across all parts of the Mach number spectrum, from low-speed subsonic flow to hypersonic flow. In our discussion of supersonic airfoils in this section, we can clearly see the importance of skin-friction drag on aerodynamic performance.

12.5 SUMMARY

In linearized supersonic flow, information is propagated along Mach lines where the Mach angle $\mu = \sin^{-1}(1/M_\infty)$. Since these Mach lines are all based on M_∞, they are straight, parallel lines which propagate away from and downstream of a body. For this reason, disturbances cannot propagate upstream in a steady supersonic flow.

The pressure coefficient, based on linearized theory, on a surface inclined at a small angle θ to the freestream is

$$C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}} \tag{12.15}$$

If the surface is inclined into the freestream, C_p is positive; if the surface is inclined away from the freestream, C_p is negative.

Based on linearized supersonic theory, the lift and wave-drag coefficients for a flat plate at an angle of attack are

$$c_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \tag{12.23}$$

and

$$c_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} \tag{12.24}$$

Equation (12.23) also holds for a thin airfoil of arbitrary shape. However, for such an airfoil, the wave-drag coefficient depends on both the shape of the mean camber line and the airfoil thickness.

12.6 PROBLEMS

12.1 Using the results of linearized theory, calculate the lift and wave-drag coefficients for an infinitely thin flat plate in a Mach 2.6 freestream at angles of attack of
(a) $\alpha = 5^\circ$ (b) $\alpha = 15^\circ$ (c) $\alpha = 30^\circ$
Compare these approximate results with those from the exact shock-expansion theory obtained in Problem 9.13. What can you conclude about the accuracy of linearized theory in this case?

12.2 For the conditions of Problem 12.1, calculate the pressures (in the form of p/p_∞) on the top and bottom surfaces of the flat plate, using linearized theory. Compare these approximate results with those obtained from exact shock-expansion theory in Problem 9.13. Make some appropriate conclusions regarding the accuracy of linearized theory for the calculation of pressures.

12.3 Consider a diamond-wedge airfoil such as shown in Figure 9.26, with a half-angle $\varepsilon = 10^\circ$. The airfoil is at an angle of attack $\alpha = 15^\circ$ to a Mach 3 freestream. Using linear theory, calculate the lift and wave-drag coefficients for the airfoil. Compare these approximate results with those from the exact shock-expansion theory obtained in Problem 9.14.

12.4 Equation (12.24), from linear supersonic theory, predicts that c_d for a flat plate decreases as M_∞ increases? Does this mean that the drag force itself

decreases as M_∞ increases? To answer this question, derive an equation for drag as a function of M_∞, and evaluate this equation.

12.5 Consider a flat plate at an angle of attack in an inviscid supersonic flow. From linear theory, what is the value of the maximum lift-to-drag ratio, and at what angle of attack does it occur?

12.6 Consider a flat plate at an angle of attack in a viscous supersonic flow; i.e. there is both skin friction drag and wave drag on the plate. Use linear theory for the lift and wave drag coefficients. Denote the total skin friction drag coefficient by C_f, and assume that it does not change with angle of attack. (a) Derive the expression for the angle of attack at which maximum lift-to-drag ratio occurs as a function of C_f and freestream Mach number. (b) Derive the expression for the maximum lift-to-drag ratio as a function of C_f and freestream Mach number M.

Answers: (a) $\alpha = (C_f)^{1/2}(M^2 - 1)^{1/4}/2$;
(b) $(c_l/c_d)_{\max} = (C_f)^{-1/2}(M^2 - 1)^{-1/4}$

12.7 Using the same flight conditions and the same value of the skin-friction coefficient from Example 12.3, and the results of Problem 12.6, calculate the maximum lift-to-drag ratio of the flat plate that is used to simulate the F-104 wing and the angle of attack at which it occurs.

12.8 The result from Problem 12.6 demonstrates that maximum lift-to-drag ratio decreases as the Mach number increases. This is a fact of nature that progressively causes designers of supersonic airplanes grief as they strive toward aerodynamically efficient airplanes at higher supersonic Mach numbers. What physics is nature using against the airplane designer in this case, and how might the designer meet this challenge?

CHAPTER 13

Introduction to Numerical Techniques for Nonlinear Supersonic Flow

Regarding computing as a straightforward routine, some theoreticians still tend to underestimate its intellectual value and challenge, while practitioners often ignore its accuracy and overrate its validity.

C. K. Chu, 1978
Columbia University

PREVIEW BOX

Our discussions of compressible flow have involved algebra (Chapters 7–10) and linear partial differential equations (Chapters 11 and 12)—all tractable mathematics. In the process, we dealt with a number of practical applications. But this is about as far as we can go with our analytical solutions allowed by tractable mathematics. For all other applications, encompassing the vast majority of all other real-world applications, the flows are governed by the more complete *nonlinear* equations of motion, for which there are no closed-form analytical solutions.

Do we throw up our hands and quit? From the title of this chapter, obviously not. Instead, we leave the world of analytical solutions and enter the relatively new world of numerical solutions. Instead of finding nice, neat equations to solve our flows, we are going to crunch numbers for our flows. But we are going to crunch numbers in an intelligent, often elegant fashion. We are going to start with the fundamental nonlinear flow equations of continuity, momentum, and energy obtained in Chapter 2, and solve them numerically for some important practical problems that could not be solved in any other way.

Two such practical problems are addressed in this chapter. The first is the design of the proper contour, the proper *shape,* of a supersonic nozzle. In Chapter 10 we studied the flow properties of nozzle flows, but the shape (i.e., the area-ratio distribution) of the nozzle was always given. Look again at the rocket engine shown in Figure 10.1. *How* was the actual shape of

that divergent nozzle actually designed? This is an extremely important question; if the contour of that divergent nozzle is not just right, undesirable shock waves may form inside the nozzle—shock waves that will reduce the performance of the rocket engine. And if you were designing the nozzle for a supersonic wind tunnel, any shock waves formed inside the nozzle are going to ruin the type of high-quality flow you want in the test section. So the design of the proper contour of a supersonic nozzle is extremely important. How do you do it? You will find the answers in this chapter.

The second problem is the solution of the flow around a blunt-nosed body in a supersonic flow. Leap ahead in this book and look at the Space Shuttle shown in Figure 14.17. What you see is a high-speed vehicle that has a blunt nose and a wing with a blunt leading edge. The fundamental reason for these blunt shapes was discussed in Section 1.1 (if you have forgotten, take a quick peak at Section 1.1). In the early days of high-speed flight, the solution of the flow around a blunt-nosed body in supersonic flight was impossible to obtain. And yet, it was, and still is, one of the most important problems in supersonic aerodynamics. We call it the "supersonic blunt body problem." Today, we can readily solve this problem. How? You will find the answer in this chapter.

The material in this chapter is serious business. It reflects modern aspects of compressible flow, and gives some idea of how solutions for complex problems are carried out today. Read on, and see what all this is about.

13.1 INTRODUCTION: PHILOSOPHY OF COMPUTATIONAL FLUID DYNAMICS

The above quotation underscores the phenomenally rapid increase in computer power available to engineers and scientists during the two decades between 1960 and 1980. This explosion in computer capability is still going on, with no specific limits in sight. As a result, an entirely new discipline in aerodynamics has evolved over the past three decades, namely, computational fluid dynamics (CFD). CFD is a new "third dimension" in aerodynamics, complementing the previous dimensions of both pure experiment and pure theory. It allows us to obtain answers to fluid dynamic problems which heretofore were intractable by classical analytical methods. Consequently, CFD is revolutionizing the airplane design process, and in many ways is modifying the way we conduct modern aeronautical research and development. For these reasons, every modern student of aerodynamics should be aware of the overall philosophy of CFD, because you are bound to be affected by it to some greater or lesser degree in your education and professional life.

The philosophy of computational fluid dynamics was introduced in Section 2.17, where it was compared with the theoretical approach leading to closed-form analytical solutions. Please stop here, return to Section 2.17, and re-read the material presented there; now that you have progressed this far and have seen a number of analytical solutions for both incompressible and compressible flows in the preceding chapters, the philosophy discussed in Section 2.17 will mean much more to you. Do this now, because the present chapter almost exclusively deals with numerical solutions with reference to Section 2.17.2 whereas Chapters 3–12 have dealt almost exclusively with analytical solutions with reference to Section 2.17.1.

In the present chapter we will experience the true essence of computational fluid dynamics for the first time in this book; we will actually see what is meant by the definition of CFD given in Section 2.17.2 as "the art of replacing the integrals or the partial derivatives (as the case may be) in the governing equations of fluid motion with discretized algebraic forms, which in turn are solved to obtain numbers for the flow field values at discrete points in time and/or space." However, because modern CFD is such a sophisticated discipline that is usually the subject of graduate level studies, and which rests squarely on the foundations of applied mathematics, we can only hope to give you an elementary treatment in the present chapter, but a treatment significant enough to represent some of the essence of CFD. For your next step in learning CFD beyond the present book, you are recommended to read Anderson, *Computational Fluid Dynamics: The Basics with Applications* (Reference 64), which the author has written to help undergraduates understand the nature of CFD before going on to more advanced studies of the discipline.

The purpose of this chapter is to provide an introduction to some of the basic ideas of CFD as applied to inviscid supersonic flows. More details are given in Reference 21. Because CFD has developed so rapidly in recent years, we can only scratch the surface here. Indeed, the present chapter is intended to give you only some basic background as well as the incentive to pursue the subject further in the modern literature.

The road map for this chapter is given in Figure 13.1. We begin by introducing the classical method of characteristics—a numerical technique that has been available in aerodynamics since 1929, but which had to wait on the modern computer for practical, everyday implementation. For this reason, the author classifies the method of characteristics under the general heading of numerical techniques,

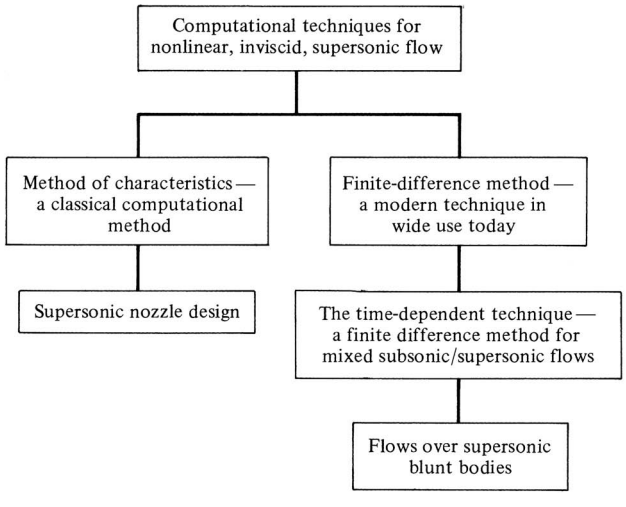


Figure 13.1 Road map for Chapter 13.

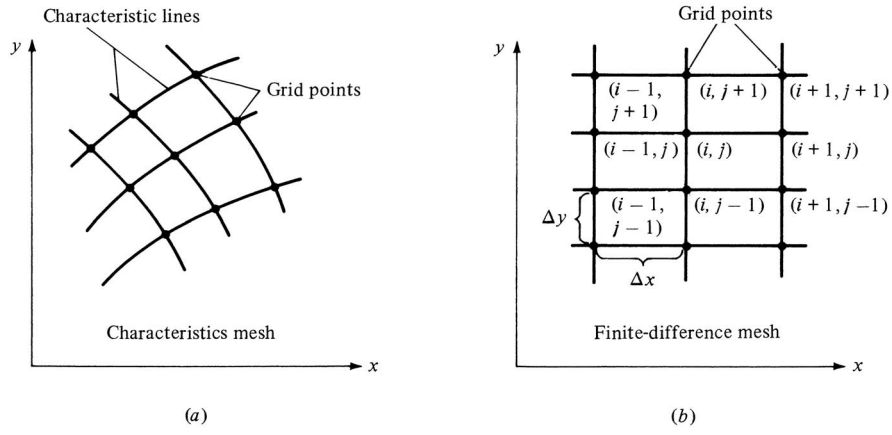


Figure 13.2 Grid points.

although others may prefer to list it under a more classical heading. We also show how the method of characteristics is applied to design the divergent contour of a supersonic nozzle. Then we move to a discussion of the finite-difference approach, which we will use to illustrate the application of CFD to nozzle flows and the flow over a supersonic blunt body.

In contrast to the linearized solutions discussed in Chapters 11 and 12, CFD represents numerical solutions to the *exact* nonlinear governing equations, that is, the equations without simplifying assumptions such as small perturbations, and which apply to all speed regimes, transonic and hypersonic as well as subsonic and supersonic. Although numerical roundoff and truncation errors are always present in any numerical representation of the governing equations, we still think of CFD solutions as being "exact solutions."

Both the method of characteristics and finite-difference methods have one thing in common: They represent a continuous flow field by a series of distinct grid points in space, as shown in Figure 13.2. The flow-field properties (u, v, p, T, etc.) are calculated at each one of these grid points. The mesh generated by these grid points is generally skewed for the method of characteristics, as shown in Figure 13.2*a*, but is usually rectangular for finite-difference solutions, as shown in Figure 13.2*b*. We will soon appreciate why these different meshes occur.

13.2 ELEMENTS OF THE METHOD OF CHARACTERISTICS

In this section, we only introduce the basic elements of the method of characteristics. A full discussion is beyond the scope of this book; see References 21, 25, and 34 for more details.

Consider a two-dimensional, steady, inviscid, supersonic flow in xy space, as given in Figure 13.2*a*. The flow variables (p, u, T, etc.) are continuous throughout

this space. However, there are certain lines in xy space along which the *derivatives* of the flow-field variables ($\partial p/\partial x$, $\partial u/\partial y$, etc.) are *indeterminate* and across which may even be discontinuous. Such lines are called *characteristic lines*. This may sound strange at first; however, let us prove that such lines exist, and let us find their precise directions in the xy plane.

In addition to the flow being supersonic, steady, inviscid, and two-dimensional, assume that it is also irrotational. The exact governing equation for such a flow is given by Equation (11.12):

$$\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial x}\right)^2\right]\frac{\partial^2\phi}{\partial x^2}+\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial y}\right)^2\right]\frac{\partial^2\phi}{\partial y^2}-\frac{2}{a^2}\frac{\partial\phi}{\partial x}\frac{\partial\phi}{\partial y}\frac{\partial^2\phi}{\partial x\,\partial y}=0 \tag{11.12}$$

[Keep in mind that we are dealing with the full velocity potential ϕ in Equation (11.12), not the perturbation potential.] Since $\partial\phi/\partial x = u$ and $\partial\phi/\partial y = v$, Equation (11.12) can be written as

$$\left(1-\frac{u^2}{a^2}\right)\frac{\partial^2\phi}{\partial x^2}+\left(1-\frac{v^2}{a^2}\right)\frac{\partial^2\phi}{\partial y^2}-\frac{2uv}{a^2}\frac{\partial^2\phi}{\partial x\,\partial y}=0 \tag{13.1}$$

The velocity potential and its derivatives are functions of x and y, for example,

$$\frac{\partial\phi}{\partial x}=f(x,y)$$

Hence, from the relation for an exact differential,

$$df=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy$$

we have

$$d\left(\frac{\partial\phi}{\partial x}\right)=du=\frac{\partial^2\phi}{\partial x^2}dx+\frac{\partial^2\phi}{\partial x\,\partial y}dy \tag{13.2}$$

Similarly,

$$d\left(\frac{\partial\phi}{\partial y}\right)=dv=\frac{\partial^2\phi}{\partial x\,\partial y}dx+\frac{\partial^2\phi}{\partial y^2}dy \tag{13.3}$$

Examine Equations (13.1) to (13.3) closely. Note that they contain the second derivatives $\partial^2\phi/\partial x^2$, $\partial^2\phi/\partial y^2$, and $\partial^2\phi/\partial x\,\partial y$. If we imagine these derivatives as "unknowns," then Equations (13.1), (13.2), and (13.3) represent three equations with three unknowns. For example, to solve for $\partial^2\phi/\partial x\,\partial y$, use Cramer's rule as follows:

$$\frac{\partial^2\phi}{\partial x\,\partial y}=\frac{\begin{vmatrix}1-\dfrac{u^2}{a^2} & 0 & 1-\dfrac{v^2}{a^2}\\ dx & du & 0\\ 0 & dv & dy\end{vmatrix}}{\begin{vmatrix}1-\dfrac{u^2}{a^2} & -\dfrac{2uv}{a^2} & 1-\dfrac{v^2}{a^2}\\ dx & dy & 0\\ 0 & dx & dy\end{vmatrix}}=\frac{N}{D} \tag{13.4}$$

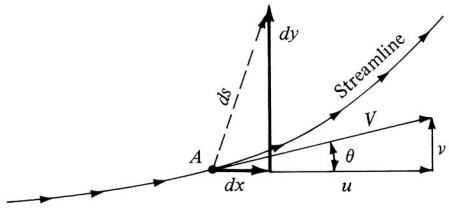


Figure 13.3 An arbitrary direction *ds* away from point *A*.

where N and D represent the numerator and denominator determinants, respectively. The physical meaning of Equation (13.4) can be seen by considering point A and its surrounding neighborhood in the flow, as sketched in Figure 13.3. The derivative $\partial^2\phi/\partial x\,\partial y$ has a specific value at point A. Equation (13.4) gives the solution for $\partial^2\phi/\partial x\,\partial y$ for an *arbitrary* choice of dx and dy. The combination of dx and dy defines an arbitrary direction ds away from point A as shown in Figure 13.3. In general, this direction is different from the streamline direction going through point A. In Equation (13.4), the differentials du and dv represent the changes in velocity that take place over the increments dx and dy. Hence, although the choice of dx and dy is arbitrary, the values of du and dv in Equation (13.4) must correspond to this choice. No matter what values of dx and dy are arbitrarily chosen, the corresponding values of du and dv will always ensure obtaining the same value of $\partial^2\phi/\partial x\,\partial y$ at point A from Equation (13.4).

The single exception to the above comments occurs when dx and dy are chosen so that $D = 0$ in Equation (13.4). In this case, $\partial^2\phi/\partial x\,\partial y$ is not defined. This situation will occur for a specific direction ds away from point A in Figure 13.3, defined for that specific combination of dx and dy for which $D = 0$. However, we know that $\partial^2\phi/\partial x\,\partial y$ has a specific defined value at point A. Therefore, the only consistent result associated with $D = 0$ is that $N = 0$, also; that is,

$$\frac{\partial^2\phi}{\partial x\,\partial y} = \frac{N}{D} = \frac{0}{0} \tag{13.5}$$

Here, $\partial^2\phi/\partial x\,\partial y$ is an indeterminate form, which is allowed to be a finite value, that is, that value of $\partial^2\phi/\partial x\,\partial y$ which we know exists at point A. *The important conclusion here is that there is some direction (or directions) through point A along which* $\partial^2\phi/\partial x\,\partial y$ *is indeterminate.* Since $\partial^2\phi/\partial x\,\partial y = \partial u/\partial y = \partial v/\partial x$, this implies that the derivatives of the flow variables are indeterminate along these lines. Hence, we have proven that lines do exist in the flow field along which derivatives of the flow variables are indeterminate; earlier, we defined such lines as *characteristic lines*.

Consider again point A in Figure 13.3. From our previous discussion, there are one or more characteristic lines through point A. *Question:* How can we calculate the precise direction of these characteristic lines? The answer can be obtained by setting $D = 0$ in Equation (13.4). Expanding the denominator determinant in

Equation (13.4), and setting it equal to zero, we have

$$\left(1-\frac{u^2}{a^2}\right)(dy)^2+\frac{2uv}{a^2}dx\,dy+\left(1-\frac{v^2}{a^2}\right)(dx)^2=0$$

or

$$\left(1-\frac{u^2}{a^2}\right)\left(\frac{dy}{dx}\right)_{\text{char}}^2+\frac{2uv}{a^2}\left(\frac{dy}{dx}\right)_{\text{char}}+\left(1-\frac{v^2}{a^2}\right)=0 \tag{13.6}$$

In Equation (13.6), dy/dx is the slope of the characteristic lines; hence, the subscript "char" has been added to emphasize this fact. Solving Equation (13.6) for $(dy/dx)_{\text{char}}$ by means of the quadratic formula, we obtain

$$\left(\frac{dy}{dx}\right)_{\text{char}}=\frac{-2uv/a^2\pm\sqrt{(2uv/a^2)^2-4(1-u^2/a^2)(1-v^2/a^2)}}{2(1-u^2/a^2)}$$

or

$$\left(\frac{dy}{dx}\right)_{\text{char}}=\frac{-uv/a^2\pm\sqrt{(u^2+v^2)/a^2-1}}{1-u^2/a^2} \tag{13.7}$$

From Figure 13.3, we see that $u = V\cos\theta$ and $v = V\sin\theta$. Hence, Equation (13.7) becomes

$$\left(\frac{dy}{dx}\right)_{\text{char}}=\frac{(-V^2\cos\theta\sin\theta)/a^2\pm\sqrt{(V^2/a^2)(\cos^2\theta+\sin^2\theta)-1}}{1-[(V^2/a^2)\cos^2\theta]} \tag{13.8}$$

Recall that the local Mach angle μ is given by $\mu = \sin^{-1}(1/M)$, or $\sin\mu = 1/M$. Thus, $V^2/a^2 = M^2 = 1/\sin^2\mu$, and Equation (13.8) becomes

$$\left(\frac{dy}{dx}\right)_{\text{char}}=\frac{(-\cos\theta\sin\theta)/\sin^2\mu\pm\sqrt{(\cos^2\theta+\sin^2\theta)/\sin^2\mu-1}}{1-(\cos^2\theta)/\sin^2\mu} \tag{13.9}$$

After considerable algebraic and trigonometric manipulation, Equation (13.9) reduces to

$$\boxed{\left(\frac{dy}{dx}\right)_{\text{char}}=\tan(\theta\mp\mu)} \tag{13.10}$$

Equation (13.10) is an important result; it states that *two* characteristic lines run through point A in Figure 13.3, namely, one line with a slope equal to $\tan(\theta-\mu)$ and the other with a slope equal to $\tan(\theta+\mu)$. The physical significance of this result is illustrated in Figure 13.4. Here, a streamline through point A is inclined at the angle θ with respect to the horizontal. The velocity at point A is V, which also makes the angle θ with respect to the horizontal. Equation (13.10) states that one characteristic line at point A is inclined *below* the streamline direction by the angle μ; this characteristic line is labeled as C_- in Figure 13.4. Equation (13.10) also states that the other characteristic line at point A is inclined *above* the streamline direction by the angle μ; this characteristic line is labeled as C_+ in Figure 13.4. Examining Figure 13.4, we see that the characteristic lines through point A are simply the left- and right-running *Mach waves* through point A. Hence,

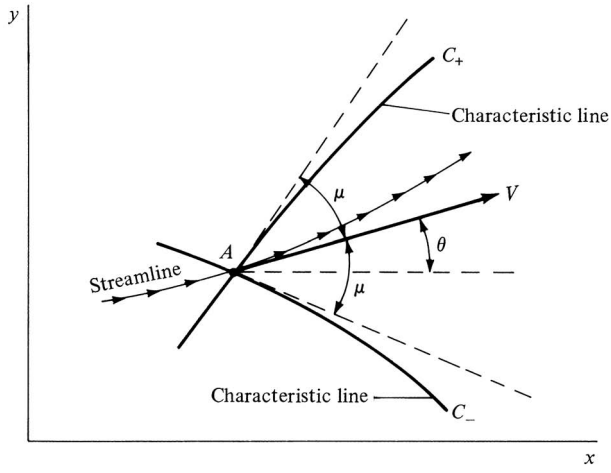


Figure 13.4 Left- and right-running characteristic lines through point *A*.

the characteristic lines are Mach lines. In Figure 13.4, the left-running Mach wave is denoted by C_+, and the right-running Mach wave is denoted by C_-. Hence, returning to Figure 13.2*a*, the characteristics mesh consists of left- and right-running Mach waves that crisscross the flow field. There are an infinite number of these waves; however, for practical calculations we deal with a finite number of waves, the intersections of which define the grid points shown in Figure 13.2*a*. Note that the characteristic lines are curved in space because (1) the local Mach angle depends on the local Mach number, which is a function of x and y, and (2) the local streamline direction θ varies throughout the flow.

The characteristic lines in Figure 13.2*a* are of no use to us by themselves. The practical consequence of these lines is that *the governing partial differential equations that describe the flow reduce to ordinary differential equations along the characteristic lines*. These equations are called the *compatibility equations,* which can be found by setting $N = 0$ in Equation (13.4), as follows. When $N = 0$, the numerator determinant yields

$$\left(1 - \frac{u^2}{a^2}\right) du\, dy + \left(1 - \frac{v^2}{a^2}\right) dx\, dv = 0$$

or

$$\frac{dv}{du} = \frac{-(1 - u^2/a^2)}{1 - v^2/a^2} \frac{dy}{dx} \tag{13.11}$$

Keep in mind that N is set to zero only when $D = 0$ in order to keep the flow-field derivatives finite, albeit of the indeterminate form $0/0$. When $D = 0$, we are restricted to considering directions *only* along the characteristic lines, as explained earlier. Hence, when $N = 0$, we are held to the same restriction. Therefore, Equation (13.11) holds only along the characteristic lines. Therefore,

in Equation (13.11),

$$\frac{dy}{dx} \equiv \left(\frac{dy}{dx}\right)_{\text{char}} \tag{13.12}$$

Substituting Equations (13.12) and (13.7) into (13.11), we obtain

$$\frac{dv}{du} = -\frac{1 - u^2/a^2}{1 - v^2/a^2} \frac{-uv/a^2 \pm \sqrt{(u^2 + v^2)/a^2 - 1}}{1 - u^2/a^2}$$

or

$$\frac{dv}{du} = \frac{uv/a^2 \mp \sqrt{(u^2 + v^2)/a^2 - 1}}{1 - v^2/a^2} \tag{13.13}$$

Recall from Figure 13.3 that $u = V \cos\theta$ and $v = V \sin\theta$. Also, $(u^2 + v^2)/a^2 = V^2/a^2 = M^2$. Hence, Equation (13.13) becomes

$$\frac{d(V \sin\theta)}{d(V \cos\theta)} = \frac{M^2 \cos\theta \sin\theta \mp \sqrt{M^2 - 1}}{1 - M^2 \sin^2\theta}$$

which, after some algebraic manipulations, reduces to

$$\boxed{d\theta = \mp\sqrt{M^2 - 1}\,\frac{dV}{V}} \tag{13.14}$$

Examine Equation (13.14). It is an *ordinary differential equation* obtained from the original governing partial differential equation, Equation (13.1). However, Equation (13.14) contains the restriction given by Equation (13.12); that is, Equation (13.14) holds *only* along the characteristic lines. Hence, Equation (13.14) gives the *compatibility relations* along the characteristic lines. In particular, comparing Equation (13.14) with Equation (13.10), we see that

$$d\theta = -\sqrt{M^2 - 1}\,\frac{dV}{V} \qquad \text{(applies along the } C_- \text{ characteristic)} \tag{13.15}$$

$$d\theta = \sqrt{M^2 - 1}\,\frac{dV}{V} \qquad \text{(applies along the } C_+ \text{ characteristic)} \tag{13.16}$$

Examine Equation (13.14) further. It should look familiar; indeed, Equation (13.14) is identical to the expression obtained for Prandtl-Meyer flow in Section 9.6, namely, Equation (9.32). Hence, Equation (13.14) can be integrated to obtain a result in terms of the Prandtl-Meyer function, given by Equation (9.42). In particular, the integration of Equations (13.15) and (13.16) yields

$$\theta + \nu(M) = \text{const} = K_- \qquad \text{(along the } C_- \text{ characteristic)} \tag{13.17}$$

$$\theta - \nu(M) = \text{const} = K_+ \qquad \text{(along the } C_+ \text{ characteristic)} \tag{13.18}$$

In Equation (13.17), K_- is a constant along a given C_- characteristic; it has different values for different C_- characteristics. In Equation (13.18), K_+ is a constant along a given C_+ characteristic; it has different values for different C_+ characteristics. Note that our compatibility relations are now given by Equations (13.17) and (13.18), which are *algebraic* equations which hold only along the characteristic

lines. In a general inviscid, supersonic, steady flow, the compatibility equations are ordinary differential equations; only in the case of two-dimensional irrotational flow do they further reduce to algebraic equations.

What is the advantage of the characteristic lines and their associated compatibility equations discussed above? Simply this—to solve the nonlinear supersonic flow, we need deal only with ordinary differential equations (or in the present case, algebraic equations) instead of the original partial differential equations. Finding the solution of such ordinary differential equations is usually much simpler than dealing with partial differential equations.

How do we use the above results to solve a practical problem? The purpose of the next section is to give such an example, namely, the calculation of the supersonic flow inside a nozzle and the determination of a proper wall contour so that shock waves do not appear inside the nozzle. To carry out this calculation, we deal with two types of grid points: (1) internal points, away from the wall, and (2) wall points. Characteristics calculations at these two sets of points are carried out as follows.

13.2.1 Internal Points

Consider the internal grid points 1, 2, and 3 as shown in Figure 13.5. Assume that we know the location of points 1 and 2, as well as the flow properties at these points. Define point 3 as the intersection of the C_- characteristic through point 1 and the C_+ characteristic through point 2. From our previous discussion, $(K_-)_1 = (K_-)_3$ because K_- is constant along a given C_- characteristic. The value of $(K_-)_1 = (K_-)_3$ is obtained from Equation (13.17) evaluated at point 1:

$$(K_-)_3 = (K_-)_1 = \theta_1 + \nu_1 \tag{13.19}$$

Similarly, $(K_+)_2 = (K_+)_3$ because K_+ is constant along a given C_+ characteristic. The value of $(K_+)_2 = (K_+)_3$ is obtained from Equation (13.18) evaluated at point 2:

$$(K_+)_3 = (K_+)_2 = \theta_2 - \nu_2 \tag{13.20}$$

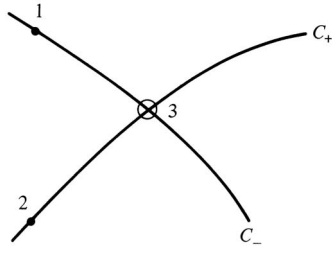


Figure 13.5 Characteristic mesh used for the location of point 3 and the calculation of flow conditions at point 3, knowing the locations and flow properties at points 1 and 2.

Now evaluate Equations (13.17) and (13.18) at point 3:

$$\theta_3 + \nu_3 = (K_-)_3 \tag{13.21}$$

and

$$\theta_3 - \nu_3 = (K_+)_3 \tag{13.22}$$

In Equations (13.21) and (13.22), $(K_-)_3$ and $(K_+)_3$ are known values, obtained from Equations (13.19) and (13.20). Hence, Equations (13.21) and (13.22) are two algebraic equations for the two unknowns θ_3 and ν_3. Solving these equations, we obtain

$$\theta_3 = \tfrac{1}{2}[(K_-)_1 + (K_+)_2] \tag{13.23}$$

$$\nu_3 = \tfrac{1}{2}[(K_-)_1 - (K_+)_2] \tag{13.24}$$

Knowing θ_3 and ν_3, all other flow properties at point 3 can be obtained as follows:

1. From ν_3, obtain the associated M_3 from Appendix C.
2. From M_3 and the known p_0 and T_0 for the flow (recall that for inviscid, adiabatic flow, the total pressure and total temperature are constants throughout the flow), find p_3 and T_3 from Appendix A.
3. Knowing T_3, compute $a_3 = \sqrt{\gamma RT_3}$. In turn, $V_3 = M_3 a_3$.

As stated earlier, point 3 is located by the intersection of the C_- and C_+ characteristics through points 1 and 2, respectively. These characteristics are curved lines; however, for purposes of calculation, we assume that the characteristics are straight-line segments between points 1 and 3 and between points 2 and 3. For example, the slope of the C_- characteristic between points 1 and 3 is assumed to be the average value between these two points, that is, $\frac{1}{2}(\theta_1 + \theta_3) - \frac{1}{2}(\mu_1 + \mu_3)$. Similarly, the slope of the C_+ characteristic between points 2 and 3 is approximated by $\frac{1}{2}(\theta_2 + \theta_3) + \frac{1}{2}(\mu_2 + \mu_3)$.

13.2.2 Wall Points

In Figure 13.6, point 4 is an internal flow point near a wall. Assume that we know all the flow properties at point 4. The C_- characteristic through point 4 intersects the wall at point 5. At point 5, the slope of the wall θ_5 is known. The flow properties at the wall point, point 5, can be obtained from the known properties at point 4 as follows. Along the C_- characteristic, K_- is constant. Hence, $(K_-)_4 = (K_-)_5$.

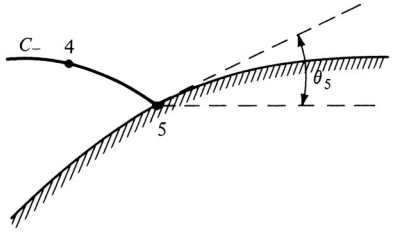


Figure 13.6 Wall point.

Moreover, the value of K_- is known from Equation (13.17) evaluated at point 4:

$$(K_-)_4 = (K_-)_5 = \theta_4 + \nu_4 \tag{13.25}$$

Evaluating Equation (13.17) at point 5, we have

$$(K_-)_5 = \theta_5 + \nu_5 \tag{13.26}$$

In Equation (13.26), $(K_-)_5$ and θ_5 are known; thus ν_5 follows directly. In turn, all other flow variables at point 5 can be obtained from ν_5 as explained earlier. The characteristic line between points 4 and 5 is assumed to be a straight-line segment with average slope given by $\frac{1}{2}(\theta_4 + \theta_5) - \frac{1}{2}(\mu_4 + \mu_5)$.

From the above discussion of both internal and wall points, we see that properties at the grid points are calculated from *known* properties at other grid points. Hence, in order to *start* a calculation using the method of characteristics, we have to know the flow properties along some *initial data* line. Then we piece together the characteristics mesh and associated flow properties by "marching downstream" from the initial data line. This is illustrated in the next section.

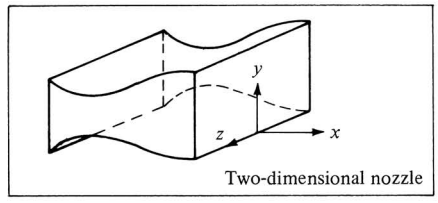
We emphasize again that the method of characteristics is an exact solution of inviscid, nonlinear supersonic flow. However, in practice, there are numerical errors associated with the finite grid; the approximation of the characteristics mesh by straight-line segments between grid points is one such example. In principle, the method of characteristics is truly exact only in the limit of an infinite number of characteristic lines.

We have discussed the method of characteristics for two-dimensional, irrotational, steady flow. The method of characteristics can also be used for rotational and three-dimensional flows, as well as unsteady flows. See Reference 21 for more details.

13.3 SUPERSONIC NOZZLE DESIGN

In Chapter 10, we demonstrated that a nozzle designed to expand a gas from rest to supersonic speeds must have a convergent-divergent shape. Moreover, the quasi-one-dimensional analysis of Chapter 10 led to the prediction of flow properties as a function of x through a nozzle of specified shape (see, e.g., Figure 10.10). The flow properties at any x station obtained from the quasi-one-dimensional analysis represent an *average* of the flow over the given nozzle cross section. The beauty of the quasi-one-dimensional approach is its simplicity. On the other hand, its disadvantages are (1) it cannot predict the details of the actual three-dimensional flow in a convergent-divergent nozzle and (2) it gives no information on the proper wall contour of such nozzles.

The purpose of the present section is to describe how the method of characteristics can supply the above information which is missing from a quasi-one-dimensional analysis. For simplicity, we treat a two-dimensional flow, as sketched in Figure 13.7. Here, the flow properties are a function of x and y. Such a two-dimensional flow is applicable to supersonic nozzles of rectangular cross section, such as sketched in the insert at the top of Figure 13.7. Two-dimensional



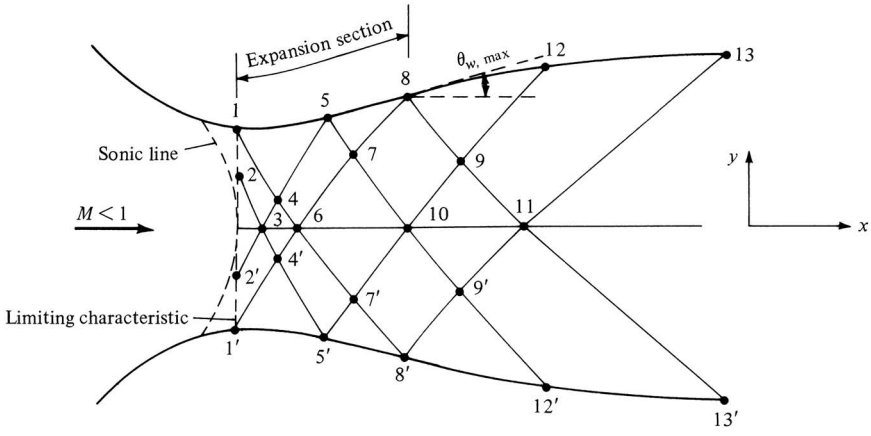


Figure 13.7 Schematic of supersonic nozzle design by the method of characteristics.

(rectangular) nozzles are used in many supersonic wind tunnels. They are also the heart of gas-dynamic lasers (see Reference 1). In addition, there is current discussion of employing rectangular exhaust nozzles on advanced military jet airplanes envisaged for the future.

Consider the following problem. We wish to design a convergent-divergent nozzle to expand a gas from rest to a given supersonic Mach number at the exit M_e. How do we design the proper contour so that we have shock-free, isentropic flow in the nozzle? The answer to this question is discussed in the remainder of this section.

For the convergent, subsonic section, there is no specific contour which is better than any other. There are rules of thumb based on experience and guided by subsonic flow theory; however, we are not concerned with the details here. We simply assume that we have a reasonable contour for the subsonic section.

Due to the two-dimensional nature of the flow in the throat region, the sonic line is generally curved, as sketched in Figure 13.7. A line called the *limiting characteristic* is sketched just downstream of the sonic line. The limiting characteristic is defined such that any characteristic line originating downstream of the limiting characteristic does not intersect the sonic line; in contrast, a characteristic line originating in the small region between the sonic line and the limiting characteristic can intersect the sonic line (for more details on the limiting characteristic, see Reference 21). To begin a method of characteristics solution, we must use an initial data line which is downstream of the limiting characteristic.

Let us assume that by independent calculation of the subsonic-transonic flow in the throat region, we know the flow properties at all points on the limiting characteristic. That is, we use the limiting characteristic as our initial data line. For example, we know the flow properties at points 1 and 2 on the limiting characteristic in Figure 13.7. Moreover, consider the nozzle contour just downstream of the throat. Letting θ denote the angle between a tangent to the wall and the horizontal, the section of the divergent nozzle where θ is increasing is called the *expansion section,* as shown in Figure 13.7. The end of the expansion section occurs where $\theta = \theta_{max}$ (point 8 in Figure 13.7). Downstream of this point, θ decreases until it equals zero at the nozzle exit. The portion of the contour where θ decreases is called the *straightening section*. The shape of the expansion section is somewhat arbitrary; typically, a circular arc of large radius is used for the expansion section of many wind-tunnel nozzles. Consequently, in addition to knowing the flow properties along the limiting characteristic, we also have an expansion section of specified shape; that is, we know θ_1, θ_5, and θ_8 in Figure 13.7. The purpose of our application of the method of characteristics now becomes the proper design of the contour of the straightening section (from points 8 to 13 in Figure 13.7).

The characteristics mesh sketched in Figure 13.7 is very coarse—this is done intentionally to keep our discussion simple. In an actual calculation, the mesh should be much finer. The characteristics mesh and the flow properties at the associated grid points are calculated as follows:

1. Draw a C_- characteristic from point 2, intersecting the centerline at point 3. Evaluating Equation (13.17) at point 3, we have

$$\theta_3 + \nu_3 = (K_-)_3$$

In the above equation, $\theta_3 = 0$ (the flow is horizontal along the centerline). Also, $(K_-)_3$ is known because $(K_-)_3 = (K_-)_2$. Hence, the above equation can be solved for ν_3.

2. Point 4 is located by the intersection of the C_- characteristic from point 1 and the C_+ characteristic from point 3. In turn, the flow properties at the internal point 4 are determined as discussed in the last part of Section 13.2.

3. Point 5 is located by the intersection of the C_+ characteristic from point 4 with the wall. Since θ_5 is known, the flow properties at point 5 are determined as discussed in Section 13.2 for wall points.

4. Points 6 through 11 are located in a manner similar to the above, and the flow properties at these points are determined as discussed before, using the internal point or wall point method as appropriate.

5. Point 12 is a wall point on the straightening section of the contour. The purpose of the straightening section is to cancel the expansion waves generated by the expansion section. Hence, there are no waves which are reflected from the straightening section. In turn, no right-running waves cross the characteristic line between points 9 and 12. As a result, the

characteristic line between points 9 and 12 is a straight line, along which θ is constant, that is, $\theta_{12} = \theta_9$. The section of the wall contour between points 8 and 12 is approximated by a straight line with an average slope of $\frac{1}{2}(\theta_8 + \theta_{12})$.

6. Along the centerline, the Mach number continuously increases. Let us assume that at point 11, the design exit Mach number M_e is reached. The characteristic line from points 11 to 13 is the last line of the calculation. Again, $\theta_{13} = \theta_{11}$, and the contour from point 12 to point 13 is approximated by a straight-line segment with an average slope of $\frac{1}{2}(\theta_{12} + \theta_{13})$.

The above description is intended to give you a "feel" for the application of the method of characteristics. If you wish to carry out an actual nozzle design, and/or if you are interested in more details, read the more complete treatments in References 21 and 34.

Note in Figure 13.7 that the nozzle flow is symmetrical about the centerline. Hence, the points below the centerline (1′, 2′, 3′, etc.) are simply mirror images of the corresponding points above the centerline. In making a calculation of the flow through the nozzle, we need to concern ourselves only with those points in the upper half of Figure 13.7, above and on the centerline.

13.4 ELEMENTS OF FINITE-DIFFERENCE METHODS

The method of characteristics described in the previous section legitimately can be considered a part of computational fluid dynamics because it uses discrete algebraic forms of the governing equations (such as Equations (13.17) and (13.18)) which are solved at discrete points in the flow (the characteristic mesh illustrated in Figure 13.5). However, most authors consider that CFD is represented by mainly finite difference and finite volume techniques, such as are discussed in Reference 64, and the method of characteristics is usually not included in the study of CFD. The purpose of this section is to give you the flavor of finite-difference techniques by describing one particular method that is readily applicable to a number of compressible flow problems. The method discussed here is representative of mainstream CFD, but it is just the tip of the iceberg of CFD. The intensive research in CFD since 1960 has produced a multitude of different algorithms and philosophies, and it is far beyond the scope of this book to go into the details of such work. See Reference 64 for an in-depth presentation of CFD at the introductory level. In addition, you are strongly encouraged to read the current literature in this regard, in particular the *AIAA Journal, Computers and Fluids,* and the *Journal of Computational Physics*. Finally, in this chapter we are dealing with numerical solutions of *inviscid supersonic* flows. See Reference 21 for an expanded discussion of finite difference methods applied to supersonic flows.

First, recall the discrete finite difference representations for partial derivatives that were derived in Section 2.17.2 using Taylors series. In particular, we recall Equations (2.168), (2.171), and (2.174), repeated and renumbered, respectively,

below for convenience:

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i+1,j} - u_{i,j}}{\Delta x} \qquad \text{(forward difference)} \tag{13.27}$$

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i,j} - u_{i-1,j}}{\Delta x} \qquad \text{(rearward difference)} \tag{13.28}$$

$$\left(\frac{\partial u}{\partial x}\right)_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x} \qquad \text{(central difference)} \tag{13.29}$$

Analogous expressions for the derivatives in the y direction are as follows:

$$\left(\frac{\partial u}{\partial y}\right)_{i,j} = \begin{cases} \dfrac{u_{i,j+1} - u_{i,j}}{\Delta y} & \text{(forward difference)} \\ \dfrac{u_{i,j} - u_{i,j-1}}{\Delta y} & \text{(rearward difference)} \\ \dfrac{u_{i,j+1} - u_{i,j-1}}{2\Delta y} & \text{(central difference)} \end{cases}$$

How do we use the finite differences obtained here? Imagine that a flow in xy space is covered by the mesh shown in Figure 13.2*b*. Assume there are N grid points. At each one of these grid points, evaluate the continuity, momentum, and energy equations with their partial derivatives replaced by the finite-difference expressions derived above. For example, replacing the derivatives in Equations (7.40), (7.42*a* and *b*), and (7.44) with finite differences, along with Equations (7.1) and (7.6*a*), we obtain a system (over all N grid points) of $6N$ simultaneous nonlinear algebraic equations in terms of the $6N$ unknowns, namely, ρ, u, v, p, T, and e, at each of the N grid points. In principle, we could solve this system for the unknown flow variables at all the grid points. In practice, this is easier said than done. There are severe problems in solving such a large number of simultaneous nonlinear equations. Moreover, we have to deal with problems associated with numerical instabilities that sometimes cause such attempted solutions to "blow up" on the computer. Finally, and most importantly, we must properly account for the boundary conditions. These considerations make all finite-difference solutions a nontrivial endeavor. As a result, a number of specialized finite-difference techniques have evolved, directed at solving different types of flow problems and attempting to increase computational efficiency and accuracy. It is beyond the scope of this book to describe these difference techniques in detail. However, one technique in particular was widely used during the 1970s and 1980s. This is an approach developed in 1969 by Robert MacCormack at the NASA Ames Research Center. Because of its widespread use and acceptance at the time, as well as its relative simplicity, we will describe MacCormack's technique in enough detail to give you a reasonable understanding of the method. This description will be carried out in the context of the following example.

Consider the two-dimensional supersonic flow through the divergent duct shown in Figure 13.8*a*. Assume the flow is supersonic at the inlet, and that all

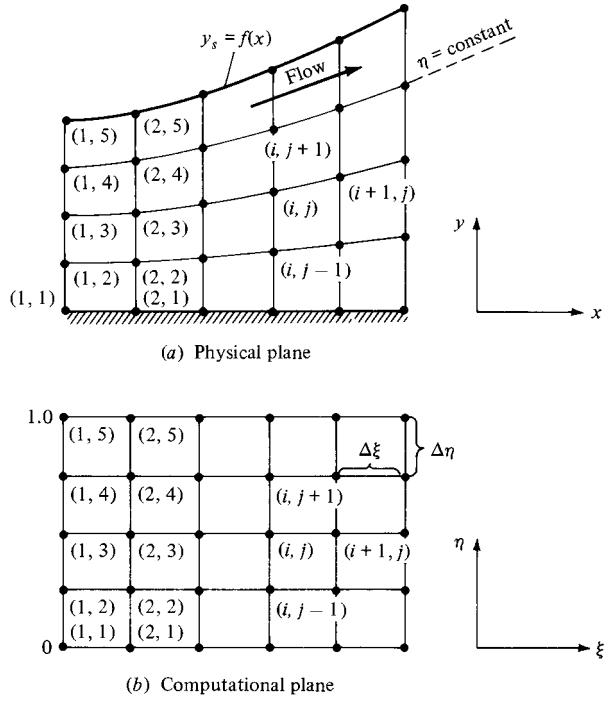


Figure 13.8 Finite-difference meshes in both the physical and computational planes.

properties are known at the inlet. That is, the flow-field variables at grid points (1, 1), (1, 2), (1, 3), (1, 4), and (1, 5) are known. The duct is formed by a flat surface at the bottom and a specified contour, $y_s = f(x)$, at the top. In addition, assume that the flow is inviscid, adiabatic, and steady, and with no body forces. It can be rotational or irrotational—the method of solution is the same. The governing equations are obtained from Equations (7.40), (7.42*a* and *b*), (7.44), (7.1), and (7.6*a*), which yield

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \tag{13.30}$$

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} \tag{13.31}$$

$$\rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} \tag{13.32}$$

$$\rho u \frac{\partial (e + V^2/2)}{\partial x} + \rho v \frac{\partial (e + V^2/2)}{\partial y} = -\frac{\partial (pu)}{\partial x} - \frac{\partial (pv)}{\partial y} \tag{13.33}$$

$$p = \rho RT \tag{13.34}$$

$$e = c_v T \tag{13.35}$$

Let us express these equations in slightly different form, as follows. Multiplying Equation (13.30) by u, and adding the results to Equation (13.31), we have

$$u\frac{\partial(\rho u)}{\partial x} + \rho u\frac{\partial u}{\partial x} + u\frac{\partial(\rho v)}{\partial y} + \rho v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x}$$

or

$$\frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} = -\frac{\partial p}{\partial x}$$

or

$$\frac{\partial}{\partial x}(\rho u^2 + p) = -\frac{\partial(\rho uv)}{\partial y} \tag{13.36}$$

Similarly, multiplying Equation (13.30) by v, and adding the result to Equation (13.32), we obtain

$$\frac{\partial(\rho uv)}{\partial x} = -\frac{\partial(\rho v^2 + p)}{\partial y} \tag{13.37}$$

Multiplying Equation (13.30) by $e + V^2/2$, and adding the result to Equation (13.33), we obtain

$$\frac{\partial}{\partial x}\left[\rho u\left(e + \frac{V^2}{2}\right) + pu\right] = -\frac{\partial}{\partial y}\left[\rho v\left(e + \frac{V^2}{2}\right) + pv\right] \tag{13.38}$$

Define the following symbols:

$$F = \rho u \tag{13.39a}$$

$$G = \rho u^2 + p \tag{13.39b}$$

$$H = \rho uv \tag{13.39c}$$

$$K = \rho u\left(e + \frac{V^2}{2}\right) + pu \tag{13.39d}$$

Then, Equations (13.30) and (13.36) to (13.38) become

$$\frac{\partial F}{\partial x} = -\frac{\partial(\rho v)}{\partial y} \tag{13.40}$$

$$\frac{\partial G}{\partial x} = -\frac{\partial(\rho uv)}{\partial y} \tag{13.41}$$

$$\frac{\partial H}{\partial x} = -\frac{\partial(\rho v^2 + p)}{\partial y} \tag{13.42}$$

$$\frac{\partial K}{\partial x} = -\frac{\partial}{\partial y}\left[\rho v\left(e + \frac{V^2}{2}\right) + pv\right] \tag{13.43}$$

Equations (13.40) to (13.43) are the continuity, x and y momentum, and energy equations, respectively—but in a slightly different form from those we are used

to seeing. The above form of these equations is frequently called the *conservation form*. Let us now treat F, G, H, and K as our primary dependent variables; these quantities are called *flux variables,* in contrast to the usual p, ρ, T, u, v, e, etc., which are called *primitive variables*. It is important to note that once the values of F, G, H, and K are known at a given grid point, the primitive variables at that point can be found from Equations (13.39*a* to *d*) and

$$p = \rho RT \tag{13.44}$$

$$e = c_v T \tag{13.45}$$

$$V^2 = u^2 + v^2 \tag{13.46}$$

That is, Equations (13.39*a* to *d*) and (13.44) to (13.46) constitute seven algebraic equations for the seven primitive variables, ρ, u, v, p, e, T, and V.

Let us return to the physical problem given in Figure 13.8*a*. Because the duct diverges, it is difficult to deal with an orthogonal, rectangular mesh; rather, a mesh which conforms to the boundary of the system will be curved, as shown in Figure 13.8*a*. On the other hand, to use our finite-difference quotients as given in Equation (13.27), (13.28), or (13.29), we desire a rectangular computational mesh. Therefore, we must *transform* the curved mesh shown in Figure 13.8*a*, known as the *physical plane,* to a rectangular mesh shown in Figure 13.8*b*, known as the *computational plane*. This transformation can be carried out as follows. Define

$$\xi = x \tag{13.47a}$$

$$\eta = \frac{y}{y_s}$$

where

$$y_s = f(x) \tag{13.47b}$$

In the above transformation, η ranges from 0 at the bottom wall to 1.0 at the top wall. In the computational plane (Figure 13.8*b*), η = constant is a straight horizontal line, whereas in the physical plane, η = constant corresponds to the curved line shown in Figure 13.8. Because we wish to apply our finite differences in the computational plane, we need the governing equations in terms of ξ and η rather than x and y. To accomplish this transformation, apply the chain rule of differentiation, using Equations (13.47*a* and *b*) as follows:

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial x} = \frac{\partial}{\partial \xi} - \frac{y}{y_s^2}\frac{dy_s}{dx}\frac{\partial}{\partial \eta}$$

or

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} - \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)\frac{\partial}{\partial \eta} \tag{13.48}$$

and

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial y} = \frac{1}{y_s}\frac{\partial}{\partial \eta} \tag{13.49}$$

Using Equations (13.48) and (13.49), we see that Equations (13.40) to (13.43) become

$$\frac{\partial F}{\partial \xi} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)\left(\frac{\partial F}{\partial \eta}\right) - \frac{1}{y_s}\frac{\partial(\rho v)}{\partial \eta} \tag{13.50}$$

$$\frac{\partial G}{\partial \xi} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)\frac{\partial G}{\partial \eta} - \frac{1}{y_s}\frac{\partial(\rho u v)}{\partial \eta} \tag{13.51}$$

$$\frac{\partial H}{\partial \xi} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)\frac{\partial H}{\partial \eta} - \frac{1}{y_s}\frac{\partial(\rho v^2 + p)}{\partial \eta} \tag{13.52}$$

$$\frac{\partial K}{\partial \xi} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)\frac{\partial K}{\partial \eta} - \frac{1}{y_s}\frac{\partial}{\partial \eta}\left[\rho v\left(e + \frac{V^2}{2}\right) + pv\right] \tag{13.53}$$

Note in the above equations that the ξ derivatives are on the left and the η derivatives are all grouped on the right.

Let us now concentrate on obtaining a numerical, finite-difference solution of the problem shown in Figure 13.8. We will deal exclusively with the computational plane, Figure 13.8*b*, where the governing continuity, x and y momentum, and energy equations are given by Equations (13.50) to (13.53), respectively. Grid points (1, 1), (2, 1), (1, 2), (2, 2), etc., in the computational plane are the same as grid points (1, 1), (2, 1), (1, 2), (2, 2), etc., in the physical plane. All the flow variables are known at the inlet, including F, G, H, K. The solution for the flow variables downstream of the inlet can be found by using MacCormack's method, which is based on Taylor's series expansions for F, G, H, and K as follows:

$$F_{i+1,j} = F_{i,j} + \left(\frac{\partial F}{\partial \xi}\right)_{\text{ave}} \Delta\xi \tag{13.54a}$$

$$G_{i+1,j} = G_{i,j} + \left(\frac{\partial G}{\partial \xi}\right)_{\text{ave}} \Delta\xi \tag{13.54b}$$

$$H_{i+1,j} = H_{i,j} + \left(\frac{\partial H}{\partial \xi}\right)_{\text{ave}} \Delta\xi \tag{13.54c}$$

$$K_{i+1,j} = K_{i,j} + \left(\frac{\partial K}{\partial \xi}\right)_{\text{ave}} \Delta\xi \tag{13.54d}$$

In Equations (13.54*a* to *d*), F, G, H, and K at point (i, j) are considered known, and these equations are used to find F, G, H, and K at point $(i + 1, j)$ assuming that we can calculate the values of $(\partial F/\partial \xi)_{\text{ave}}$, $(\partial G/\partial \xi)_{\text{ave}}$, etc. The main thrust of MacCormack's method is the calculation of these average derivatives. Examining Equations (13.54*a* to *d*), we find that this finite-difference method is clearly a "down-stream marching" method; given the flow at point (i, j) we use Equations (13.54*a* to *d*) to find the flow at point $(i + 1, j)$. Then the process is repeated to find the flow at point $(i + 2, j)$, etc. This downstream marching is similar to that performed with the method of characteristics.

The average derivatives in Equations (13.54*a* to *d*) are found by means of a straightforward "predictor-corrector" approach, outlined below. In carrying out this approach, we assume that the flow properties are known at grid point (i, j), as well as at all points directly above and below (i, j), namely, at $(i, j + 1)$, $(i, j + 2)$, $(i, j - 1)$, $(i, j - 2)$, etc.

13.4.1 Predictor Step

First, predict the value of $F_{i+1,j}$ by using a Taylor series where $\partial F/\partial \xi$ is evaluated at point (i, j). Denote this predicted value by $\bar{F}_{i+1,j}$:

$$\bar{F}_{i+1,j} = F_{i,j} + \left(\frac{\partial F}{\partial \xi}\right)_{i,j} \Delta\xi \tag{13.55}$$

In Equation (13.55), $(\partial F/\partial \xi)_{i,j}$ is obtained from the continuity equation, Equation (13.50), using *forward differences* for the η derivatives; that is,

$$\left(\frac{\partial F}{\partial \xi}\right)_{i,j} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)_{i,j} \left(\frac{F_{i,j+1} - F_{i,j}}{\Delta\eta}\right) - \frac{1}{y_s}\left[\frac{(\rho v)_{i,j+1} - (\rho v)_{i,j}}{\Delta\eta}\right] \tag{13.56}$$

In Equation (13.56), all quantities on the right-hand side are known and allow the calculation of $(\partial F/\partial \xi)_{i,j}$ which is, in turn, inserted into Equation (13.55). A similar procedure is used to find predicted values of G, H, and K, namely, $\bar{G}_{i+1,j}$, $\bar{H}_{i+1,j}$, and $\bar{K}_{i+1,j}$, using forward differences in Equations (13.51) to (13.53). In turn, predicted values of the primitive variables, $\bar{p}_{i+1,j}$, $\bar{\rho}_{i+1,j}$, etc., can be obtained from Equations (13.39*a* to *d*) and (13.44) to (13.46).

13.4.2 Corrector Step

The predicted values obtained above are used to obtain predicted values of the derivative $(\overline{\partial F/\partial \xi})_{i+1,j}$, using *rearward differences* in Equation (13.50):

$$\left(\frac{\overline{\partial F}}{\partial \xi}\right)_{i+1,j} = \left(\frac{\eta}{y_s}\frac{dy_s}{dx}\right)_{i+1,j} \frac{\bar{F}_{i+1,j} - \bar{F}_{i+1,j-1}}{\Delta\eta} - \frac{1}{y_s}\frac{(\overline{\rho v})_{i+1,j} - (\overline{\rho v})_{i+1,j-1}}{\Delta\eta} \tag{13.57}$$

In turn, the results from Equations (13.56) and (13.57) allow the calculation of the average derivative

$$\left(\frac{\partial F}{\partial \xi}\right)_{\text{ave}} = \frac{1}{2}\left[\left(\frac{\partial F}{\partial \xi}\right)_{i,j} + \left(\frac{\overline{\partial F}}{\partial \xi}\right)_{i+1,j}\right] \tag{13.58}$$

Finally, this average derivative is used in Equation (13.54*a*) to obtain the corrected value of $F_{i+1,j}$. The same process is followed to find the corrected values of $G_{i+1,j}$, $H_{i+1,j}$, and $K_{i+1,j}$ using rearward differences in Equations (13.51) to (13.53) and calculating the average derivatives $(\partial G/\partial \xi)_{\text{ave}}$, etc., in the same manner as Equation (13.58).

The above finite-difference procedure allows the step-by-step calculation of the flow field, marching downstream from some initial data line. In the flow given

in Figure 13.8, the initial data line is the inlet, where properties are considered known. Although all the calculations are carried out in the transformed, computational plane, the flow-field results obtained at points (2, 1), (2, 2), etc., in the computational plane are the same values at points (2, 1), (2, 2), etc., in the physical plane.

There are other aspects of the finite-difference solution which have not been described above. For example, what values of $\Delta\eta$ and $\Delta\xi$ in Equations (13.54*a* to *d*), (13.55), (13.56), and (13.57) are allowed in order to maintain numerical stability? How is the flow-tangency condition at the walls imposed on the finite-difference calculations? These are important matters, but we do not take the additional space to discuss them here. See chapter 11 of Reference 21 for details on these questions. Our purpose here has been to give you only a feeling for the nature of the finite-difference method.

13.5 THE TIME-DEPENDENT TECHNIQUE: APPLICATION TO SUPERSONIC BLUNT BODIES

The method of characteristics described in Section 13.2 is applicable only to supersonic flows; the characteristic lines are not defined in a practical fashion for steady, subsonic flow. Also, the particular finite-difference method outlined in Section 13.4 applies only to supersonic flows; if it were to be used in a locally subsonic region, the calculation would blow up. The reason for both of the above comments is that the method of characteristics and the steady flow, forward-marching finite-difference technique depend on the governing equations being mathematically "hyperbolic." In contrast, the equations for steady subsonic flow are "elliptic." (See Reference 21 for a description of these mathematical classifications.) The fact that the governing equations change their mathematical nature in going from locally supersonic to locally subsonic flow has historically caused theoretical aerodynamicists much grief. One problem in particular, namely, the mixed subsonic-supersonic flow over a supersonic blunt body as described in Section 9.5, was a major research area until a breakthrough was made in the late 1960s for its proper numerical solution. The purpose of this section is to describe a numerical finite-difference solution which readily allows the calculation of mixed subsonic-supersonic flows—the *time-dependent method*—and to show how it is used to solve supersonic blunt-body flows. Time-dependent techniques are very common in modern computational fluid dynamics, and as a student of aerodynamics, you should be familiar with their philosophy. These techniques are also called *time-marching techniques* because the solutions are obtained by marching in steps of time.

Consider a blunt body in a supersonic stream, as sketched in Figure 13.9*a*. The shape of the body is known and is given by $b = b(y)$. For a given freestream Mach number M_∞, we wish to calculate the shape and location of the detached shock wave, as well as the flow-field properties between the shock and the body.

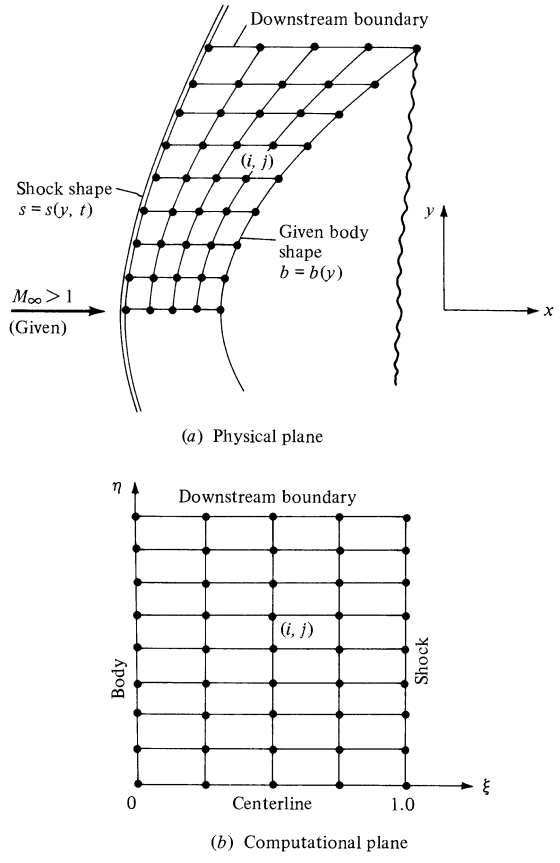


Figure 13.9 Blunt-body flow field in both the physical and computational planes.

The physical aspects of this flow field were described in Section 9.5, which you should review before progressing further.

The flow around a blunt body in a supersonic stream is rotational. Why? Examine Figure 13.10, which illustrates several streamlines around the blunt body. The flow is inviscid and adiabatic. In the uniform freestream ahead of the shock wave, the entropy is the same for each streamline. However, in crossing the shock wave, each streamline traverses a different part of the wave, and hence experiences a different increase in entropy. That is, the streamline at point a in Figure 13.10 crosses a normal shock, and hence experiences a large increase in entropy, whereas the streamline at point b crosses a weaker, oblique shock, and therefore experiences a smaller increase in entropy, $s_b < s_a$. The streamline at point c experiences an even weaker portion of the shock, and hence $s_c < s_b < s_a$. The net result is that in the flow between the shock and the body, the entropy *along* a given streamline is constant, whereas the entropy changes from one streamline

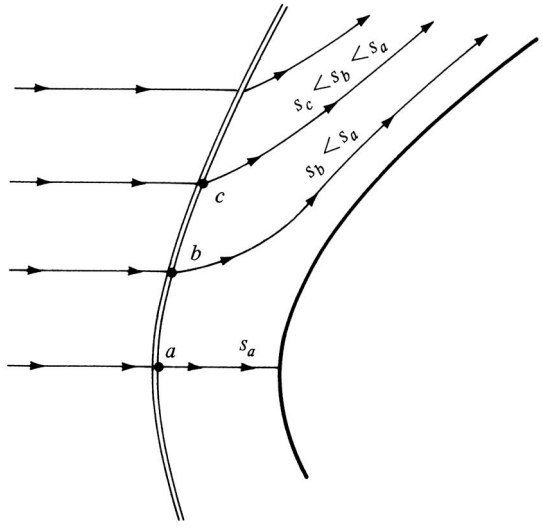


Figure 13.10 In a supersonic blunt-body flow field, the entropy is different for different streamlines.

to the next; that is, an *entropy gradient* exists normal to the streamlines. It can readily be shown (see Section 9.5.1) that an adiabatic flow with entropy gradients is *rotational*. Hence, the flow field over a supersonic blunt body is rotational.

In light of the above, we cannot use the velocity potential equation to analyze the blunt-body flow. Rather, the basic continuity, momentum, and energy equations must be employed in their fundamental form, given by Equations (7.40), (7.42*a* and *b*), and (7.44). With no body forces, these equations are

Continuity:
$$\frac{\partial \rho}{\partial t} = -\left(\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y}\right) \tag{13.59}$$

x momentum:
$$\frac{\partial u}{\partial t} = -\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{1}{\rho}\frac{\partial p}{\partial x}\right) \tag{13.60}$$

y momentum:
$$\frac{\partial v}{\partial t} = -\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{1}{\rho}\frac{\partial p}{\partial y}\right) \tag{13.61}$$

Energy:
$$\frac{\partial(e + V^2/2)}{\partial t} = -\left(u\frac{\partial(e + V^2/2)}{\partial x} + v\frac{\partial(e + V^2/2)}{\partial y} + \frac{1}{\rho}\frac{\partial(pu)}{\partial x} + \frac{1}{\rho}\frac{\partial(pv)}{\partial y}\right) \tag{13.62}$$

Notice the form of the above equations; the time derivatives are on the left, and all spatial derivatives are on the right. These equations are in the form necessary for a time-dependent finite-difference solution, as described below.

Return to Figure 13.9*a*. Recall that the body shape and freestream conditions are given, and we wish to calculate the shape and location of the shock wave as

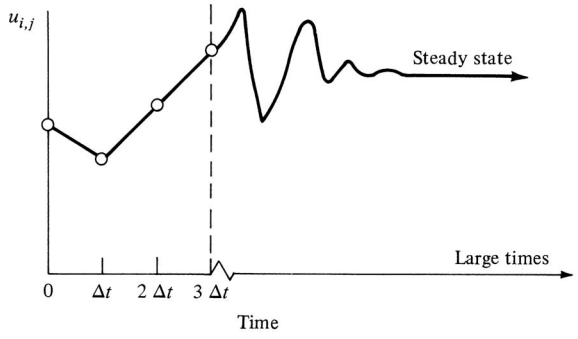


Figure 13.11 Schematic of the time variation of a typical flow variable—the time-dependent method.

well as the flow field between the shock and body. We are interested in the *steady flow* over the blunt body; however, we use a time-dependent method to obtain the steady flow. The basic philosophy of this method is as follows. First, *assume* a shock-wave shape and location. Also, cover the flow field between the shock and body with a series of grid points, as sketched in Figure 13.9*a*. At each of these grid points, *assume* values of all the flow variables, ρ, u, v, etc. These assumed values are identified as *initial conditions* at time $t = 0$. With these assumed values, the spatial derivatives on the right sides of Equations (13.59) to (13.62) are known values (obtained from finite differences). Hence, Equations (13.59) to (13.62) allow the calculation of the time derivatives $\partial\rho/\partial t$, $\partial u/\partial t$, etc. In turn, these time derivatives allow us to calculate the flow properties at each grid point at a later instant in time, say, Δt. The flow properties at time $t = \Delta t$ are different from at $t = 0$. A repetition of this cycle gives the flow-field variables at all grid points at time $t = 2\Delta t$. As this cycle is repeated many hundreds of times, the flow-field properties at each grid point are calculated as a function of time. For example, the time variation of $u_{i,j}$ is sketched in Figure 13.11. At each time step, the value of $u_{i,j}$ is different; however, at large times the changes in $u_{i,j}$ from one time step to another become small, and $u_{i,j}$ approaches a steady-state value, as shown in Figure 13.11. *It is this steady-state value that we want; the time-dependent approach is simply a means to that end.* Moreover, the shock-wave shape and location will change with time; the new shock location and shape at each time step are calculated so as to satisfy the shock relations across the wave at each of the grid points immediately behind the wave. At large times, as the flow-field variables approach a steady state, the shock shape and location also approach a steady state. Because of the time-dependent motion of the shock wave, the wave shape is a function of both t and y as shown in Figure 13.9*a*, $s = s(y, t)$.

Given this philosophy, let us examine a few details of the method. First, note that the finite-difference grid in Figure 13.9*a* is curved. We would like to apply our finite differences in a rectangular grid; hence, in Equations (13.59) to (13.62) the independent variables can be transformed as

$$\xi = \frac{x - b}{s - b} \quad \text{and} \quad \eta = y$$

where $b = b(y)$ gives the abscissa of the body and $s = s(y, t)$ gives the abscissa of the shock. The above transformation produces a rectangular grid in the computational plane, shown in Figure 13.9*b*, where the body corresponds to $\xi = 0$ and the shock corresponds to $\xi = 1$. All calculations are made in this transformed, computational plane.

The finite-difference calculations themselves can be carried out using MacCormack's method (see Section 13.4) applied as follows. The flow-field variables can be advanced in time using a Taylor series in time; for example,

$$\rho_{i,j}(t + \Delta t) = \rho_{i,j}(t) + \left[\left(\frac{\partial \rho}{\partial t}\right)_{i,j}\right]_{\text{ave}} \Delta t \tag{13.63}$$

In Equation (13.63), we know the density at grid point (i, j) at time t; that is, we know $\rho_{i,j}(t)$. Then Equation (13.63) allows us to calculate the density at the same grid point at time $t + \Delta t$, that is, $\rho_{i,j}(t + \Delta t)$, *if* we know a value of the average time derivative $[(\partial \rho/\partial t)_{i,j}]_{\text{ave}}$. This time derivative is an average between times t and $t + \Delta t$ and is obtained from a predictor-corrector process as follows.

13.5.1 Predictor Step

All the flow variables are known at time t at all the grid points. This allows us to replace the spatial derivatives on the right of Equations (13.59) to (13.62) (suitably transformed into $\xi\eta$ space) with known *forward differences*. These equations then give values of the time derivatives at time t, which are used to obtain *predicted* values of the flow-field variables at time $t + \Delta t$; for example,

$$\bar{\rho}_{i,j}(t + \Delta t) = \rho_{i,j}(t) + \left[\left(\frac{\partial \rho}{\partial t}\right)_{i,j}\right]_{t} \Delta t$$

where $\rho_{i,j}(t)$ is known, $[(\partial \rho/\partial t)_{i,j}]_t$ is obtained from the governing equation, Equation (13.59) (suitably transformed), using *forward differences* for the spatial derivatives, and $\bar{\rho}_{i,j}(t + \Delta t)$ is the predicted density at time $t + \Delta t$. Predicted values of all other flow variables $\bar{u}_{i,j}(t + \Delta t)$, etc., are obtained at all the grid points in a likewise fashion.

13.5.2 Corrector Step

Inserting the flow variables obtained above into the governing equations, Equations (13.59) to (13.62), using *rearward differences* for the spatial derivatives, predicted values of the time derivatives at $t + \Delta t$ are obtained, for example, $[(\overline{\partial \rho/\partial t})_{i,j}]_{(t+\Delta t)}$. In turn, these are averaged with the time derivatives from the predictor step to obtain; for example,

$$\left[\left(\frac{\partial \rho}{\partial t}\right)_{i,j}\right]_{\text{ave}} = \frac{1}{2}\left\{\left[\left(\frac{\partial \rho}{\partial t}\right)_{i,j}\right]_{t} + \left[\left(\overline{\frac{\partial \rho}{\partial t}}\right)_{i,j}\right]_{(t+\Delta t)}\right\} \tag{13.64}$$

Finally, the average time derivative obtained from Equation (13.64) is inserted into Equation (13.63) to yield the corrected value of density at time $t + \Delta t$. The same procedure is used for all the dependent variables, u, v, etc.

Starting from the assumed initial conditions at $t = 0$, the repeated application of Equation (13.63) along with the above predictor-corrector algorithm at each time step allows the calculation of the flow-field variables and shock shape and location as a function of time. As stated above, after a large number of time steps, the calculated flow-field variables approach a steady state, where $[(\partial\rho/\partial t)_{i,j}]_{\text{ave}} \rightarrow 0$ in Equation (13.63). Once again, we emphasize that we are interested in the steady-state answer, and the time-dependent technique is simply a means to that end.

Note that the applications of MacCormack's technique to both the steady flow calculations described in Section 13.4 and the time-dependent calculations described in the present section are analogous; in the former, we march forward in the spatial coordinate x, starting with known values along with a constant y line, whereas, in the latter, we march forward in time starting with a known flow field at $t = 0$.

Why do we bother with a time-dependent solution? Is it not an added complication to deal with an extra independent variable t in addition to the spatial variables x and y? The answers to these questions are as follows. The governing unsteady flow equations given by Equations (13.59) to (13.62) are hyperbolic with respect to time, independent of whether the flow is locally subsonic or supersonic. In Figure 13.9*a*, some of the grid points are in the subsonic region and others are in the supersonic region. However, the time-dependent solution progresses in the same manner at all these points, independent of the local Mach number. Hence, the time-dependent technique is the only approach known today which allows the uniform calculation of a mixed subsonic-supersonic flow field of arbitrary extent. For this reason, the application of the time-dependent technique, although it adds one additional independent variable, allows the straightforward solution of a flow field which is extremely difficult to solve by a purely steady-state approach.

A much more detailed description of the time-dependent technique is given in Chapter 12 of Reference 21, and especially in Reference 7, which you should study before attempting to apply this technique to a specific problem. The intent of our description here has been to give you simply a "feeling" for the philosophy and general approach of the technique.

Some typical results for supersonic blunt-body flow fields are given in Figures 13.12 to 13.15. These results were obtained with a time-dependent solution described in Reference 35. Figures 13.12 and 13.13 illustrate the behavior of a time-dependent solution during its approach to the steady state. In Figure 13.12, the time-dependent motion of the shock wave is shown for a parabolic cylinder in a Mach 4 freestream. The shock labeled $0\,\Delta t$ is the initially assumed shock wave at $t = 0$. At early times, the shock wave rapidly moves away from the body; however, after about 300 time steps, it has slowed considerably, and between 300 and 500 time steps, the shock wave is virtually motionless—it has reached its steady-state shape and location. The time variation of the stagnation point

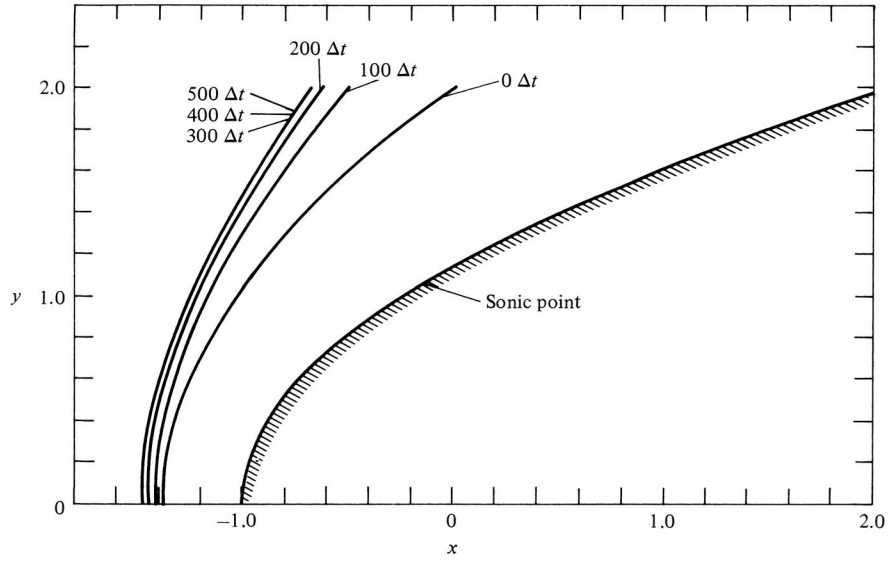


Figure 13.12 Time-dependent shock-wave motion, parabolic cylinder, $M_\infty = 4$.

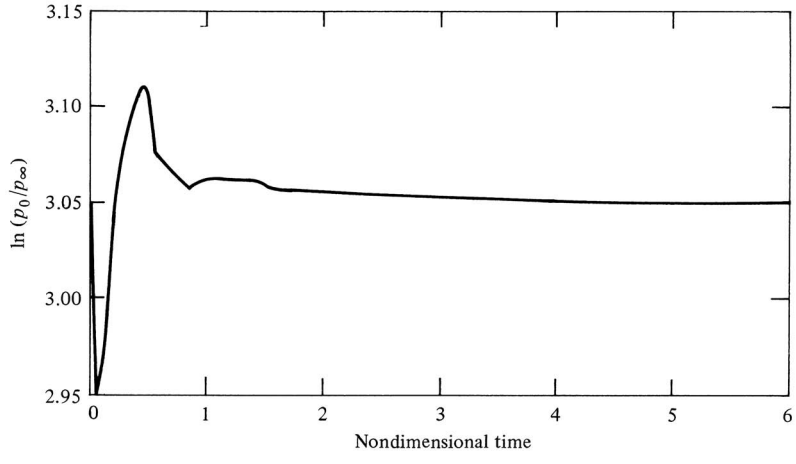


Figure 13.13 Time variation of stagnation point pressure, parabolic cylinder, $M_\infty = 4$.

pressure is given in Figure 13.13. Note that the pressure shows strong timewise oscillations at early times, but then it asymptotically approaches a steady value at large times. Again, it is this asymptotic steady state that we want, and the intermediate transient results are just a means to that end. Concentrating on just the steady-state results, Figure 13.14 gives the pressure distribution (nondimensionalized by stagnation point pressure) over the body surface for the cases of

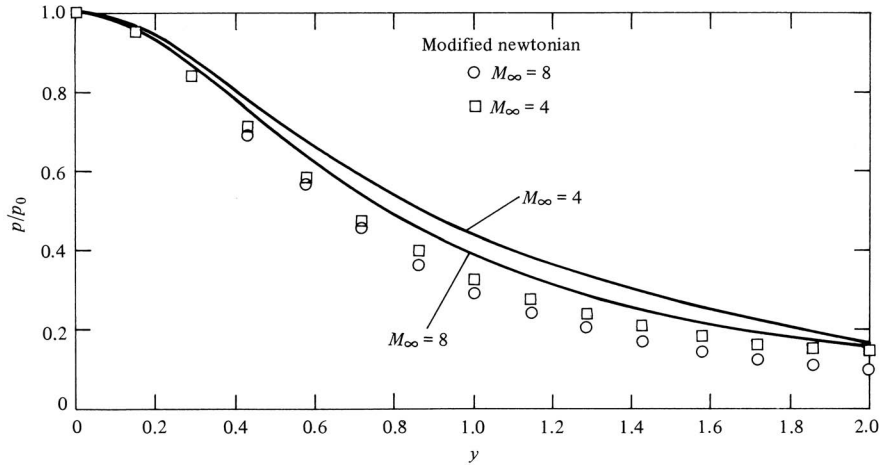


Figure 13.14 Surface pressure distributions, parabolic cylinder.

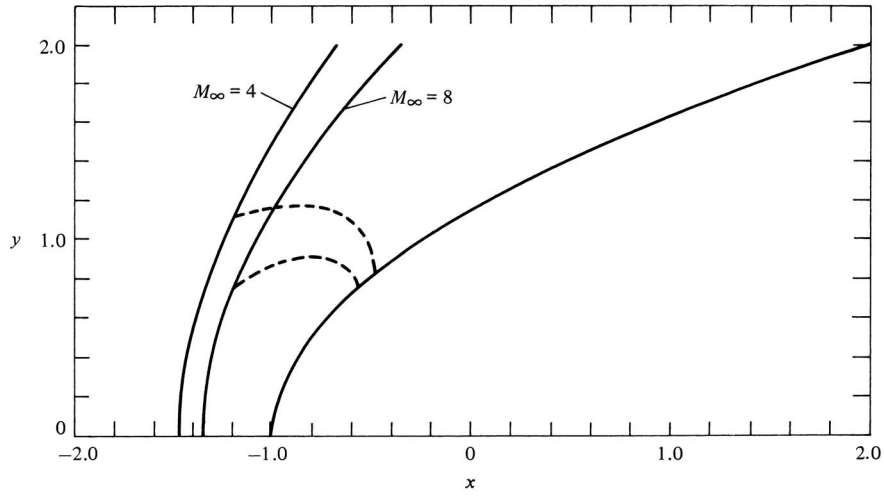


Figure 13.15 Shock shapes and sonic lines, parabolic cylinder.

both $M_\infty = 4$ and 8. The time-dependent numerical results are shown as the solid curves, whereas the open symbols are from newtonian theory, to be discussed in Chapter 14. Note that the pressure is a maximum at the stagnation point and decreases as a function of distance away from the stagnation point—a variation that we most certainly would expect based on our previous aerodynamic experience. The steady shock shapes and sonic lines are shown in Figure 13.15 for the cases of $M_\infty = 4$ and 8. Note that as the Mach number increases, the shock wave moves closer to the body.

13.6 FLOW OVER CONES

Return to Section 9.3 and review the comparison between the supersonic flow over a wedge and that over a right-circular cone at zero angle of attack. In particular, examine again Figure 9.16, which illustrates the Mach 2 flow over a 20° half-angle wedge and over a 20° half-angle cone. Recall that the flow over the cone experiences a three-dimensional relieving effect that results in a weaker shock wave, a lower surface pressure, and curved streamlines behind the shock in comparison to the purely two-dimensional flow over the wedge. This flow over the cone is a special degenerate case of three-dimensional flow, and is governed by nonlinear differential equations that must be solved numerically. For this reason, and also because of the fundamental importance of conical flow in supersonic aerodynamics, we conclude this chapter on numerical techniques with a detailed discussion of such flow. Moreover, the results of this section will carry over to the next chapter when we discuss the design of hypersonic waveriders.

Consider a body of revolution (a body generated by rotating a given planar curve about a fixed axis) at zero angle of attack as shown in Figure 13.16. A cylindrical coordinate system (r, Φ, z) is drawn, with the z axis as the axis of symmetry aligned in the direction of V_∞. By inspection of Figure 13.16, the flow field must be symmetric about the z axis; i.e., all properties are independent of Φ:

$$\frac{\partial}{\partial \phi} \equiv 0$$

The flowfield depends only on r and z. As first introduced at the end of Section 6.3, such a flow is defined as axisymmetric flow. It is a flow that takes place in three-dimensional space; however, because there are only two independent variables, r and z, axisymmetric flow is sometimes called "quasi-two-dimensional" flow.

In this section, we will further specialize to the case of a sharp right-circular cone in a supersonic flow, as sketched in Figure 13.17. This case is important for three reasons.

1. The nonlinear equations of motion lend themselves to a straightforward exact, albeit numerical, solution for this case.
2. The supersonic flow over a cone is of great practical importance in applied aerodynamics; the nose cones of many high-speed missiles and projectiles are approximately conical, as are the nose regions of the fuselages of most supersonic airplanes.

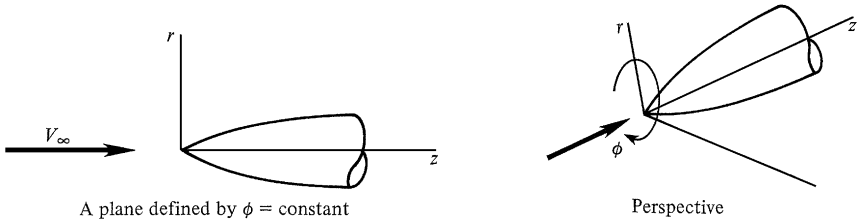


Figure 13.16 Cylindrical coordinate system for an axisymmetric body.

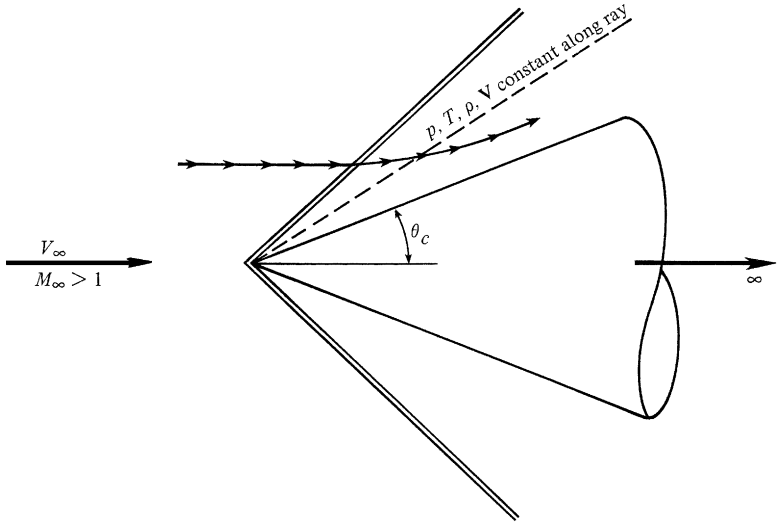


Figure 13.17 Supersonic flow over a cone.

3. The first solution for the supersonic flow over a cone was obtained by A. Busemann in 1929, long before supersonic flow became fashionable (see Reference 97). This solution was essentially graphical, and illustrated some of the important physical phenomena. A few years later, in 1933, G. I. Taylor and J. W. Macoll (see Reference 98) presented a numerical solution that is a hallmark in the evolution of compressible flow. Therefore, the study of conical flow is of historical significance. Moreover, the Taylor-Maccoll solution for the supersonic flow over a cone is a classic case in inviscid supersonic aerodynamics.

13.6.1 Physical Aspects of Conical Flow

Consider a sharp cone of semivertex angle θ_c, sketched in Figure 13.17. Assume this cone extends to infinity in the downstream direction (a semi-infinite cone). The cone is in a supersonic flow, and hence an oblique shock wave is attached at the vertex. The shape of this shock wave is also conical. A streamline from the supersonic free stream discontinuously defects as it traverses the shock, and then curves continuously downstream of the shock, becoming parallel to the cone surface asymptotically at infinity. Contrast this flow with that over a two-dimensional wedge (Figure 9.16*a*) where all streamlines behind the shock are immediately parallel to the wedge surface.

Because the cone extends to infinity, distance along the cone becomes meaningless: If the pressure were different at the 1- and 10-m stations along the surface of the cone, then what would it become at infinity? This presents a dilemma that can be reconciled only by assuming that the pressure is constant along the surface of the cone, as well as that all other flow properties are also constant. Since the

cone surface is simply a ray from the vertex, consider other such rays between the cone surface and the shock wave, as illustrated by the dashed line in Figure 13.17. It only makes sense to assume that the flow properties are constant along these rays as well. Indeed, the definition of conical flow is where all flow properties are constant along rays from a given vertex. The properties vary from one ray to the next. This aspect of conical flow has been experimentally proven. Theoretically, it results from the lack of a meaningful scale length for a semi-infinite cone.

13.6.2 Quantitative Formulation

Consider the superimposed cartesian and spherical coordinate systems sketched in Figure 13.18*a*. The z axis is the axis of symmetry for the right-circular cone, and V_∞ is oriented in the z direction. The flow is axisymmetric; properties are independent of Φ. Therefore, the picture can be reoriented as shown in Figure 13.18*b*, where r and θ are the two independent variables and V_∞ is now horizontal. At any point e in the flow field, the radial and normal components of velocity are V_r and V_θ, respectively. Our objective is to solve for the flow field between the body and the shock wave. Recall that for axisymmetric conical flow

$$\frac{\partial}{\partial\theta} \equiv 0 \qquad \text{(axisymmetric flow)}$$

$$\frac{\partial}{\partial r} \equiv 0 \qquad \text{(flow properties are constant along a ray from the vertex)}$$

The continuity equation for steady flow is Equation (2.54):

$$\nabla \cdot (\rho \mathbf{V}) = 0$$

The divergence of a vector in spherical coordinates is given by Equation 2.21. Thus, in terms of spherical coordinates, Equation (2.54) becomes

$$\nabla \cdot (\rho \mathbf{V}) = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho V_r) + \frac{1}{r \sin\theta}\frac{\partial}{\partial\theta}(\rho V_\theta \sin\theta) + \frac{1}{r \sin\theta}\frac{\partial(\rho V_\phi)}{\partial\phi} = 0 \quad (13.65)$$

Evaluating the derivatives, and applying the above conditions for axisymmetric conical flow, we see that Equation (13.65) becomes

$$\frac{1}{r^2}\left[r^2\frac{\partial(\rho V_r)}{\partial r} + \rho V_r(2r)\right] + \frac{1}{r\sin\theta}\left[\rho V_\theta \cos\theta + \sin\theta\frac{\partial(\rho V_\theta)}{\partial\theta}\right] + \frac{1}{r\sin\theta}\frac{\partial(\rho V_\theta)}{\partial\phi} = 0$$

$$\frac{2\rho V_r}{r} + \frac{\rho V_\theta}{r}\cot\theta + \frac{1}{r}\left(\rho\frac{\partial V_\theta}{\partial\theta} + V_\theta\frac{\partial\rho}{\partial\theta}\right) = 0$$

$$2\rho V_r + \rho V_\theta \cot\theta + \rho\frac{\partial V_\theta}{\partial\theta} + V_\theta\frac{\partial\rho}{\partial\theta} = 0 \quad (13.66)$$

Equation (13.66) is the continuity equation for axisymmetric conical flow.

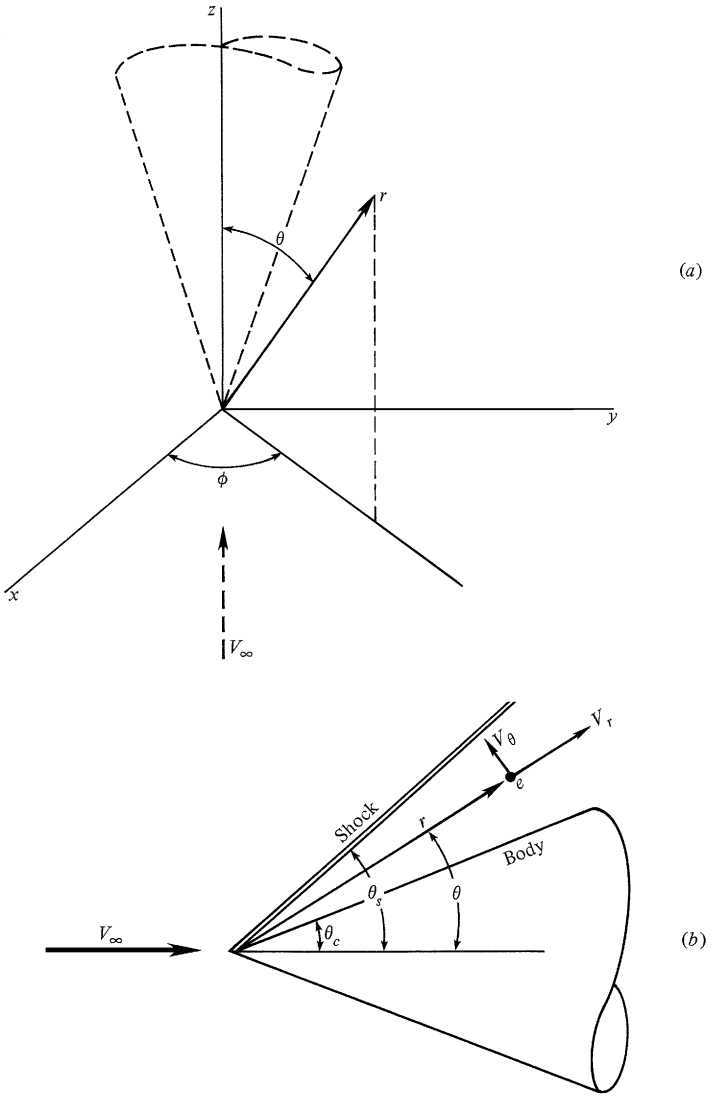


Figure 13.18 Spherical coordinate system for a cone.

Return to the conical flow field sketched in Figures 13.17 and 13.18. The shock wave is straight, and hence the increase in entropy across the shock is the same for all streamlines. Consequently, throughout the conical flow field, $\nabla s = 0$. Moreover, the flow is adiabatic and steady, and hence Equation 7.55 dictates that $\Delta h_o = 0$. In an inviscid compressible flow, entropy gradients ∇s, gradients in total enthalpy Δh_o, and vorticity $\nabla \times \mathbf{V}$, are related through *Crocco's theorem*, derived in Reference 21, and given here without proof:

$$T\nabla s = \nabla h_o - \mathbf{V} \times (\nabla \times \mathbf{V}) \tag{13.67}$$

For the conical flow considered here, since $\nabla s = \nabla h_o = 0$, Equation (13.67) gives

$$\nabla \times \mathbf{V} = 0$$

Hence, the conical flow field is *irrotational*. Crocco's theorem is derived on the basis of both the momentum and energy equations. Therefore, the relation $\nabla \times \mathbf{V} = 0$ can be used in place of either one. In spherical coordinates,

$$\nabla \times \mathbf{V} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & (r\sin\theta)\mathbf{e}_\phi \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ V_r & rV_\theta & (r\sin\theta)V_\phi \end{vmatrix} = 0 \tag{13.68}$$

where $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ are unit vectors in the r, θ, and ϕ directions, respectively. Expanded, Equation (13.68) becomes

$$\begin{aligned} \nabla \times \mathbf{V} = \frac{1}{r^2 \sin\theta} \Bigg\{ & \mathbf{e}_r \left[\frac{\partial}{\partial \theta}(rV_\phi \sin\theta) - \frac{\partial}{\partial \phi}(rV_\theta) \right] \\ & - r\mathbf{e}_\theta \left[\frac{\partial}{\partial r}(rV_\phi \sin\theta) - \frac{\partial}{\partial \phi}(V_r) \right] \\ & + (r\sin\theta)\mathbf{e}_\phi \left[\frac{\partial}{\partial r}(rV_\theta - \frac{\partial V_r}{\partial \theta} \right] \Bigg\} = 0 \end{aligned} \tag{13.69}$$

Applying the axisymmetric conical flow conditions, Equation (13.69) dramatically simplifies to

$$V_\theta \equiv \frac{\partial V_r}{\partial \theta} \tag{13.70}$$

Equation (13.70) is the irrotationality condition for axisymmetric conical flow.

Since the flow is irrotational, we can apply Euler's equation in any direction in the form of Eq. (3.12):

$$dp = -\rho V dV$$

where

$$V^2 = V_r^2 + V_\theta^2$$

Hence, Eq. (3.12) becomes

$$dp = -\rho(V_r\, dV_r + V_\theta\, dV_\theta) \tag{13.71}$$

Recall that, for isentropic flow,

$$\frac{dp}{d\rho} \equiv \left(\frac{\partial p}{\partial \rho} \right)_s = a^2$$

Thus, Eq. (13.71) becomes

$$\frac{d\rho}{\rho} = -\frac{1}{a^2}(V_r\, dV_r + V_\theta\, dV_\theta) \tag{13.72}$$

From Eq. (7.55), and defining a new reference velocity $V_{\max}$ as the maximum theoretical velocity obtainable from a fixed reservoir condition (when $V = V_{\max}$, the flow has expanded theoretically to zero temperature, hence $h = 0$), we have

$$h_o = \text{const} = h + \frac{V_2}{2} = \frac{V_{\max}^2}{2}$$

Note that $V_{\max}$ is a constant for the flow and is equal to $\sqrt{2h_o}$. For a calorically perfect gas, the above becomes

$$\frac{a^2}{\gamma - 1} + \frac{V^2}{2} = \frac{V_{\max}^2}{2}$$

or

$$a^2 = \frac{\gamma - 1}{2}\left(V_{\max}^2 - V^2\right) = \frac{\gamma - 1}{2}\left(V_{\max}^2 - V_r^2 - V_\theta^2\right) \tag{13.73}$$

Substitute Eq. (13.73) into (13.72):

$$\frac{d\rho}{\rho} = -\frac{2}{\gamma - 1}\left(\frac{V_r\, dV_r + V_\theta\, dV_\theta}{V_{\max}^2 - V_r^2 - V_\theta^2}\right) \tag{13.74}$$

Equation (13.74) is essentially Euler's equation in a form useful for studying conical flow.

Equations (13.66), (13.70), and (13.74) are three equations with three dependent variables: ρ, V_r, and V_θ. Due to the axisymmetric conical flow conditions, there is only one independent variable, namely θ. Hence, the partial derivatives in Eqs. (13.66) and (13.70) are more properly written as ordinary derivatives. From Eq. (13.66),

$$2V_r + V_\theta \cot\theta + \frac{dV_\theta}{d\theta} + \frac{V_\theta}{\rho}\frac{d\rho}{d\theta} = 0 \tag{13.75}$$

From Equation (13.74),

$$\frac{d\rho}{d\theta} = -\frac{2\rho}{\gamma - 1}\left(\frac{V_r\dfrac{dV_r}{d\theta} + V_\theta\dfrac{dV_\theta}{d\theta}}{V_{\max}^2 - V_r^2 - V_\theta^2}\right) \tag{13.76}$$

Substitute Equation (13.76) into Equation (13.75):

$$2V_r + V_\theta \cot\theta + \frac{dV_\theta}{d\theta} - \frac{2V_\theta}{\gamma - 1}\left(\frac{V_r\dfrac{dV_r}{d\theta} + V_\theta\dfrac{dV_\theta}{d\theta}}{V_{\max}^2 - V_r^2 - V_\theta^2}\right) = 0$$

or

$$\frac{\gamma-1}{2}\left(V_{\max}^2 - V_r^2 - V_\theta^2\right)\left(2V_r + V_\theta \cot\theta + \frac{dV_\theta}{d\theta}\right) - V_\theta\left(V_r\frac{dV_r}{d\theta} + V_\theta\frac{dV_\theta}{d\theta}\right) = 0 \tag{13.77}$$

Recall from Equation (13.70)

$$V_\theta = \frac{dV_r}{d\theta}$$

Hence,

$$\frac{dV_\theta}{d\theta} = \frac{d^2V_r}{d\theta^2}$$

Substituting this result into Equation (13.77), we have

$$\boxed{\begin{aligned}&\frac{\gamma-1}{2}\left[V_{\max}^2 - V_r^2 - \left(\frac{dV_r}{d\theta}\right)^2\right]\left[2V_r + \frac{dV_r}{d\theta}\cot\theta + \frac{d^2V_r}{d\theta^2}\right]\\ &\quad - \frac{dV_r}{d\theta}\left[V_r\frac{dV_r}{d\theta} + \frac{dV_r}{d\theta}\left(\frac{d^2V_r}{d\theta^2}\right)\right] = 0\end{aligned}} \tag{13.78}$$

Equation (13.78) is the Taylor-Maccoll equation for the solution of conical flows. Note that it is an ordinary differential equation, with only one dependent variable, V_r. Its solution gives $V_r = f(\theta)$; V_θ follows from Eq. (13.70), namely,

$$V_\theta = \frac{dV_r}{d\theta} \tag{13.79}$$

There is no closed-form solution to Equation (13.78); it must be solved *numerically*. To expedite the numerical solution, define the nondimensional velocity V' as

$$V' \equiv \frac{V}{V_{\max}}$$

Then, Eq. (13.78) becomes

$$\begin{aligned}&\frac{\gamma-1}{2}\left[1 - V_r'^2 - \left(\frac{dV_r'}{d\theta}\right)^2\right]\left[2V_r' + \frac{dV_r'}{d\theta}\cot\theta + \frac{d^2V_r'}{d\theta^2}\right]\\ &\quad - \frac{dV_r'}{d\theta}\left[V_r'\frac{dV_r'}{d\theta} + \frac{dV_r'}{d\theta}\left(\frac{d^2V_r'}{d\theta^2}\right)\right] = 0\end{aligned} \tag{13.80}$$

The nondimensional velocity V' is a function of Mach number only. To see this more clearly recall that

$$h + \frac{V^2}{2} = \frac{V_{\max}^2}{2}$$

$$\frac{a^2}{\gamma - 1} + \frac{V^2}{2} = \frac{V_{\max}^2}{2}$$

$$\frac{1}{\gamma - 1} + \left(\frac{a}{V}\right)^2 + \frac{1}{2} = \frac{1}{2}\left(\frac{V_{\max}}{V}\right)^2$$

$$\frac{2}{\gamma - 1} + \left(\frac{1}{M}\right)^2 + 1 = \left(\frac{V_{\max}}{V}\right)^2$$

$$\frac{V}{V_{\max}} \equiv V' = \left[\frac{2}{(\gamma - 1)M^2} + 1\right]^{-1/2} \tag{13.81}$$

Clearly, from Equation (13.81), $V' = f(M)$; given M, we can always find V', or vice versa.

13.6.3 Numerical Procedure

For the numerical solution of the supersonic flow over a right-circular cone, we will employ an inverse approach. By this, we mean that a given shock wave will be assumed, and the particular cone that supports the given shock will be calculated. This is in contrast to the direct approach, where the cone is given and the flow field and shock wave are calculated. The numerical procedure is as follows:

1. Assume a shock wave angle θ_s and a free stream Mach number M_∞, as sketched in Figure 13.19. From this, the Mach number and flow deflection angle, M_2 and δ, respectively, immediately behind the shock can be found from the oblique shock relations. Note that, contrary to our previous practice, the flow deflection angle is here denoted by δ so as not to confuse it with the polar coordinate θ.
2. From M_2 and δ, the radial and normal components of flow velocity, V_r' and V_θ', respectively, directly behind the shock can be found from the geometry of Figure 13.19. Note that V' is obtained by inserting M_2 into Equation (10.16).
3. Using the above value of V_r' directly behind the shock as a boundary value, solve Equation (13.81) for V_r' numerically in steps of θ, marching away from the shock. Here, the flow field is divided into incremental angles $\Delta\theta$, as sketched in Figure 13.19. The ordinary differential equation (13.80) can be solved at each $\Delta\theta$ using any standard numerical solution technique, such as the Runge-Kutta method.
4. At each increment in θ, the value of V_θ' is calculated from Equation (13.79). At some value of θ, namely $\theta = \theta_c$, we will find $V_\theta' = 0$. The normal component of velocity at an impermeable surface is zero. Hence, when

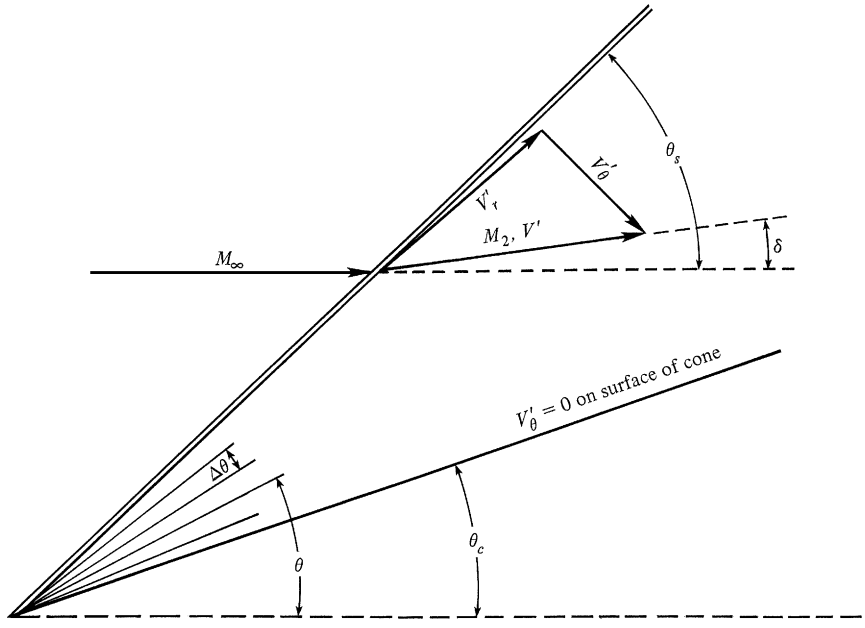


Figure 13.19 Geometry for the numerical solution of flow over a cone.

$V'_\theta = 0$ at $\theta = \theta_c$ then θ_c must represent the surface of the particular cone that supports the shock wave of given wave angle θ_s at the given Mach number M_∞ as assumed in step 1. That is, the cone angle compatible with M_∞ and θ_s is θ_c. The value of V'_r at θ_c gives the Mach number along the cone surface via Equation (13.81).

5. In the process of steps 1 through 4 here, the complete velocity flow field between the shock and the body has been obtained. Note that, at each point (or ray), $V' = \sqrt{(V'_r)^2 + (V'_\theta)^2}$ and M follow from Equation (13.81). The pressure, density, and temperature along each ray can then be obtained from the isentropic relations, Equations (8.42), (8.43), and (8.40).

If a different value of M_∞ and/or θ_s is assumed in step 1, a different flow field and cone angle θ_c will be obtained from steps 1 through 5. By a repeated series of these calculations, tables or graphs of supersonic cone properties can be generated. Such tables exist in the literature, the most common being those of Kopal (Reference 99) and Sims (Reference 100).

13.6.4 Physical Aspects of Supersonic Flow Over Cones

Some typical numerical results obtained from the solution in Section 13.6.3 are illustrated in Figure 13.20, which gives the shock wave angle θ_s as a function of cone angle θ_c with M_∞ as a parameter. Figure 13.20 for cones is analogous to Figure 9.9 for two-dimensional wedges; the two figures are qualitatively similar, but the numbers are different.

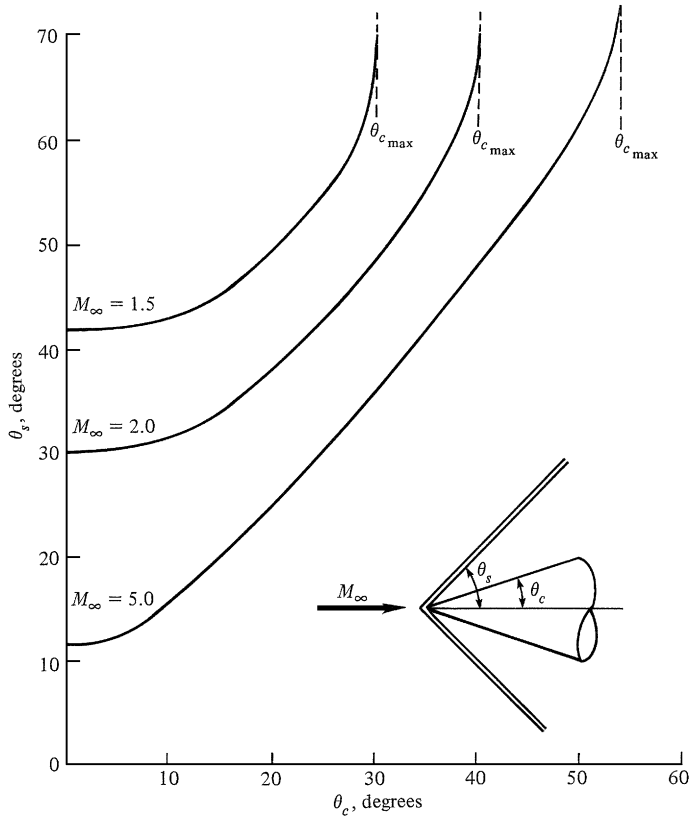


Figure 13.20 θ_c–θ_s–M disgram for cones in supersonic flow. (The top portion of the curves curls back for the strong-shock solution, which is not shown here.)

Examine Figure 13.20 closely. Note that, for a given cone angle θ_c and given M_∞, there are two possible oblique shock waves—the strong- and weak-shock solutions. This is directly analogous to the two-dimensional case discussed in Chapter 9. The weak solution is almost always observed in practice on real finite cones; however, it is possible to force the strong-shock solution by independently increasing the back pressure near the base of the cone.

Also note from Figure 13.20 that, for a given M_∞, there is a maximum cone angle $\theta_{c_{max}}$, beyond which the shock becomes detached. This is illustrated in Figure 13.21. When $\theta_c > \theta_{c_{max}}$, there exists no Taylor-Maccoll solution as given here; instead, the flow field with a detached shock must be solved by techniques such as those discussed in Section 13.5.

In comparison to the two-dimensional flow over a wedge, the three-dimensional flow over a cone has an extra dimension in which to expand. This "three-dimensional relieving effect" was discussed in Section 9.3, which should now be reviewed by the reader. In particular, recall from Figure 9.16 that the shock wave on a cone of given angle is weaker than the shock wave on a wedge of the

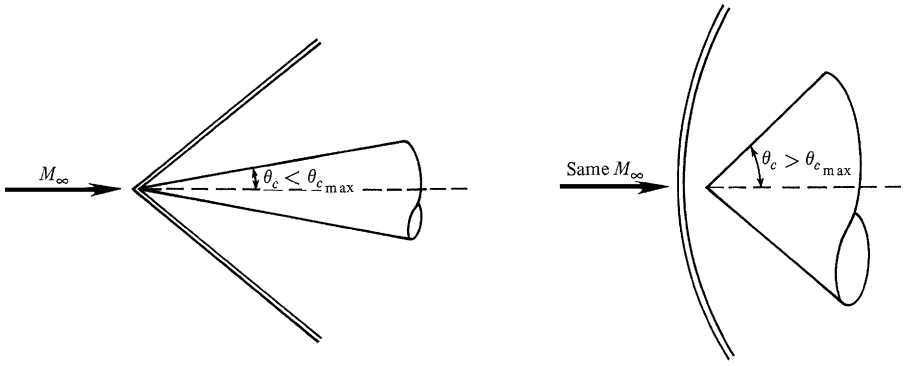


Figure 13.21 Attached and detached shock waves on cones.

same angle. It therefore follows that the cone experiences a lower surface pressure, temperature, density, and entropy than the wedge. It also follows that, for a given M_∞, the maximum allowable cone angle for an attached shock solution is greater than the maximum wedge angle. This is clearly demonstrated in Figure 13.22.

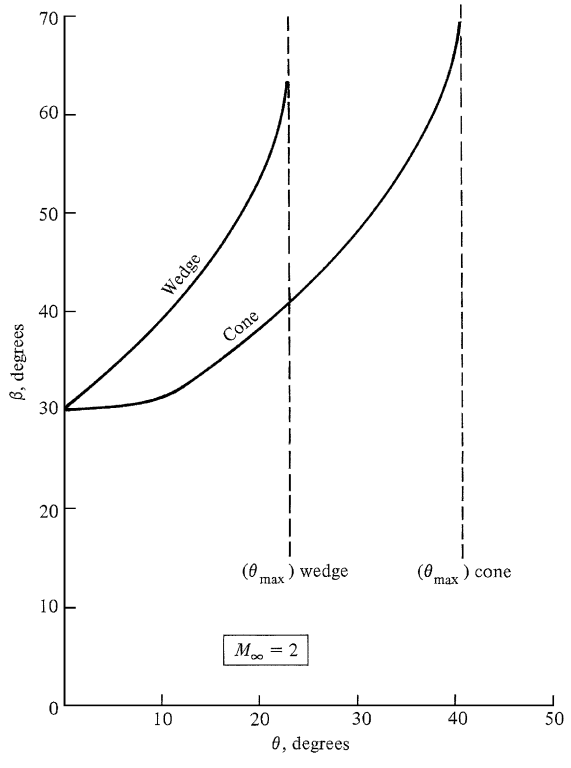


Figure 13.22 Comaparison of shock wave angles for wedges and cones at Mach 2.

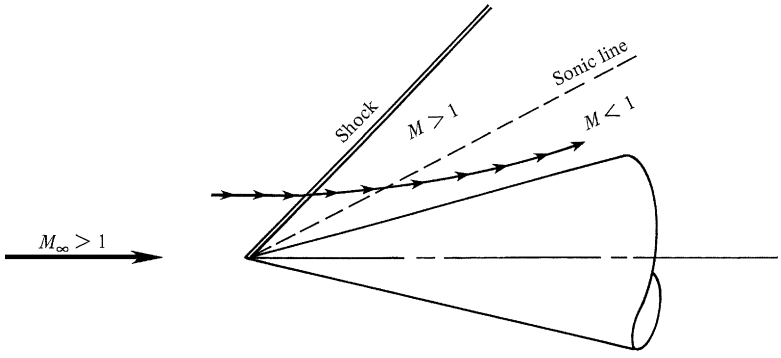


Figure 13.23 Some supersonic conical flow fields are characterized by an isentropic compression to subsonic velocities near the cone surface.

Finally, the numerical results show that any given streamline between the shock wave and cone surface is curved, as sketched in Figure 13.23, and asymptotically becomes parallel to the cone surface at infinity. Also, for most cases, the complete flow field between the shock and the cone is supersonic. However, if the cone angle is large enough, but still less than $\theta_{c_{\max}}$, there are some cases where the flow becomes subsonic near the surface. This case is illustrated in Figure 13.23, where one of the rays in the flow field becomes a sonic line. In this case, we see one of the few instances in nature where a supersonic flow field is actually isentropically compressed from supersonic to subsonic velocities. A transition from supersonic to subsonic flow is almost invariably accompanied by shock waves, as discussed in Chapter 8. However, flow over a cone can be an exception to this observation.

13.7 SUMMARY

We have now completed both branches of our road map shown in Figure 13.1. Make certain that you feel comfortable with all the material represented by this road map. A short summary of the highlights is given below:

For a steady, two-dimensional, irrotational, supersonic flow, the characteristic lines are Mach lines, and the compatibility equations which hold along these characteristic lines are

$$\theta + \nu = K_- \text{ (along a } C_- \text{ characteristic)}$$

and

$$\theta - \nu = K_+ \text{ (along a } C_+ \text{ characteristic)}$$

The numerical solution of such a flow can be carried out by solving the compatibility equation along the characteristic lines in a step-by-step fashion, starting from an appropriate initial data line.

The contour of a supersonic nozzle can be obtained by applying the method of characteristics downstream of the limiting characteristic (which is usually downstream of the geometric throat).

The essence of finite-difference methods is to replace the partial derivatives in the governing flow equations with finite-difference quotients. For supersonic steady flows, this allows us to march downstream, starting from known data along an initial data line in the supersonic flow. For the solution of mixed subsonic-supersonic flows, a time-dependent technique can be used which allows us to march forward in time, starting with assumed initial conditions at time $t = 0$ and achieving a steady-state result in the limit of large times.

A popular technique for carrying out finite-difference solutions, whether for supersonic steady flow or for a time-dependent solution of mixed subsonic and supersonic flow, is the straightforward predictor-corrector technique by MacCormack.

13.8 PROBLEM

Note: The purpose of the following problem is to provide an exercise in carrying out a unit process for the method of characteristics. A more extensive application to a complete flow field is left to your specific desires. Also, an extensive practical problem utilizing the finite-difference method requires a large number of arithmetic operations and is practical only on a digital computer. You are encouraged to set up such a problem at your leisure. The main purpose of the present chapter is to present the essence of several numerical methods, not to burden the reader with a lot of calculations or the requirement to write an extensive computer program.

13.1 Consider two points in a supersonic flow. These points are located in a cartesian coordinate system at $(x_1, y_1) = (0, 0.0684)$ and $(x_2, y_2) = (0.0121, 0)$, where the units are meters. At point (x_1, y_1): $u_1 = 639$ m/s, $v_1 = 232.6$ m/s, $p_1 = 1$ atm, $T_1 = 288$ K. At point (x_2, y_2): $u_2 = 680$ m/s, $v_2 = 0$, $p_2 = 1$ atm, $T_2 = 288$ K. Consider point 3 downstream of points 1 and 2 located by the intersection of the C_+ characteristic through point 2 and the C_- characteristic through point 1. At point 3, calculate: u_3, v_3, p_3, and T_3. Also, calculate the location of point 3, assuming the characteristics between these points are straight lines.

CHAPTER 14

Elements of Hypersonic Flow

Almost everyone has their own definition of the term hypersonic. If we were to conduct something like a public opinion poll among those present, and asked everyone to name a Mach number above which the flow of a gas should properly be described as hypersonic there would be a majority of answers round about 5 or 6, but it would be quite possible for someone to advocate, and defend, numbers as small as 3, or as high as 12.

P. L. Roe,
comment made in a lecture
at the von Karman Institute, Belgium
January 1970

PREVIEW BOX

Airbreathing hypersonic flight is held by many (including this author) to be the last frontier of air-vehicle design. Some progress has been made, but much needs to be done. The practical design of hypersonic vehicles for sustained hypersonic flight in the atmosphere will be a major challenge to the next generation of aerospace engineers.

Aeronautical history was made in March 2004 when the X-43 Hyper-X test vehicle, shown in Figure 14.1, achieved sustained flight for 11 s at Mach 6.9 powered by a supersonic combustion ramjet engine (SCRAMjet). In November, another X-43 achieved sustained flight at nearly Mach 10, making it the fastest airplane in history to date. At these speeds, aerodynamic heating becomes a major problem, and the vehicle must be fabricated from special high-temperature materials. The X-43 thermal protection design is shown in Figure 14.2. The two successful flights of the X-43 in 2004 made aerospace engineering history; for the first time a SCRAMjet engine operated successfully for a sustained period in atmospheric flight. After more than 40 years of research and technical development, enough progress had been made to allow a successful test of the design methodology. But so much more needs to be done to achieve truly practical devices and flight vehicles. That is where some of you come into the picture; perhaps you will be the ones to tackle this last frontier.



Figure 14.1 X-43 hypersonic test vehicle (NASA).

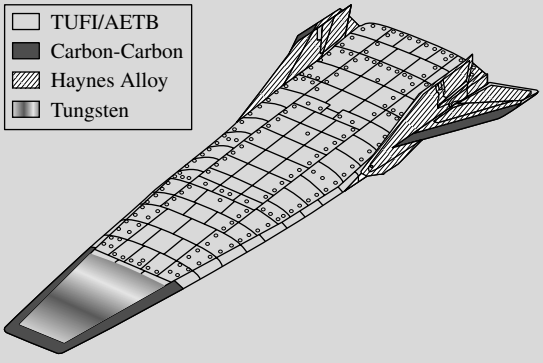


Figure 14.2 X-43 thermal protection system.

Hypersonic flight of other human-made vehicles has been a reality since 1949, when a WAC Corporal rocket, boosted to high altitudes on top of a captured German V-2 rocket, powered itself back into the atmosphere at over 5000 mph at White Sands Proving Ground in New Mexico (see Reference 55). Since then, a whole host of space vehicles, such as the space shuttle and the Apollo return module have returned to Earth after entering and flying through the atmosphere at large hypersonic Mach numbers, from Mach 26 to Mach 36. So hypersonic aerodynamics has been around for a relatively long time, but for all practical purposes it is still a young, developing discipline.

Hypersonic aerodynamics is the subject of this chapter. What is it about hypersonic flows that make them any different from supersonic flows? Why is it that they justify a separate chapter in this book? After all, hypersonic flow is flow with velocities greater than the speed of sound, which is the definition of supersonic flow, except that hypersonic flows are moving at velocities generally a lot larger than the speed of sound. Indeed, an old rule of thumb defines hypersonic flow as flow at Mach 5 or greater. However, there is nothing magic about Mach 5. If you were flying at Mach 4.99, and you accelerate to Mach 5.01, nothing new is going to happen—the flow will not change from green to red, and there will be no clap of thunder. (In contrast, if you were flying at Mach 0.99 and accelerated to Mach 1.01, the flow would "change from green to red," (i.e., the physics of the flow would change drastically as we have already seen, and there would be a "clap of thunder," the sudden occurrence of shock waves.) So why is a distinction made between hypersonic and supersonic flows? The answer is that certain physical phenomena that are not so important at supersonic speeds become dominant at hypersonic speeds. These phenomena are described in Section 14.2, which basically constitutes a four-page definition of hypersonic flow. Read on, and find out for yourself what this is all about.

Hypersonic aerodynamics is state-of-the-art aerodynamics. It is exciting, and it is fun. This chapter introduces you to some of the special aspects and analyses of hypersonic flow. It is just a beginning for you, but I predict that it will be an enjoyable beginning.

14.1 INTRODUCTION

The history of aviation has always been driven by the philosophy of "faster and higher," starting with the Wright brothers' sea level flights at 35 mi/h in 1903, and progressing exponentially to the manned space flight missions of the 1960s and 1970s. The current altitude and speed records for manned flight are the moon and 36,000 ft/s—more than 36 times the speed of sound—set by the Apollo lunar capsule in 1969. Although most of the flight of the Apollo took place in space, outside the earth's atmosphere, one of its most critical aspects was reentry

into the atmosphere after completion of the lunar mission. The aerodynamic phenomena associated with very high-speed flight, such as encountered during atmospheric reentry, are classified as *hypersonic aerodynamics*—the subject of this chapter. In addition to reentry vehicles, both manned and unmanned, there are other hypersonic applications on the horizon, such as ramjet-powered hypersonic missiles now under consideration by the military and the concept of a hypersonic transport, the basic technology of which is now being studied by NASA. Therefore, although hypersonic aerodynamics is at one extreme end of the whole flight spectrum (see Section 1.10), it is important enough to justify one small chapter in our presentation of the fundamentals of aerodynamics.

This chapter is short; its purpose is simply to introduce some basic considerations of hypersonic flow. Therefore, we have no need for a chapter road map or a summary at the end. Also, before progressing further, return to Chapter 1 and review the short discussion on hypersonic flow given in Section 1.10. For an in-depth study of hypersonic flow, see the author's book listed as Reference 55.

14.2 QUALITATIVE ASPECTS OF HYPERSONIC FLOW

Consider a 15° half-angle wedge flying at $M_\infty = 36$. From Figure 9.9, we see that the wave angle of the oblique shock is only 18°; that is, the oblique shock wave is very close to the surface of the body. This situation is sketched in Figure 14.3. Clearly, the shock layer between the shock wave and the body is very thin. Such thin shock layers are one characteristic of hypersonic flow. A practical consequence of a thin shock layer is that a major interaction frequently occurs between the inviscid flow behind the shock and the viscous boundary layer on the surface. Indeed, hypersonic vehicles generally fly at high altitudes where the density, hence Reynolds number, is low, and therefore the boundary layers are thick. Moreover, at hypersonic speeds, the boundary-layer thickness on slender bodies is approximately proportional to M_∞^2; hence, the high Mach numbers further contribute to

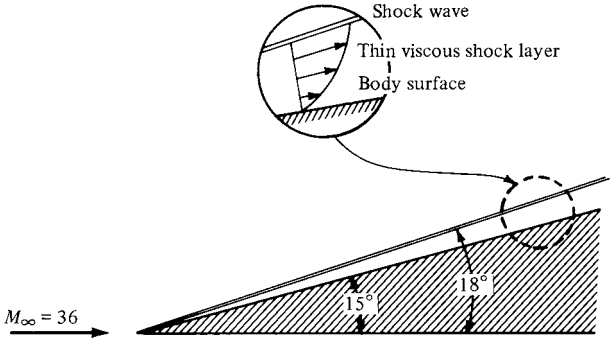
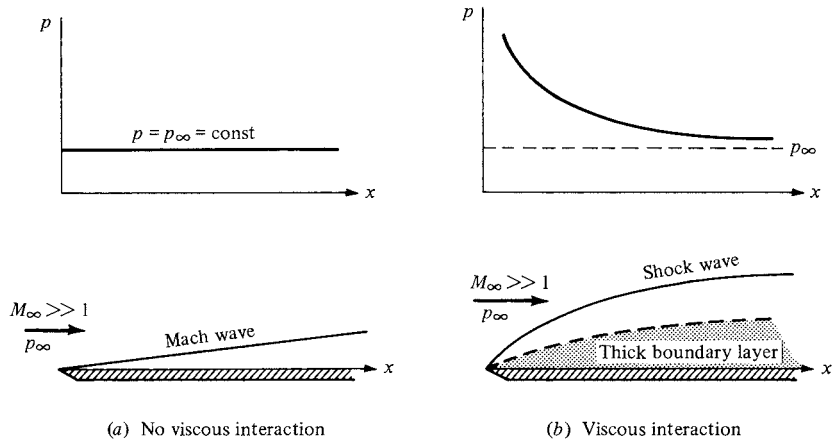


Figure 14.3 For hypersonic flow, the shock layers are thin and viscous.



(*a*) No viscous interaction (*b*) Viscous interaction

Figure 14.4 The viscous interaction on a flat plate at hypersonic speeds.

a thickening of the boundary layer. In many cases, the boundary-layer thickness is of the same magnitude as the shock-layer thickness, such as sketched in the insert at the top of Figure 14.3. Here, the shock layer is fully viscous, and the shock-wave shape and surface pressure distribution are affected by such viscous effects. These phenomena are called *viscous interaction phenomena*—where the viscous flow greatly affects the external inviscid flow, and, of course, the external inviscid flow affects the boundary layer. A graphic example of such viscous interaction occurs on a flat plate at hypersonic speeds, as sketched in Figure 14.4. If the flow were completely inviscid, then we would have the case shown in Figure 14.4*a*, where a Mach wave trails downstream from the leading edge. Since there is no deflection of the flow, the pressure distribution over the surface of the plate is constant and equal to p_∞. In contrast, in real life there is a boundary layer over the flat plate, and at hypersonic conditions this boundary layer can be thick, as sketched in Figure 14.4*b*. The thick boundary layer deflects the external, inviscid flow, creating a comparably strong, curved shock wave which trails downstream from the leading edge. In turn, the surface pressure from the leading edge is considerably higher than p_∞, and only approaches p_∞ far downstream of the leading edge, as shown in Figure 14.4*b*. In addition to influencing the aerodynamic force, such high pressures increase the aerodynamic heating at the leading edge. Therefore, hypersonic viscous interaction can be important, and this has been one of the major areas of modern hypersonic aerodynamic research.

There is a second and frequently more dominant aspect of hypersonic flow, namely, high temperatures in the shock layer, along with large aerodynamic heating of the vehicle. For example, consider a blunt body reentering the atmosphere at Mach 36, as sketched in Figure 14.5. Let us calculate the temperature in the shock layer immediately behind the normal portion of the bow shock wave. From Appendix B, we find that the static temperature ratio across a normal shock wave

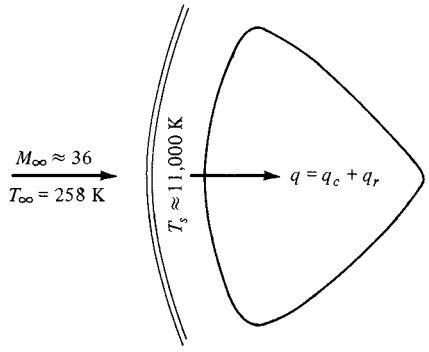


Figure 14.5 High-temperature shock layer.

with $M_\infty = 36$ is 252.9; this is denoted by T_s/T_∞ in Figure 14.5. Moreover, at a standard altitude of 59 km, $T_\infty = 258$ K. Hence, we obtain $T_s = 65{,}248$ K—an incredibly high temperature, which is more than six times hotter than the surface of the sun! This is, in reality, an incorrect value, because we have used Appendix B which is good only for a calorically perfect gas with $\gamma = 1.4$. However, at high temperatures, the gas will become chemically reacting; γ will no longer equal 1.4 and will no longer be constant. Nevertheless, we get the impression from this calculation that the temperature in the shock layer will be very high, albeit something less than 65,248 K. Indeed, if a proper calculation of T_s is made taking into account the chemically reacting gas, we would find that $T_s \approx 11{,}000$ K—still a very high value. Clearly, high-temperature effects are very important in hypersonic flow.

Let us examine these high-temperature effects in more detail. If we consider air at $p = 1$ atm and $T = 288$ K (standard sea level), the chemical composition is essentially 20 percent O_2 and 80 percent N_2 by volume. The temperature is too low for any significant chemical reaction to take place. However, if we were to increase T to 2000 K, we would observe that the O_2 begins to dissociate; that is,

$$O_2 \rightarrow 2O \qquad 2000 \text{ K} < T < 4000 \text{ K}$$

If the temperature were increased to 4000 K, most of the O_2 would be dissociated, and N_2 dissociation would commence:

$$N_2 \rightarrow 2N \qquad 4000 \text{ K} < T < 9000 \text{ K}$$

If the temperature were increased to 9000 K, most of the N_2 would be dissociated, and ionization would commence:

$$\begin{aligned} N &\rightarrow N^+ + e^- \\ O &\rightarrow O^+ + e^- \end{aligned} \qquad T > 9000 \text{ K}$$

Hence, returning to Figure 14.5, the shock layer in the nose region of the body is a partially ionized plasma, consisting of the atoms N and O, the ions N^+ and O^+, and electrons, e^-. Indeed, the presence of these free electrons in the shock layer

is responsible for the "communications blackout" experienced over portions of the trajectory of a reentry vehicle.

One consequence of these high-temperature effects is that all our equations and tables obtained in Chapters 7 to 13 that depended on a constant $\gamma = 1.4$ are no longer valid. Indeed, the governing equations for the high-temperature, chemically reacting shock layer in Figure 14.5 must be solved numerically, taking into account the proper physics and chemistry of the gas itself. The analysis of aerodynamic flows with such real physical effects is discussed in detail in Chapters 16 and 17 of Reference 21; such matters are beyond the scope of this book.

Associated with the high-temperature shock layers is a large amount of heat transfer to the surface of a hypersonic vehicle. Indeed, for reentry velocities, aerodynamic heating dominates the design of the vehicle, as explained at the end of Section 1.1. (Recall that the third historical example discussed in Section 1.1 was the evolution of the blunt-body concept to reduce aerodynamic heating; review this material before progressing further.) The usual mode of aerodynamic heating is the transfer of energy from the hot shock layer to the surface by means of thermal conduction at the surface; that is, if $\partial T/\partial n$ represents the temperature gradient in the gas normal to the surface, then $q_c = -k(\partial T/\partial n)$ is the heat transfer into the surface. Because $\partial T/\partial n$ is a flow-field property generated by the flow of the gas over the body, q_c is called *convective heating*. For reentry velocities associated with ICBMs (about 28,000 ft/s), this is the only meaningful mode of heat transfer to the body. However, at higher velocities, the shock-layer temperature becomes even hotter. From experience, we know that all bodies emit thermal radiation, and from physics you know that blackbody radiation varies as T^4; hence, radiation becomes a dominant mode of heat transfer at high temperatures. (For example, the heat you feel by standing beside a fire in a fireplace is radiative heating from the flames and the hot walls.) When the shock layer reaches temperatures on the order of 11,000 K, as for the case given in Figure 14.5, thermal radiation from the hot gas becomes a substantial portion of the total heat transfer to the body surface. Denoting radiative heating by q_r, we can express the total aerodynamic heating q as the sum of convective and radiative heating; $q = q_c + q_r$. For Apollo reentry, $q_r/q \approx 0.3$, and hence radiative heating was an important consideration in the design of the Apollo heat shield. For the entry of a space probe into the atmosphere of Jupiter, the velocities will be so high and the shock-layer temperatures so large that the convective heating is negligible, and in this case $q \approx q_r$. For such a vehicle, radiative heating becomes the dominant aspect in its design. Figure 14.6 illustrates the relative importance of q_c and q_r for a typical manned reentry vehicle in the earth's atmosphere; note how rapidly q_r dominates the aerodynamic heating of the body as velocities increase above 36,000 ft/s. The details of shock-layer radiative heating are interesting and important; however, they are beyond the scope of this book. For a thorough survey of the engineering aspects of shock-layer radiative heat transfer, see Reference 36.

In summary, the aspects of thin shock-layer viscous interaction and high-temperature, chemically reacting and radiative effects distinguish hypersonic flow

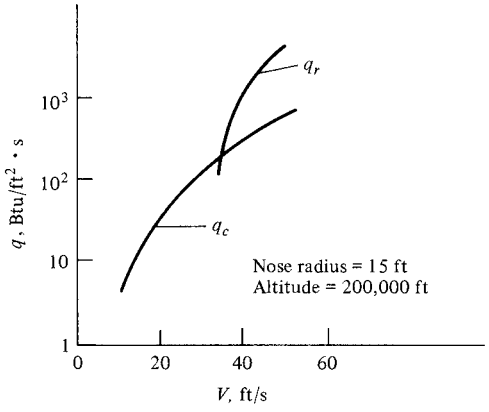


Figure 14.6 Convective and radiative heating rates of a blunt reentry vehicle as a function of flight velocity. (*Source: Anderson, Reference 36.*)

from the more moderate supersonic regime. Hypersonic flow has been the subject of several complete books; see, for example, References 37 to 41. In particular, see Reference 55 for a modern textbook on the subject.

14.3 NEWTONIAN THEORY

Return to Figure 14.3; note how close the shock wave lies to the body surface. This figure is redrawn in Figure 14.7 with the streamlines added to the sketch. When viewed from afar, the straight, horizontal streamlines in the freestream appear to almost impact the body, and then move tangentially along the body. Return to Figure 1.6, which illustrates Isaac Newton's model for fluid flow, and compare it with the hypersonic flow field shown in Figure 14.7; they have certain distinct similarities. (Also, review the discussion surrounding Figure 1.6 before progressing further.) Indeed, the thin shock layers around hypersonic bodies are the closest example in fluid mechanics to Newton's model. Therefore, we might expect that results based on Newton's model would have some applicability in hypersonic flows. This is indeed the case; newtonian theory is used frequently to estimate the pressure distribution over the surface of a hypersonic body. The purpose of this section is to derive the famous newtonian sine-squared law first mentioned in Section 1.1 and to show how it is applied to hypersonic flows.

Consider a surface inclined at the angle θ to the freestream, as sketched in Figure 14.8. According to the newtonian model, the flow consists of a large number of individual particles which impact the surface and then move tangentially to the surface. During collision with the surface, the particles lose their component of momentum normal to the surface, but the tangential component is preserved. The time rate of change of the normal component of momentum equals the force exerted on the surface by the particle impacts. To quantify this model, examine

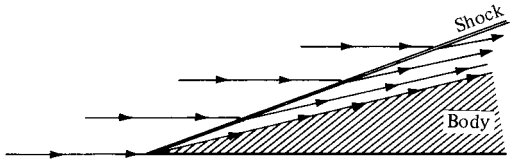


Figure 14.7 Streamlines in a hypersonic flow.

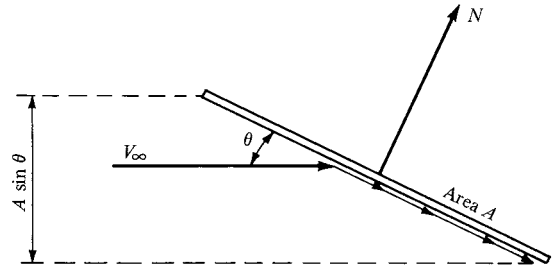


Figure 14.8 Schematic for newtonian impact theory.

Figure 14.8. The component of the freestream velocity normal to the surface is $V_\infty \sin\theta$. If the area of the surface is A, the mass flow incident on the surface is $\rho_\infty (A \sin\theta) V_\infty$. Hence, the time rate of change of momentum is

$$\text{Mass flow} \times \text{change in normal component of velocity}$$

or

$$(\rho_\infty V_\infty A \sin\theta)(V_\infty \sin\theta) = \rho_\infty V_\infty^2 A \sin^2\theta$$

In turn, from Newton's second law, the force on the surface is

$$N = \rho_\infty V_\infty^2 A \sin^2\theta \tag{14.1}$$

This force acts along the same line as the time rate of change of momentum (i.e., normal to the surface), as sketched in Figure 14.8. From Equation (14.1), the normal force per unit area is

$$\frac{N}{A} = \rho_\infty V_\infty^2 \sin^2\theta \tag{14.2}$$

Let us now interpret the physical meaning of the normal force per unit area in Equation (14.2), N/A, in terms of our modern knowledge of aerodynamics. Newton's model assumes a stream of individual particles all moving in straight, parallel paths toward the surface; that is, the particles have a completely directed, rectilinear motion. There is no random motion of the particles—it is simply a stream of particles such as pellets from a shotgun. In terms of our modern concepts, we know that a moving gas has molecular motion that is a composite of random motion of the molecules as well as a directed motion. Moreover, we know that the freestream static pressure p_∞ is simply a measure of the purely *random motion* of the molecules. Therefore, when the purely *directed motion* of the particles in Newton's model results in the normal force per unit area, N/A in Equation (14.2),

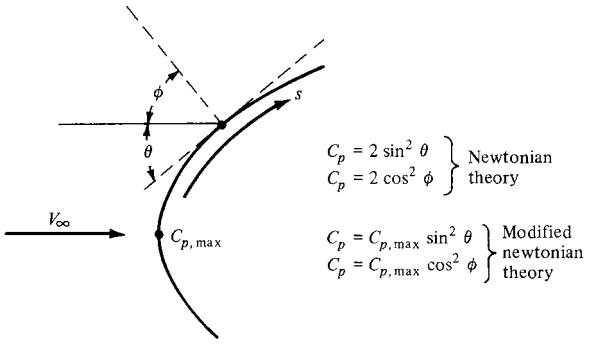


Figure 14.9 Definition of angles for newtonian theory.

this normal force per unit area must be construed as the *pressure difference* above p_∞, namely, $p - p_\infty$ on the surface. Hence, Equation (14.2) becomes

$$p - p_\infty = \rho_\infty V_\infty^2 \sin^2\theta \tag{14.3}$$

Equation (14.3) can be written in terms of the pressure coefficient $C_p = (p-p_\infty)/\frac{1}{2}\rho_\infty V_\infty^2$, as follows

$$\frac{p - p_\infty}{\frac{1}{2}\rho_\infty V_\infty^2} = 2\sin^2\theta$$

or

$$\boxed{C_p = 2\sin^2\theta} \tag{14.4}$$

Equation (14.4) is Newton's sine-squared law; it states that the pressure coefficient is proportional to the sine square of the angle between a tangent to the surface and the direction of the freestream. This angle θ is illustrated in Figure 14.9. Frequently, the results of newtonian theory are expressed in terms of the angle between a normal to the surface and the freestream direction, denoted by ϕ as shown in Figure 14.9. In terms of ϕ, Equation (14.4) becomes

$$C_p = 2\cos^2\phi \tag{14.5}$$

which is an equally valid expression of newtonian theory.

Consider the blunt body sketched in Figure 14.9. Clearly, the maximum pressure, hence the maximum value of C_p, occurs at the stagnation point, where $\theta = \pi/2$ and $\phi = 0$. Equation (14.4) predicts $C_p = 2$ at the stagnation point. Contrast this hypersonic result with the result obtained for incompressible flow theory in Chapter 3, where $C_p = 1$ at a stagnation point. Indeed, the stagnation pressure coefficient increases continuously from 1.0 at $M_\infty = 0$ to 1.28 at $M_\infty =$ 1.0 to 1.86 for $\gamma = 1.4$ as $M_\infty \to \infty$. (Prove this to yourself.)

The result that the maximum pressure coefficient approaches 2 at $M_\infty \to \infty$ can be obtained independently from the one-dimensional momentum equation, namely, Equation (8.6). Consider a normal shock wave at hypersonic speeds, as

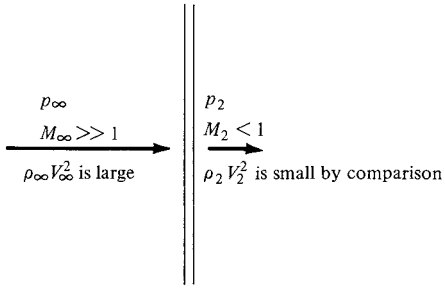


Figure 14.10 Hypersonic flow across a normal shock wave.

sketched in Figure 14.10. For this flow, Equation (8.6) gives

$$p_\infty + \rho_\infty V_\infty^2 = p_2 + \rho_2 V_2^2 \tag{14.6}$$

Recall that across a normal shock wave the flow velocity decreases, $V_2 < V_\infty$; indeed, the flow behind the normal shock is subsonic. This change becomes more severe as M_∞ increases. Hence, at hypersonic speeds, we can assume that $(\rho_\infty V_\infty^2) \gg (\rho_2 V_2^2)$, and we can neglect the latter term in Equation (14.6). As a result, Equation (14.6) becomes, at hypersonic speeds in the limiting case as $M_\infty \to \infty$,

$$p_2 - p_\infty = \rho_\infty V_\infty^2$$

or

$$C_p = \frac{p_2 - p_\infty}{\frac{1}{2}\rho_\infty V_\infty^2} = 2$$

thus confirming the newtonian results from Equation (14.4).

As stated above, the result that $C_p = 2$ at a stagnation point is a limiting value as $M_\infty \to \infty$. For large but finite Mach numbers, the value of C_p at a stagnation point is less than 2. Return again to the blunt body shown in Figure 14.9. Considering the distribution of C_p as a function of distance s along the surface, the largest value of C_p will occur at the stagnation point. Denote the stagnation point value of C_p by $C_{p,\max}$, as shown in Figure 14.9. $C_{p,\max}$ for a given M_∞ can be readily calculated from normal shock-wave theory. [If $\gamma = 1.4$, then $C_{p,\max}$ can be obtained from $p_{0,2}/p_1 = p_{0,2}/p_\infty$, tabulated in Appendix B. Recall from Equation (11.22) that $C_{p,\max} = (2/\gamma M_\infty^2)(p_{0,2}/p_\infty - 1)$.] Downstream of the stagnation point, C_p can be assumed to follow the sine-squared variation predicted by newtonian theory; that is,

$$\boxed{C_p = C_{p,\max} \sin^2 \theta} \tag{14.7}$$

Equation (14.7) is called the *modified* newtonian law. For the calculation of the C_p distribution around blunt bodies, Equation (14.7) is more accurate than Equation (14.4).

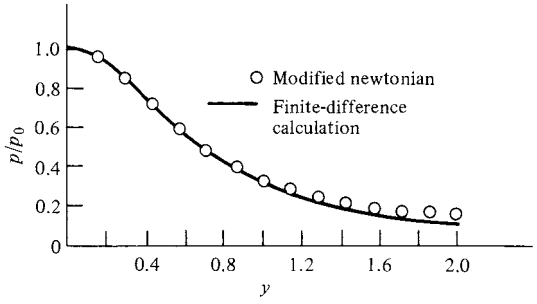


Figure 14.11 Surface pressure distribution, paraboloid, $M_\infty = 4$. Comparison of modified newtonian theory and time-dependent finite-difference calculations.

Return to Figure 13.14, which gives the numerical results for the pressure distributions around a blunt, parabolic cylinder at $M_\infty = 4$ and 8. The open symbols in this figure represent the results of modified newtonian theory, namely, Equation (14.7). For this two-dimensional body, modified newtonian theory is reasonably accurate only in the nose region, although the comparison improves at the higher Mach numbers. It is generally true that newtonian theory is more accurate at larger values of both M_∞ and θ. The case for an axisymmetric body, a paraboloid at $M_\infty = 4$, is given in Figure 14.11. Here, although M_∞ is relatively low, the agreement between the time-dependent numerical solution (see Chapter 13) and newtonian theory is much better. It is generally true that newtonian theory works better for three-dimensional bodies. In general, the modified newtonian law, Equation (14.7), is sufficiently accurate that it is used very frequently in the preliminary design of hypersonic vehicles. Indeed, extensive computer codes have been developed to apply Equation (14.7) to three-dimensional hypersonic bodies of general shape. Therefore, we can be thankful to Isaac Newton for supplying us with a law which holds reasonably well at hypersonic speeds, although such an application most likely never crossed his mind. Nevertheless, it is fitting that three centuries later, Newton's fluid mechanics has finally found a reasonable application.

14.4 THE LIFT AND DRAG OF WINGS AT HYPERSONIC SPEEDS: NEWTONIAN RESULTS FOR A FLAT PLATE AT ANGLE OF ATTACK

Question: At *subsonic* speeds, how do the lift coefficient C_L and drag coefficient C_D for a wing vary with angle of attack α?

Answer: As shown in Chapter 5, we know that:

1. The lift coefficient varies *linearly* with angle of attack, at least up to the stall; see, for example, Figure 5.24.

2. The drag coefficient is given by the drag polar, as expressed in Equation (5.63), repeated below:

$$C_D = c_d + \frac{C_L^2}{\pi e \text{AR}} \tag{5.63}$$

Since C_L is proportional to α, then C_D varies as the *square* of α.
Question: At *supersonic* speeds, how do C_L and C_D for a wing vary with α?
Answer: In Chapter 12, we demonstrated for an airfoil at supersonic speeds that:

1. Lift coefficient varies *linearly* with α, as seen from Equation (12.23), repeated below:

$$c_l = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \tag{12.23}$$

2. Drag coefficient varies as the *square* of α, as seen from Equation (12.24) for the flat plate, repeated below:

$$c_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} \tag{12.24}$$

The characteristics of a finite wing at supersonic speeds follow essentially the same functional variation with the angle of attack, namely, C_L is proportional to α and C_D is proportional to α^2.

Question: At *hypersonic* speeds, how do C_L and C_D for a wing vary with α? We have shown that C_L is proportional to α for both subsonic and supersonic speeds—does the same proportionality hold for hypersonic speeds? We have shown that C_D is proportional to α^2 for both subsonic and supersonic speeds—does the same proportionality hold for hypersonic speeds? The purpose of the present section is to address these questions.

In an approximate fashion, the lift and drag characteristics of a wing in hypersonic flow can be modeled by a flat plate at an angle of attack, as sketched in Figure 14.12. The exact flow field over the flat plate involves a series of expansion

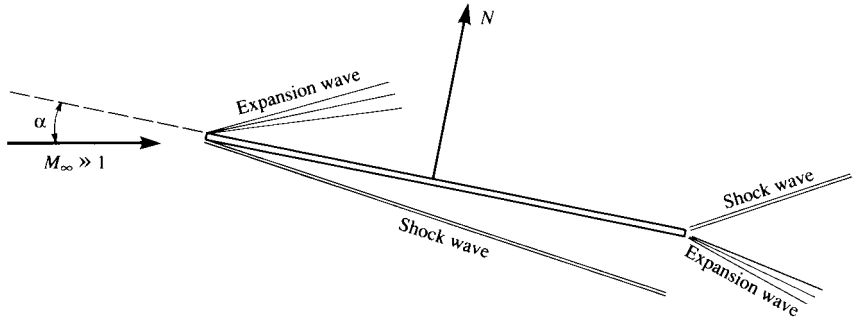


Figure 14.12 Wave system on a flat plate in hypersonic flow.

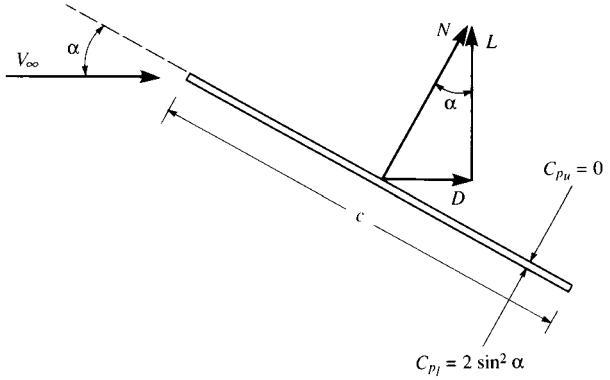


Figure 14.13 Flat plate at angle of attack. Illustration of aerodynamic forces.

and shock waves as shown in Figure 14.12; the exact lift- and wave-drag coefficients can be obtained from the shock-expansion method as described in Section 9.7. However, for hypersonic speeds, the lift- and wave-drag coefficients can be further approximated by the use of newtonian theory, as described in this equation.

Consider Figure 14.13. Here, a two-dimensional flat plate with chord length c is at an angle of attack α to the freestream. Since we are not including friction, and because surface pressure always acts normal to the surface, the resultant aerodynamic force is perpendicular to the plate; that is, in this case, the normal force N is the resultant aerodynamic force. (For an infinitely thin flat plate, this is a general result that is not limited to newtonian theory, or even to hypersonic flow.) In turn, N is resolved into lift and drag, denoted by L and D, respectively, as shown in Figure 14.13. According to newtonian theory, the pressure coefficient on the lower surface is

$$C_{p,l} = 2 \sin^2 \alpha \tag{14.8}$$

The upper surface of the flat plate shown in Figure 14.13, in the spirit of newtonian theory, receives no direct "impact" of the freestream particles; the upper surface is said to be in the "shadow" of the flow. Hence, consistent with the basic model of newtonian flow, only freestream pressure acts on the upper surface, and we have

$$C_{p,u} = 0 \tag{14.9}$$

Returning to the discussion of aerodynamic force coefficients in Section 1.5, we note that the normal force coefficient is given by Equation (1.15). Neglecting friction, this becomes

$$c_n = \frac{1}{c} \int_0^c (C_{p,l} - C_{p,u})\, dx \tag{14.10}$$

where x is the distance along the chord from the leading edge. (*Please note:* In this section, we treat a flat plate as an airfoil section; hence, we will use lowercase letters to denote the force coefficients, as first described in Chapter 1.) Substituting Equations (14.8) and (14.9) into (14.10), we obtain

$$c_n = \frac{1}{c}(2\sin^2\alpha)c$$

or

$$= 2\sin^2\alpha \tag{14.11}$$

From the geometry of Figure 14.13, we see that the lift and drag coefficients, defined as $c_l = L/q_\infty S$ and $c_d = D/q_\infty S$, respectively, where $S = (c)(l)$, are given by

$$c_l = c_n \cos\alpha \tag{14.12}$$

and

$$c_d = c_n \sin\alpha \tag{14.13}$$

Substituting Equation (14.11) into Equations (14.12) and (14.13), we obtain

$$c_l = 2\sin^2\alpha\cos\alpha \tag{14.14}$$

$$c_d = 2\sin^3\alpha \tag{14.15}$$

Finally, from the geometry of Figure 14.13, the lift-to-drag ratio is given by

$$\frac{L}{D} = \cot\alpha \tag{14.16}$$

[*Note:* Equation (14.16) is a general result for inviscid supersonic or hypersonic flow over a flat plate. For such flows, the resultant aerodynamic force is the normal force N. From the geometry shown in Figure 14.13, the resultant aerodynamic force makes the angle α with respect to lift, and clearly, from the right triangle between L, D, and N, we have $L/D = \cot\alpha$. Hence, Equation (14.16) is not limited to newtonian theory.]

The aerodynamic characteristics of a flat plate based on newtonian theory are shown in Figure 14.14. Although an infinitely thin flat plate, by itself, is not a practical aerodynamic configuration, its aerodynamic behavior at hypersonic speeds is consistent with some of the basic characteristics of other hypersonic shapes. For example, consider the variation of c_l shown in Figure 14.14. First, note that, at a small angle of attack, say, in the range of α from 0 to 15°, c_l varies in a *nonlinear* fashion; that is, the slope of the lift curve is *not* constant. This is in direct contrast to the subsonic case we studied in Chapters 4 and 5, where the lift coefficient for an airfoil or a finite wing was shown to vary linearly with α at small angles of attack, up to the stalling angle. This is also in contrast with the results from linearized supersonic theory as itemized in Section 12.3, leading to Equation (12.23) where a linear variation of c_l with α for a flat plate is indicated. However, the nonlinear lift curve shown in Figure 14.14 is *totally consistent* with the results discussed in Section 11.3, where hypersonic flow was shown to be governed by the *nonlinear* velocity potential equation, *not* by the linear equation expressed by Equation (11.18). In that section, we noted that both transonic and

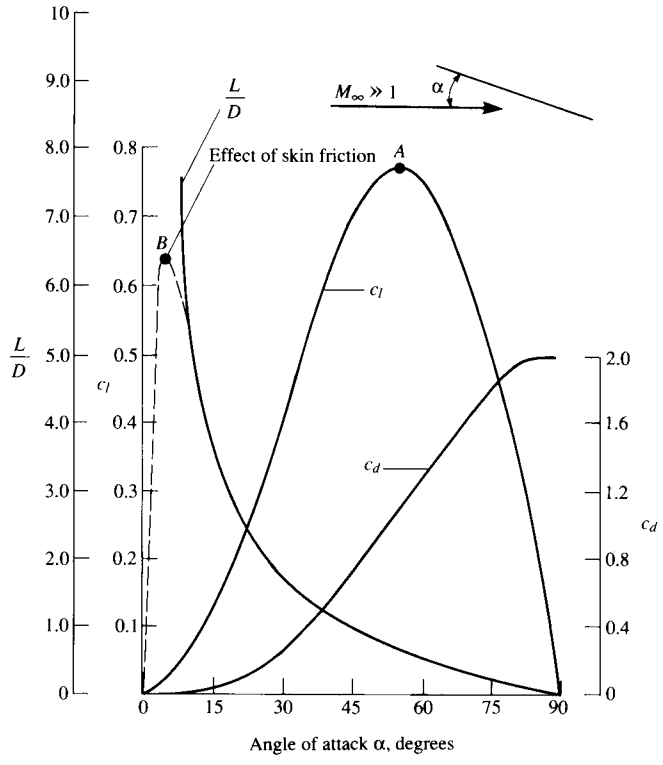


Figure 14.14 Aerodynamic properties of a flat plate based on newtonian theory.

hypersonic flow cannot be described by a linear theory—both these flows are inherently nonlinear regimes, even for low angles of attack. Once again, the flat-plate lift curve shown in Figure 14.14 certainly demonstrates the nonlinearity of hypersonic flow.

Also, note from the lift curve in Figure 14.14 that c_l first increases as α increases, reaches a maximum value at an angle of attack of about 55° (54.7° to be exact), and then decreases, reaching zero at $\alpha = 90°$. However, the attainment of $c_{l,\max}$ (point A) in Figure 14.14 is *not* due to any viscous, separated flow phenomenon analogous to that which occurs in subsonic flow. Rather, in Figure 14.14, the attainment of a maximum c_l is purely a geometric effect. To understand this better, return to Figure 14.13. Note that, as α increases, C_p continues to increase via the newtonian expression

$$C_p = 2 \sin^2 \alpha$$

That is, C_p reaches a maximum value at $\alpha = 90°$. In turn, the *normal force* N shown in Figure 14.13 continues to increase as α increases, also reaching a maximum value at $\alpha = 90°$. However, recall from Equation (14.12) that the vertical

component of the aerodynamic force, namely, the lift, is given by

$$L = N \cos \alpha \tag{14.17}$$

Hence, as α increases to 90°, although N continues to increase monotonically, the value of L reaches a maximum value around $\alpha = 55°$, and then begins to decrease at higher α due to the effect of the cosine variation shown in Equation (14.17)—strictly a geometric effect. In other words, in Figure 14.13, although N is increasing with α, it eventually becomes inclined enough relative to the vertical that its vertical component (lift) begins to decrease gradually. It is interesting to note that a large number of practical hypersonic configurations achieve a maximum C_L at an angle of attack in the neighborhood of that shown in Figure 14.14, namely, around 55°.

The maximum lift coefficient for a hypersonic flat plate, and the angle at which it occurs, is easily quantified using newtonian theory. Differentiating Equation (14.14) with respect to α, and setting the derivative equal to zero (for the condition of maximum c_l), we have

$$\frac{dc_l}{d\alpha} = (2 \sin^2 2)(-\sin \alpha) + 4 \cos^2 \alpha \sin \alpha = 0$$

or

$$\sin^2 \alpha = 2 \cos^2 \alpha = 2(1 - \sin^2 \alpha)$$

or

$$\sin^2 \alpha = \tfrac{2}{3}$$

Hence,

$$\alpha = 54.7°$$

This is the angle of attack at which c_l is a maximum. The maximum value of c_l is obtained by substituting the above result for α into Equation (14.14):

$$c_{l,\max} = 2 \sin^2(54.7°) \cos(54.7°) = 0.77$$

Note, although c_l increases over a wide latitude in the angle of attack (c_l increases in the range from $\alpha = 0$ to $\alpha = 54.7°$), its rate of increase is small (that is, the effective lift slope is small). In turn, the resulting value for the maximum lift coefficient is relatively small—at least in comparison to the much higher $c_{l,\max}$ values associated with low-speed flows (see Figures 4.25 and 4.28). Returning to Figure 14.14, we now note the *precise* values associated with the peak of the lift curve (point A), namely, the peak value of c_l is 0.77, and it occurs at an angle of attack of 54.7°.

Examining the variation of drag coefficient c_d in Figure 14.14, we note that it monotonically increases from zero at $\alpha = 0$ to a maximum of 2 at $\alpha = 90°$. The newtonian result for drag is essentially *wave drag* at hypersonic speeds because we are dealing with an inviscid flow, hence no friction drag. The variation of c_d with α for the low angle of attack in Figure 14.14 is essentially a *cubic* variation, in contrast to the result from linearized supersonic flow, namely, Equation (12.24), which shows that c_d varies as the square angle of attack. The hypersonic result that c_d varies as α^3 is easily obtained from Equation (14.15), which for

small α becomes

$$c_d = 2\alpha^3 \tag{14.18}$$

The variation of the lift-to-drag ratio as predicted by newtonian theory is also shown in Figure 14.14. The solid curve is the pure newtonian result; it shows that L/D is infinitely large at $\alpha = 0$ and monotonically decreases to zero at $\alpha = 90°$. The infinite value of L/D at $\alpha = 0$ is purely fictional—it is due to the neglect of skin friction. When skin friction is added to the picture, denoted by the dashed curve in Figure 14.14, L/D reaches a maximum value at a small angle of attack (point B in Figure 14.14) and is equal to zero at $\alpha = 0$. (At $\alpha = 0$, no lift is produced, but there is a finite drag due to friction; hence, $L/D = 0$ at $\alpha = 0$.)

Let us examine the conditions associated with $(L/D)_{\max}$ more closely. The value of $(L/D)_{\max}$ and the angle of attack at which it occurs (i.e., the coordinates of point B in Figure 14.14) are strictly a function of the zero-lift drag coefficient, denoted by $c_{d,0}$. The zero-lift drag coefficient is simply due to the integrated effect of skin friction over the plate surface at zero angle of attack. At small angles of attack, the skin friction exerted on the plate should be essentially that at zero angle of attack; hence, we can write the total drag coefficient [referring to Eq. (14.15)] as

$$c_d = 2\sin^3\alpha + c_{d,0} \tag{14.19}$$

Furthermore, when α is small, we can write Equations (14.14) and (14.19) as

$$c_l = 2\alpha^2 \tag{14.20}$$

and

$$c_d = 2\alpha^3 + c_{d,0} \tag{14.21}$$

Dividing Equation (14.20) by (14.21), we have

$$\frac{c_l}{c_d} = \frac{2\alpha^2}{2\alpha^3 + c_{d,0}} \tag{14.22}$$

The conditions associated with maximum lift-to-drag ratio can be found by differentiating Equation (14.22) and setting the result equal to zero:

$$\frac{d(c_l/c_d)}{d\alpha} = \frac{(2\alpha^3 + c_{d,0})4\alpha - 2\alpha^2(6\alpha^2)}{(2\alpha^3 + c_{d,0})} = 0$$

or

$$8\alpha^4 + 4\alpha c_{d,0} - 12\alpha^4 = 0$$

$$4\alpha^3 = 4c_{d,0}$$

Hence,

$$\boxed{\alpha = (c_{d,0})^{1/3}} \tag{14.23}$$

Substituting Equation (14.23) into Equation (14.21), we obtain

$$\left(\frac{c_l}{c_d}\right)_{\max} = \frac{2(c_{d,0})^{2/3}}{2c_{d,0} + c_{d,0}} = \frac{2/3}{(c_{d,0})^{1/3}}$$

or
$$\left(\frac{L}{D}\right)_{\max} = \left(\frac{c_l}{c_d}\right)_{\max} = \boxed{0.67/(c_{d,0})^{1/3}} \tag{14.24}$$

Equations (14.23) and (14.24) are important results. They clearly state that the coordinates of the maximum L/D point in Figure 14.14, when friction is included (point B in Figure 14.14), are strictly a function of $c_{d,0}$. In particular, note the expected trend that $(L/D)_{\max}$ decreases as $c_{d,0}$ increases—the higher the friction drag, the lower is L/D. Also, the angle of attack at which maximum L/D occurs increases as $c_{d,0}$ increases. There is yet another interesting aerodynamic condition that holds at $(L/D)_{\max}$, derived as follows. Substituting Equation (14.23) into (14.21), we have

$$c_d = 2c_{d,0} + c_{d,0} = 3c_{d,0} \tag{14.25}$$

Since the total drag coefficient is the sum of the wave-drag coefficient $c_{d,w}$ and the friction-drag coefficient $c_{d,0}$ we can write

$$c_d = c_{d,w} + c_{d,0} \tag{14.26}$$

However, at the point of maximum L/D (point B in Figure 14.14), we know from Equation (14.25) that $c_d = 3c_{d,0}$. Substituting this result into Equation (14.26), we obtain

$$3c_{d,0} = c_{d,w} + c_{d,0}$$

or
$$\boxed{c_{d,w} = 2c_{d,0}} \tag{14.27}$$

This clearly shows that, for the hypersonic flat plate using newtonian theory, at the flight condition associated with maximum lift-to-drag ratio, wave drag is twice the friction drag.

This brings to an end our short discussion of the lift and drag of wings at hypersonic speeds as modeled by the newtonian flat-plate problem. The quantitative and qualitative results presented here are reasonable representations of the hypersonic aerodynamic characteristics of a number of practical hypersonic vehicles; the flat-plate problem is simply a straightforward way of demonstrating these characteristics.

14.4.1 Accuracy Considerations

How accurate is newtonian theory in the prediction of pressure distributions over hypersonic bodies? The comparison shown in Figure 14.11 indicates that Equation (14.7) leads to a reasonably accurate pressure distribution over the surface of a *blunt body*. Indeed, for "back-of-the-envelope" estimates of the pressure distributions over blunt bodies at hypersonic speeds, modified newtonian is quite satisfactory. However, what about relatively thin bodies at small angles of attack? We can provide an answer by using the newtonian flat-plate relations derived in the present section, and compare these results with exact shock-expansion theory (Section 9.7), for flat plates at small angles of attack. This is the purpose of the following worked example.

EXAMPLE 14.1

Consider an infinitely thin flat plate at an angle of attack of 15° in a Mach 8 flow. Calculate the pressure coefficients on the top and bottom surface, the lift and drag coefficients, and the lift-to-drag ratio using (*a*) exact shock-expansion theory, and (*b*) newtonian theory. Compare the results.

■ Solution

(*a*) Using the diagram in Figure 9.35 showing a flat plate at angle of attack, and following the shock-expansion technique given in Example 9.11, we have for the upper surface, for $M_1 = 8$ and $\nu_1 = 95.62°$,

$$\nu_2 = \nu_1 + \theta = 95.62 + 15 = 110.62°$$

From Appendix C, interpolating between entries,

$$M_2 = 14.32$$

From Appendix A, for $M_1 = 8$, $p_{0_1}/p_1 = 0.9763 \times 10^4$, and for $M_2 = 14.32$, $p_{0_2}/p_2 = 0.4808 \times 10^6$. Since $p_{0_1} = p_{0_2}$,

$$\frac{p_2}{p_1} = \frac{p_{0_1}}{p_1} \Big/ \frac{p_{0_2}}{p_2} = \frac{0.9763 \times 10^4}{0.4808 \times 10^6} = 0.0203$$

The pressure coefficient is given by Equation (11.22), and the freestream static pressure in Figure 9.28 is denoted by p_1. Hence

$$C_{p_2} = \frac{2}{\gamma M_1^2}\left(\frac{p_2}{p_1} - 1\right) = \frac{2}{(1.4)(8)^2}(0.0203 - 1) = \boxed{-0.0219}$$

To obtain the pressure coefficient on the bottom surface from the oblique shock theory, we have from the θ-β-M for $M_1 = 8$ and $\theta = 15°$, $\beta = 21°$:

$$M_{n,1} = M_1 \sin\beta = 8 \sin 21° = 2.87$$

Interpolating from Appendix B, for $M_{n,1} = 2.87$, $p_3/p_1 = 9.443$. Hence the pressure coefficient on the bottom surface is

$$C_{p_3} = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_2} - 1\right) = \frac{2}{(1.4)(8)^2}(9.443 - 1) = \boxed{0.1885}$$

The lift coefficient can be obtained from the pressure coefficients via Equations (1.15), (1.16), and (1.18).

$$c_n = \frac{1}{c}\int_0^c (C_{p,\ell} - C_{p,u})\,dx = C_{p_3} - C_{p_2} = 0.1885 - (-0.0219) = 0.2104$$

The axial force on the plate is zero, because the pressure acts only perpendicular to the plate. On a formal basis, dy/dx in Equation (1.16) is zero for a flat plate. Hence, from Equation (1.18),

$$c_\ell = c_n \cos\alpha = 0.2104 \cos 15° = \boxed{0.2032}$$

From Equation (1.19),

$$c_d = c_n \sin\alpha = 0.2104 \sin 15^\circ = \boxed{0.0545}$$

Hence,

$$\frac{L}{D} = \frac{c_\ell}{c_d} = \frac{0.2032}{0.0545} = \boxed{3.73}$$

(*b*) From newtonian theory, the pressure coefficient is given by Equation (14.4), where $\theta \equiv \alpha$. This is the pressure coefficient on the lower surface, hence

$$C_{p_3} = 2\sin^2\alpha = 2\sin^2 15^\circ = \boxed{0.134}$$

From Equation (14.9), we have for the upper surface

$$C_{p_2} = \boxed{0}$$

Hence,

$$c_\ell = (C_{p_3} - C_{p_2})\cos\alpha = 0.134\cos 15^\circ = \boxed{0.1294}$$

and

$$c_d = (C_{p_3} - C_{p_2})\sin\alpha = 0.134\sin 15^\circ = \boxed{0.03468}$$

and

$$\frac{L}{D} = \frac{c_\ell}{c_d} = \frac{0.1294}{0.3468} = \boxed{3.73}$$

Discussion. From the above worked example, we see that newtonian theory *underpredicts* the pressure coefficient on the bottom surface by 29 percent, and of course predicts a value of zero for the pressure coefficient on the upper surface in comparison to -0.0219 from exact theory—an error of 100 percent. Also, newtonian theory underpredicts c_ℓ and c_d by 36.6 percent. However, the value of L/D from newtonian theory is *exactly correct*. This is no surprise, for two reasons. First, the newtonian values of c_ℓ and c_d are both underpredicted by the same amount, hence their ratio is not affected. Second, the value of L/D for supersonic or hypersonic inviscid flow over a flat plate, no matter what theory is used to obtain the pressures on the top and bottom surfaces, is simply a matter of *geometry*. Because the pressure acts normal to the surface, the resultant aerodynamic force is perpendicular to the plate (i.e., the resultant force *is* the normal force N). Examining Figure 1.16, we see that when this is the case, the vectors **R** and **N** are the same vectors, and L/D is geometrically given by

$$\frac{L}{D} = \cot\alpha$$

For the above worked example, where $\alpha = 15^\circ$, we have

$$\frac{L}{D} = \cot 15^\circ = 3.73$$

which agrees with the above calculations where c_ℓ and c_d were first obtained, and L/D is found from the ratio, $L/D = c_\ell/c_d$. So, Equation (14.16), derived in our discussion of newtonian theory applied to a flat plate, is not unique to newtonian theory; it is a general result when the resultant aerodynamic force is perpendicular to the plate.

We induce from Example 14.1 the general fact that the newtonian sine-squared law, Equation (14.4), does not accurately predict the hypersonic pressure distribution on the surface of two-dimensional bodies with local tangent lines that are at small or moderate angles to the flow, such as the bi-convex airfoil shape shown in Figure 12.3. On the other hand, it generally turns out that the newtonian prediction of the lift-to-drag ratio for slender shapes at small to moderate angles of attack is reasonably accurate. These statements apply to a gas with the ratio of specific heats substantially greater than one, such as the case of air with $\gamma = 1.4$ treated in Example 14.1. In the next section, we will see that newtonian theory becomes more accurate as $M_\infty \to \infty$ and $\gamma \to 1$. For more information on the accuracy of newtonian theory applied to two-dimensional slender shapes, see Reference 77 which is a study of this specific matter.

Finally, we note that newtonian theory does a better job of predicting the pressure on axisymmetric slender bodies, such as the 15° half-angle cone shown in Figure 14.15.

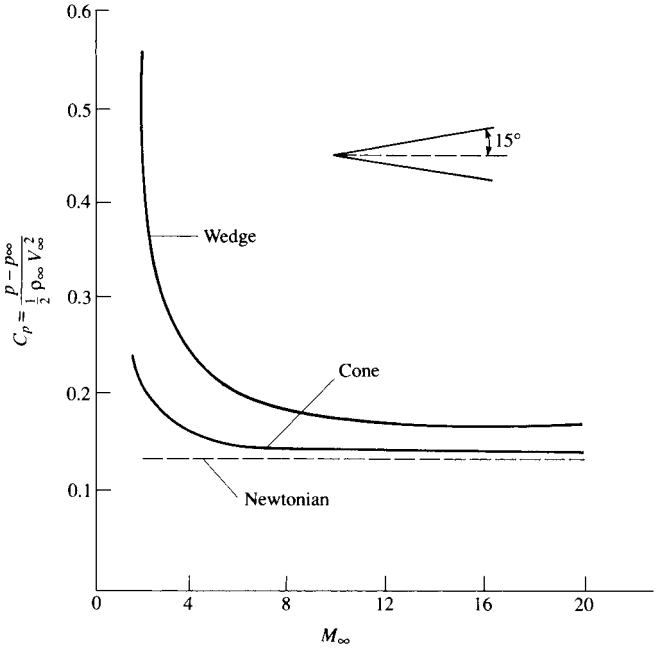


Figure 14.15 Comparison between newtonian and exact results for the pressure coefficient on a sharp wedge and a sharp cone. Also, an illustration of Mach number independence at high Mach numbers.

14.5 HYPERSONIC SHOCK-WAVE RELATIONS AND ANOTHER LOOK AT NEWTONIAN THEORY

The basic oblique shock relations are derived and discussed in Chapter 9. These are exact shock relations and hold for all Mach numbers greater than unity, supersonic or hypersonic (assuming a calorically perfect gas). However, some interesting approximate and simplified forms of these shock relations are obtained in the limit of a high Mach number. These limiting forms are called the hypersonic shock relations; they are obtained below.

Consider the flow through a straight oblique shock wave. (See, e.g., Figure 9.2.) Upstream and downstream conditions are denoted by subscripts 1 and 2, respectively. For a calorically perfect gas, the classical results for changes across the shock are given in Chapter 9. To begin with, the exact oblique shock relation for pressure ratio across the wave is given by Equation (9.16). Since $M_{n,1} = M_1 \sin\beta$, this equation becomes

Exact:
$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}\left(M_1^2 \sin^2\beta - 1\right) \tag{14.28}$$

where β is the wave angle. In the limit as M_1 goes to infinity, the term $M_1^2 \sin^2\beta \gg 1$, and hence Equation (14.28) becomes

as $M_1 \to \infty$:
$$\boxed{\frac{p_2}{p_1} = \frac{2\gamma}{\gamma+1} M_1^2 \sin^2\beta} \tag{14.29}$$

In a similar vein, the density and temperature ratios are given by Equations (9.15) and (9.17), respectively. These can be written as follows:

Exact:
$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2 \sin^2\beta}{(\gamma-1)M_1^2 \sin^2\beta + 2} \tag{14.30}$$

as $M_1 \to \infty$:
$$\boxed{\frac{\rho_2}{\rho_1} = \frac{\gamma+1}{\gamma-1}} \tag{14.31}$$

$$\frac{T_2}{T_1} = \frac{(p_2/p_1)}{(\rho_2/\rho_1)} \qquad \text{(from the equation of state: } p = \rho RT\text{)}$$

as $M_1 \to \infty$:
$$\boxed{\frac{T_2}{T_1} = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2} M_1^2 \sin^2\beta} \tag{14.32}$$

The relationship among Mach number M_1, shock angle β, and deflection angle θ is expressed by the so-called θ–β–M relation given by Equation (9.23), repeated below:

Exact:
$$\tan\theta = 2\cot\beta \left[\frac{M_1^2 \sin^2\beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2}\right] \tag{9.23}$$

This relation is plotted in Figure 9.9, which is a standard plot of the wave angle versus the deflection angle, with the Mach number as a parameter. Returning to Figure 9.9, we note that, in the hypersonic limit, where θ is small, β is also small. Hence, in this limit, we can insert the usual small-angle approximation into Equation (9.23):

$$\sin\beta \approx \beta$$
$$\cos 2\beta \approx 1$$
$$\tan\theta \approx \sin\theta \approx \theta$$

resulting in

$$\theta = \frac{2}{\beta}\left[\frac{M_1^2\beta^2 - 1}{M_1^2(\gamma+1)+2}\right] \tag{14.33}$$

Applying the high Mach number limit to Equation (14.33), we have

$$\theta = \frac{2}{\beta}\left[\frac{M_1^2\beta^2}{M_1^2(\gamma+1)}\right] \tag{14.34}$$

In Equation (14.34), M_1 cancels, and we finally obtain in both the small-angle and hypersonic limits,

as $M_1 \to \infty$ *and* θ, *hence* β *is small:*

$$\boxed{\frac{\beta}{\theta} = \frac{\gamma+1}{2}} \tag{14.35}$$

Note that, for $\gamma = 1.4$,

$$\boxed{\beta = 1.2\theta} \tag{14.36}$$

It is interesting to observe that, in the hypersonic limit for a slender wedge, the wave angle is only 20 percent larger than the wedge angle—a graphic demonstration of a thin shock layer in hypersonic flow.

In aerodynamics, pressure distributions are usually quoted in terms of the nondimensional pressure coefficient C_p, rather than the pressure itself. The pressure coefficient is defined as

$$C_p = \frac{p_2 - p_1}{q_1} \tag{14.37}$$

where p_1 and q_1 are the upstream (freestream) static pressure and dynamic pressure, respectively. Recall from Section 11.3 that Equation (14.37) can also be written as Equation (11.22), repeated below:

$$C_p = \frac{2}{\gamma M_1^2}\left(\frac{p_2}{p_1} - 1\right) \tag{11.22}$$

Combining Equations (11.22) and (14.28), we obtain an exact relation for C_p behind an oblique shock wave as follows:

Exact:

$$C_p = \frac{4}{\gamma+1}\left(\sin^2\beta - \frac{1}{M_1^2}\right) \tag{14.38}$$

In the hypersonic limit,

as $M_1 \to \infty$:

$$\boxed{C_p = \left(\frac{4}{\gamma + 1}\right) \sin^2 \beta} \tag{14.39}$$

Pause for a moment, and review our results. We have obtained limiting forms of the oblique shock equations, valid for the case when the upstream Mach number becomes very large. These limiting forms, called the hypersonic shock-wave relations, are given by Equations (14.29), (14.31), and (14.32), which yield the pressure ratio, density ratio, and temperature ratio across the shock when $M_1 \to \infty$. Furthermore, in the limit of both $M_1 \to \infty$ and small θ (such as the hypersonic flow over a slender airfoil shape), the limiting relation for the wave angle as a function of the deflection angle is given by Equation (14.35). Finally, the form of the pressure coefficient behind an oblique shock is given in the limit of hypersonic Mach numbers by Equation (14.39). Note that the limiting forms of the equations are always simpler than their corresponding exact counterparts.

In terms of actual *quantitative* results, it is always recommended that the *exact* oblique shock equations be used, even for hypersonic flow. This is particularly convenient because the exact results are tabulated in Appendix B. The value of the relations obtained in the hypersonic limit (as described above) is more for theoretical analysis rather than for the calculation of actual numbers. For example, in this section, we use the hypersonic shock relations to shed additional understanding of the significance of newtonian theory. In the next section, we will examine the same hypersonic shock relations to demonstrate the principle of Mach number independence.

Newtonian theory was discussed at length in Sections 14.3 and 14.4. For our purposes here, temporarily discard any thoughts of newtonian theory, and simply recall the exact oblique shock relation for C_p as given by Equation (14.38), repeated below (with freestream conditions now denoted by a subscript ∞ rather than a subscript 1, as used earlier):

$$C_p = \frac{4}{\gamma + 1}\left[\sin^2 \beta - \frac{1}{M_\infty^2}\right] \tag{14.38}$$

Equation (14.39) gave the limiting value of C_p as $M_\infty \to \infty$, repeated below:

as $M_\infty \to \infty$:

$$C_p \to \frac{4}{\gamma + 1} \sin^2 \beta \tag{14.39}$$

Now take the additional limit of $\gamma \to 1.0$. From Equation (14.39), in both limits as $M_\infty \to \infty$ and $\gamma \to 1.0$, we have

$$C_p \to 2 \sin^2 \beta \tag{14.40}$$

Equation (14.40) is a result from exact oblique shock theory; it has nothing to do with newtonian theory (as yet). Keep in mind that β in Equation (14.40) is the wave angle, not the deflection angle.

Let us go further. Consider the exact oblique shock relation for the density ratio, ρ/ρ_∞, given by Equation (14.30), repeated below (again with a subscript ∞

replacing the subscript 1):

$$\frac{\rho_2}{\rho_\infty} = \frac{(\gamma + 1)M_\infty^2 \sin^2 \beta}{(\gamma - 1)M_\infty^2 \sin^2 \beta + 2} \tag{14.41}$$

Equation (14.31) was obtained as the limit where $M_\infty \to \infty$, namely,

$$\textit{as } M_\infty \to \infty: \qquad \frac{\rho_2}{\rho_\infty} \to \frac{\gamma + 1}{\gamma - 1} \tag{14.42}$$

In the additional limit as $\gamma \to 1$, we find

$$\textit{as } \gamma \to 1 \textit{ and } M_\infty \to \infty: \qquad \boxed{\frac{\rho_2}{\rho_\infty} \to \infty} \tag{14.43}$$

that is, the density behind the shock is infinitely large. In turn, mass flow considerations then dictate that the shock wave is coincident with the body surface. This is further substantiated by Equation (14.35), which is good for $M_\infty \to \infty$ and small deflection angles:

$$\frac{\beta}{\theta} \to \frac{\gamma + 1}{2} \tag{14.35}$$

In the additional limit as $\gamma \to 1$, we have

$$\textit{as } \gamma \to 1 \textit{ and } M_\infty \to \infty \textit{ and } \theta \textit{ and } \beta \textit{ are small:} \qquad \boxed{\beta = \theta}$$

that is, the shock wave lies on the body. In light of this result, Equation (14.40) is written as

$$\boxed{C_p = 2 \sin^2 \theta} \tag{14.44}$$

Examine Equation (14.44). It is a result from exact oblique shock theory, taken in the combined limit of $M_\infty \to \infty$ and $\gamma \to 1$. However, it is also precisely the newtonian results given by Equation (14.4). Therefore, we make the following conclusion. The closer the actual hypersonic flow problem is to the limits $M_\infty \to \infty$ and $\gamma \to 1$, the closer it should be physically described by newtonian flow. In this regard, we gain a better appreciation of the true significance of newtonian theory. We can also state that the application of newtonian theory to practical hypersonic flow problems, where γ is always greater than unity, is theoretically not proper, and the agreement that is frequently obtained with experimental data has to be viewed as somewhat fortuitous. Nevertheless, the simplicity of newtonian theory along with its (sometimes) reasonable results (no matter how fortuitous) has made it a widely used and popular engineering method for the estimation of surface pressure distributions, hence lift- and wave-drag coefficients, for hypersonic bodies.

14.6 MACH NUMBER INDEPENDENCE

Examine again the hypersonic shock-wave relation for pressure ratio as given by Equation (14.29); note that, as the freestream Mach number approaches infinity, the pressure ratio itself also becomes infinitely large. On the other hand, the *pressure coefficient* behind the shock, given in the hypersonic limit by Equation (14.39), *is a constant value at high values of the Mach number*. This hints strongly of a situation where certain aspects of a hypersonic flow do not depend on Mach number, as long as the Mach number is sufficiently high. This is a type of "independence" from the Mach number, formally called the *hypersonic Mach number independence principle*. From the above argument, C_p clearly demonstrates Mach number independence. In turn, recall that the lift- and wave-drag coefficients for a body shape are obtained by integrating the local C_p, as shown by Equations (1.15), (1.16), (1.18), and (1.19). These equations demonstrate that, since C_p is independent of the Mach number at high values of M_∞, the lift and drag coefficients are also Mach number independent. Keep in mind that these conclusions are theoretical, based on the limiting form of the hypersonic shock relations.

Let us examine an example that clearly illustrates the Mach number independence principle. In Figure 14.15, the pressure coefficients for a 15° half-angle wedge and a 15° half-angle cone are plotted versus freestream Mach number for $\gamma = 1.4$. The exact wedge results are obtained from Equation (14.38), and the exact cone results are obtained from the solution of the classical Taylor-Maccoll equation. (See Reference 21 for a detailed discussion of the solution of the supersonic flow over a cone. There, you will find that the governing continuity, momentum, and energy equations for a conical flow cascade into a single differential equation called the Taylor-Maccoll equation. In turn, this equation allows the *exact* solution of this conical flow field.) Both sets of results are compared with newtonian theory, $C_p = 2 \sin^2 \theta$, shown as the dashed line in Figure 14.15. This comparison demonstrates two general aspects of newtonian results:

1. The accuracy of the newtonian results improves as M_∞ increases. This is to be expected from our discussion in Section 14.5. Note from Figure 14.15 that below $M_\infty = 5$ the newtonian results are not even close, but the comparison becomes much closer as M_∞ increases above 5.
2. Newtonian theory is usually more accurate for three-dimensional bodies (e.g., the cone) than for two-dimensional bodies (e.g., the wedge). This is clearly evident in Figure 14.15 where the newtonian result is much closer to the cone results than to the wedge results.

However, more to the point of Mach number independence, Figure 14.15 also shows the following trends. For both the wedge and the cone, the exact results show that, at low supersonic Mach numbers, C_p decreases rapidly as M_∞ is increased. However, at hypersonic speeds, the rate of decrease diminishes considerably, and C_p appears to reach a plateau as M_∞ becomes large; that is, C_p becomes relatively independent of M_∞ at high values of the Mach number. This is

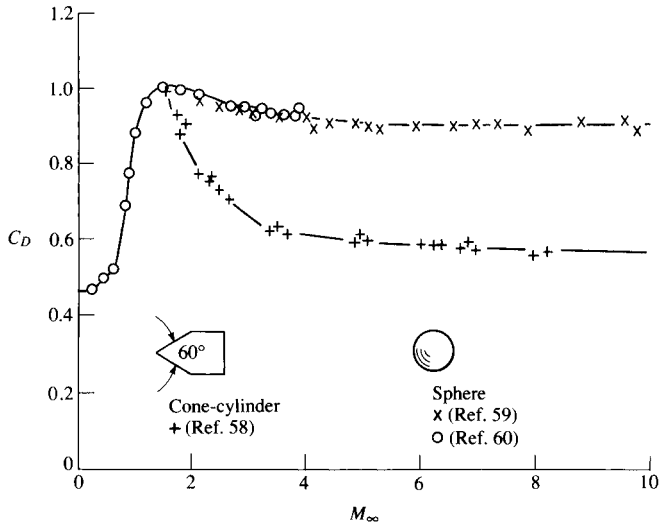


Figure 14.16 Drag coefficient for a sphere and a cone cylinder from ballistic range measurements; an example of Mach number independence at hypersonic speeds. (*Data Source: Cox and Crabtree, Reference 61.*)

the essence of the Mach number independence principle; at high Mach numbers, certain aerodynamic quantities such as pressure coefficient, lift- and wave-drag coefficients, and flow-field structure (such as shock-wave shapes and Mach wave patterns) become essentially independent of the Mach number. Indeed, newtonian theory gives results that are totally independent of the Mach number, as clearly demonstrated by Equation (14.4).

Another example of Mach number independence is shown in Figure 14.16. Here, the measured drag coefficients for spheres and for a large-angle cone cylinder are plotted versus the Mach number, cutting across the subsonic, supersonic, and hypersonic regimes. Note the large drag rise in the subsonic regime associated with the drag-divergence phenomenon near Mach 1 and the decrease in C_D in the supersonic regime beyond Mach 1. Both of these variations are expected and well understood. For our purposes in the present section, note, in particular, the variation of C_D in the hypersonic regime; for both the sphere and cone cylinder, C_D approaches a plateau and becomes relatively independent of the Mach number as M_∞ becomes large. Note also that the sphere data appear to achieve "Mach number independence" at lower Mach numbers than the cone cylinder.

Keep in mind from the above analysis that it is the nondimensional variables that become Mach number independent. Some of the dimensional variables, such as p, are not Mach number independent; indeed, $p \to \infty$ and $M_\infty \to \infty$.

Finally, the Mach number independence principle is well grounded mathematically. The governing inviscid flow equations (the Euler equations) expressed in terms of suitable nondimensional quantities, along with the boundary conditions

for the limiting hypersonic case, do not have the Mach number appearing in them—hence, by definition, the solution to these equations is independent of the Mach number. See References 21 and 55 for more details.

14.7 HYPERSONICS AND COMPUTATIONAL FLUID DYNAMICS

The design of hypersonic vehicles today is greatly dependent on the use of computational fluid dynamics, much more so than the design of vehicles for any other flight regime. The primary reason for this is the lack of experimental ground test facilities that can simultaneously simulate the Mach numbers, Reynolds numbers, and high-temperature levels associated with hypersonic flight. For such simulation, CFD is the primary tool. Reflecting once again on the philosophy illustrated in Figure 2.46, in the realm of hypersonic flow the three partners are not quite equal. Pure experimental work in hypersonics usually involves tests at either the desired Mach number, the desired Reynolds number, or the desired temperature level, but not all at the same time nor in the same test facilities. As a result, experimental data for the design of hypersonic vehicles is a patchwork of different data taken in different facilities under different conditions. Moreover, the data are usually incomplete, especially for the high-temperature effects, which are difficult to simulate in a wind tunnel. The designer must then do his or her best to piece together the information for the specified design conditions. The next partner shown in Figure 2.46, pure theory, is greatly hampered by the nonlinear nature of hypersonic flow, hence making mathematical solutions intractable. In addition, the proper inclusion of high-temperature chemically reacting flows in any pure theory is extremely difficult. For these reasons, the third partner shown in Figure 2.46, computational fluid dynamics, takes on a dominant role. The numerical calculation of both inviscid and viscous hypersonic flows, including all the high-temperature effects discussed in Section 14.2, has been a major thrust of CFD research and design application since the 1960s. Indeed, hypersonics has paced the development of CFD since its beginning.

As an example of CFD applied to a hypersonic flight vehicle appropriate to this chapter, consider the space shuttle shown in Figure 14.17. A numerical solution of the three-dimensional inviscid flow field around the shuttle was carried out by Maus et al. in Reference 78. They made two sets of calculations, one for a perfect gas with $\gamma = 1.4$, and one assuming chemically reacting air in local chemical equilibrium. The freestream Mach number was 23 in both cases. The CFD technique used for these calculations involved a time-dependent solution of the flow in the blunt nose region, patterned after our discussion in Section 13.5, and starting beyond the sonic line a downstream marching approach patterned after our discussion in Section 13.4. The calculated surface pressure distributions along the windward centerline of the space shuttle for both the perfect gas case (the circles) and the chemically reacting case (the triangles) are shown in Figure 14.18. The expansion around the nose, the pressure plateau over the relatively flat bottom surface,

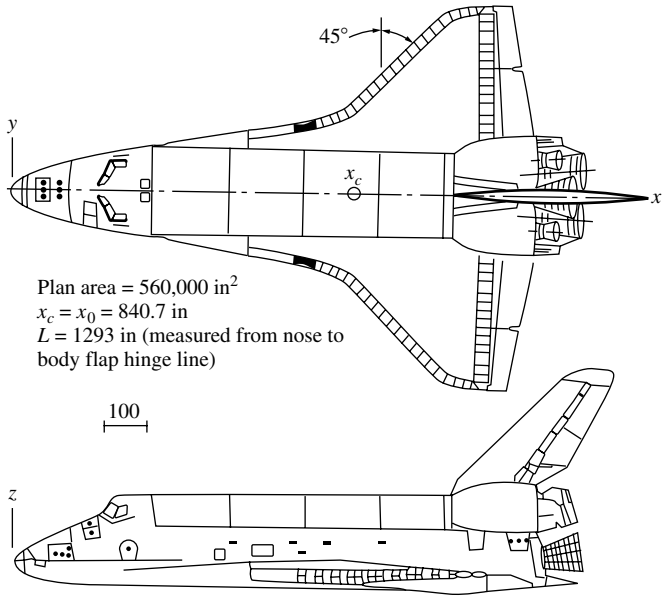


Figure 14.17 Space shuttle geometry.

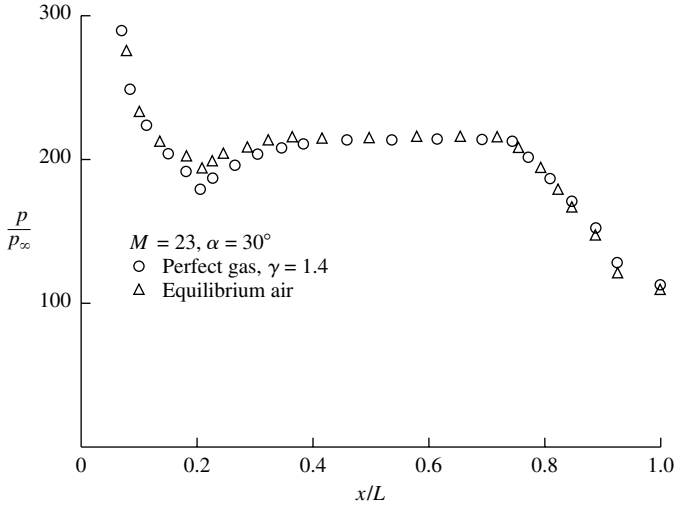


Figure 14.18 Pressure distribution along the windward centerline of the space shuttle; comparison between a calorically perfect gas and chemically reacting equilibrium air calculations. (*Data Source: Maus et al., Reference 78.*)

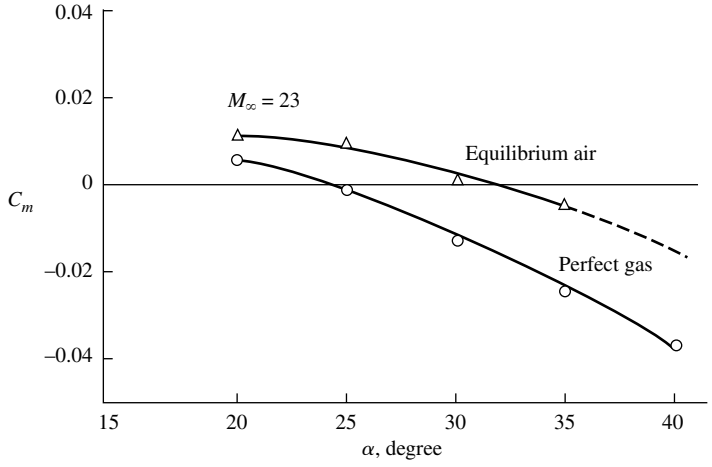


Figure 14.19 Predicted pitching moment coefficient for the space shuttle; comparison between a calorically perfect gas and equilibrium air calculations. (*Data Source: Maus et al., Reference 78.*)

and the further expansion over the slightly inclined back portion of the body, are all quite evident. Also note that there is little difference in the pressure distributions between the two cases; this is an example of the more general result that pressure is usually the flow variable *least* affected by chemically reacting effects.

It is interesting to note, however, that a flight characteristic as mundane as the vehicle pitching moment coefficient is affected by chemically reacting flow effects. Close examination of Figure 14.18 shows that, for the chemically reacting flow, the pressures are slightly higher on the forward part of the shuttle, and slightly lower on the rearward part. This results in a more positive pitching moment. Since the moment is the integral of the pressure through a moment arm, a slight change in pressure can substantially affect the moment. This is indeed the case here, as shown in Figure 14.19, which is a plot of the resulting calculated pitching moment as a function of angle of attack for the space shuttle. Clearly, the pitching moment is substantially greater for the chemically reacting case. The work by Maus et al. was the first to point out this effect on pitching moment, and it serves to reinforce the importance of high-temperature flows on hypersonic aerodynamics. It also serves to reinforce the importance of CFD in the analysis of hypersonic flows. The predicted pitching moment used for the space shuttle design came from "cold-flow" wind tunnel tests which did not simulate the high-temperature effects, that is, the designers used data for a perfect gas with $\gamma = 1.4$ obtained in the wind tunnel. This is represented by the lower curve in Figure 14.19. The early flight experience with the shuttle indicated a much higher pitching moment at hypersonic speeds than predicted, which required that the body flap deflection for trim to be more than twice that predicted—an alarming situation. The reason for this is now known; the actual flight environment encountered by the shuttle at

high Mach numbers was that of a high-temperature chemically reacting flow—the situation reflected in the upper curve in Figure 14.19. The difference in the pitching moment between the two curves in Figure 14.19 is enough to account for the unexpected extra body flap deflection required to trim the shuttle. Although these CFD results were obtained well after the design of the shuttle, they serve to underscore the importance of CFD to present and future hypersonic vehicle designs.

14.8 HYPERSONIC VISCOUS FLOW: AERODYNAMIC HEATING

Aerodynamic heating can become so severe at hypersonic speeds that it is the dominant design consideration for hypersonic vehicles. Indeed, for the reason discussed at the end of Section 1.1 and sketched in Figures 1.8 and 1.9, the nose and wing leading edges of hypersonic vehicles *must be blunt* rather than sharp, or else the vehicle will be destroyed by aerodynamic heating. In the history of flight, the most unfortunate example of such destruction occurred on February 1, 2003, when the space shuttle *Columbia* disintegrated over Texas during entry into the earth's atmosphere. Several of the thermal protection tiles near the leading edge of the left wing had been damaged by debris during launch. This allowed hot gases to penetrate the surface and destroy the internal wing structure.

The physical mechanisms that create atmospheric heating, both thermal conduction and radiation, are briefly discussed at the end of Section 14.2. In the present section we will present some engineering methods for predicting aerodynamic heating, and apply them to some hypersonic flow examples. Aerodynamic heating is a major subject in its own right, and is well beyond the scope of this book. (See Reference 55 for an in-depth discussion of aerodynamic heating applied to hypersonic flows.) However, its importance to the design of hypersonic vehicles demands that we examine a few aspects in the present chapter.

14.8.1 Aerodynamic Heating and Hypersonic Flow—the Connection

What is it about hypersonic flight that makes aerodynamic heating so severe? We address this question by reaching ahead to Chapter 16 and Equation (16.55), which introduces a dimensionless heat transfer coefficient called the Stanton number C_H, defined as

$$C_H \equiv \frac{\dot{q}_w}{\rho_e u_e (h_{aw} - h_w)} \tag{14.45}$$

In Equation (14.45), $\dot{q}_w$ is the heat transfer rate per unit area at a given point on the body surface. In the English engineering system the units of $\dot{q}_w$ are ft-lb/(s $\cdot$ ft^2); in the international system the units are W/m^2. Also in Equation (14.45), ρ_e is the local density at the edge of the boundary layer at the given point, u_e is the local velocity at the edge of the boundary layer, h_w is the enthalpy of the gas at the

wall, and h_{aw} is the adiabatic wall enthalpy defined as the enthalpy of the gas at the wall when the wall temperature is the adiabatic wall temperature—the wall temperature when the wall becomes so hot that no more energy is conducted into the wall from the gas adjacent to the wall.

Consider the hypersonic flow over a flat plate at zero angle of attack, where $\rho_e = \rho_\infty$ and $u_e = V_\infty$ (ignoring any viscous interaction effect as described in Section 14.2). For high Mach number laminar flow over a flat plate, T_{aw} is about 12 percent less than the total temperature in the freestream. For our purposes here, we make the approximation that $T_{aw} \approx T_o$, and hence in Equation (14.45)

$$h_{aw} \approx h_o \tag{14.46}$$

where h_o is the total enthalpy of the freestream. From Equation (7.54), we can write

$$h_o = h_\infty + \frac{V_\infty^2}{2} \tag{14.47}$$

At hypersonic speeds, V_∞ is very large. Also the ambient air far ahead of the vehicle is relatively cool; hence $h_\infty = c_p T_\infty$ is relatively small. Thus, at high speeds, from Equation (14.47),

$$h_o \approx \frac{V_\infty^2}{2} \tag{14.48}$$

The surface temperature of the plate, although it may be hot by normal standards, still must be maintained at below the melting or decomposition temperature of the surface, which is usually much smaller than the total temperature at high Mach numbers. Thus we can easily make the assumption that

$$h_o \gg h_w \tag{14.49}$$

It follows from Equations (14.46), (14.48), and (14.49) that

$$h_{aw} - h_w \approx h_o - h_w \approx h_o \approx \frac{V_\infty^2}{2} \tag{14.50}$$

Equation (14.45), written for a flat plate, is

$$C_H = \frac{\dot{q}_w}{\rho_\infty V_\infty (h_{aw} - h_w)}$$

Invoking the approximation given by Equation (14.50), we have

$$C_H \approx \frac{\dot{q}_w}{\rho_\infty V_\infty (V_\infty^2/2)}$$

or

$$\dot{q}_w \approx \frac{1}{2} \rho_\infty V_\infty^3 C_H \tag{14.51}$$

Equation (14.51) states that *the aerodynamic heating rate varies as the cube of the velocity*. This is in contrast to aerodynamic drag, which varies only as the square of the velocity. For this reason, at very high velocities, aerodynamic heating

becomes a dominant aspect of hypersonic vehicle design. This is the connection between aerodynamic heating and hypersonic flow.

14.8.2 Blunt versus Slender Bodies in Hypersonic Flow

We have made the claim that the nose and wing leading edges of hypersonic vehicles must be blunt rather than sharp in order to reduce the aerodynamic heating in those regions. In this section we will begin to demonstrate this fact quantitatively.

In Section 14.8.1 we focused on the local heat transfer rate per unit area at a point on the surface of the vehicle, $\dot{q}_w$. Here we expand our view to the total heat transferred to the vehicle per unit time, dQ/dt, which is equal to the local heat transfer rate integrated over the whole surface area of the vehicle. We can define an integrated overall Stanton number $\overline{C}_H$ by an equation similar to Equation (14.45),

$$\overline{C}_H \equiv \frac{dQ/dt}{\rho_\infty V_\infty (h_o - h_w) S} \tag{14.52}$$

where S is a reference area (planform area of a wing, cross-sectional area of a spherical entry vehicle, or the like) in the same spirit as in the definition of the lift or drag coefficients for a vehicle. Using the approximations made in Section 14.8.1, Equation (14.52) can be approximated by an expression similar to Equation (14.51); i.e.,

$$\frac{dQ}{dt} = \frac{1}{2} \rho_\infty V_\infty^3 S \overline{C}_H \tag{14.53}$$

Again we borrow a result from Chapters 16 and 18, namely that there exists an analogy between skin friction and aerodynamic heating, called Reynolds' analogy, expressed for a laminar flow by Equation (18.50), repeated below:

$$\frac{C_H}{C_f} = \frac{1}{2} \mathrm{Pr}^{-2/3} \tag{18.50}$$

where C_f is the local skin friction coefficient as first defined in Section 1.5, and Pr is the Prandtl number defined in Section 15.6. For our analysis here, it is safe to assume that $\mathrm{Pr} = 1$. Also, Reynolds analogy expressed by Eq. (18.50) can be written in terms of the integrated heat transfer and skin friction coefficients, $\overline{C}_H$ and C_f respectively. Hence, we have

$$\frac{\overline{C}_H}{C_f} = \frac{1}{2} \tag{14.54}$$

Inserting Equation (14.54) into (14.53) gives

$$\frac{dQ}{dt} = \frac{1}{4} \rho_\infty V_\infty^3 S C_f \tag{14.55}$$

Let us consider the case of a hypersonic vehicle entering the atmosphere at very high Mach number from a mission in space. The force that slows this vehicle

during entry is aerodynamic drag. From Newton's second law, we have

$$F = D = -m\frac{dV_\infty}{dt} \tag{14.56}$$

where m is the mass of the vehicle and the minus sign is necessary because dV_∞/dt is negative; i.e., the vehicle is decelerating. From Equation (14.56),

$$\frac{dV_\infty}{dt} = -\frac{D}{m} = -\frac{1}{2m}\rho_\infty V_\infty^2 S C_D \tag{14.57}$$

where C_D is the drag coefficient of the vehicle. Mathematically, we can write dQ/dt as $(dQ/dV_\infty)(dV_\infty/dt)$, where dV_∞/dt is given by Equation (14.57).

$$\frac{dQ}{dt} = \frac{dQ}{dV_\infty}\left(-\frac{1}{2m}\rho_\infty V_\infty^2 S C_D\right) \tag{14.58}$$

Equating Equations (14.55) and (14.58),

$$\frac{dQ}{dV_\infty}\left(-\frac{1}{2m}\rho_\infty V_\infty^2 S C_D\right) = \frac{1}{4}\rho_\infty V_\infty^3 S C_f$$

or

$$\frac{dQ}{dV_\infty} = -\frac{1}{2} m V_\infty \frac{C_f}{C_D}$$

or

$$dQ = -\frac{1}{2} m \frac{C_f}{C_D}\frac{dV_\infty^2}{2} \tag{14.59}$$

Integrate Equation (14.59) from the beginning of entry to the atmosphere, where $Q = 0$ and $V_\infty = V_E$, to the end of entry where $Q = Q_{\text{total}}$ and $V_\infty = 0$:

$$\int_0^{Q_{\text{total}}} dQ = -\frac{1}{2}\frac{C_f}{C_D}\int_{V_E}^0 d\left(m\frac{V_\infty^2}{2}\right)$$

or

$$\boxed{Q_{\text{total}} = \frac{1}{2}\frac{C_f}{C_D}\left(\frac{1}{2} m V_E^2\right)} \tag{14.60}$$

Equation (14.60) gives the total heat input Q_{total} to the vehicle. It reflects two vital conclusions:

1. The quantity $\frac{1}{2}mV_E^2$ is the initial kinetic energy of the vehicle as it first enters the atmosphere. Equation (14.60) says that the total heat input is directly proportional to this initial kinetic energy.
2. Total heat input is directly proportional to the ratio of skin friction drag to the total drag, C_f/C_D.

The second of these conclusions is of direct relevance to our discussion. Recall from Section 1.5 that the aerodynamic drag on a vehicle is the sum of drag due to the pressure distribution exerted over its surface, called pressure drag

D_p, and the drag due to shear stress exerted over its surface, called skin friction drag D_f. In terms of the pressure drag coefficient C_{D_p} and the skin friction drag coefficient C_f we have:

$$C_D = C_{D_p} + C_f$$

From Equation (14.60), to minimize the total aerodynamic heating, we need to minimize the ratio

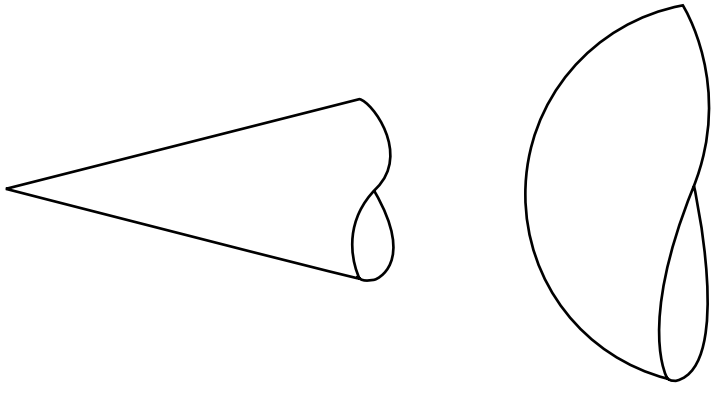
$$\frac{C_f}{C_{D_p} + C_f}$$

Now consider two extremes of aerodynamic configurations: a sharp-nosed *slender body* such as the cone shown in Figure 14.20*a* and the *blunt body* shown in Figure 14.20*b*. For a slender body, the skin friction drag is large in comparison to the pressure drag, hence $C_D \approx C_f$ and

$$\frac{C_f}{C_D} \approx 1 \qquad \text{slender body}$$

On the other hand, for a blunt body the pressure drag is large in comparison to the skin friction drag, hence $C_D \approx C_{D_p}$ and

$$\frac{C_f}{C_D} \ll 1 \qquad \text{blunt body}$$



Slender Body

Large C_f

Small C_{D_p}

$C_D \approx C_f$

Drag mainly friction drag

(*a*)

Blunt Body

Large C_{D_p}

Small C_f

$C_D \approx C_{D_p}$

Drag mainly pressure drag

(*b*)

Figure 14.20 Comparison of blunt and slender bodies.

In light of Equation (14.60), this leads to the following vital conclusion:

To minimize aerodynamic heating, the vehicle must be a blunt body, i.e., have a blunt nose.

For this reason, all successful entry vehicles in practice, from intercontinental ballistic missiles (ICBMs), to the Apollo lunar return capsule, to the space shuttle, have utilized rounded noses and rounded leading edges. Even vehicles designed for sustained hypersonic atmospheric flight such as the X-43 shown in Figures 14.1 and 14.2 have rounded noses and leading edges, although the radii of curvature is small because the minimization of drag, hence the maximization of lift-to-drag ratio, also becomes important in the design of such vehicles.

14.8.3 Aerodynamic Heating to a Blunt Body

The concept that a blunt body would reduce aerodynamic heating in comparison to a slender body was first advanced by H. Julian "Harvey" Allen in 1951, as discussed in Section 1.1. From that time on, the calculation of blunt body aerodynamic heating has been of paramount importance in the design of hypersonic vehicles. In this section we examine the calculation of aerodynamic heating to the stagnation point of a blunt body because the stagnation point is frequently (but not always) the point of maximum heat transfer rate to a hypersonic vehicle.

The boundary layer in the region of a stagnation point is laminar, and lends itself to the exact solution discussed in Section 18.5. When you read Chapter 18 as you progress through this book, you will have the opportunity to enjoy this rather elegant solution. In the present section, however, we wish to highlight only one result from Section 18.5, namely that given by Equation (18.83), repeated below.

$$\dot{q}_w \propto \frac{1}{\sqrt{R}} \tag{18.83}$$

where R is the nose radius at the stagnation point. This states that stagnation point heating varies *inversely* with the square root of the nose radius; hence to reduce the heating, increase the nose radius. Here we have absolute mathematical proof that a blunt body reduces aerodynamic heating.

The laminar boundary layer solution for the stagnation point case is described in Section 18.5. This solution yields a detailed result for stagnation point aerodynamic heating, namely Equations (18.65) for a circular cylinder and (18.70) for a sphere, that gives the aerodynamic heating as a function of the detailed flow field properties at the stagnation point. A much simpler engineering formula for aerodynamic heating is given by Tauber and Meneses (Ref. 101) in a generalized form of Equation (14.51) as

$$\dot{q}_w \rho_\infty^N V_\infty^M C \tag{14.61}$$

where, for the stagnation point,

$$M = 3,\ N = 0.5,\ C = 1.83 \times 10^{-8}\ R^{-1/2}\left(1 - \frac{h_w}{h_o}\right)$$

when the units for $\dot{q}_w$, V_∞, and ρ_∞ and R are respectively, W/cm^2, m/s, kg/m^3, and m. Thus, for the stagnation point, using Equation (14.61), we have

$$\dot{q}_w = \rho_\infty^{0.5} V_\infty^3 (1.83 \times 10^{-8}\ R^{-0.5}) \left(1 - \frac{h_w}{h_o}\right) \tag{14.62}$$

In Equation (14.62) we see the now familiar result that aerodynamic heating varies with the cube of the velocity, and that the stagnation point heating varies inversely with the square root of the nose radius. Equation (14.62) also shows that $\dot{q}_w$ varies with the square root of the density, which at first glance appears not to be consistent with Equation (14.51), repeated below

$$\dot{q}_w = \frac{1}{2}\rho_\infty V_\infty^3 C_H \tag{14.51}$$

This relation appears to show that $\dot{q}_w$ is proportional to density to the first power. However, once again drawing on results from Chapter 18, and specifically from Equation (18.54), we see that the Stanton number itself for laminar flow is inversely proportional to the square root of the Reynolds number. Since the Reynolds number by definition is in turn proportional to ρ_∞, we can state that

$$C_H \propto \frac{1}{\sqrt{\text{Re}}} \propto \frac{1}{\sqrt{\rho_\infty}}$$

and then from Equation (14.51),

$$\dot{q}_w \propto \sqrt{\rho_\infty}$$

This is consistent with Equation (14.62).

EXAMPLE 14.2

During the entry of the space shuttle into the earth's atmosphere, maximum stagnation point heating occurs at the trajectory point corresponding to an altitude of 68.9 km, where $\rho_\infty = 1.075 \times 10^{-4}$ kg/m^3, and a flight velocity of 6.61 km/s. At this point on its entry trajectory, the shuttle is at a 40.2 degree angle of attack, which presents an effective nose radius at the stagnation point of 1.29 m. If the wall temperature is $T_w = 1110$ K, calculate the stagnation point heating rate.

■ Solution

Equation (14.62), repeated here, is

$$\dot{q}_w = \rho_\infty^{0.5} V_\infty^3 (1.83 \times 10^{-8} R^{-0.5}) \left(1 - \frac{h_w}{h_o}\right)$$

To evaluate the ratio h_w/h_o, we have from Equation (14.48)

$$h_o \approx \frac{V_\infty^2}{2} = \frac{(6610)^2}{2} = 2.185 \times 10^7 \text{ J/(kg} \cdot \text{K)}$$

For h_w, when the wall temperature is 1110 K, we can reasonably assume a calorically perfect gas. As calculated in Example 7.1, for calorically perfect air, $C_p = 1004.5$ J/(kg · K). Thus,

$$h_w = c_p T_w = (1004.5)(1110) = 1.115 \times 10^6 \text{ J/(kg} \cdot \text{K)}$$

Thus,

$$\frac{h_w}{h_o} = \frac{1.115 \times 10^6}{2.185 \times 10^7} = 0.051$$

The stagnation point heat transfer is, from Equation (14.62),

$$\begin{aligned} \dot{q}_w &= \rho_\infty^{0.5} V_\infty^3 (1.83 \times 10^{-8} R^{-0.5}) \left(1 - \frac{h_w}{h_0}\right) \\ &= (1.075 \times 10^{-4})^{0.5} (6610)^3 (1.83 \times 10^{-8}) \times (1.29)^{-0.5} (1 - 0.051) \\ &= \boxed{45.78 \text{ W/cm}^2} \end{aligned}$$

Zoby (Reference 102) quotes a maximum stagnation point heating of 45 W/cm^2 based on experimental data obtained for the space shuttle at the given altitude and velocity on the entry trajectory. Our calculated result from Equation (14.62) agrees very well with the experimental data.

In Example 14.2 we calculated the heat transfer rate to the stagnation point. This is the point on the body of maximum aerodynamic heating rate. Along the windward centerline (on the bottom surface), the heating rate rapidly decreases with distance downstream from the stagnation point. Figure 14.21 gives experimental data for the local aerodynamic heating rate as reported in Reference 102. Note the qualitative similarity between the variation of $\dot{q}_w$ with distance as shown in Figure 14.21 and the variation of pressure with distance as shown in Figure 14.18. Although the results shown in these two figures are for slightly different angles of attack, this comparison illustrates a qualitative trend that is frequently seen in hypersonic aerodynamics, namely that the distribution of the aerodynamic heating rate over a surface tends to qualitatively follow the distribution of pressure over the surface.

14.9 APPLIED HYPERSONIC AERODYNAMICS: HYPERSONIC WAVERIDERS

The maximum lift-to-drag ratio $(L/D)_{\max}$ for a flight vehicle is a measure of its aerodynamic efficiency. Unfortunately, for supersonic and hypersonic flight vehicles, as the freestream Mach number increases, $(L/D)_{\max}$ decreases rather dramatically. This is just a fact of nature, brought about by the rapidly increasing

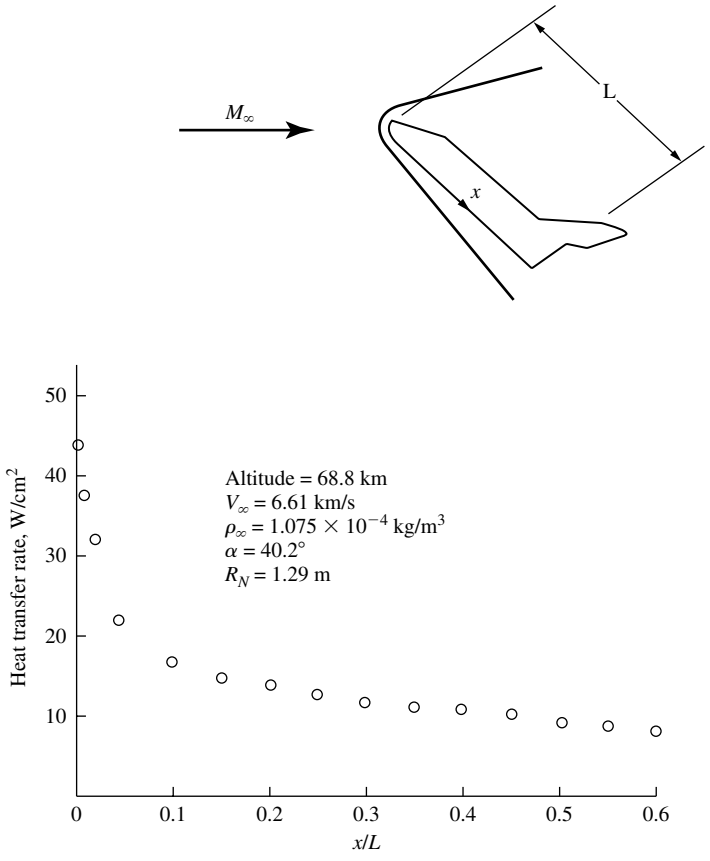


Figure 14.21 Experimental data for the local aerodynamic heating rate along the windward centerline of the space shuttle. Data from Zoby (Reference 102).

shock-wave strength as Mach number increases, with consequent large increases in wave drag. Return to the variation of L/D versus angle of attack for a flat plate shown in Figure 14.14. The solid curve is from the newtonian analysis discussed in Section 14.4. This is an inviscid flow result, and shows that L/D theoretically approaches infinity as the angle of attack α approaches zero. In reality, the viscous shear stress acting on the plate surface causes L/D to peak at a low value of α and to go to zero as $\alpha \rightarrow 0$. This is illustrated by the dashed curve in Figure 14.14, which shows the variation of L/D modified by skin friction as predicted by the reference temperature method discussed in Section 18.4. The skin-friction calculation is for laminar flow at Mach 10 and a Reynolds number of 3×10^6. Note that $(L/D)_{\max}$ for the flat plate is about 6.5. By comparison, $(L/D)_{\max}$ for a Boeing 747 at normal cruising conditions near Mach 1 is about 20. So the $(L/D)_{\max}$ for a hypersonic flat plate, as shown in Figure 14.14, is a low value, reflecting the characteristically low lift-to-drag ratios generated by hypersonic vehicles. And the infinitely thin flat plate is the most efficient lifting surface aerodynamically

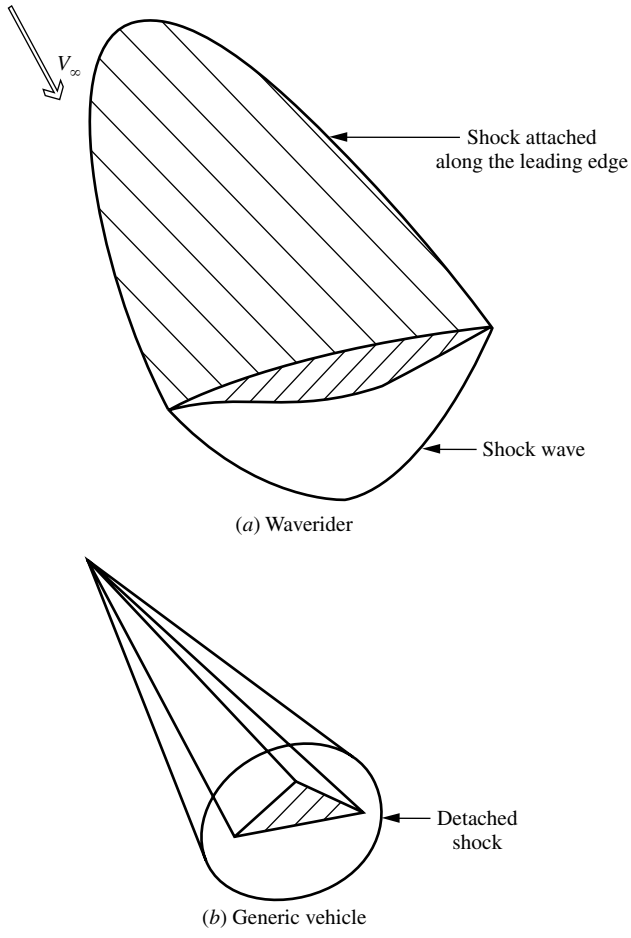


Figure 14.22 Comparison of waverider and generic hypersonic configurations.

compared to other hypersonic shapes with finite thickness. *Conclusion*: The L/D value of vehicles at hypersonic Mach numbers is low. This is particularly bothersome for future hypersonic vehicles designed for sustained flight in the atmosphere. Current design practice for such vehicles is illustrated by the X-43 shown in Figures 14.1 and 14.2.

There is a class of hypersonic vehicle shapes, however, that generates higher value of L/D than other shapes—waveriders. A waverider is a supersonic or hypersonic vehicle that has an attached shock wave all along its leading edge, as sketched in Figure 14.22*a*. Because of this, the vehicle appears to be riding on top of its shock wave, hence the term "waverider." This is in contrast to a more conventional hypersonic vehicle, where the shock wave is usually detached from the leading edge, as sketched in Figure 14.22*b*. The aerodynamic advantage of the waverider in Figure 14.22*a* is that the high pressure behind the shock wave under the vehicle does not "leak" around the leading edge to the top surface; the flow

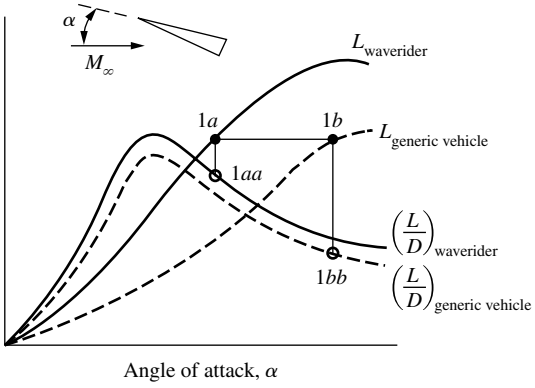
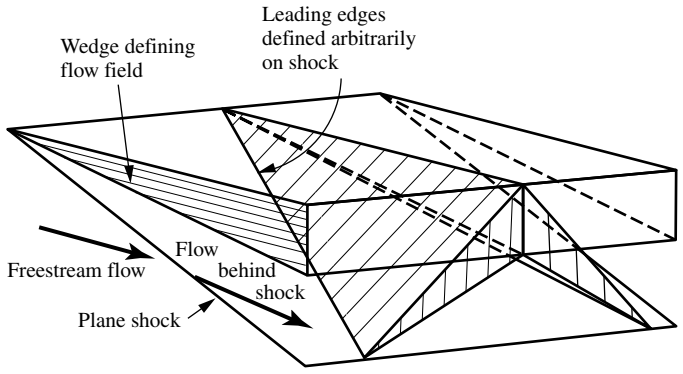


Figure 14.23 Comparison of lift and L/D curves between waverider and a generic hypersonic vehicle.

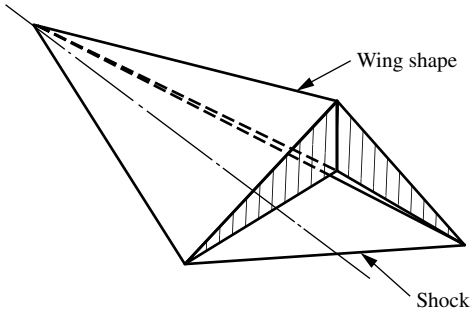
field over the bottom surface is contained, and the high pressure is preserved, and therefore more lift is generated on the vehicle. In contrast, for the vehicle shown in Figure 14.22*b*, there is communication between the flows over the bottom and top surfaces; the pressure tends to leak around the leading edge, and the general integrated pressure level on the bottom surface is reduced, resulting in less lift. Because of this, the generic vehicle in Figure 14.22*b* must fly at a larger angle of attack α to produce the same lift as the waverider in Figure 14.22a. This is illustrated in Figure 14.23, where the lift curves (L versus α) are sketched for the two vehicles shown in Figure 14.22. Note that the lift curve for the waverider is considerably higher because of the pressure containment compared to that for the generic vehicle. At the same lift, points 1*a* and 1*b* in Figure 14.23 represent the waverider and generic vehicles, respectively. Also shown in Figure 14.23 are typical variations of L/D versus α, which for slender hypersonic vehicles are not too different for the shapes in Figures 14.22*a* and *b*. (Although the lift of the waverider at a given angle of attack is increased by the pressure containment on the bottom surface, so is the wave drag; hence, the L/D ratio at a given angle of attack for the waverider is better, but not greatly so, than that for the generic vehicle.) However, note that because the waverider generates the same lift at a smaller α (point 1*a* in Figure 14.23) than does the generic vehicle, which must fly at a large α (point 1*b* in Figure 14.23), the L/D for the waverider is considerably higher (point 1*aa*) than that for the generic shape point (1*bb*). Therefore, for sustained hypersonic cruising flight in the atmosphere the waverider configuration has a definite advantage.

Question: How do you design a vehicle shape such that the shock wave is attached all along its leading edge; that is, how do you design a waverider?

One answer is as follows. Consider the simple flow field generated by a wedge in a supersonic or hypersonic freestream as discussed in Section 9.3. Imagine that the top surface of the wedge is parallel to the freestream, and hence the only wave in the flow is the planar shock wave propagating below the wedge, as sketched at the top of Figure 14.24. Now imagine two straight lines arbitrarily traced on the surface of the shock wave, coming to a point at the front of the shock. Consider all of the streamlines of the flow behind the shock that emanate from these



Construction from known flow field



Resulting wing and shock

Figure 14.24 Nonweiler or "caret" wing.

arbitrarily traced lines. Taken together, these streamlines form a stream surface that can be considered the surface of a vehicle with its leading edges defined by the two arbitrarily traced lines on the shock wave. Because the flow field behind a planar shock wave is uniform with parallel streamlines, these stream surfaces are flat surfaces that trace out a vehicle shape with a caret cross section as shown in Figure 14.24, named after the caret symbol $\wedge$. If you now mentally strip away the imaginary generating flow field shown at the top of Figure 14.24, you have left the caret-shaped vehicle shown at the bottom of Figure 14.24. Concentrating on the vehicle shape at the bottom of Figure 14.24, the planar surfaces on the bottom of the vehicle are stream surfaces that exist behind a planar oblique shock wave—stream surfaces that are generated by streamlines that begin on the shock surface itself. Hence, the shock wave is, by definition, attached to the leading edge of the vehicle; this planar attached shock is shown stretching between the two straight leading edges of the vehicle sketched at the bottom of Figure 14.24. By definition, therefore, this vehicle is a waverider. *Caution:* The waverider is in principle a point-designed vehicle. The generating oblique shock sketched at the top of Figure 14.24 pertains to a given freestream Mach number M_∞ and a given

flow deflection angle of the imaginary wedge that generates the oblique shock. Nevertheless, if you construct the vehicle shape shown at the bottom of Figure 14.24 and put it in a freestream at the given M_∞ and at an angle of attack such that the flow deflection angle of the vehicle bottom surface is the same as that of the imaginary generating wedge, then nature will make certain that the shock wave is attached all along the vehicle's leading edge; that is, the vehicle will be a waverider. Note that in Figure 14.24 we have oriented the imaginary generating wedge such that its top surface is parallel to the freestream; hence, there is no wave over the top surface of the wedge. Consequently, the top surfaces of the resulting caret waverider shown at the bottom of Figure 14.24 are aligned with the freestream, and there is no wave above the waverider.

In principle any shape can be used for the imaginary body producing the flow field from which a waverider shape is carved. The simplest case is to use a wedge for the imaginary body as just described. This has the advantage that a wedge produces a simple known flow field that is easily calculated, as treated in Chapter 9. You do not need a CFD solution for this flow. The flow over a cone at zero angle of attack in a supersonic or hypersonic flow is similarly a known flow field that can be used to generate waverider shapes. Because this conical flowfield is quasi-three-dimensional, it provides more flexibility in the generation of waverider shapes. The idea is the same. Consider the supersonic or hypersonic conical flow field over a right-circular cone at zero angle of attack as sketched at the top of Figure 14.25. The exact numerical solution of this flow field is discussed in Section 13.6. The flow field is obtained from a solution of the Taylor-Maccoll Equation, Equation (13.78), and tabulated results given in References 99 and 100. In short, this is a known flow field. At the top of Figure 14.25, we see a conical shock wave attached at the vertex of the right-circular cone. This cone is simply the imaginary body generating the flow field.

Consider the dashed curve drawn on the bottom surface of the conical shock wave as sketched at the top of Figure 14.25. All of the streamlines flowing through this dashed curve constitute a stream surface. In turn this stream surface defines the bottom surface of a waverider with a leading edge traced out by the dashed curve, as sketched at the bottom of Figure 14.25. Any curve can be traced on the conical shock; hence, any stream surface of the conical flow field downstream of the shock can be used as the surface of a waverider. When this is done, the shock wave will be attached all along the leading edge of the waverider, as shown in Figure 14.25. Moreover, the attached shock wave on this resulting waverider will, of course, be a segment of the conical shock wave shown at the top of Figure 14.25.

The waverider concept was first introduced by Nonweiler (Reference 103) in 1959, who generated caret-shaped waveriders from the two-dimensional flowfield behind a planar oblique shock wave generated by a wedge, as described earlier. Nonweiler was interested in such waveriders as lifting atmospheric entry bodies. The first extension of Nonweiler's concept to the use of a conical flow as generating flowfield by Jones (Reference 104) in 1963, and further extensions to other axisymmetric generating flows are discussed by Jones et al. in Reference 105. An excellent and authoritative survey of waverider research up to 1979 is given by

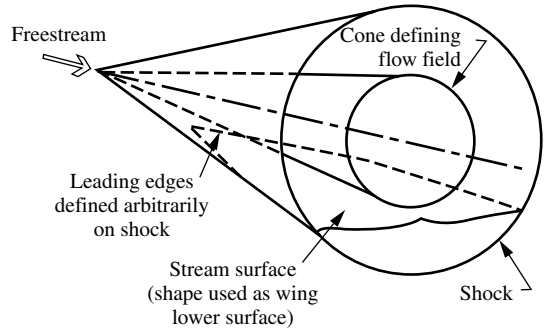


Figure 14.25 Cone flow wing.

Townend in (Reference 106). In the early 1980s Rasmussen and his colleagues at the University of Oklahoma (for example, see References 107–109) utilized hypersonic small-disturbance theory to design waveriders from flow fields over right-circular cones as well as elliptic cones. In a manner consistent with his use of analytical solutions of the waverider flows, Rasmussen was also able to use the classic calculus of variations to optimize the waverider shapes utilizing the inviscid properties of the flow.

14.9.1 Viscous-Optimized Waveriders

In the work just described, the waverider configurations were designed (and sometimes optimized) on the basis of inviscid flow fields, not including the effect of skin-friction drag. In turn, the drag predicted by such inviscid analyses was simply wave drag, and the resulting values of the inviscid L/D looked promising. However, waveriders tend to have large wetted surface areas, and the skin-friction drag, always added to the waverider aerodynamics after the fact, tended to greatly decrease the predicted inviscid lift-to-drag ratio. This made the waverider a less interesting prospect and led to a temporary lack of interest, indeed outright skepticism by researchers and vehicle designers in the waverider as a viable hypersonic configuration. Beginning in 1987, the author and his students at the University of Maryland took a different tack. New families of waveriders were generated wherein the skin-friction drag was included within an optimization routine to calculate waveriders with maximum L/D. In this fashion, the trade-offs between

wave drag and friction drag were accounted for during the optimization process, and the resulting family of waveriders had a shape and wetted surface area so as to optimize L/D. This family of waveriders is called viscous-optimization hypersonic waveriders, and subsequent CFD calculations and wind-tunnel tests have proven their viability, thus greatly enhancing modern interest in the waverider concept.

The design process for viscous-optimized waveriders was first published in References 110 and 111. This work, beginning in the late 1980s, led to a new class of waveriders where the optimization process is trying to reduce the wetted surface area, hence reducing skin-friction drag, while maximizing L/D. Because detailed viscous effects cannot be couched in simple analytical forms, the formal optimization methods based on the calculus of variations cannot be used. Instead, a numerical optimization technique was used based on the simplex method of Nelder and Mead (Reference 112). By using a numerical optimization technique, other real configuration aspects could be included in the analysis in addition to viscous effects, such as blunted leading edges and an expansion upper surface (in contrast to the standard assumption of a freestream upper surface, i.e., an upper surface with all generators parallel to the freestream direction). The results of the study by Bowcutt et al. led to a new class of waveriders, namely, viscous-optimized waveriders. Moreover, these waveriders produced relatively high values of L/D, as will be discussed later.

For the viscous-optimized waverider configurations, the following philosophy was followed:

1. The lower (compression) surface was generated by a stream surface behind a conical shock wave. The inviscid conical flow field was obtained from the numerical solution of the Taylor-Maccoll equation, derived in Section 13.6.
2. The upper surface was treated as an expansion surface, generated in a manner similar to that for the inviscid flow about a tapered, axisymmetric cylinder at zero angle of attack, and calculated by means of the axisymmetric method of characteristics.
3. The viscous effects were calculated by means of an integral boundary-layer analysis following surface streamlines, including transition from laminar to turbulent flow.
4. Blunt leading edges were included to the extent of determining the maximum leading-edge radius required to yield acceptable leading-edge surface temperatures, and then the leading-edge drag was estimated by modified newtonian theory.
5. The final waverider configuration, optimized for maximum L/D at a given Mach number and Reynolds number with body fineness ratio as a constraint, was obtained from the numerical simplex method taking into account all of the effects itemized in steps 1–4 within the optimization process itself.

The following discussion provides some insight into the optimization process. First, assume a given conical shock wave in a flow at a given Mach number, say, a conical shock wave angle of $\theta_s = 11$ deg. at Mach 6. As discussed previously,

now trace a curve on the surface of the shock wave. The stream surface generated from this curve is a bottom surface of a waverider, and the curve itself forms the leading edge of the waverider. An infinite number of such curves can be traced on the conical shock wave, generating an infinite number of waverider shapes using the conical shock with $\theta_s = 11$ deg. at $M_\infty = 6$. Indeed, some of these leading-edge curves are shown in Figure 14.26. The optimization procedure progresses through a series of these leading-edge shapes, each one generating a new waverider with a certain lift-to-drag ratio, and finally settling on that particular leading-edge shape that yields the maximum value of L/D. This is the optimum waverider for the given generating conical shock wave angle of $\theta_s = 11$ deg. This resulting $(L/D)_{\max}$ is then plotted as a point in Figure 14.27 for the conical shock wave angle $\theta_s = 11$ deg. Figure 14.27 also gives the corresponding value of lift coefficient C_L and volumetric efficiency $\eta = V^{2/3}/S_p$, where V is the vehicle volume and S_p is the planform area. Now choose another conical shock angle for generating the flow field, say, $\theta_s = 12$ deg., and repeat the preceding procedure, finding that leading-edge shape that yields the waverider shape that produces the highest L/D. This result is now plotted in Figure 14.27 for $\theta_s = 12$ deg. Then another conical shock wave angle, say $\theta_s = 13$ deg., is chosen, and the process is repeated again, finding that particular waverider shape that produces the highest L/D. This point is now plotted in Figure 14.27 for $\theta_s = 13$ deg. And so forth. The front views of these optimized waverider shapes are shown in Figure 14.28, each one labeled according to its generating conical shock-wave angle. These same optimized waveriders are shown in perspective in Figure 14.29. Returning to Figure 14.27, note that the curve of L/D versus θ_s itself has a maximum value

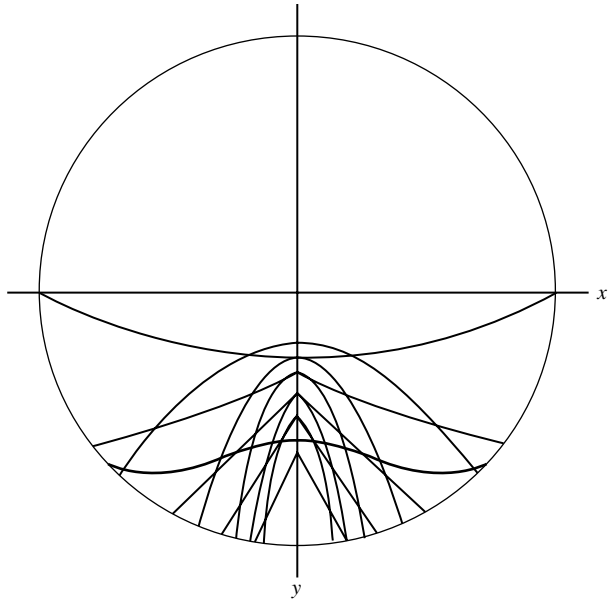


Figure 14.26 Examples of initial and optimized waverider leading-edge shapes.

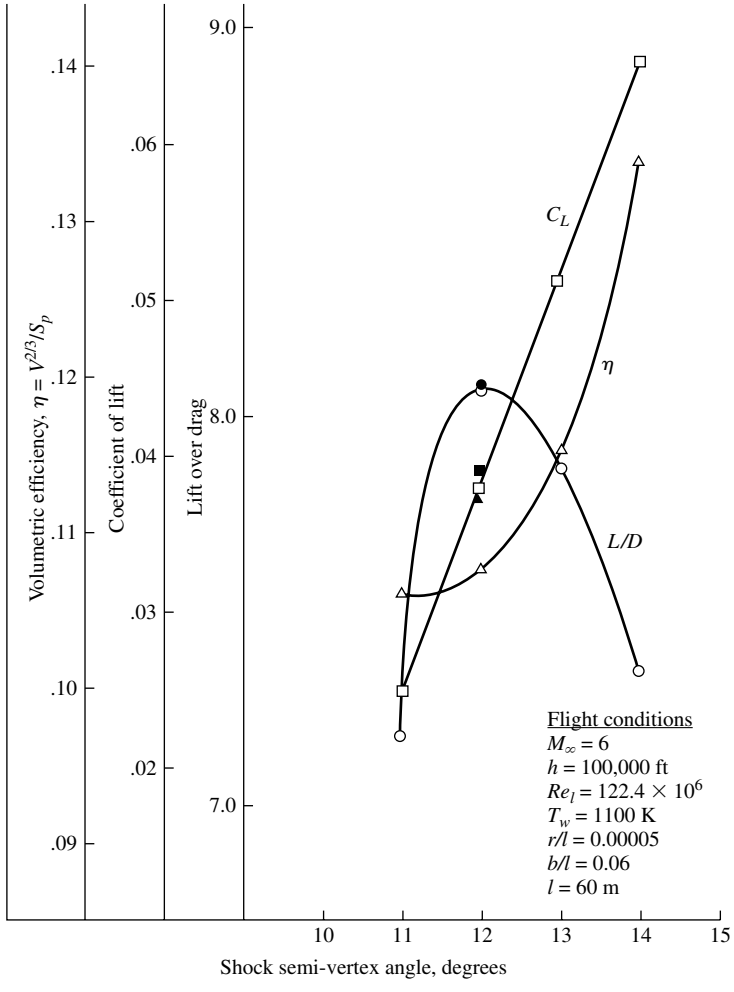


Figure 14.27 Results for a series of optimized waveriders at Mach 6; $l =$ length of waverider, $b/l =$ body fineness ratio, and $r =$ leading edge radius.

of L/D, occurring in this case for $\theta_s = 12$ deg. This yields an "optimum of the optimums" and defines the final viscous optimized waverider at $M_\infty = 6$ for the flight conditions shown in Figure 14.27. Noted on Figure 14.27 is the body fineness ratio, b/l, where b is the wing span and l is the length of the waverider. Recall that fineness ratio is taken as a constraint in the optimization process. For this case, $b/l = 0.06$. Finally, a summary three-view of the best optimum (the optimum of the optimums) waverider, which here corresponds to $\theta_s = 12$ deg., is given in Figure 14.30. Also, in Figures 14.28 to 14.30 the lines on the upper and lower surfaces of the waveriders are inviscid streamlines. Note in these figures that the shape of the optimum waverider changes considerably with θ_s. Moreover, examining (for example) Figure 14.30, note the rather complex curvature of the leading edge in both the planform and front views; the optimization program is

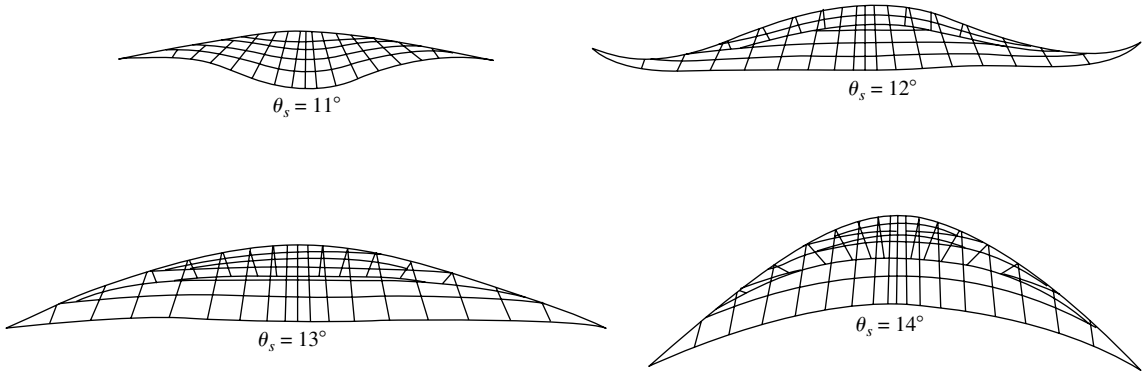


Figure 14.28 Results for a series of optimized waveriders at Mach 6.

shaping the waverider to adjust both wave drag and skin-friction drag so that the overall L/D is a maximum. Indeed, it was observed that the best optimum shape at any given M_∞ results in the magnitudes of wave drag and skin-friction drag being approximately the same, never differing by more than a factor of 2. For conical shock angles below the best optimum (for example, $\theta_s = 11°$ in Figures 14.28 and 14.29), skin-friction drag is greater than wave drag; in contrast, for conical shock angles above the optimum (for example $\theta_s = 13°$ and $14°$ in Figures 14.28 and 14.29), skin-friction drag is less than wave drag. [*Note*: In Section 14.4, using newtonian theory for a flat plate at angle of attack, at the flight condition associated with the maximum lift-to-drag ratio, wave drag is twice the friction drag, as proved by Equation (14.27).]

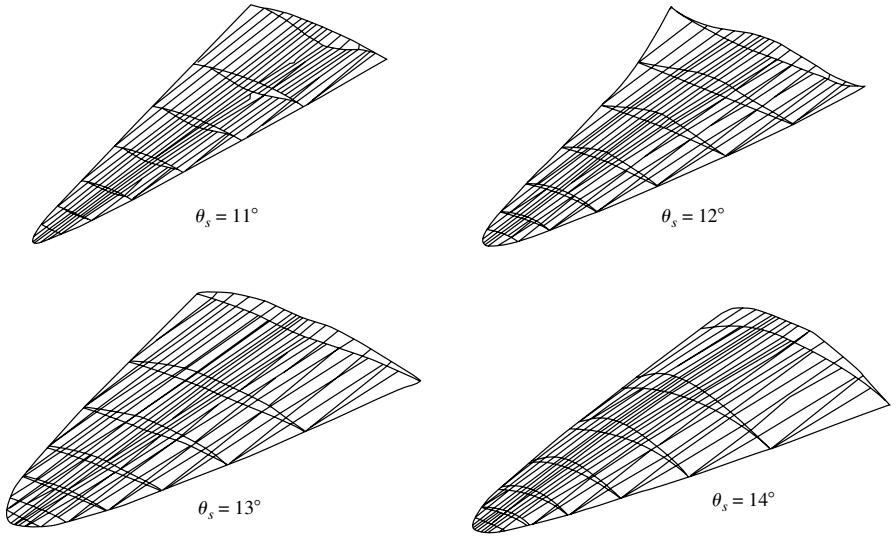


Figure 14.29 Perspective views of a series of optimized waveriders at Mach 6.

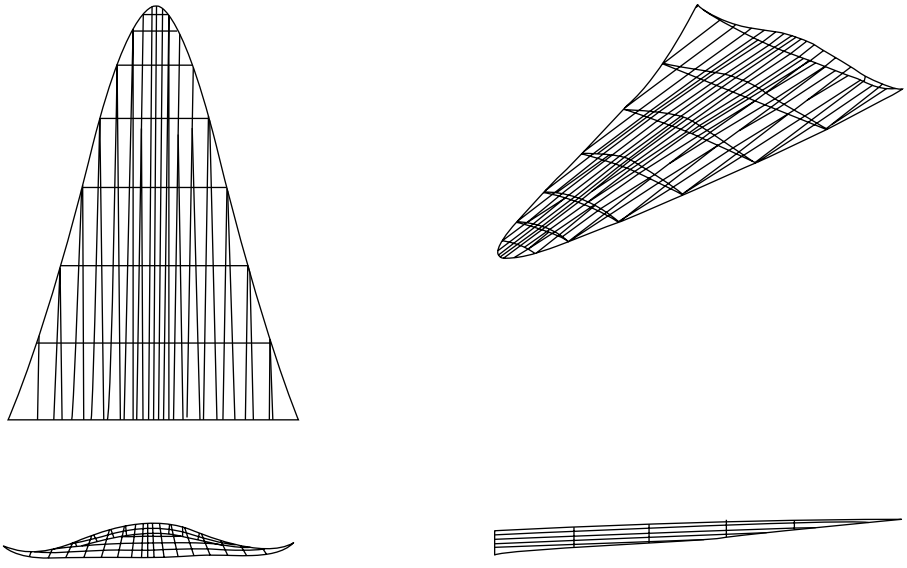


Figure 14.30 Three-view and perspective of the best optimized waverider at Mach 6.

The results in Figures 14.27 to 14.30 pertain to $M_\infty = 6$. Using the same procedure, the best optimum waverider shape at any supersonic or hypersonic Mach number can be obtained. For example, the shape of the best optimized waverider for $M_\infty = 25$ is given in Figure 14.31. Comparing the optimum configuration of $M_\infty = 6$ (Figure 14.30), note that the Mach 25 shape has more wing sweep. This pertains to a conical flow field with a smaller wave angle, both of which are intuitively expected at higher Mach number. However, the body slenderness ratio at $M_\infty = 6$ is constrained to be $b/l = 0.06$ (analogous to a supersonic transport such as the Concorde), but that $b/l = 0.09$ is the constraint chosen at $M_\infty = 25$ (analogous to a hydrogen-fueled hypersonic airplane). The two different slenderness ratios are chosen on the basis of reality for two different aircraft with two different missions at either extreme of the hypersonic flight spectrum.

For supersonic and hypersonic vehicles, L/D markedly decreases as M_∞ increases. Indeed, Kuchemann (Reference 70) gives the following general empirical correlation for $(L/D)_{\max}$ based on actual flight-vehicle experience:

$$(L/D)_{\max} = \frac{4(M_\infty + 3)}{M_\infty}$$

This variation is shown as the solid curve in Figure 14.32. This figure is important to our present discussion; it brings home the importance of viscous optimized waveriders. The Kuchemann curve (the solid curve) in Figure 14.32 represents a type of "L/D barrier" for conventional vehicles, which is difficult to break. The open circles in Figure 14.32, which form an almost shotgun scatter of points, are data for a variety of conventional vehicles representing various wind-tunnel and flight tests. (Precise identification of the sources for these points is given in

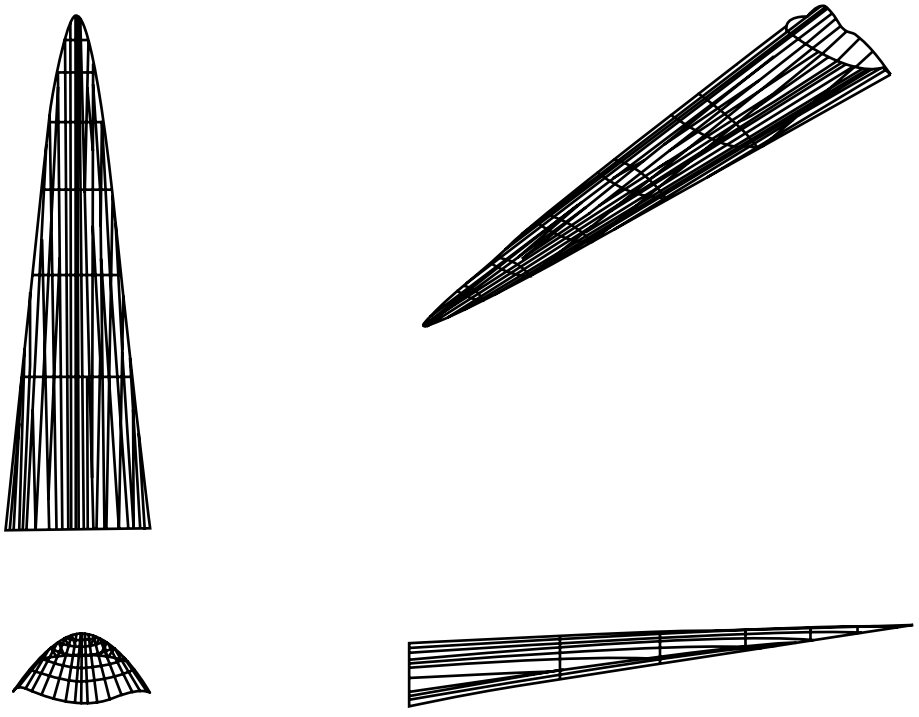


Figure 14.31 Three-view and perspective of the best optimized waverider at Mach 25.

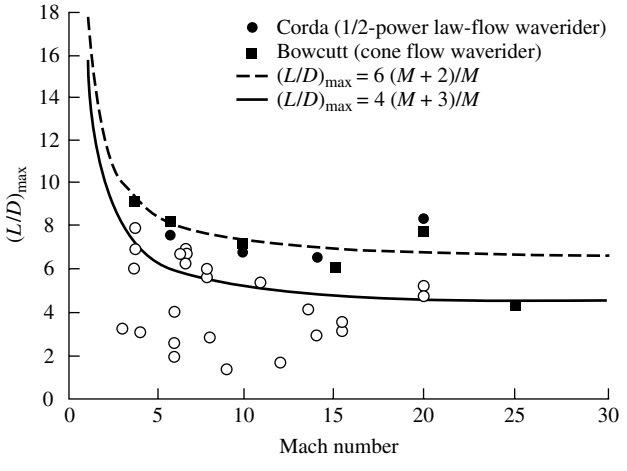


Figure 14.32 Maximum lift-to-drag ratio comparison for various hypersonic configurations.

Reference 113). The solid symbols pertain to the various optimized hypersonic waveriders discussed here. The solid squares are results for the waveriders based on conical generating flows as described below. The solid circles are results for another family of viscous optimized waveriders based on the shock wave and downstream streamsurfaces generated by a one-half power law ogive-shaped body, obtained by Corda and Anderson (Reference 114). From Figure 14.32 we see that the viscous optimized waveriders break the L/D barrier, that is, they give $(L/D)_{\max}$ values that lie above the Kuchemann curve. Indeed, the L/D variation of the viscous optimized waveriders is more closely given by

$$(L/D)_{\max} = \frac{6(M_\infty + 2)}{M_\infty}$$

This variation is shown as the dotted curve in Figure 14.32. The importance of the viscous optimized waveriders is established by the results shown in Figure 14.32. These results have been confirmed by various wind-tunnel tests. They are the reason for renewed interest in the waverider configuration as a hypersonic vehicle, particularly for sustained cruising in the atmosphere.

The physical aspects that define the hypersonic flow regime were discussed in Section 14.2. The influence of viscous interaction effects, high-temperature flows, and aerodynamic heating on waverider design is discussed at length in the second edition of Reference 55 (see pages 361–374, 409–413, and 644–646 of the second edition of Reference 55). Also, hypersonic vehicle design is sensitive to the location of the transition from laminar to turbulent flow, and the design of hypersonic waveriders is no exception. Numerical experiments carried out at $M_\infty = 10$ wherein the transition location was varied over a wide latitude, ranging from all-laminar flow on one hand, to almost all-turbulent flow on the other hand, with various cases in between, are discussed in References 55, 111, and 113. Although these physical phenomena have an effect on the optimized shape of viscous optimized hypersonic waveriders, the resulting values of $(L/D)_{\max}$ are not greatly changed.

Even with these real physical phenomena included in the optimization process, the viscous optimized hypersonic waverider remains a viable configuration for future hypersonic vehicle design. Indeed, the Air Force–sponsored and Boeing-designed X-51, shown in Figure 14.33, is a viscous-optimized waverider.



Figure 14.33 X51 waverider. (*Source: U.S. Air Force.*)

SCRAMjet-powered, and designed for flight at Mach 5 to 6, the X-51 will provide the technology for future atmospheric cruise missiles. At the time of writing, its first flight is scheduled for late 2009 or early 2010. If successful, it will be the first hypersonic waverider to achieve sustained flight in the atmosphere, and only the second SCRAMjet-powered hypersonic vehicle of any type to achieve sustained atmospheric flight, the first being the X-43 in 2004 (Figures 14.1 and 14.2).

14.10 SUMMARY

Only a few of the basic elements of hypersonic flow are presented here, with special emphasis on newtonian flow results. Useful information on hypersonic flows can be extracted from such results. We have derived the basic newtonian sine-squared law:

$$C_p = 2 \sin^2 \theta \tag{14.4}$$

and used this result to treat the case of a hypersonic flat plate in Section 14.4. We also obtained the limiting form of the oblique shock relations as $M_\infty \to \infty$, that is, the hypersonic shock relations. From these relations, we were able to examine the significance of newtonian theory more thoroughly, namely, Equation (14.4) becomes an exact relation for a hypersonic flow in the combined limit of $M_\infty \to \infty$ and $\gamma \to 1$. Finally, these hypersonic shock relations illustrate the existence of the Mach number independence principle.

14.11 PROBLEMS

14.1 Repeat Problem 9.13 using

a. Newtonian theory

b. Modified newtonian theory

Compare these results with those obtained from exact shock-expansion theory (Problem 9.13). From this comparison, what comments can you make about the accuracy of newtonian and modified newtonian theories at low supersonic Mach numbers?

14.2 Consider a flat plate at $\alpha = 20^\circ$ in a Mach 20 freestream. Using straight newtonian theory, calculate the lift- and wave-drag coefficients. Compare these results with exact shock-expansion theory.

14.3 Consider a hypersonic vehicle with a spherical nose flying at Mach 20 at a standard altitude of 150,000 ft, where the ambient temperature and pressure are 500°R and 3.06 lb/ft^2, respectively. At the point on the surface of the nose located 20° away from the stagnation point, estimate the: (*a*) pressure, (*b*) temperature, (*c*) Mach number, and (*d*) velocity of the flow.

PART 4

Viscous Flow

In Part 4, we deal with flows that are dominated by viscosity and thermal conduction—viscous flows. We will treat both incompressible and compressible viscous flows. ■

CHAPTER 15

Introduction to the Fundamental Principles and Equations of Viscous Flow

I do not see then, I admit, how one can explain the resistance of fluids by the theory in a satisfactory manner. It seems to me on the contrary that this theory, dealt with and studied with profound attention gives, at least in most cases, resistance absolutely zero: a singular paradox which I leave to geometricians to explain.

Jean LeRond d'Alembert, 1768

PREVIEW BOX

The real life of aerodynamics—that is what the present and remaining chapters are all about. Except for the few earlier sections on viscous flow, most of our previous discussions in this book have dealt with inviscid flows. Do not get the wrong impression; a large number of practical aerodynamic applications are appropriately treated by assuming inviscid flow, as we have already seen. Thank goodness for this, because inviscid flows are usually easier to analyze than viscous flows. But some aspects of aerodynamics are inherently viscous in nature, such as skin-friction drag, aerodynamic heating, and flow separation. To deal with these important aspects, we have to undertake the study of viscous flow, which is the subject of the remainder of this book.

The present chapter is all about the fundamental aspects of viscous flow. Here you will find new definitions, new concepts, and new equations, including the derivation of the Navier-Stokes equations, which are nothing more than the continuity, momentum, and energy equations for a viscous flow. Although some of the basic concepts of viscous flow were first introduced in Chapter 1, the present chapter goes far beyond that discussed in Chapter 1. As you read through this chapter, as well as the subsequent chapters, you will find a small amount of repetition of a few thoughts

introduced in earlier chapters. This is intentional. I want to make Part 4 of this book a stand-alone, almost self-contained presentation of viscous flow. Besides, a little bit of repetition never hurt anybody, and in the educational process it can help a lot. So jump into this chapter and immerse yourself in the ideas and thoughts of viscous flow. Come into the real world.

15.1 INTRODUCTION

In the above quotation, the "theory" referred to by d'Alembert is inviscid, incompressible flow theory; we have seen in Chapter 3 that such theory leads to a prediction of zero drag on a closed two-dimensional body—this is d'Alembert's paradox. In reality, there is always a finite drag on any body immersed in a moving fluid. Our earlier predictions of zero drag are a result of the inadequacy of the theory rather than some fluke of nature. With the exception of induced drag and supersonic wave drag, which can be obtained from inviscid theory, the calculation of all other forms of drag must explicitly take into account the presence of viscosity, which has not been included in our previous inviscid analyses. The purpose of the remaining chapters in this book is to discuss the basic aspects of viscous flows, thus "rounding out" our overall presentation of the fundamentals of aerodynamics. In so doing, we address the predictions of aerodynamic drag and aerodynamic heating. To help put our current discussion in perspective, return to the block diagram of flow categories given in Figure 1.45. All of our previous discussions have focused on blocks *D*, *E*, and *F*—inviscid, incompressible and compressible flows. Now, for the remaining six chapters, we move to the left branch in Figure 1.45, and deal with blocks *C*, *E*, and *F*—*viscous,* incompressible and compressible flows.

Our treatment of viscous flows will be intentionally brief—our purpose is to present enough of the fundamental concepts and equations to give you the flavor of viscous flows. A thorough presentation of viscous flow theory would double the size of this book (at the very least) and is clearly beyond our scope. A study of viscous flow is an essential part of any serious study of aerodynamics. Many books have been exclusively devoted to the presentation of viscous flows; References 42 and 43 are two good examples. You are encouraged to examine these references closely.

There have been sections on viscous flow topics earlier in this book, namely, Sections 1.11, 4.12, 9.10, 10.6, and 12.4. These are stand-alone sections dealing with viscous flow aspects pertinent to the chapters in which they appear. You, the reader, have had two choices: (1) to read these viscous flow sections in order to discover how friction has an impact on some of the ideal inviscid flows discussed in the main body of the chapter, or (2) to by-pass these sections in order to preserve the intellectual continuity of a study of inviscid flows that is, after all, the main thrust of Parts 2 and 3 of this book. Now, we are at Part 4 dealing exclusively with viscous flow. Readers who took the first choice will find some slight repetition in

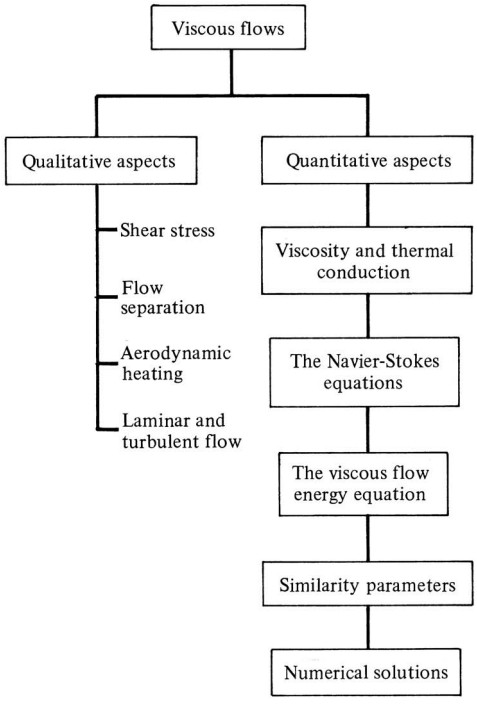


Figure 15.1 Road map for Chapter 15.

Part 4, but repetition is a good thing in learning a new subject. Readers who took the second choice will find Part 4 a totally self-contained discussion of viscous flow that does not depend on reading the earlier viscous flow sections; however, at appropriate stages of our discussions you will find direct references to specific earlier sections that fit nicely into the continuity of Part 4.

The road map for the present chapter is given in Figure 15.1. Our course is to first examine some qualitative aspects of viscous flows as shown on the left branch of Figure 15.1. Then we quantify some of these aspects as given on the right branch. In the process, we obtain the governing equations for a general viscous flow—in particular, the Navier-Stokes equations (the momentum equations) and the viscous flow energy equation. Finally, we examine a numerical solution to these equations.

15.2 QUALITATIVE ASPECTS OF VISCOUS FLOW

What is a viscous flow? *Answer:* A flow where the effects of viscosity, thermal conduction, and mass diffusion are important. The phenomenon of mass diffusion is important in a gas with gradients in its chemical species, for example, the flow of air over a surface through which helium is being injected or the chemically reacting flow through a jet engine or over a high-speed reentry body. In this book,

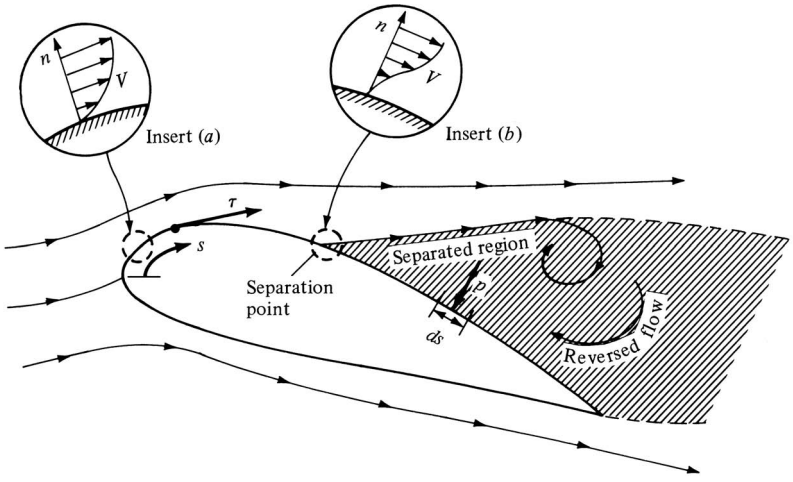


Figure 15.2 Effect of viscosity on a body in a moving fluid: shear stress and separated flow.

we are not concerned with the effects of diffusion, and therefore we treat a viscous flow as one where only viscosity and thermal conduction are important.

First, consider the influence of viscosity. Imagine two solid surfaces slipping over each other, such as this book being pushed across a table. Clearly, there will be a frictional force between these objects which will retard their relative motion. The same is true for the flow of a fluid over a solid surface; the influence of friction between the surface and the fluid adjacent to the surface acts to create a frictional force which retards the relative motion. This has an effect on both the surface and the fluid. The surface feels a "tugging" force in the direction of the flow, tangential to the surface. This tangential force per unit area is defined as the *shear stress* τ, first introduced in Section 1.5 and illustrated in Figure 15.2. As an equal and opposite reaction, the fluid adjacent to the surface feels a retarding force which decreases its local flow velocity, as shown in insert *a* of Figure 15.2. Indeed, the influence of friction is to create $V = 0$ right at the body surface—this is called the *no-slip* condition which dominates viscous flow. In any real continuum fluid flow over a solid surface, the flow velocity is zero at the surface. Just above the surface, the flow velocity is finite, but retarded, as shown in insert *a*. If n represents the coordinate normal to the surface, then in the region near the surface, $V = V(n)$, where $V = 0$ at $n = 0$, and V increases as n increases. The plot of V versus n as shown in insert *a* is called a *velocity profile*. Clearly, the region of flow near the surface has velocity gradients, $\partial V/\partial n$, which are due to the frictional force between the surface and the fluid.

In addition to the generation of shear stress, friction also plays another (but related) role in dictating the flow over the body in Figure 15.2. Consider a fluid element moving in the viscous flow near a surface, as sketched in Figure 15.3. Assume that the flow is in its earliest moments of being started. At the station s_1,

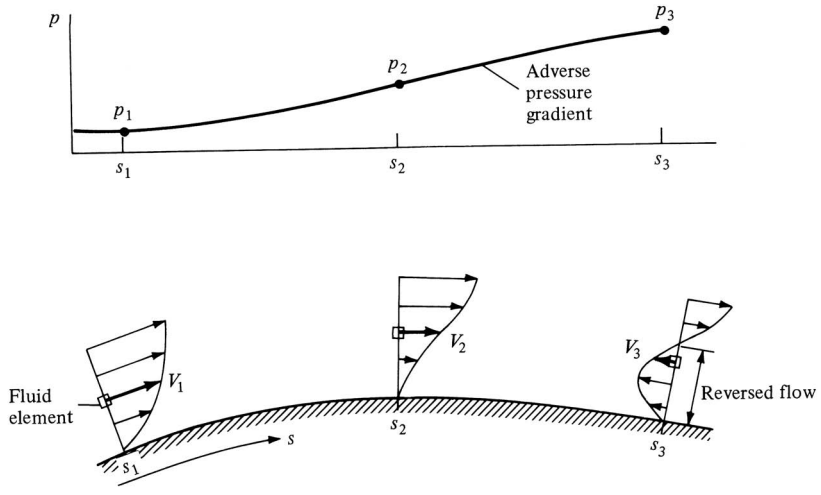


Figure 15.3 Separated flow induced by an adverse pressure gradient. This picture corresponds to the early evolution of the flow; once the flow separates from the surface between points 2 and 3, the fluid element shown at s_3 is in reality different from that shown at s_1 and s_2 because the primary flow moves away from the surface, as shown in Figure 15.2.

the velocity of the fluid element is V_1. Assume that the flow over the surface produces an increasing pressure distribution in the flow direction (i.e., assume $p_3 > p_2 > p_1$). Such a region of increasing pressure is called an *adverse pressure gradient*. Now follow the fluid element as it moves downstream. The motion of the element is already retarded by the effect of friction; in addition, it must work its way along the flow against an increasing pressure, which tends to further reduce its velocity. Consequently, at station 2 along the surface, its velocity V_2 is less than V_1. As the fluid element continues to move downstream, it may completely "run out of steam," come to a stop, and then, under the action of the adverse pressure gradient, actually reverse its direction and start moving back upstream. This "reversed flow" is illustrated at station s_3 in Figure 15.3, where the fluid element is now moving upstream at the velocity V_3. The picture shown in Figure 15.3 is meant to show the flow details very near the surface at the very initiation of the flow. In the bigger picture of this flow at later times shown in Figure 15.2, the consequence of such reversed-flow phenomena is to cause the flow to *separate from the surface* and create a large wake of recirculating flow downstream of the surface. The point of separation on the surface in Figure 15.2 occurs where $\partial V/\partial n = 0$ at the surface, as sketched in insert *b* of Figure 15.2. Beyond this point, reversed flow occurs. Therefore, in addition to the generation of shear stress, the influence of friction can cause the flow over a body to separate from the surface. When such separated flow occurs, the pressure distribution over the surface is greatly altered. The primary flow over the body in Figure 15.2 no longer sees the complete body shape; rather, it sees the body shape upstream

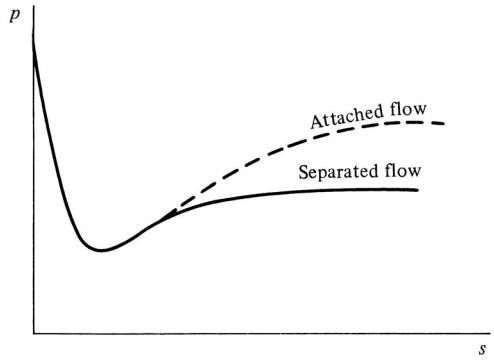


Figure 15.4 Schematic of the pressure distributions for attached and separated flow over the upper surface of the body illustrated in Figure 15.2.

of the separation point, but downstream of the separation point it sees a greatly deformed "effective body" due to the large separated region. The net effect is to create a pressure distribution over the actual body surface which results in an integrated force in the flow direction, that is, a drag. To see this more clearly, consider the pressure distribution over the upper surface of the body as sketched in Figure 15.4. If the flow were attached, the pressure over the downstream portion of the body would be given by the dashed curve. However, for separated flow, the pressure over the downstream portion of the body is smaller, given by the solid curve in Figure 15.4. Now return to Figure 15.2. Note that the pressure over the upper rearward surface contributes a force in the negative drag direction; that is, p acting over the element of surface ds shown in Figure 15.2 has a horizontal component in the upstream direction. If the flow were inviscid, subsonic, and attached and the body were two-dimensional, the forward-acting components of the pressure distribution shown in Figure 15.2 would exactly cancel the rearward-acting components due to the pressure distribution over other parts of the body such that the net, integrated pressure distribution would give zero drag. This would be d'Alembert's paradox discussed in Chapter 3. However, for the viscous, separated flow, we see that p is reduced in the separated region; hence, it can no longer fully cancel the pressure distribution over the remainder of the body. The net result is the production of drag; this is called the *pressure drag due to flow separation* and is denoted by D_p.

In summary, we see that the effects of viscosity are to produce two types of drag as follows:

D_f is the skin friction drag, that is, the component in the drag direction of the integral of the shear stress τ over the body.

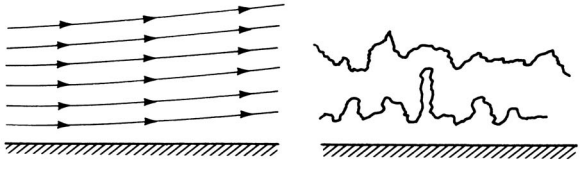
D_p is the pressure drag due to separation, that is, the component in the drag direction of the integral of the pressure distribution over the body.

D_p is sometimes called *form drag*. The sum $D_f + D_p$ is called the *profile drag* of a two-dimensional body. For a three-dimensional body such as a complete airplane, the sum $D_f + D_p$ is frequently called *parasite drag*. (See Reference 2 for a more extensive discussion of the classification of different drag contributions.)

The occurrence of separated flow over an aerodynamic body not only increases the drag but also results in a substantial loss of lift. Such separated flow is the cause of airfoil stall as discussed in Section 4.3. For these reasons, the study, understanding, and prediction of separated flow is an important aspect of viscous flow.

Let us turn our attention to the influence of thermal conduction—another overall physical characteristic of viscous flow in addition to friction. Again, let us draw an analogy from two solid bodies slipping over each other, such as the motion of this book over the top of a table. If we would press hard on the book, and vigorously rub it back and forth over the table, the cover of the book as well as the table top would soon become warm. Some of the energy we expend in pushing the book over the table will be dissipated by friction, and this shows up as a form of heating of the bodies. The same phenomenon occurs in the flow of a fluid over a body. The moving fluid has a certain amount of kinetic energy; in the process of flowing over a surface, the flow velocity is decreased by the influence of friction, as discussed earlier, and hence the kinetic energy is decreased. This lost kinetic energy reappears in the form of internal energy of the fluid, hence causing the temperature to rise. This phenomenon is called *viscous dissipation* within the fluid. In turn, when the fluid temperature increases, there is an overall temperature difference between the warmer fluid and the cooler body. We know from experience that heat is transferred from a warmer body to a cooler body; therefore, heat will be transferred from the warmer fluid to the cooler surface. This is the mechanism of *aerodynamic heating* of a body. Aerodynamic heating becomes more severe as the flow velocity increases, because more kinetic energy is dissipated by friction, and hence the overall temperature difference between the warm fluid and the cool surface increases. As discussed in Chapter 14, at hypersonic speeds, aerodynamic heating becomes a dominant aspect of the flow.

All the aspects discussed above—shear stress, flow separation, aerodynamic heating, etc.—are dominated by a single major question in viscous flow, namely, Is the flow laminar or turbulent? Consider the viscous flow over a surface as sketched in Figure 15.5. If the path lines of various fluid elements are smooth and



(*a*) Laminar flow (*b*) Turbulent flow

Figure 15.5 Path lines for laminar and turbulent flows.

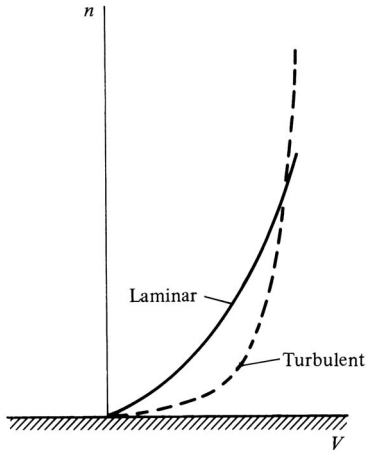


Figure 15.6 Schematic of velocity profiles for laminar and turbulent flows.

regular, as shown in Figure 15.5*a*, the flow is called *laminar flow*. In contrast, if the motion of a fluid element is very irregular and tortuous, as shown in Figure 15.5*b*, the flow is called *turbulent flow*. Because of the agitated motion in a turbulent flow, the higher-energy fluid elements from the outer regions of the flow are pumped close to the surface. Hence, the average flow velocity near a solid surface is larger for a turbulent flow in comparison with laminar flow. This comparison is shown in Figure 15.6, which gives velocity profiles for laminar and turbulent flow. Note that immediately above the surface, the turbulent flow velocities are much larger than the laminar values. If $(\partial V/\partial n)_{n=0}$ denotes the velocity gradient at the surface, we have

$$\left[\left(\frac{\partial V}{\partial n}\right)_{n=0}\right]_{\text{turbulent}} > \left[\left(\frac{\partial V}{\partial n}\right)_{n=0}\right]_{\text{laminar}}$$

Because of this difference, the frictional effects are more severe for a turbulent flow; both the shear stress and aerodynamic heating are larger for the turbulent flow in comparison with laminar flow. However, turbulent flow has a major redeeming value; because the energy of the fluid elements close to the surface is larger in a turbulent flow, a turbulent flow does not separate from the surface as readily as a laminar flow. If the flow over a body is turbulent, it is less likely to separate from the body surface, and if flow separation does occur, the separated region will be smaller. As a result, the pressure drag due to flow separation D_p will be smaller for turbulent flow.

This discussion points out one of the great compromises in aerodynamics. For the flow over a body, is laminar or turbulent flow preferable? There is no pat answer; it depends on the shape of the body. In general, if the body is slender, as sketched in Figure 15.7*a*, the friction drag D_f is much greater than D_p. For

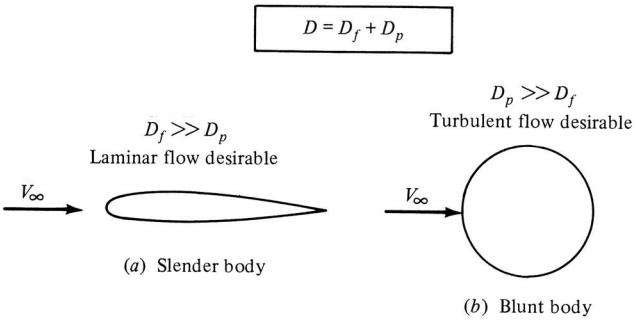


Figure 15.7 Drag on slender and blunt bodies.

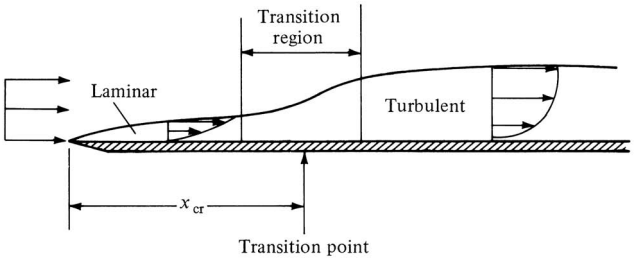


Figure 15.8 Transition from laminar to turbulent flow.

this case, because D_f is smaller for laminar than for turbulent flow, laminar flow is desirable for slender bodies. In contrast, if the body is blunt, as sketched in Figure 15.7*b*, D_p is much greater than D_f. For this case, because D_p is smaller for turbulent than for laminar flow, turbulent flow is desirable for blunt bodies. The above comments are not all-inclusive; they simply state general trends, and for any given body, the aerodynamic virtues of laminar versus turbulent flow must always be assessed.

Although, from the above discussion, laminar flow is preferable for some cases, and turbulent flow for other cases, in reality we have little control over what actually happens. Nature makes the ultimate decision as to whether a flow will be laminar or turbulent. There is a general principle in nature that a system, when left to itself, will always move toward its state of maximum disorder. To bring order to the system, we generally have to exert some work on the system or expend energy in some manner. (This analogy can be carried over to daily life; a room will soon become cluttered and disordered unless we exert some effort to keep it clean.) Since turbulent flow is much more "disordered" than laminar flow, nature will always favor the occurrence of turbulent flow. Indeed, in the vast majority of practical aerodynamic problems, turbulent flow is usually present.

Let us examine this phenomenon in more detail. Consider the viscous flow over a flat plate, as sketched in Figure 15.8. The flow immediately upstream of

the leading edge is uniform at the freestream velocity. However, downstream of the leading edge, the influence of friction will begin to retard the flow adjacent to the surface, and the extent of this retarded flow will grow higher above the plate as we move downstream, as shown in Figure 15.8. To begin with, the flow just downstream of the leading edge will be laminar. However, after a certain distance, instabilities will appear in the laminar flow; these instabilities rapidly grow, causing transition to turbulent flow. The transition from laminar to turbulent flow takes place over a finite region, as sketched in Figure 15.8. However, for purposes of analysis, we frequently model the transition region as a single point, called the *transition point,* upstream of which the flow is laminar and downstream of which the flow is turbulent. The distance from the leading edge to the transition point is denoted by x_{cr}. The value of x_{cr} depends on a whole host of phenomena. For example, some characteristics which encourage transition from laminar to turbulent flow, and hence reduce x_{cr}, are:

1. *Increased surface roughness.* Indeed, to promote turbulent flow over a body, rough grit can be placed on the surface near the leading edge to "trip" the laminar flow into turbulent flow. This is a frequently used technique in wind-tunnel testing. Also, the dimples on the surface of a golf ball are designed to encourage turbulent flow, thus reducing D_p. In contrast, in situations where we desire large regions of laminar flow, such as the flow over the NACA six-series laminar-flow airfoils, the surface should be as smooth as possible. The main reason why such airfoils do not produce in actual flight the large regions of laminar flow observed in the laboratory is that manufacturing irregularities and bug spots (believe it or not) roughen the surface and promote early transition to turbulent flow.
2. *Increased turbulence in the freestream.* This is particularly a problem in wind-tunnel testing; if two wind tunnels have different levels of freestream turbulence, then data generated in one tunnel are not repeatable in the other.
3. *Adverse pressure gradients.* In addition to causing flow-field separation as discussed earlier, an adverse pressure gradient strongly favors transition to turbulent flow. In contrast, strong favorable pressure gradients (where p decreases in the downstream direction) tend to preserve initially laminar flow.
4. *Heating of the fluid by the surface.* If the surface temperature is warmer than the adjacent fluid, such that heat is transferred to the fluid from the surface, the instabilities in the laminar flow will be amplified, thus favoring early transition. In contrast, a cold wall will tend to encourage laminar flow.

There are many other parameters which influence transition; see Reference 42 for a more extensive discussion. Among these are the similarity parameters of the flow, principally Mach number and Reynolds number. High values of M_∞ and low values of Re tend to encourage laminar flow; hence, for high-altitude hypersonic flight, laminar flow can be quite extensive. The Reynolds number itself is a dominant factor in transition to turbulent flow. Referring to Figure 15.8,

we define a *critical* Reynolds number, Re_{cr}, as

$$\mathrm{Re}_{cr} \equiv \frac{\rho_\infty V_\infty x_{cr}}{\mu_\infty}$$

The value of Re_{cr} for a given body under specified conditions is difficult to predict; indeed, the analysis of transition is still a very active area of modern aerodynamic research. As a rule of thumb in practical applications, we frequently take $\mathrm{Re}_{cr} \approx 500{,}000$; if the flow at a given x station is such that $\mathrm{Re} = \rho_\infty V_\infty x/\mu_\infty$ is considerably below 500,000, then the flow at that station is most likely laminar, and if the value of Re is much larger than 500,000, then the flow is most likely turbulent.

To obtain a better feeling for Re_{cr}, let us imagine that the flat plate in Figure 15.8 is a wind-tunnel model. Assume that we carry out an experiment under standard sea level conditions [$\rho_\infty = 1.23$ kg/m^3 and $\mu_\infty = 1.79 \times 10^{-5}$ kg/(m · s)] and *measure* x_{cr} for a certain freestream velocity; for example, say that $x_{cr} = 0.05$ m when $V_\infty = 120$ m/s. In turn, this measured value of x_{cr} determines the measured Re_{cr} as

$$\mathrm{Re}_{cr} = \frac{\rho_\infty V_\infty x_{cr}}{\mu_\infty} = \frac{1.23(120)(0.05)}{1.79 \times 10^{-5}} = 412{,}000$$

Hence, for the given flow conditions and the surface characteristics of the flat plate, transition will occur whenever the local Re exceeds 412,000. For example, if we double V_∞, that is, $V_\infty = 240$ m/s, then we will observe transition to occur at $x_{cr} = 0.05/2 = 0.025$ m, such that Re_{cr} remains the same value of 412,000.

This brings to an end our introductory qualitative discussion of viscous flow. The physical principles and trends discussed in this section are very important, and you should study them carefully and feel comfortable with them before progressing further.

15.3 VISCOSITY AND THERMAL CONDUCTION

The basic physical phenomena of viscosity and thermal conduction in a fluid are due to the transport of momentum and energy via random molecular motion. Each molecule in a fluid has momentum and energy, which it carries with it when it moves from one location to another in space before colliding with another molecule. The transport of molecular momentum gives rise to the macroscopic effect we call viscosity, and the transport of molecular energy gives rise to the macroscopic effect we call thermal conduction. This is why viscosity and thermal conduction are labeled as *transport phenomena*. A study of these transport phenomena at the molecular level is part of kinetic theory, which is beyond the scope of this book. Instead, in this section we simply state the macroscopic results of such molecular motion.

Consider the flow sketched in Figure 15.9. For simplicity, we consider a one-dimensional shear flow, that is, a flow with horizontal streamlines in the x direction but with gradients in the y direction of velocity, $\partial u/\partial y$, and temperature, $\partial T/\partial y$.

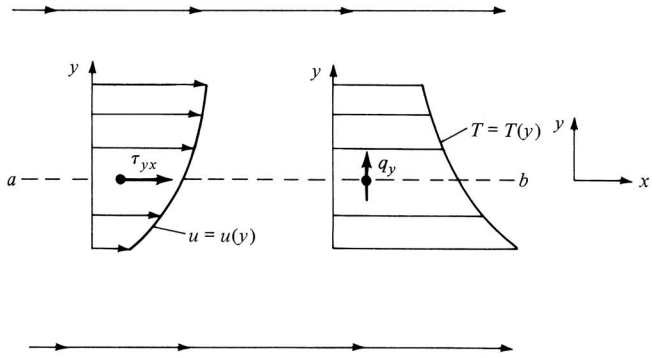


Figure 15.9 Relationship of shear stress and thermal conduction to velocity and temperature gradients, respectively.

Consider a plane ab perpendicular to the y axis, as shown in Figure 15.9. The shear stress exerted on plane ab by the flow is denoted by τ_{yx} and is proportional to the velocity gradient in the y direction, $\tau_{yx} \propto \partial u/\partial y$. The constant of proportionality is defined as the *viscosity coefficient* μ. Hence,

$$\tau_{yx} = \mu \frac{\partial u}{\partial y} \tag{15.1}$$

The subscripts on τ_{yx} denote that the shear stress is acting in the x direction and is being exerted on a plane perpendicular to the y axis. The velocity gradient $\partial u/\partial y$ is also taken perpendicular to this plane (i.e., in the y direction). The dimensions of μ are mass/length $\times$ time, as originally stated in Section 1.7 and as can be seen from Equation (15.1). In addition, the time rate of heat conducted per unit area across plane ab in Figure 15.9 is denoted by $\dot{q}_y$ and is proportional to the temperature gradient in the y direction, $\dot{q}_y \propto \partial T/\partial y$. The constant of proportionality is defined as the *thermal conductivity* k. Hence,

$$\dot{q}_y = -k \frac{\partial T}{\partial y} \tag{15.2}$$

where the minus sign accounts for the fact that the heat is transferred from a region of high temperature to a region of lower temperature; that is, $\dot{q}_y$ is in the opposite direction of the temperature gradient. The dimensions of k are mass $\times$ length/($s^2 \cdot K$), which can be obtained from Equation (15.2) keeping in mind that $\dot{q}_y$ is energy per second per unit area.

Both μ and k are physical properties of the fluid and, for most normal situations, are functions of temperature only. A conventional relation for the temperature variation of μ for air is given by Sutherland's law,

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0}\right)^{3/2} \frac{T_0 + 110}{T + 110} \tag{15.3}$$

where T is in kelvin and μ_0 is a reference viscosity at a reference temperature, T_0. For example, if we choose reference conditions to be standard sea level values, then $\mu_0 = 1.7894 \times 10^{-5}$ kg/(m · s) and $T_0 = 288.16$ K. The temperature variation of k is analogous to Equation (15.3) because the results of elementary kinetic theory show that $k \propto \mu c_p$; for air at standard conditions,

$$k = 1.45\mu c_p \tag{15.4}$$

where $c_p = 1000$ J/(kg · K).

Equations (15.3) and (15.4) are only approximate and do not hold at high temperatures. They are given here as representative expressions which are handy to use. For any detailed viscous flow calculation, you should consult the published literature for more precise values of μ and k.

In order to simplify our introduction of the relation between shear stress and viscosity, we considered the case of a one-dimensional shear flow in Figure 15.9. In this picture, the y and z components of velocity, v and w, respectively, are zero. However, in a general three-dimensional flow, u, v, and w are finite, and this requires a generalization of our treatment of stress in the fluid. Consider the fluid element sketched in Figure 15.10. In a three-dimensional flow, each face of the fluid element experiences both tangential and normal stresses. For example, on face *abcd*, τ_{xy} and τ_{xz} are the tangential stresses, and τ_{xx} is the normal stress. As before, the nomenclature τ_{ij} denotes a stress in the j direction exerted on a plane perpendicular to the i axis. Similarly, on face *abfe*, we have the tangential stresses τ_{yx} and τ_{yz}, and the normal stress τ_{yy}. On face *adge*, we have the tangential stresses τ_{zx} and τ_{zy}, and the normal stress τ_{zz}. Now recall the discussion in the last part of Section 2.12 concerning the *strain* of a fluid element, that is, the change in the angle κ shown in Figure 2.33. What is the force which causes this deformation shown in Figure 2.33? Returning to Figure 15.10, we have to say that the strain

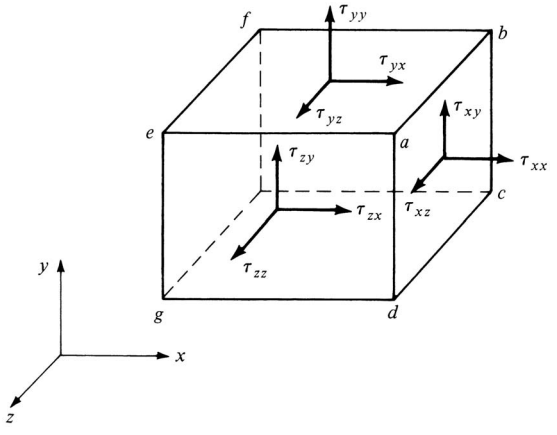


Figure 15.10 Shear and normal stresses caused by viscous action on a fluid element.

is caused by the tangential shear stress. However, in contrast to solid mechanics where stress is proportional to strain, in fluid mechanics the stress is proportional to the *time rate of strain*. The time rate of strain in the xy plane was given in Section 2.12 as Equation (2.135a):

$$\varepsilon_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \tag{2.135a}$$

Examining Figure 15.10, the strain in the xy plane must be carried out by τ_{xy} and τ_{yx}. Moreover, we assume that moments on the fluid element in Figure 15.10 are zero; hence, $\tau_{xy} = \tau_{yx}$. Finally, from the above, we know that $\tau_{xy} = \tau_{yx} \propto \varepsilon_{xy}$. The proportionality constant is the viscosity coefficient μ. Hence, from Equation (2.135a), we have

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \tag{15.5}$$

which is a generalization of Equation (15.1), extended to the case of multidimensional flow. For the shear stresses in the other planes, Equations (2.135b and c) yield

$$\tau_{yz} = \tau_{zy} = \mu \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) \tag{15.6}$$

and

$$\tau_{zx} = \tau_{xz} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \tag{15.7}$$

The normal stresses τ_{xx}, τ_{yy}, and τ_{zz} shown in Figure 15.10 may at first seem strange. In our previous treatments of inviscid flow, the only force normal to a surface in a fluid is the pressure force. However, if the gradients in velocity $\partial u/\partial x$, $\partial v/\partial y$, and $\partial w/\partial z$ are *extremely large* on the faces of the fluid element, there can be a meaningful viscous-induced normal force on each face which acts *in addition to* the pressure. These normal stresses act to compress or expand the fluid element, hence changing its volume. Recall from Section 2.12 that the derivatives $\partial u/\partial x$, $\partial v/\partial y$, and $\partial w/\partial z$ are related to the dilatation of a fluid element, that is, to $\nabla \cdot \mathbf{V}$. Hence, the normal stresses should in turn be related to these derivatives. Indeed, it can be shown that

$$\tau_{xx} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial u}{\partial x} \tag{15.8}$$

$$\tau_{yy} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial v}{\partial y} \tag{15.9}$$

$$\tau_{zz} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial w}{\partial z} \tag{15.10}$$

In Equations (15.8) to (15.10), λ is called the *bulk viscosity coefficient,* sometimes identified as the second viscosity coefficient. In 1845, the Englishman George Stokes hypothesized that

$$\lambda = -\tfrac{2}{3}\mu \tag{15.11}$$

To this day, the correct expression for the bulk viscosity is still somewhat controversial, and so we continue to use the above expression given by Stokes. Once again, the normal stresses are important only where the derivatives $\partial u/\partial x$, $\partial v/\partial y$, and $\partial w/\partial z$ are very large. For most practical flow problems, τ_{xx}, τ_{yy}, and τ_{zz} are small, and hence the uncertainty regarding λ is essentially an academic question. An example where the normal stress is important is inside the internal structure of a shock wave. Recall that, in real life, shock waves have a finite but small thickness. If we consider a normal shock wave across which large changes in velocity occur over a small distance (typically 10^{-5} cm), then clearly $\partial u/\partial x$ will be very large, and τ_{xx} becomes important inside the shock wave.

To this point in our discussion, the transport coefficients μ and k have been considered molecular phenomena, involving the transport of momentum and energy by random molecular motion. This molecular picture prevails in a laminar flow. The values of μ and k are physical properties of the fluid; that is, their values for different gases can be found in standard reference sources, such as the *Handbook of Chemistry and Physics* (The Chemical Rubber Co.). In contrast, for a turbulent flow the transport of momentum and energy can also take place by random motion of large turbulent eddies, or globs of fluid. This turbulent transport gives rise to effective values of viscosity and thermal conductivity defined as *eddy viscosity* ε and *eddy thermal conductivity* κ, respectively. (Please do not confuse this use of the symbols ε and κ with the time rate of strain and strain itself, as used earlier.) These turbulent transport coefficients ε and κ can be much larger (typically 10 to 100 times larger) than the respective molecular values μ and k. Moreover, ε and κ predominantly depend on characteristics of the flow field, such as velocity gradients; they are not just a molecular property of the fluid such as μ and k. The proper calculation of ε and κ for a given flow has remained a state-of-the-art research question for the past 80 years; indeed, the attempt to model the complexities of turbulence by defining an eddy viscosity and thermal conductivity is even questionable. The details and basic understanding of turbulence remain one of the greatest unsolved problems in physics today. For our purpose here, we simply adopt the ideas of eddy viscosity and eddy thermal conductivity, and for the transport of momentum and energy in a turbulent flow, we replace μ and k in Equations (15.1) to (15.10) by the combination $\mu + \varepsilon$ and $k + \kappa$; that is,

$$\tau_{yx} = (\mu + \varepsilon)\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$$

$$\dot{q}_y = -(k + \kappa)\frac{\partial T}{\partial y}$$

An example of the calculation of ε and κ is as follows. In 1925, Prandtl suggested that

$$\varepsilon = \rho l^2 \left|\frac{\partial u}{\partial y}\right| \tag{15.12}$$

for a flow where the dominant velocity gradient is in the y direction. In Equation (15.12), l is called the *mixing length,* which is different for different applications; it is an empirical constant which must be obtained from experiment. Indeed, *all* turbulence models require the input of empirical data; no self-contained purely theoretical turbulence model exists today. Prandtl's mixing length theory, embodied in Equation (15.12), is a simple relation which appears to be adequate for a number of engineering problems. For these reasons, the mixing length model for ε has been used extensively since 1925. In regard to κ, a relation similar to Equation (15.4) can be assumed (using 1.0 for the constant); that is,

$$\kappa = \varepsilon c_p \tag{15.13}$$

The comments on eddy viscosity and thermal conductivity are purely introductory. The modern aerodynamicist has a whole stable of turbulence models to choose from, and before tackling the analysis of a turbulent flow, you should be familiar with the modern approaches described in such books as References 42 to 45.

15.4 THE NAVIER-STOKES EQUATIONS

In Chapter 2, Newton's second law was applied to obtain the fluid-flow momentum equation in both integral and differential forms. In particular, recall Equations (2.13*a* to *c*), where the influence of viscous forces was expressed simply by the generic terms $(\mathcal{F}_x)_{\text{viscous}}$, $(\mathcal{F}_y)_{\text{viscous}}$, and $(\mathcal{F}_z)_{\text{viscous}}$. The purpose of this section is to obtain the analogous forms of Equations (2.13*a* to *c*) where the viscous forces are expressed explicitly in terms of the appropriate flow-field variables. The resulting equations are called the *Navier-Stokes equations*—probably the most pivotal equations in all of theoretical fluid dynamics.

In Section 2.3, we discussed the philosophy behind the derivation of the governing equations, namely, certain physical principles are applied to a suitable *model* of the fluid flow. Moreover, we saw that such a model could be either a finite control volume (moving or fixed in space) or an infinitesimally small element (moving or fixed in space). In Chapter 2, we chose the fixed, finite control volume for our model and obtained integral forms of the continuity, momentum, and energy equations directly from this model. Then, indirectly, we went on to extract partial differential equations from the integral forms. Before progressing further, it would be wise for you to review these matters from Chapter 2.

For the sake of variety, let us not use the fixed, finite control volume employed in Chapter 2; rather, in this section, let us adopt an infinitesimally small moving fluid element of fixed mass as our model of the flow, as sketched in Figure 15.11. To this model let us apply Newton's second law in the form $\mathbf{F} = m\mathbf{a}$. Moreover, for the time being consider only the x component of Newton's second law:

$$F_x = ma_x \tag{15.14}$$

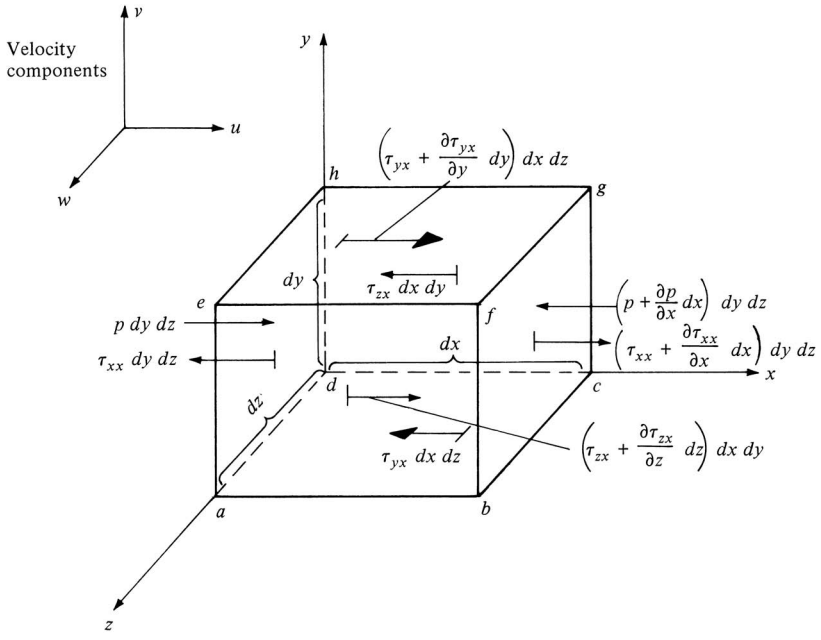


Figure 15.11 Infinitesimally small, moving fluid element. Only the forces in the x direction are shown.

In Equation (15.14), F_x is the sum of all the body and surface forces acting on the fluid element in the x direction. Let us ignore body forces; hence, the net force acting on the element in Figure 15.11 is simply due to the pressure and viscous stress distributions over the surface of the element. For example, on face *abcd*, the only force in the x direction is that due to shear stress, $\tau_{yx}\, dx\, dz$. Face *efgh* is a distance dy above face *abcd*; hence, the shear force in the x direction on face *efgh* is $[\tau_{yx} + (\partial \tau_{yx}/\partial y)\, dy]\, dx\, dz$. Note the directions of the shear stress on faces *abcd* and *efgh*; on the bottom face, τ_{yx} is to the left (the negative x direction), whereas on the top face, $\tau_{yx} + (\partial \tau_{yx}/\partial y)\, dy$ is to the right (the positive x direction). These directions are due to the convention that positive increases in all three components of velocity, u, v, and w, occur in the positive directions of the axes. For example, in Figure 15.11, u increases in the positive y direction. Therefore, concentrating on face *efgh*, u is higher just above the face than on the face; this causes a "tugging" action which tries to pull the fluid element in the positive x direction (to the right) as shown in Figure 15.11. In turn, concentrating on face *abcd*, u is lower just beneath the face than on the face; this causes a retarding or dragging action on the fluid element, which acts in the negative x direction (to the left), as shown in Figure 15.11. The directions of all the other viscous stresses shown in Figure 15.11, including τ_{xx}, can be justified in a like fashion. Specifically, on face *dcgh*, τ_{zx} acts in the negative x direction, whereas on face *abfe*, $\tau_{zx} + (\partial \tau_{zx}/\partial z)\, dz$ acts in the positive x direction. On face *adhe*,

which is perpendicular to the x axis, the only forces in the x direction are the pressure force $p\,dy\,dz$, which always acts in the direction *into* the fluid element, and $\tau_{xx}\,dy\,dz$, which is in the negative x direction. In Figure 15.11, the reason why τ_{xx} on face *adhe* is to the left hinges on the convention mentioned earlier for the direction of increasing velocity. Here, by convention, a positive increase in u takes place in the positive x direction. Hence, the value of u just to the left of face *adhe* is smaller than the value of u on the face itself. As a result, the viscous action of the normal stress acts as a "suction" on face *adhe*; that is, there is a dragging action toward the left that wants to retard the motion of the fluid element. In contrast, on face *bcgf*, the pressure force $[p + (\partial p/\partial x)\,dx]\,dy\,dz$ presses inward on the fluid element (in the negative x direction), and because the value of u just to the right of face *bcgf* is larger than the value of u on the face, there is a "suction" due to the viscous normal stress which tries to pull the element to the right (in the positive x direction) with a force equal to $[\tau_{xx} + (\partial \tau_{xx}/\partial x)\,dx]\,dy\,dz$.

Return to Equation (15.14). Examining Figure 15.11 in light of our previous discussion, we can write for the net force in the x direction acting on the fluid element:

$$F_x = \left[p - \left(p + \frac{\partial p}{\partial x}dx\right)\right] dy\,dz + \left[\left(\tau_{xx} + \frac{\partial \tau_{xx}}{\partial x}dx\right) - \tau_{xx}\right] dy\,dz$$
$$+ \left[\left(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y}dy\right) - \tau_{yx}\right] dx\,dz + \left[\left(\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z}dz\right) - \tau_{zx}\right] dx\,dy$$

or

$$F_x = \left(-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right) dx\,dy\,dz \tag{15.15}$$

Equation (15.15) represents the left-hand side of Equation (15.14). Considering the right-hand side of Equation (15.14), recall that the mass of the fluid element is fixed and is equal to

$$m = \rho\,dx\,dy\,dz \tag{15.16}$$

Also, recall that the acceleration of the fluid element is the time rate of change of its velocity. Hence, the component of acceleration in the x direction, denoted by a_x, is simply the time rate of change of u; since we are following a moving fluid element, this time rate of change is given by the *substantial derivative* (see Section 2.9 for a review of the meaning of substantial derivative). Thus,

$$a_x = \frac{Du}{Dt} \tag{15.17}$$

Combining Equations (15.14) to (15.17), we obtain

$$\boxed{\rho\frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}} \tag{15.18a}$$

which is the x component of the momentum equation for a viscous flow. In a similar fashion, the y and z components can be obtained as

$$\boxed{\rho\frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z}} \tag{15.18b}$$

$$\boxed{\rho\frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z}} \tag{15.18c}$$

Equations (15.18a to c) are the momentum equations in the x, y, and z directions, respectively. They are scalar equations and are called the *Navier-Stokes* equations in honor of two men—the Frenchman M. Navier and the Englishman G. Stokes—who independently obtained the equations in the first half of the nineteenth century.

With the expressions for $\tau_{xy} = \tau_{yx}$, $\tau_{yz} = \tau_{zy}$, $\tau_{zx} = \tau_{xz}$, τ_{xx}, τ_{yy}, and τ_{zz} from Equations (15.5) to (15.10), the Navier-Stokes equations, Equations (15.18a to c), can be written as

$$\boxed{\begin{aligned}\rho\frac{\partial u}{\partial t} + \rho u\frac{\partial u}{\partial x} + \rho v\frac{\partial u}{\partial y} + \rho w\frac{\partial u}{\partial z} &= -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x}\left(\lambda\nabla\cdot\mathbf{V} + 2\mu\frac{\partial u}{\partial x}\right)\\ &+ \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)\right]\end{aligned}} \tag{15.19a}$$

$$\boxed{\begin{aligned}\rho\frac{\partial v}{\partial t} + \rho u\frac{\partial v}{\partial x} + \rho v\frac{\partial v}{\partial y} + \rho w\frac{\partial v}{\partial z} &= -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)\right]\\ &+ \frac{\partial}{\partial y}\left(\lambda\nabla\cdot\mathbf{V} + 2\mu\frac{\partial v}{\partial y}\right) + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)\right]\end{aligned}} \tag{15.19b}$$

$$\boxed{\begin{aligned}\rho\frac{\partial w}{\partial t} + \rho u\frac{\partial w}{\partial x} + \rho v\frac{\partial w}{\partial y} + \rho w\frac{\partial w}{\partial z} &= -\frac{\partial p}{\partial z} + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)\right]\\ &+ \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)\right] + \frac{\partial}{\partial z}\left(\lambda\nabla\cdot\mathbf{V} + 2\mu\frac{\partial w}{\partial z}\right)\end{aligned}} \tag{15.19c}$$

Equations (15.19a to c) represent the complete Navier-Stokes equations for an unsteady, compressible, three-dimensional viscous flow. To analyze incompressible viscous flow, Equations (15.19a to c) and the continuity equation [say, Equation (2.52)] are sufficient. However, for a compressible flow, we need an additional equation, namely, the energy equation to be discussed in the next section.

In the above form, the Navier-Stokes equations are suitable for the analysis of laminar flow. For a turbulent flow, the flow variables in Equations (15.19*a* to *c*) can be assumed to be time-mean values over the turbulent fluctuations, and μ can be replaced by $\mu + \varepsilon$, as discussed in Section 15.3. For more details, see References 42 and 43.

15.5 THE VISCOUS FLOW ENERGY EQUATION

The energy equation was derived in Section 2.7, where the first law of thermodynamics was applied to a finite control volume fixed in space. The resulting integral form of the energy equation was given by Equation (2.95), and differential forms were obtained in Equations (2.96) and (2.114). In these equations, the influence of viscous effects was expressed generically by such terms as $\dot{Q}'_{\text{viscous}}$ and $\dot{W}'_{\text{viscous}}$. It is recommended that you review Section 2.7 before progressing further.

In the present section, we derive the energy equation for a viscous flow using as our model an infinitesimal moving fluid element. This will be in keeping with our derivation of the Navier-Stokes equation in Section 15.4, where the infinitesimal element was shown in Figure 15.11. In the process, we obtain explicit expressions for $\dot{Q}'_{\text{viscous}}$ and $\dot{W}'_{\text{viscous}}$ in terms of the flow-field variables. That is, we once again derive Equation (2.114), except the viscous terms are now displayed in detail.

Consider again the moving fluid element shown in Figure 15.11. To this element, apply the first law of thermodynamics, which states

	Rate of change of energy inside fluid element	=	net flux of heat into element	+	rate of work done on element due to pressure and stress forces on surface	
or	A	=	B	+	C	(15.20)

where A, B, and C denote the respective terms above.

Let us first evaluate C; that is, let us obtain an expression for the rate of work done on the moving fluid element due to the pressure and stress forces on the surface of the element. (Note that we are neglecting body forces in this derivation.) These surface forces are illustrated in Figure 15.11, which for simplicity shows only the forces in the x direction. Recall from Section 2.7 that the rate of doing work by a force exerted on a moving body is equal to the product of the force and the component of velocity in the direction of the force. Hence, the rate of work done on the moving fluid element by the forces in the x direction shown in Figure 15.11 is simply the x component of velocity u multiplied by the forces; for example, on face *abcd* the rate of work done by $\tau_{yx}\,dx\,dz$ is $u\tau_{yx}\,dx\,dz$, with similar expressions for the other faces. To emphasize these energy considerations, the moving fluid element is redrawn in Figure 15.12, where the rate of work done on each face by forces in the x direction is shown explicitly. Study this figure

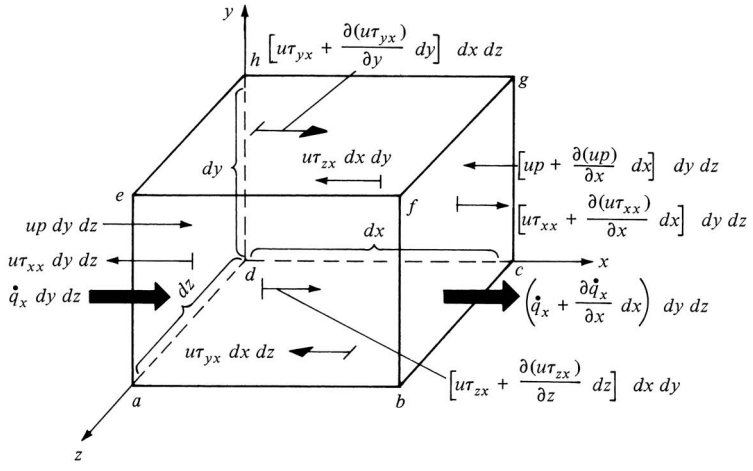


Figure 15.12 Energy fluxes associated with an infinitesimally small, moving fluid element. For simplicity, only the fluxes in the x direction are shown.

carefully, referring frequently to its companion in Figure 15.11, until you feel comfortable with the work terms given in each face. To obtain the *net* rate of work done on the fluid element by the forces in the x direction, note that forces in the positive x direction do positive work and that forces in the negative x direction do negative work. Hence, comparing the pressure forces on faces *adhe* and *bcgf* in Figure 15.12, the net rate of work done by pressure in the x direction is

$$\left[up - \left(up + \frac{\partial(up)}{\partial x}dx\right)\right] dy\,dz = -\frac{\partial(up)}{\partial x}dx\,dy\,dz$$

Similarly, the net rate of work done by the shear stresses in the x direction on faces *abcd* and *efgh* is

$$\left[\left(u\tau_{yx} + \frac{\partial(u\tau_{yx})}{\partial y}dy\right) - u\tau_{yx}\right] dx\,dz = \frac{\partial(u\tau_{yx})}{\partial y}dx\,dy\,dz$$

Considering all the forces shown in Figure 15.12, the net rate of work done on the moving fluid element is simply

$$\left[-\frac{\partial(up)}{\partial x} + \frac{\partial(u\tau_{xx})}{\partial x} + \frac{\partial(u\tau_{yx})}{\partial y} + \frac{\partial(u\tau_{zx})}{\partial z}\right] dx\,dy\,dz$$

The above expression considers only forces in the x direction. When the forces in the y and z directions are also included, similar expressions are obtained (draw some pictures and obtain these expressions yourself). In total, the net rate of work done on the moving fluid element is the sum of all contributions in the x, y, and

z directions; this is denoted by C in Equation (15.20) and is given by

$$C = \left[-\left(\frac{\partial(up)}{\partial x} + \frac{\partial(vp)}{\partial y} + \frac{\partial(wp)}{\partial z} \right) + \frac{\partial(u\tau_{xx})}{\partial x} + \frac{\partial(u\tau_{yx})}{\partial y} + \frac{\partial(u\tau_{zx})}{\partial z} + \frac{\partial(v\tau_{xy})}{\partial x} + \frac{\partial(v\tau_{yy})}{\partial y} + \frac{\partial(v\tau_{zy})}{\partial z} + \frac{\partial(w\tau_{xz})}{\partial x} + \frac{\partial(w\tau_{yz})}{\partial y} + \frac{\partial(w\tau_{zz})}{\partial z} \right] dx\, dy\, dz \tag{15.21}$$

Note in Equation (15.21) that the term in large parentheses is simply $\nabla \cdot p\mathbf{V}$.

Let us turn our attention to B in Equation (15.20), that is, the net flux of heat into the element. This heat flux is due to (1) volumetric heating such as absorption or emission of radiation and (2) heat transfer across the surface due to temperature gradients (i.e., thermal conduction). Let us treat the volumetric heating the same as was done in Section 2.7; that is, define $\dot{q}$ as the rate of volumetric heat addition per unit mass. Noting that the mass of the moving fluid element in Figure 15.12 is $\rho\, dx\, dy\, dz$, we obtain

$$\text{Volumetric heating of element} = \rho\dot{q}\, dx\, dy\, dz \tag{15.22}$$

Thermal conduction was discussed in Section 15.3. In Figure 15.12, the heat transferred by thermal conduction into the moving fluid element across face *adhe* is $\dot{q}_x\, dy\, dz$, and the heat transferred out of the element across face *bcgf* is $[\dot{q}_x + (\partial\dot{q}_x/\partial x)\, dx]\, dy\, dz$. Thus, the net heat transferred in the x direction into the fluid element by thermal conduction is

$$\left[\dot{q}_x - \left(\dot{q}_x + \frac{\partial \dot{q}_x}{\partial x} dx \right) \right] dy\, dz = -\frac{\partial \dot{q}_x}{\partial x} dx\, dy\, dz$$

Taking into account heat transfer in the y and z directions across the other faces in Figure 15.12, we obtain

$$\begin{matrix}\text{Heating of fluid element} \\ \text{by thermal conduction}\end{matrix} = -\left(\frac{\partial \dot{q}_x}{\partial x} + \frac{\partial \dot{q}_y}{\partial y} + \frac{\partial \dot{q}_z}{\partial z} \right) dx\, dy\, dz \tag{15.23}$$

The term B in Equation (15.20) is the sum of Equations (15.22) and (15.23). Also, recalling that thermal conduction is proportional to temperature gradient, as exemplified by Equation (15.2), we have

$$B = \left[\rho\dot{q} + \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z} \right) \right] dx\, dy\, dz \tag{15.24}$$

Finally, the term A in Equation (15.20) denotes the time rate of change of energy of the fluid element. In Section 2.7, we stated that the energy of a moving fluid per unit mass is the sum of the internal and kinetic energies, for example, $e + V^2/2$. Since we are following a moving fluid element, the time rate of change of energy per unit mass is given by the substantial derivative (see Section 2.9).

Since the mass of the fluid element is $\rho\, dx\, dy\, dz$, we have

$$A = \rho \frac{D}{Dt}\left(e + \frac{V^2}{2}\right) dx\, dy\, dz \tag{15.25}$$

The final form of the energy equation for a viscous flow is obtained by substituting Equations (15.21), (15.24), and (15.25) into Equation (15.20), obtaining

$$\begin{aligned}\rho \frac{D(e + V^2/2)}{Dt} = {} & \rho \dot{q} + \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right) \\ & + \frac{\partial}{\partial z}\left(k \frac{\partial T}{\partial z}\right) - \nabla \cdot p\mathbf{V} + \frac{\partial (u\tau_{xx})}{\partial x} + \frac{\partial (u\tau_{yx})}{\partial y} \\ & + \frac{\partial (u\tau_{zx})}{\partial z} + \frac{\partial (v\tau_{xy})}{\partial x} + \frac{\partial (v\tau_{yy})}{\partial y} + \frac{\partial (v\tau_{zy})}{\partial z} \\ & + \frac{\partial (w\tau_{xz})}{\partial x} + \frac{\partial (w\tau_{yz})}{\partial y} + \frac{\partial (w\tau_{zz})}{\partial z}\end{aligned} \tag{15.26}$$

Equation (15.26) is the general energy equation for unsteady, compressible, three-dimensional, viscous flow. Compare Equation (15.26) with Equation (2.105); the viscous terms are now explicitly spelled out in Equation (15.26). [Note that the body force term in Equation (15.26) has been neglected.] Moreover, the normal and shear stresses that appear in Equation (15.26) can be expressed in terms of the velocity field via Equations (15.5) to (15.10). This substitution will not be made here because the resulting equation would simply occupy too much space.

Reflect on the viscous flow equations obtained in this chapter—the Navier-Stokes equations given by Equations (15.19*a* to *c*) and the energy equation given by Equation (15.26). These equations are obviously more complex than the inviscid flow equations dealt with in previous chapters. This underscores the fact that viscous flows are inherently more difficult to analyze than inviscid flows. This is why, in the study of aerodynamics, the student is first introduced to the concepts associated with inviscid flow. Moreover, this is why we attempt to model a number of practical aerodynamic problems in real life as inviscid flows—simply to allow a reasonable analysis of such flows. However, there are many aerodynamic problems, especially those involving the prediction of drag and flow separation, which must take into account viscous effects. For the analysis of such problems, the basic equations derived in this chapter form a starting point.

Question: What is the form of the continuity equation for a viscous flow? To answer this question, review the derivation of the continuity equation in Section 2.4. You will note that the consideration of the viscous or inviscid nature of the flow never enters the derivation—the continuity equation is simply a statement that mass is conserved, which is independent of whether the flow is viscous or inviscid. Hence, Equation (2.52) holds in general.

15.6 SIMILARITY PARAMETERS

In Section 1.7, we introduced the concept of dimensional analysis, from which sprung the similarity parameters necessary to ensure the dynamic similarity between two or more different flows (see Section 1.8). In the present section, we revisit the governing similarity parameters, but cast them in a slightly different light.

Consider a steady two-dimensional, viscous, compressible flow. The x-momentum equation for such a flow is given by Equation (15.19a), which for the present case reduces to

$$\rho u\frac{\partial u}{\partial x}+\rho v\frac{\partial u}{\partial y}=-\frac{\partial p}{\partial x}+\frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\right] \tag{15.27}$$

In Equation (15.27), ρ, u, p, etc., are the actual dimensional variables, say, $[\rho]=$ kg/m^3, etc. Let us introduce the following dimensionless variables:

$$\rho'=\frac{\rho}{\rho_\infty}\quad u'=\frac{u}{V_\infty}\quad v'=\frac{v}{V_\infty}\quad p'=\frac{p}{p_\infty}$$
$$\mu'=\frac{\mu}{\mu_\infty}\quad x'=\frac{x}{c}\quad y'=\frac{y}{c}$$

where ρ_∞, V_∞, p_∞, and μ_∞ are reference values (say, e.g., freestream values) and c is a reference length (say, the chord of an airfoil). In terms of these dimensionless variables, Equation (15.27) becomes

$$\rho' u'\frac{\partial u'}{\partial x'}+\rho' v'\frac{\partial u'}{\partial y'}=-\left(\frac{p_\infty}{\rho_\infty V_\infty^2}\right)\frac{\partial p'}{\partial x'}+\left(\frac{\mu_\infty}{\rho_\infty V_\infty c}\right)\frac{\partial}{\partial y'}\left[\mu'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right] \tag{15.28}$$

Noting that

$$\frac{p_\infty}{\rho_\infty V_\infty^2}=\frac{\gamma p_\infty}{\gamma\rho_\infty V_\infty^2}=\frac{a_\infty^2}{\gamma V_\infty^2}=\frac{1}{\gamma M_\infty^2}$$

and

$$\frac{\mu_\infty}{\rho_\infty V_\infty c}=\frac{1}{\mathrm{Re}_\infty}$$

where M_∞ and Re_∞ are the freestream Mach and Reynolds numbers, respectively, Equation (15.28) becomes

$$\rho' u'\frac{\partial u'}{\partial x'}+\rho' v'\frac{\partial u'}{\partial y'}=-\frac{1}{\gamma M_\infty^2}\frac{\partial p'}{\partial x'}+\frac{1}{\mathrm{Re}_\infty}\frac{\partial}{\partial y'}\left[\mu'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right] \tag{15.29}$$

Equation (15.29) tells us something important. Consider two different flows over two bodies of different shapes. In one flow, the ratio of specific heats, Mach number, and Reynolds number are γ_1, $M_{\infty 1}$, and $\mathrm{Re}_{\infty 1}$, respectively; in the other flow, these parameters have different values, γ_2, $M_{\infty 2}$, and $\mathrm{Re}_{\infty 2}$. Equation (15.29) is valid for both flows. It can, in principle, be solved to obtain u' as a function of x' and y'. However, since γ, M_∞, and Re_∞ are different for the two cases,

the coefficients of the derivatives in Equation (15.29) will be different. This will ensure, if

$$u' = f_1(x', y')$$

represents the solution for one flow and

$$u' = f_2(x', y')$$

represents the solution for the other flow, that

$$f_1 \neq f_2$$

However, consider now the case where the two different flows have the *same* values of γ, M_∞, and Re_∞. Now the coefficients of the derivatives in Equation (15.29) will be the same for both flows; that is, Equation (15.29) is *numerically identical* for the two flows. In addition, assume the two bodies are geometrically similar, so that the boundary conditions in terms of the nondimensional variables are the same. Then, the solutions of Equation (15.29) for the two flows in terms of $u' = f_1(x', y')$ and $u' = f_2(x', y')$ must be identical; that is,

$$f_1(x', y') \equiv f_2(x', y') \tag{15.30}$$

Recall the definition of dynamically similar flows given in Section 1.8. There, we stated in part that two flows are dynamically similar if the distributions of V/V_∞, p/p_∞, etc., are the same throughout the flow field when plotted against common nondimensional coordinates. This is precisely what Equation (15.30) is saying—that u' as a function of x' and y' is the same for the two flows. That is, the variation of the *nondimensional* velocity as a function of the *nondimensional* coordinates is the same for the two flows. How did we obtain Equation (15.30)? Simply by saying that γ, M_∞, and Re_∞ are the same for the two flows and that the two bodies are geometrically similar. *These are precisely the criteria for two flows to be dynamically similar,* as originally stated in Section 1.8.

What we have seen in the above derivation is a formal mechanism to identify governing similarity parameters for a flow. By couching the governing flow equations in terms of nondimensional variables, we find that the coefficients of the derivatives in these equations are dimensionless similarity parameters or combinations thereof.

To see this more clearly, and to extend our analysis further, consider the energy equation for a steady, two-dimensional, viscous, compressible flow, which from Equation (15.26) can be written as (assuming no volumetric heating and neglecting the normal stresses)

$$\rho u \frac{\partial(e + V^2/2)}{\partial x} + \rho v \frac{\partial(e + V^2/2)}{\partial y} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right) - \frac{\partial(up)}{\partial x} - \frac{\partial(vp)}{\partial y} + \frac{\partial(v\tau_{xy})}{\partial x} + \frac{\partial(u\tau_{yx})}{\partial y} \tag{15.31}$$

Substituting Equation (15.5) into (15.31), we have

$$\rho u\frac{\partial(e+V^2/2)}{\partial x}+\rho v\frac{\partial(e+V^2/2)}{\partial y}=\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)-\frac{\partial(up)}{\partial x}-\frac{\partial(vp)}{\partial y}+\frac{\partial}{\partial x}\left[\mu v\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\right]+\frac{\partial}{\partial y}\left[\mu u\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)\right] \quad (15.32)$$

Using the same nondimensional variables as before, and introducing

$$e'=\frac{e}{c_vT_\infty}\quad k'=\frac{k}{k_\infty}\quad V'^2=\frac{V^2}{V_\infty^2}=\frac{u^2+v^2}{V_\infty^2}=(u')^2+(v')^2$$

Equation (15.32) can be written as

$$\begin{aligned}&\frac{\rho_\infty V_\infty c_vT_\infty}{c}\left(\rho'u'\frac{\partial e'}{\partial x'}+\rho'v'\frac{\partial e'}{\partial y'}\right)\\&=-\frac{\rho_\infty V_\infty^3}{2c}\left[\rho'u'\frac{\partial}{\partial x'}(u'^2+v'^2)+\rho'v'\frac{\partial}{\partial y'}(u'^2+v'^2)\right]\\&\quad+\frac{k_\infty T_\infty}{c^2}\left[\frac{\partial}{\partial x'}\left(k'\frac{\partial T'}{\partial x'}\right)+\frac{\partial}{\partial y'}\left(k'\frac{\partial T'}{\partial y'}\right)\right]-\frac{V_\infty p_\infty}{c}\left(\frac{\partial(u'p')}{\partial x'}+\frac{\partial(v'p')}{\partial y'}\right)\\&\quad+\frac{\mu_\infty V_\infty^2}{c^2}\left\{\frac{\partial}{\partial x'}\left[\mu'v'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right]+\frac{\partial}{\partial y'}\left[\mu'u'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right]\right\}\end{aligned}$$

or

$$\rho'u'\frac{\partial e'}{\partial x'}+\rho'v'\frac{\partial e'}{\partial y'}=\frac{V_\infty^2}{2c_vT_\infty}\left[\rho'u'\frac{\partial}{\partial x'}(u'^2+v'^2)+\rho'v'\frac{\partial}{\partial y'}(u'^2+v'^2)\right]+\frac{k_\infty}{c\rho_\infty V_\infty c_v}\left[\frac{\partial}{\partial x'}\left(k'\frac{\partial T'}{\partial x'}\right)+\frac{\partial}{\partial y'}\left(k'\frac{\partial T'}{\partial y'}\right)\right]-\frac{p_\infty}{\rho_\infty c_vT_\infty}\left(\frac{\partial(u'p')}{\partial x'}+\frac{\partial(v'p')}{\partial y'}\right)+\frac{\mu_\infty V_\infty}{c\rho_\infty c_vT_\infty}\left\{\frac{\partial}{\partial x'}\left[\mu'v'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right]+\frac{\partial}{\partial y'}\left[\mu'u'\left(\frac{\partial v'}{\partial x'}+\frac{\partial u'}{\partial y'}\right)\right]\right\} \quad (15.32a)$$

Examining the coefficients of each term on the right-hand side of Equation (15.32*a*), we find, consecutively,

$$\frac{V_\infty^2}{c_v T_\infty} = \frac{(\gamma-1)V_\infty^2}{RT_\infty} = \frac{\gamma(\gamma-1)V_\infty^2}{\gamma RT_\infty} = \frac{\gamma(\gamma-1)V_\infty^2}{a_\infty^2} = \gamma(\gamma-1)M_\infty^2$$

$$\frac{k_\infty}{c\rho_\infty V_\infty c_v} = \frac{k_\infty \gamma \mu_\infty}{c\rho_\infty V_\infty c_p \mu_\infty} = \frac{\gamma}{\mathrm{Pr}_\infty \mathrm{Re}_\infty}$$

Note: In the above, we have introduced a new dimensionless parameter, the *Prandtl number,* $\mathrm{Pr}_\infty \equiv \mu_\infty c_p / k_\infty$, the significance of which will be discussed later:

$$\frac{p_\infty}{\rho_\infty c_v T_\infty} = \frac{(\gamma-1)p_\infty}{\rho_\infty RT_\infty} = \frac{(\gamma-1)p_\infty}{p_\infty} = \gamma - 1$$

$$\frac{\mu_\infty V_\infty}{c\rho_\infty c_v T_\infty} = \frac{\mu_\infty}{\rho_\infty V_\infty c}\left(\frac{V_\infty^2}{c_v T_\infty}\right) = \frac{1}{\mathrm{Re}_\infty}(\gamma-1)\frac{V_\infty^2}{RT_\infty} = \gamma(\gamma-1)\frac{M_\infty^2}{\mathrm{Re}_\infty}$$

Hence, Equation (15.32) can be written as

$$\begin{aligned}
&\rho' u' \frac{\partial e'}{\partial x'} + \rho' v' \frac{\partial e'}{\partial y'} \qquad (15.33)\\
&= \frac{\gamma(\gamma-1)}{2} M_\infty^2 \left[\rho' u' \frac{\partial}{\partial x'}(u'^2 + v'^2) + \rho' v' \frac{\partial}{\partial y'}(u'^2 + v'^2)\right]\\
&\quad + \frac{\gamma}{\mathrm{Pr}_\infty \mathrm{Re}_\infty}\left[\frac{\partial}{\partial x'}\left(k'\frac{\partial T'}{\partial x'}\right) + \frac{\partial}{\partial y'}\left(k'\frac{\partial T'}{\partial y'}\right)\right]\\
&\quad - (\gamma-1)\left(\frac{\partial(u'p')}{\partial x'} + \frac{\partial(v'p')}{\partial y'}\right)\\
&\quad + \gamma(\gamma-1)\frac{M_\infty^2}{\mathrm{Re}_\infty}\left\{\frac{\partial}{\partial x'}\left[\mu' v'\left(\frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'}\right)\right]\right.\\
&\qquad \left. + \frac{\partial}{\partial y'}\left[\mu' u'\left(\frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'}\right)\right]\right\}
\end{aligned}$$

Examine Equation (15.33). It is a nondimensional equation which, in principle, can be solved for $e' = f(x', y')$. If we have two different flows, but with the same values γ, M_∞, Re_∞, and Pr_∞, Equation (15.33) will be numerically identical for the two flows, and if we are considering geometrically similar bodies, then the solution $e' = f(x', y')$ will be identical for the two flows.

Reflecting upon Equations (15.29) and (15.33), which are the nondimensional x-momentum and energy equations, respectively, we clearly see that the governing similarity parameters for a viscous, compressible flow are γ, M_∞, Re_∞, and Pr_∞. If the above parameters are the same for two different flows with geometrically similar bodies, then the flows are dynamically similar. We obtained these results by considering the x-momentum equation and the energy equation, both

in two dimensions. The same results would have occurred if we had considered three-dimensional flow and the y- and z-momentum equations.

Note that the similarity parameters γ, M_∞, and Re_∞ were obtained from the momentum equation. When the energy equation is considered, an additional similarity parameter is introduced, namely, the Prandtl number. On a physical basis, the Prandtl number is an index which is proportional to the ratio of energy dissipated by friction to the energy transported by thermal conduction; that is,

$$\mathrm{Pr} = \frac{\mu_\infty c_p}{k} \propto \frac{\text{frictional dissipation}}{\text{thermal conduction}}$$

In the study of compressible, viscous flow, Prandtl number is just as important as γ, Re_∞, or M_∞. For air at standard conditions, $\mathrm{Pr}_\infty = 0.71$. Note that Pr_∞ is a property of the gas. For different gases, Pr_∞ is different. Also, like μ and k, Pr_∞ is, in general, a function of temperature; however, for air over a reasonable temperature range (up to $T_\infty = 600$ K), it is safe to assume $\mathrm{Pr}_\infty = \text{constant} = 0.71$.

15.7 SOLUTIONS OF VISCOUS FLOWS: A PRELIMINARY DISCUSSION

The governing continuity, momentum, and energy equations for a general unsteady, compressible, viscous, three-dimensional flow are given by Equations (2.52), (15.19*a* to *c*), and (15.26), respectively. Examine these equations closely. They are nonlinear, coupled, partial differential equations. Moreover, they have additional terms—namely, the viscous terms—in comparison to the analogous equations for an inviscid flow treated in Part 3. Since we have already seen that the nonlinear inviscid flow equations do not lend themselves to a general analytical solution, we can certainly expect the viscous flow equations also not to have any general solutions (at least, at the time of this writing, no general analytical solutions have been found). This leads to the following question: How, then, can we make use of the viscous flow equations in order to obtain some practical results? The answer is much like our approach to the solution of inviscid flows. We have the following options:

1. There are a few viscous flow problems which, by their physical and geometrical nature, allow many terms in the Navier-Stokes solutions to be precisely zero, with the resulting equations being simple enough to solve, either analytically or by simple numerical methods. Sometimes this class of solutions is called "exact solutions" of the Navier-Stokes equations, because no simplifying *approximations* are made to reduce the equations—just *precise* conditions are applied to reduce the equations. Chapter 16 is devoted to this class of solutions; an example is Couette flow (to be defined later).
2. We can simplify the equations by treating certain classes of physical problems for which some terms in the viscous flow equations are small and can be neglected. This is an approximation, not a precise condition. The boundary-layer equations developed and discussed in Chapter 17 are a case

in point. However, as we will see, the boundary-layer equations may be simpler than the full viscous flow equations, but they are still nonlinear.

3. We can tackle the solution of the full viscous flow equations by modern numerical techniques. For example, some of the computational fluid dynamic algorithms discussed in Chapter 13 in conjunction with "exact" solutions for the inviscid flow equations carry over to exact solutions for the viscous flow equations. These matters will be discussed in Chapter 20.

There are some inherent very important differences between the analysis of viscous flows and the study of inviscid flows that were presented in Parts 2 and 3. The remainder of this section highlights these differences.

First, we have already demonstrated in Example 2.5 that viscous flows are *rotational flows*. Therefore, a velocity potential cannot be defined for a viscous flow, thus losing the attendant advantages that were discussed in Sections 2.15 and 11.2. On the other hand, a stream function can be defined, because the stream function satisfies the continuity equation and has nothing to do with the flow being rotational or irrotational (see Section 2.14).

Second, the boundary condition at a solid surface for a viscous flow is the *no-slip* condition. Due to the presence of friction between the surface material and the adjacent layer of fluid, the fluid velocity right at the surface is zero. This no-slip condition was discussed in Section 15.2. For example, if the surface is located at $y = 0$ in a cartesian coordinate system, then the no-slip boundary condition on velocity is

$$\text{At } y = 0: \qquad u = 0 \qquad v = 0 \qquad w = 0$$

This is in contrast to the analogous boundary condition for an inviscid flow, namely, the flow-tangency condition at a surface as discussed in Section 3.7, where only the component of the velocity normal to the surface is zero. Also, recall that for an inviscid flow, there is no boundary condition on the temperature; the temperature of the gas adjacent to a solid surface in an inviscid flow is governed by the physics of the flow field and has no connection whatsoever with the actual wall temperature. However, for a viscous flow, the mechanism of thermal conduction ensures that the temperature of the fluid immediately adjacent to the surface is the same as the temperature of the material surface. In this respect, the no-slip condition is more general than that applied to the velocity; in addition to $u = v = 0$ at the wall, we also have $T = T_w$ at the wall, where T is the gas temperature immediately adjacent to the wall and T_w is the temperature of the surface material. Thus,

$$\text{At } y = 0: \qquad T = T_w \tag{15.34}$$

In many problems, T_w is specified and held constant; this boundary condition is easily applied. However, consider the following, more general case. Imagine a viscous flow over a surface where heat is being transferred from the gas to the surface, or vice versa. Also, assume that the surface is at a certain temperature, T_w, when the flow first starts, but that T_w changes as a function of time as the surface is either heated or cooled by the flow [i.e., $T_w = T_w(t)$]. Because this timewise

variation is dictated in part by the flow which is being calculated, T_w becomes an unknown in the problem and must be calculated along with the solution of the viscous flow. For this general case, the boundary condition at the surface is obtained from Equation (15.2) applied at the wall; that is,

$$\text{At } y = 0: \qquad \dot{q}_w = -\left(k\frac{\partial T}{\partial y}\right)_w \tag{15.35}$$

Here, the surface material is responding to the heat transfer to the wall $\dot{q}_w$, hence changing T_w, which in turn affects $\dot{q}_w$. This general, unsteady heat transfer problem must be solved by treating the viscous flow and the thermal response of the material simultaneously. This problem is beyond the scope of the present book.

Finally, let us imagine the above, unsteady case carried out to the limit of large times. That is, imagine a wind-tunnel model which is at room temperature suddenly inserted in a supersonic or hypersonic stream. At early times, say, for the first few seconds, the surface temperature remains relatively cool, and the assumption of constant wall temperature T_w is reasonable [Equation (15.34)]. However, due to the heat transfer to the model [Equation (15.35)], the surface temperature soon starts to increase and becomes a function of time, as discussed in the previous paragraph. However, as T_w increases, the heating rate decreases. Finally, at large times, T_w increases to a high enough value that the net heat transfer rate to the surface becomes zero, that is, from Equation (15.35),

$$\dot{q}_w = -\left(k\frac{\partial T}{\partial y}\right)_w = 0$$

or

$$\left(\frac{\partial T}{\partial y}\right)_w = 0 \tag{15.36}$$

When the situation of zero heat transfer is achieved, a state of equilibrium exists; the wall temperature at which this occurs is, by definition, the equilibrium wall temperature, or, as it is more commonly denoted, the *adiabatic wall temperature,* T_{aw}. Hence, for the case of an *adiabatic wall* (no heat transfer), the wall boundary condition is given by Equation (15.36).

In summary, for the wall boundary condition associated with the solution of the energy equation [Equation (15.26)], we have three possible cases:

1. *Constant temperature wall, where T_w is a specified constant* [Equation (15.34)]. For this given wall temperature, the temperature gradient at the wall $(\partial T/\partial y)_w$ is obtained as part of the flow-field solution and allows the direct calculation of the aerodynamic heating to the wall via Equation (15.35).
2. *The general, unsteady case, where the heat transfer to the wall $\dot{q}_w$ causes the wall temperature T_w to change, which in turn causes $\dot{q}_w$ to change* Here, both T_w and $(\partial T/\partial y)_w$ change as a function of time, and the problem must be solved by treating jointly the viscous flow as well as the thermal response of the wall material (which usually implies a separate thermal conduction heat transfer numerical analysis).

3. *The adiabatic wall case (zero heat transfer), where* $(\partial T/\partial y)_w = 0$ [Equation (15.36)]. Here, the boundary condition is applied to the temperature *gradient* at the wall, not to the wall temperature itself. Indeed, the wall temperature for this case is defined as the adiabatic wall temperature T_{aw} and is obtained as part of the flow-field solution.

Finally, we emphasize again that, from the point of view of applied aerodynamics, the practical results obtained from a viscous flow analysis are the skin friction and heat transfer at the surface. However, to obtain these quantities, we usually need a complete solution of the viscous flow field; among the data obtained from such a solution are the velocity and temperature gradients at the wall. These, in turn, allow the direct calculation of τ_w and $\dot{q}_w$ from

$$\tau_w = \mu \left(\frac{\partial u}{\partial y} \right)_w$$

and

$$\dot{q}_w = -k \left(\frac{\partial T}{\partial y} \right)_w$$

Another practical result provided by a viscous flow analysis is the prediction and calculation of flow separation; we have discussed numerous cases in the preceding chapters where the pressure field around an aerodynamic body can be greatly changed by flow separation; the flows over cylinders and spheres (see Sections 3.18 and 6.6) are cases in point.

Clearly, the study of viscous flow is important within the entire scope of aerodynamics. The purpose of the following chapters is to provide an introduction to such flows. We will organize our study following the three options itemized at the beginning of this section; that is, we will treat, in turn, certain specialized "exact" solutions of the Navier-Stokes equations, boundary-layer solutions, and then "exact" numerical solutions of Navier-Stokes equations. In so doing, we hope that the reader will gain an overall, introductory picture of the whole area of viscous flow. Entire books have been written on this subject, see, for example, References 42 and 43. We cannot possibly present such detail here; rather, our objective is simply to provide a "feel" for and a basic understanding of the material. Let us proceed.

15.8 SUMMARY

We have now completed the road map given in Figure 15.1. The main results of this chapter are summarized below:

Shear stress and flow separation are two major ramifications of viscous flow. Shear stress is the cause of skin friction drag D_f, and flow separation is the source of pressure drag D_p, sometimes called form drag. Transition from laminar to turbulent flow causes D_f to increase and D_p to decrease.

Shear stress in a flow is due to velocity gradients: for example, $\tau_{yx} = \mu\, \partial u/\partial y$ for a flow with gradients in the y direction. Similarly, heat conduction is due to temperature gradients; for example, $\dot{q}_y = -k\, \partial T/\partial y$, etc. Both μ and k are physical properties of the gas and are functions of temperature.

The general equations of viscous flow are

x momentum:
$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \tag{15.18a}$$

y momentum:
$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \tag{15.18b}$$

z momentum:
$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \tag{15.18c}$$

Energy:
$$\begin{aligned} \rho \frac{D(e + V^2/2)}{Dt} = {} & \rho \dot{q} + \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right) \\ & + \frac{\partial}{\partial z}\left(k \frac{\partial T}{\partial z}\right) - \nabla \cdot p\mathbf{V} + \frac{\partial (u\tau_{xx})}{\partial x} + \frac{\partial (u\tau_{yx})}{\partial y} \\ & + \frac{\partial (u\tau_{zx})}{\partial z} + \frac{\partial (v\tau_{xy})}{\partial x} + \frac{\partial (v\tau_{yy})}{\partial y} + \frac{\partial (v\tau_{zy})}{\partial z} \\ & + \frac{\partial (w\tau_{xz})}{\partial x} + \frac{\partial (w\tau_{yz})}{\partial y} + \frac{\partial (w\tau_{zz})}{\partial z} \end{aligned} \tag{15.26}$$

where
$$\tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)$$

$$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)$$

$$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)$$

$$\tau_{xx} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial u}{\partial x}$$

$$\tau_{yy} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial v}{\partial y}$$

$$\tau_{zz} = \lambda(\nabla \cdot \mathbf{V}) + 2\mu \frac{\partial w}{\partial z}$$

The similarity parameters for a flow can be obtained by nondimensionalizing the governing equations; the coefficients in front of the nondimensionalized derivatives give the similarity parameters or combinations thereof. For a viscous, compressible flow, the main similarity parameters are γ, M_∞, Re_∞, and Pr_∞.

Exact analytical solutions of the complete Navier-Stokes equations exist for only a few very specialized cases. Instead, the equations are frequently simplified by making appropriate approximations about the flow. In modern times, exact solutions of the complete Navier-Stokes equations for many practical problems can be obtained numerically, using various techniques of computational fluid dynamics.

15.9 PROBLEMS

15.1 Consider the incompressible viscous flow of air between two infinitely long parallel plates separated by a distance h. The bottom plate is stationary, and the top plate is moving at the constant velocity u_e in the direction of the plate. Assume that no pressure gradient exists in the flow direction.

a. Obtain an expression for the variation of velocity between the plates.

b. If $T = \text{constant} = 320$ K, $u_e = 30$ m/s, and $h = 0.01$ m, calculate the shear stress on the top and bottom plates.

15.2 Assume that the two parallel plates in Problem 15.1 are both stationary but that a constant pressure gradient exists in the flow direction (i.e., $dp/dx = \text{constant}$).

a. Obtain an expression for the variation of velocity between the plates.

b. Obtain an expression for the shear stress on the plates in terms of dp/dx.

CHAPTER 16

A Special Case: Couette Flow

The resistance arising from the want of lubricity in the parts of a fluid is, other things being equal, proportional to the velocity with which the parts of the fluid are separated from one another.

Isaac Newton, 1687,
from Section IX of Book II
of his Principia

PREVIEW BOX

An old but wise expression states that you must learn to walk before you can run. General applications of viscous flow are frequently complex and challenging—you are constantly "running" in order to obtain their solutions. This chapter teaches you to walk first. Here we treat a special viscous flow problem that lends itself to rather straightforward solutions (all we have to do is "walk" to obtain them). Yet these solutions illustrate some of the most important aspects of viscous flows in general, obtaining and highlighting the parameters that dictate skin friction and aerodynamic heating. Here we will learn some new ideas with strange-sounding words such as "recovery factor" and "Reynolds analogy." We will be able to see some of the basic physics of viscous flow, stripped of the extra geometrical complexities that go along with more complex flow applications. Although this chapter involves a peculiar-sounding flow, *Couette flow,* the results are far from peculiar. Indeed, in this chapter we are going to take a walk through some of the most important ideas surrounding the basic analyses of viscous flow. Get going, and enjoy your walk.

16.1 INTRODUCTION

The general equations of viscous flow were derived and discussed in Chapter 15. In particular, the viscous flow momentum equations were treated in Section 15.4 and are given in partial differential equation form by Equations (15.19*a* to *c*)—the Navier-Stokes equations. These, along with the viscous flow energy equation,

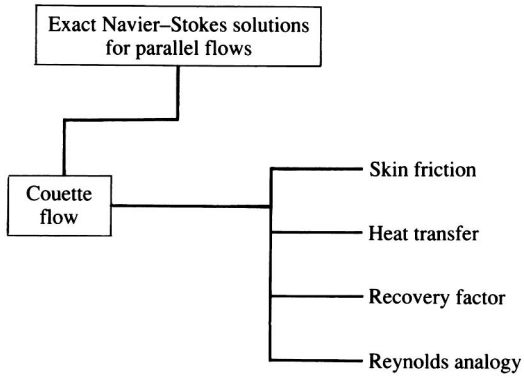


Figure 16.1 Road map for Chapter 16.

Equation (15.26), derived in Section 15.5, are the theoretical tools for the study of viscous flows. However, examine these equations closely; as discussed in Section 15.7, they are a system of coupled, nonlinear partial differential equations—equations which contain more terms and which are inherently more elaborate than the inviscid flow equations treated in Parts 2 and 3 of this book. Three classes of solutions of these equations were itemized in Section 15.5. The first itemized class was that of "exact" solutions of the Navier-Stokes equations for a few specific physical problems which, by their physical and geometrical nature, allow many terms in the governing equations to be precisely zero, resulting in a system of equations simple enough to solve, either analytically or by simple numerical methods. Such exact problems are the subject of this chapter.

The road map for this chapter is given in Figure 16.1. The type of flow considered here is generally labeled as *parallel flow* because the streamlines are straight and parallel to each other. We will consider one of these flows, Couette flow, which will be defined in due course. In addition to representing exact solutions of the Navier-Stokes equations, this flow illustrates some of the important practical facets of any viscous flow, as itemized on the right side of the road map. In a clear, uncomplicated fashion, we will be able to calculate and study the surface skin friction and heat transfer. We will also use the results to define the recovery factor and Reynolds analogy—two practical engineering tools that are frequently used in the analysis of skin friction and heat transfer.

16.2 COUETTE FLOW: GENERAL DISCUSSION

Consider the flow model shown in Figure 16.2. Here we see a viscous fluid contained between two parallel plates separated by a distance D. The upper plate is moving to the right at velocity u_e. Due to the no-slip condition, there can be no relative motion between the plate and the fluid; hence, at $y = D$ the flow velocity is $u = u_e$ and is directed toward the right. Similarly, the flow velocity at

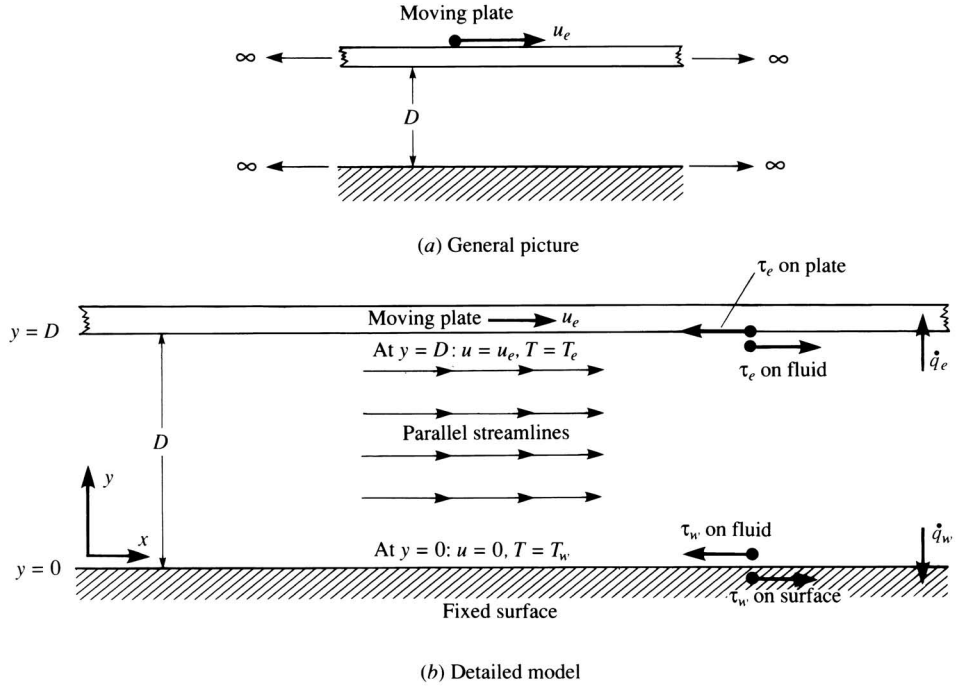


Figure 16.2 Model for Couette flow.

$y = 0$, which is the surface of the stationary lower plate, is $u = 0$. In addition, the two plates may be at different temperatures; the upper plate is at temperature T_e and the lower plate is at temperature T_w. Again, due to the no-slip condition as discussed in Section 15.7, the fluid temperature at $y = D$ is $T = T_e$ and that at $y = 0$ is $T = T_w$.

Clearly, there is a flow field between the two plates; the driving force for this flow is the motion of the upper plate, dragging the flow along with it through the mechanism of friction. The upper plate is exerting a shear stress, τ_e, acting toward the right on the fluid at $y = D$, thus causing the fluid to move toward the right. By an equal and opposite reaction, the fluid is exerting a shear stress τ_e on the upper plate acting toward the left, tending to retard its motion. We assume that the upper plate is being driven by some external force that is sufficient to overcome the retarding shear stress and to allow the plate to move at the constant velocity u_e. Similarly, the lower plate is exerting a shear stress τ_w acting toward the left on the fluid at $y = 0$. By an equal and opposite reaction, the fluid is exerting a shear stress τ_w acting toward the right on the lower plate. (In all subsequent diagrams dealing with viscous flow, the only shear stresses shown will be those due to the fluid acting on the surface, unless otherwise noted.)

In addition to the velocity field induced by the relative motion of the two plates, there will also be a temperature field induced by the following two

mechanisms:

1. The plates in general will be at different temperatures, thus causing temperature gradients in the flow.
2. The kinetic energy of the flow will be partially dissipated by the influence of friction and will be transformed into internal energy within the fluid. These changes in internal energy will be reflected by changes in temperature. This phenomenon is called *viscous dissipation*.

Consequently, temperature gradients will exist within the flow; in turn, these temperature gradients result in the transfer of heat through the fluid. Of particular interest is the heat transfer at the upper and lower surfaces, denoted by $\dot{q}_e$ and $\dot{q}_w$, respectively. These heat transfers are shown in Figure 16.2; the directions for $\dot{q}_e$ and $\dot{q}_w$ show heat being transferred from the fluid to the wall in both cases. When heat flows from the fluid to the wall, this is called a *cold wall* case, such as sketched in Figure 16.2. When heat flows from the wall into the fluid, this is called a *hot wall* case. Keep in mind that the heat flux through the fluid at any point is given by the Fourier law expressed by Equation (15.2); that is, the heat flux in the y direction is expressed as

$$\dot{q}_y = -k\frac{\partial T}{\partial y} \tag{15.2}$$

where the minus sign accounts for the fact that heat is transferred from a region of high temperature to a region of lower temperature; that is, $\dot{q}_y$ is in the opposite direction of the temperature gradient.

Let us examine the geometry of Couette flow as illustrated in Figure 16.2. An x-y cartesian coordinate system is oriented with the x axis in the direction of the flow and the y axis perpendicular to the flow. Since the two plates are parallel, the only possible flow pattern consistent with this picture is that of straight, parallel streamlines. Moreover, since the plates are infinitely long (i.e., stretching to plus and minus infinity in the x direction), then the flow properties cannot change with x. (If the properties did change with x, then the flow-field properties would become infinitely large or infinitesimally small at large values of x—a physical inconsistency.) Thus, all partial derivatives with respect to x are zero. The only changes in the flow-field variables take place in the y direction. Moreover, the flow is steady, so that all time derivatives are zero. With this geometry in mind, return to the governing Navier-Stokes equations given by Equations (15.19a to c) and Equation (15.26). In these equations, for Couette flow,

$$v = w = 0 \quad \frac{\partial u}{\partial x} = \frac{\partial T}{\partial x} = \frac{\partial p}{\partial x} = 0$$

Hence, from Equations (15.19a to c) and Equation (15.26), we have

x-momentum equation:
$$\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right) = 0 \tag{16.1}$$

y-momentum equation:
$$\frac{\partial p}{\partial y} = 0 \tag{16.2}$$

Energy equation:
$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial y}\left(\mu u\frac{\partial u}{\partial y}\right)=0 \tag{16.3}$$

Equations (16.1) to (16.3) are the governing equations for Couette flow. Note that these equations are exact forms of the Navier-Stokes equations applied to the geometry of Couette flow—no approximations have been made. Also, note from Equation (16.2) that the variation of pressure in the y direction is zero; this in combination with the earlier result that $\partial p/\partial x = 0$ implies that the *pressure is constant* throughout the entire flow field. Couette flow is a constant pressure flow. It is interesting to note that all the previous flow problems discussed in Parts 2 and 3, being *inviscid* flows, were established and maintained by the existence of *pressure gradients* in the flow. In these problems, the pressure gradient was nature's mechanism of grabbing hold of the flow and making it move. However, in the problem we are discussing now—being a viscous flow—shear stress is another mechanism by which nature can exert a force on a flow. For Couette flow, the shear stress exerted by the moving plate on the fluid is the exclusive driving mechanism that maintains the flow; clearly, no pressure gradient is present, nor is it needed.

This section has presented the general nature of Couette flow. Note that we have made no distinction between incompressible and compressible flow; all aspects discussed here apply to both cases. Also, we note that, although Couette flow appears to be a rather academic problem, the following sections illustrate, in a simple fashion, many of the important characteristics of practical viscous flows in real engineering applications.

The next two sections will treat the separate cases of incompressible and compressible Couette flow. Incompressible flow will be discussed first because of its relative simplicity; this is the subject of Section 16.3. Then compressible Couette flow, and how it differs from the incompressible case, will be examined in Section 16.4.

As a final note in this section, it is obvious from our general discussion of Couette flow that the flow-field properties vary only in the y direction; all derivatives in the x direction are zero. Therefore, as a matter of mathematical preciseness, all the partial derivatives in Equations (16.1) to (16.3) can be written as ordinary derivatives. For example, Equation (16.1) can be written as

$$\frac{d}{dy}\left(\mu\frac{du}{dy}\right)=0$$

However, our discussion of Couette flow is intended to serve as a straightforward example of a viscous flow problem, "breaking the ice" so-to-speak for the more practical but more complex problems to come—problems which involve changes in both the x and y directions, and which are described by *partial* differential equations. Therefore, on pedagogical grounds, we choose to continue the partial differential notation here, simply to make the reader feel more comfortable when we extend these concepts to the boundary layer and full Navier-Stokes solutions in Chapters 17 and 20, respectively.

16.3 INCOMPRESSIBLE (CONSTANT PROPERTY) COUETTE FLOW

In the study of viscous flows, a flow field in which ρ, μ, and k are treated as constants is sometimes labeled as "constant property" flow. This assumption is made in the present section. On a physical basis, this means that we are dealing with an incompressible flow, where ρ is constant. Also, since μ and k are functions of temperature (see Section 15.3), constant property flow implies that T is constant also. (We will relax this assumption slightly at the end of this section.)

The governing equations for Couette flow were derived in Section 16.2. In particular, the y-momentum equation, Equation (16.2), along with the geometrical property that $\partial p/\partial x = 0$, states that the pressure is constant throughout the flow. Consequently, all the information about the velocity field comes from the x-momentum equation, Equation (16.1), repeated below:

$$\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right) = 0 \tag{16.1}$$

For constant μ, this becomes

$$\frac{\partial^2 u}{\partial y^2} = 0 \tag{16.4}$$

Integrating with respect to y twice, we obtain

$$u = ay + b \tag{16.5}$$

where a and b are constants of integration. These constants can be obtained from the boundary conditions illustrated in Figure 16.2, as follows:

At $y = 0$, $u = 0$; hence, $b = 0$.

At $y = D$, $u = u_e$; hence, $a = u_e/D$.

Thus, the variation of velocity for incompressible Couette flow is given by Equation (16.5) as

$$\boxed{u = u_e\left(\frac{y}{D}\right)} \tag{16.6}$$

Note the important result that the velocity varies *linearly* across the flow. This result is sketched in Figure 16.3.

Once the velocity profile is obtained, we can obtain the shear stress at any point in the flow from Equation (15.1), repeated below (the subscript yx is dropped here because we know the only shear stress acting in this problem is that in the x direction):

$$\tau = \mu\frac{\partial u}{\partial y} \tag{16.7}$$

From Equation (16.6),

$$\frac{\partial u}{\partial y} = \frac{u_e}{D} \tag{16.8}$$

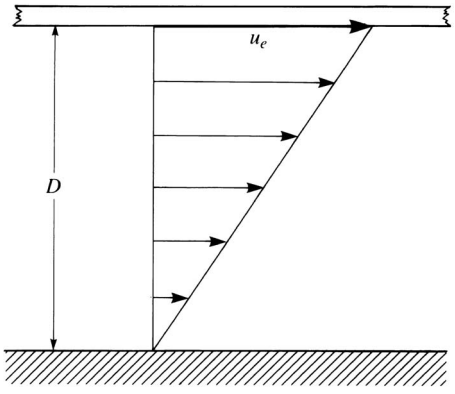


Figure 16.3 Velocity profile for incompressible Couette flow.

Hence, from Equations (16.7) and (16.8), we have

$$\boxed{\tau = \mu \left(\frac{u_e}{D} \right)} \tag{16.9}$$

Note that the shear stress is *constant* throughout the flow. Moreover, the straightforward result given by Equation (16.9) illustrates two important physical trends—trends that we will find to be almost universally present in all viscous flows:

1. As u_e increases, the shear stress increases. From Equation (16.9), τ increases linearly with u_e; this is a specific result germane to Couette flow. For other problems, the increase is not necessarily linear.
2. As D increases, the shear stress decreases; that is, as the thickness of the viscous shear layer increases, all other things being equal, the shear stress becomes smaller. From Equation (16.9), τ is inversely proportional to D—again a result germane to Couette flow. For other problems, the decrease in τ is not necessarily in *direct* inverse proportion to the shear-layer thickness.

With the above results in mind, reflect for a moment on the quotation from Isaac Newton's *Principia* given at the beginning of this chapter. Here, the "want of lubricity" is, in modern terms, interpreted as the shear stress. This want of lubricity is, according to Newton, "proportional to the velocity with which the parts of the fluid are separated from one another," that is, in the context of the present problem proportional to u_e/D. This is precisely the statement contained in Equation (16.9). In more recent times, Newton's statement is generalized to the form given by Equation (16.7), and even more generalized by Equation (15.1). For this reason, Equations (15.1) and (16.7) are frequently called the newtonian shear stress law, and fluids which obey this law are called *newtonian fluids*. [There

are some specialized fluids which do not obey Equation (15.1) or (16.7); they are called non-newtonian fluids—some polymers and blood are two such examples.] By far, the vast majority of aeronautical applications deal with air or other gases, which are newtonian fluids. In hydrodynamics, water is the primary medium, and it is a newtonian fluid. Therefore, we will deal exclusively with newtonian fluids in this book. Consequently, the quote given at the beginning of this chapter is one of Newton's most important contributions to fluid mechanics—it represents the first time in history where shear stress is recognized as being proportional to a velocity gradient.

Let us now turn our attention to heat transfer in a Couette flow. Here, we continue our assumptions of constant ρ, μ, and k, but at the same time, we will allow a temperature gradient to exist in the flow. In an exact sense, this is inconsistent; if T varies throughout the flow, then ρ, μ, and k also vary. However, for this application, we assume that the temperature variations are small—indeed, small enough such that ρ, μ, and k are *approximately* constant—and treat them so in the equations. On the other hand, the temperature changes, although small on an absolute basis, are sufficient to result in meaningful heat flux through the fluid. The results obtained will reflect some of the important trends in aerodynamic heating associated with high-speed flows, to be discussed in subsequent chapters.

For Couette flow with heat transfer, return to Figure 16.2. Here, the temperature of the upper plate is T_e and that of the lower plate is T_w. Hence, we have as boundary conditions for the temperature of the fluid:

$$\text{At } y = 0: \qquad T = T_w$$

$$\text{At } y = D: \qquad T = T_e$$

The temperature profile in the flow is governed by the energy equation, Equation (16.3). For constant μ and k, this equation is written as

$$\frac{k}{\mu}\left(\frac{\partial^2 T}{\partial y^2}\right) + \frac{\partial}{\partial y}\left(u\frac{\partial u}{\partial y}\right) = 0 \tag{16.10}$$

Also, since μ is assumed to be constant, Equations (16.10) and (16.1) are *totally uncoupled*. That is, for the constant property flow considered here, the solution of the momentum equation [Equation (16.1)] is totally separate from the solution of the energy equation [Equation (16.10)]. Therefore, in this problem, although the temperature is allowed to vary, the velocity field is still given by Equation (16.6), as sketched in Figure 16.3.

In dealing with flows where energy concepts are important, the enthalpy h is frequently a more fundamental variable than temperature; we have seen much evidence of this in Part 3, where energy changes were a vital consideration. In the present problem, where the temperature changes are small enough to justify the assumptions of constant ρ, μ, and k, this is not quite the same situation. However, because we will need to solve Equation (16.10), which is an energy equation for a flow (no matter how small the energy changes), and because we are using Couette flow as an example to set the stage for more complex problems, it is

instructional (but by no means necessary) to couch Equation (16.10) in terms of enthalpy. Assuming constant specific heat, we have

$$h = c_p T \tag{16.11}$$

Equation (16.11) is valid for the Couette flow of any fluid with constant heat capacity; here, the germane specific heat is that at constant pressure c_p because the entire flow field is at constant pressure. In this sense, Equation (16.11) is a result of applying the first law of thermodynamics to a constant pressure process and recalling the fundamental definition of heat capacity as the heat added per unit change in temperature, $\delta q/dT$. Of course, if the fluid is a calorically perfect gas, then Equation (16.11) is a basic thermodynamic property of such a gas quite independent of what the process may be [see Section 7.2 and Equation (7.6*b*)]. Inserting Equation (16.11) into Equation (16.10), we have

$$\frac{k}{\mu c_p}\frac{\partial^2 h}{\partial y^2} + \frac{\partial}{\partial y}\left(u\frac{\partial u}{\partial y}\right) = 0 \tag{16.12}$$

Recall the definition of the Prandtl number from Section 15.6, namely,

$$\mathrm{Pr} = \frac{\mu c_p}{k}$$

Equation (16.12) can be written in terms of the Prandtl number as

$$\frac{1}{\mathrm{Pr}}\frac{\partial^2 h}{\partial y^2} + \frac{\partial}{\partial y}\left(u\frac{\partial u}{\partial y}\right) = 0$$

or

$$\frac{\partial^2 h}{\partial y^2} + \frac{\mathrm{Pr}}{2}\frac{\partial}{\partial y}\left(\frac{\partial u^2}{\partial y}\right) = 0 \tag{16.13}$$

Integrating twice in the y direction, we find that Equation (16.13) yields

$$h + \left(\frac{\mathrm{Pr}}{2}\right)u^2 = ay + b \tag{16.14}$$

where a and b are constants of integration [different from the a and b in Equation (16.5)]. Expressions for a and b are found by applying Equation (16.14) at the boundaries, as follows:

$At\ y = 0$: $\quad h = h_w \quad$ and $\quad u = 0$

$At\ y = D$: $\quad h = h_e \quad$ and $\quad u = u_e$

Hence, from Equation (16.14) at the boundaries,

$$b = h_w$$

and

$$a = \frac{h_e - h_w + (\mathrm{Pr}/2)u_e^2}{D}$$

Inserting these values into Equation (16.14) and rearranging, we have

$$h = h_w + \left[h_e - h_w + \left(\frac{\mathrm{Pr}}{2}\right)u_e^2\right]\frac{y}{D} - \left(\frac{\mathrm{Pr}}{2}\right)u^2 \tag{16.15}$$

Inserting Equation (16.6) for the velocity profile in Equation (16.15) yields

$$\boxed{h = h_w + \left[h_e - h_w + \left(\frac{\Pr}{2}\right) u_e^2\right] \frac{y}{D} - \left(\frac{\Pr}{2}\right) u_e^2 \left(\frac{y}{D}\right)^2} \tag{16.16}$$

Note that h varies *parabolically* with y/D across the flow. Since $T = h/c_p$, then the temperature profile across the flow is also parabolic. The precise shape of the parabolic curve depends on h_w (or T_w), h_e (or T_e), and Pr. Also note that, as expected from our discussion of the viscous flow similarity parameters in Section 15.6, the Prandtl number is clearly a strong player in the results; Equation (16.16) is one such example.

Once the enthalpy (or temperature) profile is obtained, we can obtain the heat flux at any point in the flow from Equation (15.2), repeated below (the subscript y is dropped here because we know the only direction of heat transfer is in the y direction for this problem):

$$\dot{q} = -k \frac{\partial T}{\partial y} \tag{16.17}$$

Equation (16.17) can be written as

$$\dot{q} = -\frac{k}{c_p} \frac{\partial h}{\partial y} \tag{16.18}$$

In Equation (16.18), the enthalpy gradient is obtained by differentiating Equation (16.16) as follows:

$$\frac{\partial h}{\partial y} = \left[h_e - h_w + \left(\frac{\Pr}{2}\right) u_e^2\right] \frac{1}{D} - \Pr u_e^2 \frac{y}{D^2} \tag{16.19}$$

Inserting Equation (16.19) into Equation (16.18), and writing k/c_p as $\mu/\Pr$, we have

$$\dot{q} = -\mu \left(\frac{h_e - h_w}{\Pr} + \frac{u_e^2}{2}\right) \frac{1}{D} + \mu u_e^2 \frac{y}{D^2} \tag{16.20}$$

From Equation (16.20), note that $\dot{q}$ is *not* constant across the flow, unlike the shear stress discussed earlier. Rather, $\dot{q}$ varies linearly with y. The physical reason for the variation of $\dot{q}$ is *viscous dissipation* which takes place within the flow, and which is associated with the shear stress in the flow. Indeed, the last term in Equation (16.20), in light of Equations (16.6) and (16.9), can be written as

$$\mu u_e^2 \frac{y}{D^2} = \tau u_e \left(\frac{y}{D}\right) = \tau u$$

Hence, Equation (16.20) becomes

$$\dot{q} = -\mu \left(\frac{h_e - h_w}{\Pr} + \frac{u_e^2}{2}\right) \frac{1}{D} + \tau u \tag{16.21}$$

The variation of $\dot{q}$ across the flow is due to the last term in Equation (16.21), and this term involves shear stress multiplied by velocity. The term τu is *viscous*

dissipation; it is the time rate of heat generated at a point in the flow by one streamline at a given velocity "rubbing" against an adjacent streamline at a slightly different velocity—analogous to the heat you feel when rubbing your hands together vigorously. Note that, if u_e is negligibly small, then the viscous dissipation is small and can be neglected; that is, in Equation (16.20) the last term can be neglected (u_e is small), and in Equation (16.21) the last term can be neglected (τ is small if u_e is small). In this case, the heat flux becomes constant across the flow, simply equal to

$$\dot{q} \approx -\frac{\mu}{\text{Pr}} \left(\frac{h_e - h_w}{D} \right) \tag{16.22}$$

In this case, the "driving potential" for heat transfer across the flow is simply the enthalpy difference ($h_e - h_w$) or, in other words, the temperature difference ($T_e - T_w$) across the flow. However, as we have emphasized, if u_e is not negligible, then viscous dissipation becomes another factor that drives the heat transfer across the flow.

Of particular practical interest is the heat flux at the walls—the *aerodynamic heating* as we label it here. We denote the heat transfer at a wall as $\dot{q}_w$. Moreover, it is conventional to quote aerodynamic heating at a wall without any sign convention. For example, if the heat transfer from the fluid to the wall is 10 W/cm^2, or, if in reverse the heat transfer from the wall to the fluid is 10 W/cm^2, it is simply quoted as such; in both cases, $\dot{q}_w$ is given as 10 W/cm^2 without any sign convention. In this sense, we write Equation (16.18) as

$$\dot{q}_w = \frac{k}{c_p} \left| \frac{\partial h}{\partial y} \right|_w = \frac{\mu}{\text{Pr}} \left| \frac{\partial h}{\partial y} \right|_w \tag{16.23}$$

where the subscript w implies conditions at the wall. The direction of the net heat transfer at the wall, whether it is from the fluid to the wall or from the wall to the fluid, is easily seen from the temperature gradient at the wall; if the wall is cooler than the adjacent fluid, heat is transferred into the wall, and if the wall is hotter than the adjacent fluid, heat is transferred into the fluid. Another criterion is to compare the wall temperature with the adiabatic wall temperature, to be defined shortly.

Return to the picture of Couette flow in Figure 16.2. To calculate the heat transfer at the lower wall, use Equation (16.23) with the enthalpy gradient given by Equation (16.19) evaluated at $y = 0$:

$$\boxed{At\ y = 0: \qquad \dot{q}_w = \frac{\mu}{\text{Pr}} \left| \frac{h_e - h_w + \frac{1}{2} \text{Pr}\, u_e^2}{D} \right|} \tag{16.24}$$

To calculate the heat transfer at the upper wall, use Equation (16.23) with the enthalpy gradient given by Equation (16.19) evaluated at $y = D$. In this case,

Equation (16.19) yields

$$\frac{\partial h}{\partial y} = \frac{h_e - h_w + \frac{1}{2}\,\mathrm{Pr}\,u_e^2}{D} - \frac{\mathrm{Pr}\,u_e^2}{D} = \frac{h_e - h_w - \frac{1}{2}\,\mathrm{Pr}\,u_e^2}{D}$$

In turn, from Equation (16.23)

$$\boxed{At\ y = D: \qquad \dot{q}_w = \frac{\mu}{\mathrm{Pr}} \left| \frac{h_e - h_w - \frac{1}{2}\,\mathrm{Pr}\,u_e^2}{D} \right|} \tag{16.25}$$

Let us examine the above results for three different scenarios, namely, (1) *negligible viscous dissipation,* (2) *equal wall temperature,* and (3) *adiabatic wall conditions* (no heat transfer to the wall). In the process, we define three important concepts in the analysis of aerodynamic heating: (1) *adiabatic wall temperature,* (2) *recovery factor,* and (3) *Reynolds analogy*.

16.3.1 Negligible Viscous Dissipation

To some extent, we have already discussed this case in regard to the local heat flux at any point within the flow. If u_e is very small, hence τ is very small, then the amount of viscous dissipation is negligibly small, and Equation (16.21) becomes

$$\dot{q} = -\frac{\mu}{\mathrm{Pr}} \left(\frac{h_e - h_w}{D} \right) \tag{16.26}$$

Clearly, for this case, the heat flux is constant across the flow. Moreover, the enthalpy profile given by Equation (16.16) becomes

$$h = h_w + (h_e - h_w)\frac{y}{D} \tag{16.27}$$

Since $h = c_p T$, the temperature profile is identical to the enthalpy profile:

$$T = T_w + (T_e - T_w)\frac{y}{D} \tag{16.28}$$

Note that the temperature varies *linearly* across the flow, as sketched in Figure 16.4. The case shown here is for the upper wall at a higher temperature than the

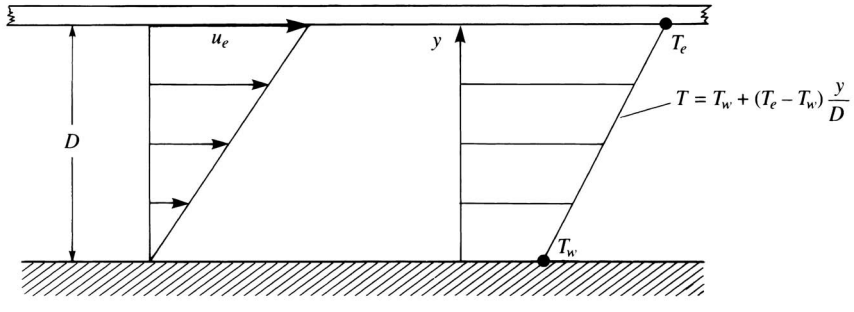


Figure 16.4 Couette flow temperature profile for negligible viscous dissipation.

lower wall. The heat transfer at the lower wall is obtained from Equation (16.24) with a negligible u_e:

$$At\ y = 0: \qquad \dot{q}_w = \frac{\mu}{\mathrm{Pr}} \left| \frac{h_e - h_w}{D} \right| \tag{16.29}$$

The heat transfer at the upper wall is similarly obtained as

$$At\ y = D: \qquad \dot{q}_w = \frac{\mu}{\mathrm{Pr}} \left| \frac{h_e - h_w}{D} \right| \tag{16.30}$$

Equations (16.29) and (16.30) are identical; this is no surprise, since we have already shown that the heat flux is constant across the flow, as shown by Equation (16.26), and therefore the heat transfer at both walls should be the same. Equations (16.29) and (16.30) can also be written in terms of temperature as

$$\dot{q}_w = k \left| \frac{T_e - T_w}{D} \right| \tag{16.31}$$

Examining Equations (16.29) to (16.31), we can make some conclusions which can be generalized to most viscous flow problems, as follows:

1. Everything else being equal, the larger the temperature difference across the viscous layer, the greater the heat transfer at the wall. The temperature difference ($T_e - T_w$) or the enthalpy difference ($h_e - h_w$) takes on the role of a "driving potential" for heat transfer. For the special case treated here, the heat transfer at the wall is directly proportional to this driving potential.
2. Everything else being equal, the thicker the viscous layer (the larger D is) the smaller the heat transfer is at the wall. For the special case treated here, $\dot{q}_w$ is inversely proportional to D.
3. Heat flows from a region of high temperature to low temperature. For negligible viscous dissipation, if the temperature at the top of the viscous layer is higher than that at the bottom, heat flows from the top to the bottom. In the case sketched in Figure 16.4, heat is transferred from the upper plate into the fluid, and then is transferred from the fluid to the lower plate.

16.3.2 Equal Wall Temperatures

Here we assume that $T_e = T_w$; that is, $h_e = h_w$. The enthalpy profile for this case, from Equation (16.16), is

$$h = h_w + \frac{1}{2}\mathrm{Pr}\,u_e^2 \left(\frac{y}{D}\right) - \frac{1}{2}\mathrm{Pr}\,u_e^2 \left(\frac{y}{D}\right)^2 \tag{16.32}$$

or

$$= h_w + \frac{1}{2}\mathrm{Pr}\,u_e^2 \left[\frac{y}{D} - \left(\frac{y}{D}\right)^2\right]$$

In terms of temperature, this becomes

$$T = T_w + \frac{\mathrm{Pr}\,u_e^2}{2c_p} \left[\frac{y}{D} - \left(\frac{y}{D}\right)^2\right] \tag{16.33}$$

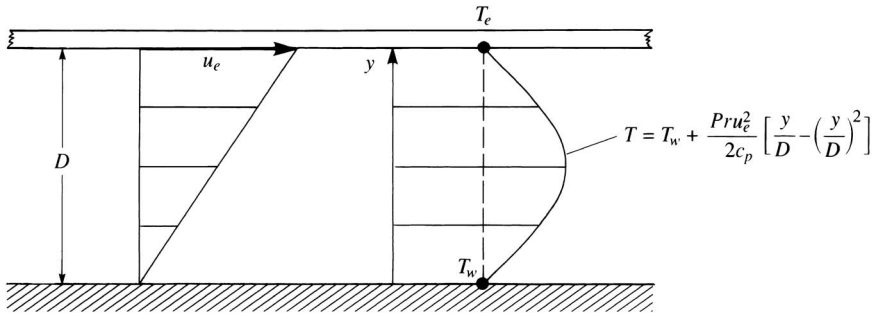


Figure 16.5 Couette flow temperature profile for equal wall temperature with viscous dissipation.

Note that the temperature varies parabolically with y, as sketched in Figure 16.5. The maximum value of temperature occurs at the midpoint, $y = D/2$. This maximum value is obtained by evaluating Equation (16.33) at $y = D/2$.

$$T_{\max} = T_w + \frac{\Pr u_e^2}{8c_p} \tag{16.34}$$

The heat transfer at the walls is obtained from Equations (16.24) and (16.25) as

At $y = 0$:
$$\dot{q}_w = \frac{\mu}{2}\frac{u_e^2}{D} \tag{16.35}$$

At $y = D$:
$$\dot{q}_w = \frac{\mu}{2}\frac{u_e^2}{D} \tag{16.36}$$

Equations (16.35) and (16.36) are identical; the heat transfers at the upper and lower walls are equal. In this case, as can be seen by inspecting the temperature distribution shown in Figure 16.5, the upper and lower walls are both cooler than the adjacent fluid. Hence, at both the upper and lower walls, heat is transferred from the fluid to the wall.

Question: Since the walls are at equal temperature, where is the heat transfer coming from? *Answer: Viscous dissipation.* The local temperature increase in the flow as sketched in Figure 16.5 is due *solely* to viscous dissipation within the fluid. In turn, both walls experience an aerodynamic heating effect due to this viscous dissipation. This is clearly evident in Equations (16.35) and (16.36), where $\dot{q}_w$ depends on the velocity u_e. Indeed, $\dot{q}_w$ is directly proportional to the *square* of u_e. In light of Equation (16.9), Equations (16.35) and (16.36) can be written as

$$\dot{q}_w = \tau\left(\frac{u_e}{2}\right) \tag{16.37}$$

which further emphasizes that $\dot{q}_w$ is due entirely to the action of shear stress in the flow. From Equations (16.35) to (16.37), we can make the following conclusions that reflect general properties of most viscous flows:

1. Everything else being equal, aerodynamic heating increases as the flow velocity increases. This is why aerodynamic heating becomes an important

design factor in high-speed aerodynamics. Indeed, for most hypersonic vehicles, you can begin to appreciate that viscous dissipation generates extreme temperatures within the boundary layer adjacent to the vehicle surface and frequently makes aerodynamic heating the dominant design factor. In the Couette flow case shown here—a far cry from hypersonic flow—we see that $\dot{q}_w$ varies directly as u_e^2.

2. Everything else being equal, aerodynamic heating decreases as the thickness of the viscous layer increases. For the case considered here, $\dot{q}_w$ is inversely proportional to D. This conclusion is the same as that made for the above case of negligible viscous dissipation but with unequal wall temperature.

16.3.3 Adiabatic Wall Conditions (Adiabatic Wall Temperature)

Let us imagine the following situation. Assume that the flow illustrated in Figure 16.5 is established. We have the parabolic temperature profile established as shown, and we have heat transfer into the walls as just discussed. However, both wall temperatures are considered *fixed,* and both are equal to the same constant value. *Question:* How can the wall temperature remain fixed at the same time that heat is transferred into the wall? *Answer:* There must be some independent mechanism that conducts heat away from the wall at the same rate that the aerodynamic heating is pumping heat into the wall. This is the only way for the wall temperature to remain fixed at some cooler temperature than the adjacent fluid. For example, the wall can be some vast heat sink that can absorb heat without any appreciable change in temperature, or possibly there are cooling coils within the plate that can carry away the heat, much like the water coils that keep the engine of your automobile cool. In any event, to have the picture shown in Figure 16.5 with a constant wall temperature independent of time, some exterior mechanism must carry away the heat that is transferred from the fluid to the walls. Now imagine that, at the lower wall, this exterior mechanism is suddenly shut off. The lower wall will now begin to grow hotter in response to $\dot{q}_w$, and T_w will begin to increase with time. At any given instant during this transient process, the heat transfer to the lower wall is given by Equation (16.24), repeated below:

$$\dot{q}_w = \frac{\mu}{\text{Pr}} \left| \frac{h_e - h_w + \frac{1}{2}\,\text{Pr}\, u_e^2}{D} \right| \tag{16.24}$$

At time $t = 0$, when the exterior cooling mechanism is just shut off, $h_w = h_e$, and $\dot{q}_w$ is given by Equation (16.35), namely,

At time $t = 0$: $$\dot{q}_w = \frac{\mu}{2} \frac{u_e^2}{D}$$

However, as time now progresses, T_w (and therefore h_w) increases. From Equation (16.24), as h_w increases, the numerator decreases in magnitude, and hence $\dot{q}_w$ decreases. That is,

At $t > 0$: $$\dot{q}_w < \frac{\mu}{2} \frac{u_e^2}{D}$$

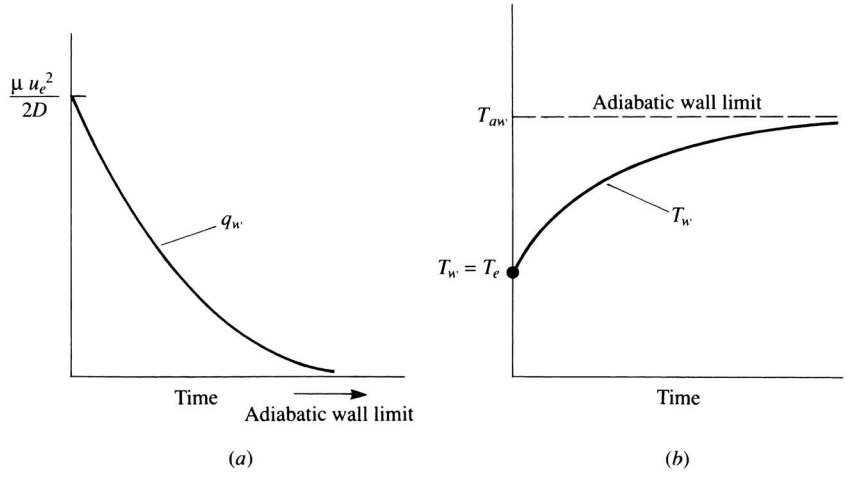


Figure 16.6 Illustration for the definition of an adiabatic wall and the adiabatic wall temperature.

Hence, as time progresses from when the exterior cooling mechanism was first cut off at the lower wall, the wall temperature increases, and the aerodynamic heating to the wall decreases. This in turn slows the rate of increase of T_w as time progresses. The transient variations of both $\dot{q}_w$ and T_w are sketched in Figure 16.6. In Figure 16.6*a*, we see that, as time increases to large values, the heat transfer to the wall approaches zero—this is defined as the *equilibrium, or the adiabatic wall condition*. For an *adiabatic* wall, the heat transfer is, by definition, equal to *zero*. Simultaneously, the wall temperature T_w approaches asymptotically a limiting value defined as the *adiabatic wall temperature* T_{aw}, and the corresponding enthalpy is defined as the *adiabatic wall enthalpy* h_{aw}.

The purpose of this discussion is to define an adiabatic wall condition; the example involving a timewise approach to this condition was just for convenience and edification. Let us now assume that the lower wall in our Couette flow is an adiabatic wall. For this case, we already know the value of heat transfer to the wall—by definition, it is zero. The question now becomes, What is the value of the adiabatic wall enthalpy h_{aw}, and in turn the adiabatic wall temperature T_{aw}? The answer is given by Equation (16.23), where $\dot{q}_w = 0$ for an adiabatic wall.

Adiabatic wall:
$$\dot{q}_w = 0 \rightarrow \left(\frac{\partial h}{\partial y}\right)_w = \left(\frac{\partial T}{\partial y}\right)_w = 0 \tag{16.38}$$

Therefore, from Equation (16.19), with $\partial h/\partial y = 0$, $y = 0$, and $h_w = h_{aw}$, by definition

$$h_e - h_{aw} + \tfrac{1}{2}\,\mathrm{Pr}\,u_e^2 = 0$$

or
$$\boxed{h_{aw} = h_e + \mathrm{Pr}\,\frac{u_e^2}{2}} \tag{16.39}$$

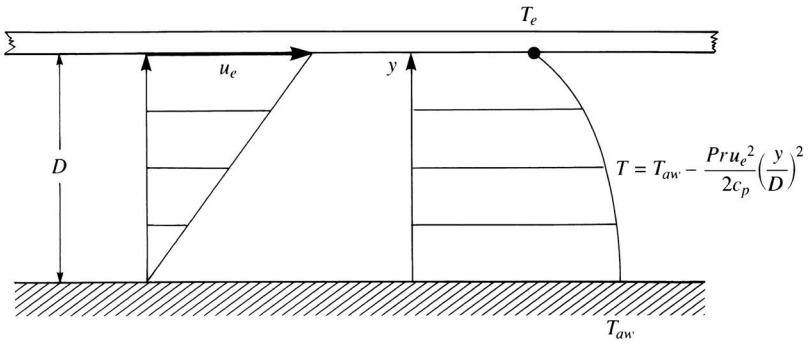


Figure 16.7 Couette flow temperature profile for an adiabatic lower wall.

In turn, the adiabatic wall temperature is given by

$$\boxed{T_{aw} = T_e + \Pr \frac{u_e^2}{2c_p}} \tag{16.40}$$

Clearly, the higher the value of u_e, the higher is the adiabatic wall temperature.

The enthalpy profile across the flow for this case is given by a combination of Equations (16.16) and (16.40), as follows. Setting $h_w = h_{aw}$ in Equation (16.16), we obtain

$$h = h_{aw} + \left(h_e + h_{aw} + \Pr \frac{u_e^2}{2}\right) \frac{y}{D} - \frac{\Pr}{2} u_e^2 \left(\frac{y}{D}\right)^2 \tag{16.41}$$

From Equation (16.39),

$$h_e - h_{aw} = -\Pr \frac{u_e^2}{2} \tag{16.42}$$

Inserting Equation (16.42) into (16.41), we have

$$h = h_{aw} - \Pr \frac{u_e^2}{2} \left(\frac{y}{D}\right)^2 \tag{16.43}$$

Equation (16.43) gives the enthalpy profile across the flow. The temperature profile follows from Equation (16.43) as

$$T = T_{aw} - \Pr \frac{u_e^2}{2c_p} \left(\frac{y}{D}\right)^2 \tag{16.44}$$

This variation of T is sketched in Figure 16.7. Note that T_{aw} is the maximum temperature in the flow. Moreover, the temperature curve is perpendicular at the plate for $y = 0$; that is, the temperature gradient at the lower plate is zero, as expected for an adiabatic wall. This result is also obtained by differentiating Equation (16.44):

$$\frac{\partial T}{\partial y} = -\Pr \frac{u_e^2}{c_p D} \left(\frac{y}{D}\right)$$

which gives $\partial T/\partial y = 0$ at $y = 0$.

16.3.4 Recovery Factor

As a corollary to the above case for the adiabatic wall, we take this opportunity to define the recovery factor—a useful engineering parameter in the analysis of aerodynamic heating. The total enthalpy of the flow at the upper plate (which represents the upper boundary on a viscous shear layer) is, by definition,

$$h_0 = h_e + \frac{u_e^2}{2} \tag{16.45}$$

(The significance and definition of total enthalpy are discussed in Section 7.5.) Compare Equation (16.45), which is a general definition, with Equation (16.39), repeated below, which is for the special case of Couette flow:

$$h_{aw} = h_e + \text{Pr}\,\frac{u_e^2}{2} \tag{16.39}$$

Note that h_{aw} is different from h_0, the difference provided by the value of Pr as it appears in Equation (16.39). We now *generalize* Equation (16.39) to a form which holds for any viscous flow, as follows:

$$\boxed{h_{aw} = h_e + r\frac{u_e^2}{2}} \tag{16.46a}$$

Similarly, Equation (16.40) can be generalized to

$$\boxed{T_{aw} = T_e + r\frac{u_e^2}{2c_p}} \tag{16.46b}$$

In Equations (16.46*a* and *b*), r is defined as the *recovery factor*. It is the factor that tells us how close the adiabatic wall enthalpy is to the total enthalpy at the upper boundary of the viscous flow. If $r = 1$, then $h_{aw} = h_0$. An alternate expression for the recovery factor can be obtained by combining Equations (16.46) and (16.45) as follows. From Equation (16.46),

$$r = \frac{h_{aw} - h_e}{u_e^2/2} \tag{16.47}$$

From Equation (16.45),

$$\frac{u_e^2}{2} = h_0 - h_e \tag{16.48}$$

Inserting Equation (16.48) into (16.47), we have

$$\boxed{r = \frac{h_{aw} - h_e}{h_0 - h_e} = \frac{T_{aw} - T_e}{T_0 - T_e}} \tag{16.49}$$

where T_0 is the total temperature. Equation (16.49) is frequently used as an alternate definition of the recovery factor.

In the special case of Couette flow, by comparing Equation (16.39) or (16.40) with Equation (16.46*a*) or (16.46*b*), we find that

$$r = \text{Pr} \tag{16.50}$$

For Couette flow, the recovery factor is simply the Prandtl number. Note that, if $\text{Pr} < 1$, then $h_{aw} < h_0$; conversely, if $\text{Pr} > 1$, then $h_{aw} > h_0$.

In more general viscous flow cases, the recovery factor is not simply the Prandtl number; however, in general, for incompressible viscous flows, we will find that the recovery *factor* is some *function* of Pr. Hence, the Prandtl number is playing its role as an important viscous flow parameter. As expected from Section 15.6, for a compressible viscous flow, the recovery factor is a function of Pr along with the Mach number and the ratio of specific heats.

16.3.5 Reynolds Analogy

Another useful engineering relation for the analysis of aerodynamic heating is Reynolds analogy, which can easily be introduced within the context of our discussion of Couette flow. Reynolds analogy is a relation between the skin friction coefficient and the heat transfer coefficient. The skin friction coefficient c_f was first introduced in Section 1.5. In our context here, we define the skin friction coefficient as

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_e u_e^2} \tag{16.51}$$

From Equation (16.9), we have, for Couette flow,

$$\tau_w = \mu \left(\frac{u_e}{D} \right) \tag{16.52}$$

Combining Equations (16.51) and (16.52), we have

$$c_f = \frac{\mu(u_e/D)}{\frac{1}{2}\rho_e u_e^2} = \frac{2\mu}{\rho_e u_e D} \tag{16.53}$$

Let us define the Reynolds number for Couette flow as

$$\text{Re} = \frac{\rho_e u_e D}{\mu}$$

Then, Equation (16.53) becomes

$$c_f = \frac{2}{\text{Re}} \tag{16.54}$$

Equation (16.54) is interesting in its own right. It demonstrates that the skin friction coefficient is a function of just the Reynolds number—a result which applies in general for other incompressible viscous flows [although the function is not necessarily the same as given in Equation (16.54)].

Now let us define a *heat transfer coefficient* as

$$C_H = \frac{\dot{q}_w}{\rho_e u_e (h_{aw} - h_w)} \tag{16.55}$$

In Equation (16.55), C_H is called the *Stanton number;* it is one of several different types of heat transfer coefficient that is used in the analysis of aerodynamic heating. It is a dimensionless quantity, in the same vein as the skin-friction coefficient. For Couette flow, from Equation (16.24), and dropping the absolute value signs for convenience, we have

$$\dot{q}_w = \frac{\mu}{\text{Pr}} \left(\frac{h_e - h_w + \frac{1}{2} \text{Pr}\, u_e^2}{D} \right) \tag{16.56}$$

Inserting Equation (16.39) into (16.56), we have for Couette flow

$$\dot{q}_w = \frac{\mu}{\text{Pr}} \left(\frac{h_{aw} - h_w}{D} \right) \tag{16.57}$$

Inserting Equation (16.57) into (16.55), we obtain

$$C_H = \frac{(\mu/\text{Pr})[(h_{aw} - h_w)/D]}{\rho_e u_e (h_{aw} - h_w)} = \frac{\mu/\text{Pr}}{\rho_e u_e D} = \frac{1}{\text{Re Pr}} \tag{16.58}$$

Equation (16.58) is interesting in its own right. It demonstrates that the Stanton number is a function of the Reynolds number and Prandtl number—a result that applies generally for other incompressible viscous flows [although the function is not necessarily the same as given in Equation (16.58)].

We now combine the results for c_f and C_H obtained above. From Equations (16.54) and (16.58), we have

$$\frac{C_H}{c_f} = \left(\frac{1}{\text{Re Pr}} \right) \frac{\text{Re}}{2}$$

or

$$\boxed{\frac{C_H}{c_f} = \frac{1}{2} \text{Pr}^{-1}} \tag{16.59}$$

Equation (16.59) is Reynolds analogy as applied to Couette flow. Reynolds analogy is, in general, a relation between the heat transfer coefficient and the skin friction coefficient. For Couette flow, this relation is given by Equation (16.59). Note that the ratio C_H/c_f is simply a function of the Prandtl number—a result that applies usually for other incompressible viscous flows, although not necessarily the same function as given in Equation (16.59).

16.3.6 Interim Summary

In this section, we have studied incompressible Couette flow. Although it is a somewhat academic flow, it has all the trappings of many practical viscous flow problems, with the added advantage of lending itself to a simple, straightforward solution. We have taken this advantage, and have discussed incompressible

Couette flow in great detail. Our major purpose in this discussion is to make the reader familiar with many concepts used in general in the analysis of viscous flows without clouding the picture with more fluid dynamic complexities. In the context of our study of Couette flow, we have one additional question to address, namely, What is the effect of compressibility? This question is addressed in the next section.

EXAMPLE 16.1

Consider the geometry sketched in Figure 16.2. The velocity of the upper plate is 200 ft/s, and the two plates are separated by a distance of 0.01 in. The fluid between the plates is air. Assume incompressible flow. The temperature of both plates is the standard sea level value of 519°R.

(*a*) Calculate the velocity in the middle of the flow.
(*b*) Calculate the shear stress.
(*c*) Calculate the maximum temperature in the flow.
(*d*) Calculate the heat transfer to either wall.
(*e*) If the lower wall is suddenly made adiabatic, calculate its temperature.

■ Solution

Assume that μ is constant throughout the flow, and that it is equal to its value of 3.7373×10^{-7} slug/ft/s at the standard sea level temperature of 519°R.

(*a*) From Equation (16.6),

$$u = u_e \left(\frac{y}{D}\right)$$

$$u = (200)\left(\tfrac{1}{2}\right) = \boxed{100 \text{ ft/s}}$$

(*b*) From Equation (16.9),

$$\tau_w = \mu \frac{u_e}{D} \qquad \text{where } D = 0.01 \text{ in} = 8.33 \times 10^{-4} \text{ ft}$$

$$\tau_w = \frac{(3.7373 \times 10^{-7})(200)}{8.33 \times 10^{-4}} = \boxed{0.09 \text{ lb/ft}^2}$$

Note that the shear stress is relatively small—less than a tenth of a pound acting over a square foot.

(*c*) From Equation (16.34), for equal wall temperatures, the maximum temperature, which occurs at $y/D = 0.5$, is

$$T = T_w + \frac{\text{Pr}}{c_p}\frac{u_e^2}{2}\left[\frac{y}{D} - \left(\frac{y}{D}\right)^2\right] = T_w + \frac{\text{Pr}\, u_e^2}{8c_p}$$

For air at standard conditions, Pr = 0.71 and $c_p = 6006$ (ft · lb)/(slug · °R). Hence,

$$T = 519 + \frac{(0.71)(200)^2}{8(6006)} = 519 + 0.6 = \boxed{519.6°\text{R}}$$

Notice that the maximum temperature in the flow is only six-tenths of a degree above the wall temperature—viscous dissipation for this relatively low-speed case is very small.

This certainly justifies our assumption of constant ρ, μ, and k in this section, and gives us a feeling for the energy changes associated with an essentially incompressible flow—they are very small.

(*d*) From Equation (16.35),

$$\dot{q}_w = \frac{\mu}{2}\left(\frac{u_e^2}{D}\right) = \frac{(3.7373 \times 10^{-7})(200)^2}{(2)(8.33 \times 10^{-4})} = \boxed{8.97 \text{ (ft} \cdot \text{lb)/(ft}^2\text{/s)}}$$

Since there are 778 ft · lb to a Btu (British thermal unit), then

$$\dot{q}_w = 8.97 \text{ (ft} \cdot \text{lb)/(ft}^2\text{/s)} = 0.0115 \text{ Btu/(ft}^2\text{/s)}$$

(*e*) From Equation (16.40),

$$T_{aw} = T_e + \frac{\text{Pr}}{c_p}\left(\frac{u_e^2}{2}\right) = 519 + \frac{(0.71)(200)^2}{(2)(6006)}$$

$$= 519 + 2.36 = \boxed{521.36^\circ\text{R}}$$

Note in the above example that the adiabatic wall temperature is higher than the maximum flow temperature calculated in part (*c*) for the cold wall case. In general, for cold wall cases, the viscous dissipation in the flow is not sufficient to heat the gas anywhere in the flow to a temperature as high as the adiabatic wall temperature. Also, we again note the comparatively low temperature increase—T_{aw} is only 2.36° higher than the upper wall temperature. In contrast, for the compressible flow to be treated in the next section, the temperature increases can be substantial—this is one of the major aspects that distinguishes compressible viscous flow from incompressible viscous flow. Note that, in the present problem, the Mach number of the upper plate is

$$M_e = \frac{u_e}{a_e} = \frac{u_e}{\sqrt{\gamma R T_e}} = \frac{200}{\sqrt{(1.4)(1716)(519)}} = 0.18$$

Again, this certainly justifies our assumption of incompressible flow for this problem.

16.4 COMPRESSIBLE COUETTE FLOW

Return to Figure 16.2, which is our general model of Couette flow. We now assume that u_e is large enough; hence, the changes in temperature within the flow are substantial enough, so that we must treat ρ, μ, and k as variables—this is compressible Couette flow. Since $T = T(y)$, then $\mu = \mu(y)$ and $k = k(y)$. Also, since $\partial p/\partial x = 0$ from the geometry and $\partial p/\partial y = 0$ from Equation (16.2), then the pressure is constant throughout the compressible Couette flow, just as in the incompressible case discussed in Section 16.3. From the equation of state, we have $\rho = p/RT$; because $T = T(y)$, ρ is also a function of y, varying inversely with temperature.

The governing equations for compressible Couette flow are Equations (16.1) to (16.3), with μ and k as variables. Let us arrange these equations in a form convenient for solution. From Equation (16.1),

$$\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right) = \frac{\partial \tau}{\partial y} = 0 \tag{16.60}$$

or

$$\tau = \text{const} \tag{16.61}$$

Hence, just as in the incompressible case, the shear stress is constant across the flow. However, keep in mind that $\mu = \mu(y)$, and, from $\tau = \mu(\partial u/\partial y)$, clearly the velocity gradient, $\partial u/\partial y$, is *not* constant across the flow—this is an essential difference between compressible and incompressible flows. With all this in mind, Equation (16.3), repeated below

$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial y}\left(\mu u\frac{\partial u}{\partial y}\right) = 0 \tag{16.3}$$

can be written as

$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \tau\frac{\partial u}{\partial y} = 0 \tag{16.62}$$

The temperature variation of μ is accurately given by Sutherland's law, Equation (15.3), for the temperature range of interest in this book. Hence, from Equation (15.3) and recalling that it is written in the International System of Units, we have

$$\tau = \mu\frac{\partial u}{\partial y} = \mu_0\left(\frac{T}{T_0}\right)^{3/2}\frac{T_0 + 110}{T + 110}\left(\frac{\partial u}{\partial y}\right) \tag{16.63}$$

The solution for compressible Couette flow requires a numerical solution of Equation (16.62). Note that, with μ and k as variables, Equation (16.62) is a *nonlinear* differential equation, and for the conditions stated, it does not have a neat, closed-form, analytic solution. Recognizing the need for a numerical solution, let us write Equation (16.62) in terms of the ordinary differential equation that it really is. (Recall that we have been using the partial differential notation only as a carry-over from the Navier-Stokes equations and to make the equations for our study of Couette flow look more familiar when treating the two-dimensional and three-dimensional viscous flows discussed in Chapters 17 to 20—just a pedagogical ploy on our part):

$$\frac{d}{dy}\left(k\frac{\partial T}{\partial y}\right) + \tau\frac{du}{dy} = 0 \tag{16.64}$$

Equation (16.64) must be solved between $y = 0$, where $T = T_w$, and $y = D$, where $T = T_e$. Note that the boundary conditions must be satisfied at two different locations in the flow, namely, at $y = 0$ and $y = D$; this is called a *two-point boundary value problem*. We present two approaches to the numerical solution of this problem. Both approaches are used for the solutions of more complex

viscous flows to be discussed in Chapters 17 through 20, and that is why they are presented here in the context of Couette flow—simply to "break the ice" for our subsequent discussions.

16.4.1 Shooting Method

This method is a classic method for the solution of the boundary-layer equations to be discussed in Chapter 17. For the solution of compressible Couette flow, the same philosophy follows as that to be applied to boundary-layer solutions, and that is why we discuss it now. The method involves a double iteration, that is, two minor iterations nested within a major iteration. The scheme is as follows:

1. *Assume* a value for τ in Equation (16.64). A reasonable assumption to start with is the incompressible value, $\tau = \mu(u_e/D)$. Also, assume that the variation of $u(y)$ is given by the incompressible result from Equation (16.6).
2. Starting at $y = 0$ with the known boundary condition $T = T_w$, integrate Equation (16.64) across the flow until $y = D$. Use any standard numerical technique for ordinary differential equations, such as the well-known Runge-Kutta method (see, e.g., Reference 52). However, to start this numerical integration, because Equation (16.64) is second order, *two* boundary conditions must be specified at $y = 0$. We only have one physical condition, namely, $T = T_w$. Therefore, we have to *assume* a second condition; let us assume a value for the temperature gradient at the wall, that is, assume a value for $(dT/dy)_w$. A value based on the incompressible flow solution discussed in Section 16.3 would be a reasonable assumption. With the *assumed* $(dT/dy)_w$ and the *known* T_w at $y = 0$, then Equation (16.64) is integrated numerically away from the wall, starting at $y = 0$ and moving in small increments, Δy in the direction of increasing y. Values of T at each increment in y are produced by the numerical algorithm.
3. Stop the numerical integration when $y = D$ is reached. Check to see if the numerical value of T at $y = D$ equals the specified boundary condition, $T = T_e$. Most likely, it will not because we have had to assume a value for $(dT/dy)_w$ in step 2. Hence, return to step 2, assume another value of $(dT/dy)_w$, and repeat the integration. Continue to repeat steps 2 and 3 until convergence is obtained, that is, until a value of $(dT/dy)_w$ is found such that, after the numerical integration, $T = T_e$ at $y = D$. From the converged temperature profile obtained by repetition of steps 2 and 3, we now have numerical values for T as a function of y that satisfy both boundary conditions; that is, $T = T_w$ at the lower wall and $T = T_e$ at the upper wall. However, do not forget that this converged solution was obtained for the *assumed* value of τ and the assumed velocity profile $u(y)$ in step 1. Therefore, the converged profile for T is not necessarily the correct profile. We must continue further; this time to find the correct value for τ.
4. From the converged temperature profile obtained by the repetitive iteration in steps 2 and 3, we can obtain $\mu = \mu(y)$ from Equation (15.3).

5. From the definition of shear stress,

$$\tau = \mu \frac{du}{dy}$$

we have

$$\frac{du}{dy} = \frac{\tau}{\mu} \tag{16.65}$$

Recall from the solution of the momentum equation, Equation (16.60), that τ is a constant. Using the assumed value of τ from step 1, and the values of $\mu = \mu(y)$ from step 4, numerically integrate Equation (16.65) starting at $y = 0$ and using the known boundary condition $u = 0$ at $y = 0$. Since Equation (16.65) is first order, this single boundary condition is sufficient to initiate the numerical integration. Values of u at each increment in y, Δy, are produced by the numerical algorithm.

6. Stop the numerical integration when $y = D$ is reached. Check to see if the numerical value of u at $y = D$ equals the specified boundary condition, $u = u_e$. Most likely, it will not, because we have had to *assume* a value of τ and $u(y)$ all the way back in step 1, which has carried through to this point in our iterative solution. Hence, return to step 5, assume another value for τ, and repeat the integration of Equation (16.65). Continue to repeat steps 5 and 6 [using the same values of $\mu = \mu(y)$ from step 4] until convergence is obtained, that is, until a value of τ is found that, after the numerical integration of Equation (16.65), $u = u_e$ at $y = D$. From the converged velocity profile obtained by repetition of steps 5 and 6, we now have numerical values for u as a function of y that satisfy both boundary conditions; that is, $u = 0$ at $y = 0$ and $u = u_e$ at $y = D$. However, do not forget that this converged solution was obtained using $\mu = \mu(y)$ from step 4, which was obtained using the initially assumed τ and $u(y)$ from step 1. Therefore, the converged profile for u obtained here is not necessarily the correct profile. We must continue one big step further.

7. Return to step 2, using the new value of τ and the new $u(y)$ obtained from step 6. Repeat steps 2 through 7 until total convergence is obtained. When this double iteration is completed, then the profile for $T = T(y)$ obtained at the last cycle of step 3, the profile for $u = u(y)$ obtained at the last cycle of step 6, and the value of τ obtained at the last cycle of step 7 are all the correct values for the given boundary conditions. The problem is solved!

Looking over the shooting method as described above, we see two minor iterations nested within a major iteration. Steps 2 and 3 constitute the first minor iteration and provide ultimately the temperature profile. Steps 5 and 6 are the second minor iteration and provide ultimately the velocity profile. Steps 2 to 7 constitute the major iteration and ultimately result in the proper value of τ.

The shooting method described above for the solution of compressible Couette flow is carried over almost directly for the solution of the boundary-layer equations to be described in Chapter 18. In the same vein, there is another

completely different approach to the solution of compressible Couette flow which carries over directly for the solution of the Navier-Stokes equations to be described in Chapter 20. This is the time-dependent, finite-difference method, first discussed in Chapter 13 and applied to the inviscid flow over a supersonic blunt body in Section 13.5. In order to prepare ourselves for Chapter 20, we briefly discuss the application of this method to the solution of compressible Couette flow.

16.4.2 Time-Dependent Finite-Difference Method

Return to the picture of Couette flow in Figure 16.2. Imagine, for a moment, that the space between the upper and lower plates is filled with a flow field which is *not* a Couette flow; for example, imagine some arbitrary flow field with gradients in both the x and y directions, including gradients in pressure. We can imagine such a flow existing at some instant during the start-up process just after the upper plate is set into motion. This would be a transient flow field, where u, T, ρ, etc., would be functions of time t as well as of x and y. Finally, after enough time elapses, the flow will approach a steady state, and this steady state will be the Couette flow solution discussed above. Let us track this picture numerically. That is, starting from an assumed initial flow field at time $t = 0$, let us solve the unsteady Navier-Stokes equations in steps of time until a steady flow is obtained at large times. As discussed in Section 13.5, the time-asymptotic steady flow is the desired result; the time-dependent approach is just a means to that end. At this stage in our discussion, it would be well for you to review the philosophy (not the details) presented in Section 13.5 before progressing further.

The Navier-Stokes equations are given by Equations (15.18*a* to *c*) and (15.26). For an unsteady, two-dimensional flow, they are

Continuity:

$$\frac{\partial p}{\partial t} = -\frac{\partial(\rho u)}{\partial x} - \frac{\partial(\rho v)}{\partial y} \tag{16.66}$$

x momentum:

$$\frac{\partial u}{\partial t} = -u\frac{\partial u}{\partial x} - v\frac{\partial u}{\partial y} - \frac{1}{\rho}\left[\frac{\partial p}{\partial x} - \frac{\partial \tau_{xx}}{\partial x} - \frac{\partial \tau_{yz}}{\partial y}\right] \tag{16.67}$$

y momentum:

$$\frac{\partial v}{\partial t} = -u\frac{\partial v}{\partial x} - v\frac{\partial v}{\partial y} - \frac{1}{\rho}\left[\frac{\partial p}{\partial y} - \frac{\partial \tau_{xy}}{\partial x} - \frac{\partial \tau_{yy}}{\partial y}\right] \tag{16.68}$$

Energy:

$$\begin{aligned}\frac{\partial(e + V^2/2)}{\partial t} = &-u\frac{\partial(e + V^2/2)}{\partial x} - v\frac{\partial(e + V^2/2)}{\partial y} \\ &+ \frac{1}{\rho}\left\{\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) - \frac{\partial(pu)}{\partial x} - \frac{\partial(pv)}{\partial y}\right. \\ &\left. + \frac{\partial(u\tau_{xx})}{\partial x} + \frac{\partial(u\tau_{yx})}{\partial y} + \frac{\partial(v\tau_{xy})}{\partial x} + \frac{\partial(v\tau_{yy})}{\partial y}\right\}\end{aligned} \tag{16.69}$$

Note that Equations (16.66) to (16.69) are written with the time derivatives on the left-hand side and spatial derivatives on the right-hand side. These are analogous to the form of the Euler equations given by Equations (13.59) to (13.62). In Equations (16.67) to (16.69), τ_{xy}, τ_{xx}, and τ_{yy} are given by Equations (15.5), (15.8), and (15.9), respectively.

The above equations can be solved by means of MacCormack's method as described in Chapter 13. This is a predictor-corrector approach, and its arrangement for the time-dependent method is described in Section 13.5. The application to compressible Couette flow is outlined as follows:

1. Divide the space between the two plates into a finite-difference grid, as sketched in Figure 16.8*a*. The length L of the grid is somewhat arbitrary, but it must be longer than a certain minimum, to be described shortly.

2. At $x = 0$ (the inflow boundary), specify some inflow conditions for u, v, ρ, and T (hence, e, since $e = c_v T$). The incompressible solution for Couette flow makes reasonable inflow boundary conditions.

3. At all the remaining grid points, arbitrarily assign values for all the flow-field variables, u, v, ρ, and T. This arbitrary flow field, which constitutes the initial conditions at $t = 0$, can have finite values of v, and can include pressure gradients.

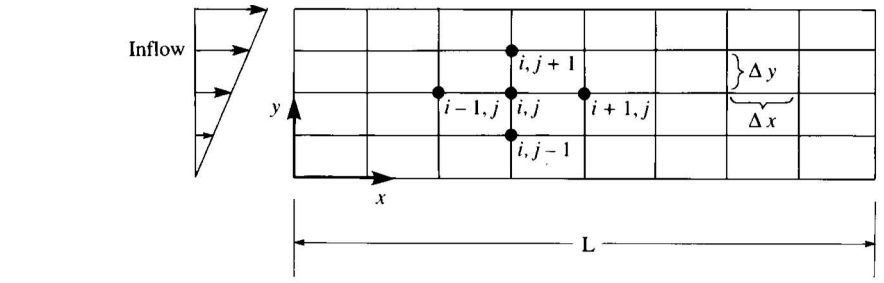
4. Starting with the initial flow field established in step 3, solve Equations (16.66) to (16.69) in steps of time. For example, consider the x-momentum equation in the form of Equation (16.67). MacCormack's predictor-corrector method, applied to this equation, is as follows.

Predictor: Assume that we know the complete flow field at time t, and we wish to advance the flow-field variables to time $t + \Delta t$. Replace the spatial derivatives with forward differences:

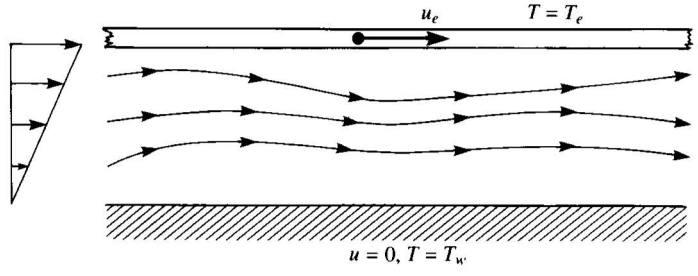
$$\left(\frac{\partial u}{\partial t}\right)_{i,j} = -u_{i,j}\left(\frac{u_{i+1,j} - u_{i,j}}{\Delta x}\right) - v_{i,j}\left(\frac{u_{i,j+1} - u_{i,j}}{\Delta y}\right) - \frac{1}{\rho_{i,j}}\left\{\frac{p_{i+1,j} - p_{i,j}}{\Delta x} - \left[\frac{(\tau_{xx})_{i+1,j} - (\tau_{xx})_{i,j}}{\Delta x}\right] - \left[\frac{(\tau_{yx})_{i,j+1} - (\tau_{yx})_{i,j}}{\Delta y}\right]\right\} \tag{16.70}$$

All the quantities on the right-hand side are known at time t; we want to advance the flow-field values to the next time, $t + \Delta t$. That is, the right-hand side of Equation (16.70) is a known number at time t. Form the *predicted* value of $u_{i,j}$ at time $t + \Delta t$, denoted by $\bar{u}_{i,j}$ from the first two terms of a Taylor's series as

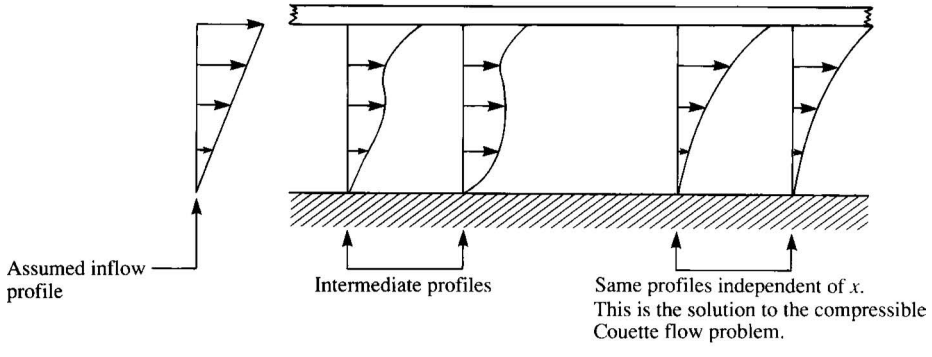
$$\bar{u}_{i,j} = \underbrace{u_{i,j}}_{\substack{\text{Known at} \\ \text{time } t}} + \underbrace{\left(\frac{\partial u}{\partial t}\right)_{i,j}}_{\substack{\text{Calculated} \\ \text{from} \\ \text{Equation (16.70)}}} \Delta t \tag{16.71}$$



(*a*) Finite-difference grid



(*b*) Transient flow



(*c*) Steady-state flow

Figure 16.8 Illustration of the finite-difference grid, and characteristics of the flow during its transient approach to the steady state.

Calculate predicted values for ρ, v, and e, namely, $\bar{\rho}_{i,j}$, $\bar{v}_{i,j}$, and $\bar{e}_{i,j}$, by the same approach applied to Equations (16.66), (16.68), and (16.69), respectively. Do this for all the grid points in Figure 16.8*a*.

Corrector: Return to Equation (16.67), and replace the spatial derivatives with rearward differences using the predicted (barred) quantities

obtained from the predictor step:

$$\left(\overline{\frac{\partial u}{\partial t}}\right)_{i,j} = -\bar{u}_{i,j}\left(\frac{\bar{u}_{i,j}-\bar{u}_{i-1,j}}{\Delta x}\right) - \bar{v}_{i,j}\left(\frac{\bar{u}_{i,j}-\bar{u}_{i,j-1}}{\Delta y}\right) \qquad (16.72)$$

$$-\frac{1}{\bar{\rho}_{i,j}}\left\{\frac{\bar{p}_{i,j}-\bar{p}_{i-1,j}}{\Delta x} - \left[\frac{(\bar{\tau}_{xx})_{i,j}-(\bar{\tau}_{xx})_{i-1,j}}{\Delta x}\right]\right.$$

$$\left. - \left[\frac{(\bar{\tau}_{yx})_{i,j}-(\bar{\tau}_{yx})_{i,j-1}}{\Delta y}\right]\right\}$$

Finally, calculate the corrected value of $u_{i,j}$ at time $t + \Delta t$, denoted by $u_{i,j}^{t+\Delta t}$, from the first two terms of a Taylor's series using an average time derivative obtained from Equations (16.70) and (16.72). That is,

$$u_{i,j}^{t+\Delta t} = u_{i,j}^{t} + \frac{1}{2}\left[\underbrace{\left(\frac{\partial u}{\partial t}\right)_{i,j}}_{\substack{\text{From}\\ \text{Equation (16.70)}}} + \underbrace{\left(\overline{\frac{\partial u}{\partial t}}\right)_{i,j}}_{\substack{\text{From}\\ \text{Equation (16.72)}}}\right]\Delta t \qquad (16.73)$$

Carry out the same process using Equations (16.66), (16.68), and (16.69) to obtain $\rho_{i,j}^{t+\Delta t}$, $v_{i,j}^{t+\Delta t}$, and $e_{i,j}^{t+\Delta t}$. The complete flow field at time $t + \Delta t$ is now obtained.

5. Repeat step 4, except starting with the newly calculated flow-field variables at the previous time. The flow-field variables will change from one time step to the next. This transient flow field will not even have parallel streamlines (i.e., there will be finite values of v throughout the flow). This is sketched in Figure 16.8*b*. Make the calculations for a large number of time steps; as we go out to large times, the changes in the flow-field variables from one time step to another will become smaller. Finally, if we go out to a large enough time (hundreds, sometimes even thousands, of time steps in some problems), the flow-field variables will not change anymore—a steady flow will be achieved, as sketched in Figure 16.8*c*. Moving from left to right in Figure 16.8*c*, we see a developing flow near the entrance, influenced by the assumed inflow profile. However, at the right of Figure 16.8*c*, the history of the inflow has died out, and the flow-field profiles become independent of distance. Indeed, we have chosen L to be a sufficient length for this to occur. The flow field near the exit is the desired solution to the compressible Couette flow problem.

The value of Δt in Equations (16.71) and (16.73) is not arbitrary. The steps outlined above constitute an *explicit* finite-difference method, and hence there is a stability bound on Δt. The value of Δt must be less than some prescribed maximum, or else the numerical solution will become unstable and "blow up"

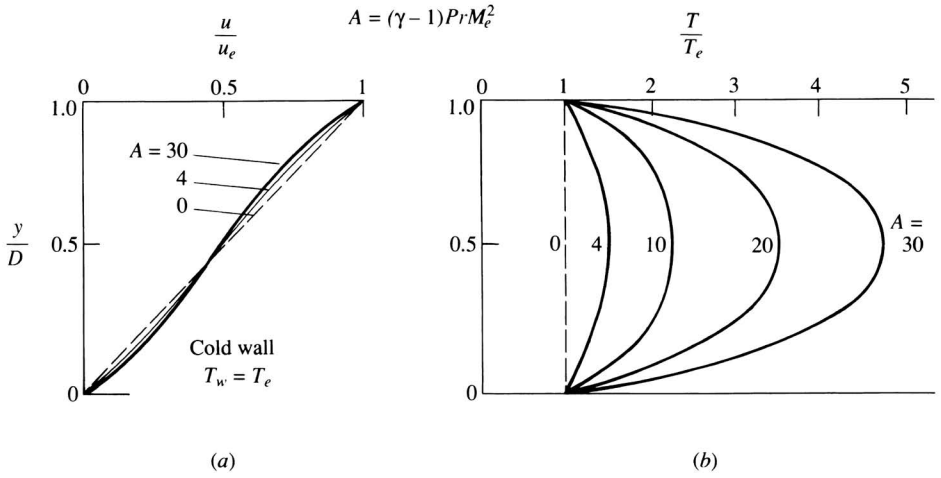


Figure 16.9 Velocity and temperature profiles for compressible Couette flow. Cold wall cases. (*Data Source:* White, Reference 43.)

on the computer. A useful expression for Δt is the Courant-Friedrichs-Lewy (CFL) criterion, which states that Δt should be the minimum of Δt_x and Δt_y, where

$$\Delta t_x = \frac{\Delta x}{u + a} \qquad \Delta t_y = \frac{\Delta y}{v + a} \tag{16.74}$$

In Equation (16.74), a is the local speed of sound. Equation (16.74) is evaluated at every grid point, and the minimum value is used to advance the whole flow field.

The time-dependent technique described above is a common approach to the solution of the compressible Navier-Stokes equations, and for that reason, it has been outlined here. Our purpose has been not so much to outline the solution of Couette flow by means of this technique, but rather to present the technique as a precursor to our later discussions on Navier-Stokes solutions.

16.4.3 Results for Compressible Couette Flow

Some typical results for compressible Couette flow are shown in Figure 16.9 for a cold wall case, and in Figure 16.10 for an adiabatic lower wall case. These results are obtained from White (Reference 43); they assume a viscosity-temperature relation of $\mu/\mu_{\max} = (T/T_{\text{ref}})^{2/3}$, which is not quite as accurate for a gas as is Sutherland's law [Equation (15.3)]. Recall from Section 15.6 that a compressible viscous flow is governed by the following similarity parameters: the Mach number, the Prandtl number, and the ratio of specific heats, γ. Therefore, we expect the results for compressible Couette flow to be governed by the same parameters. Such is the case, as illustrated in Figures 16.9 and 16.10. Here we see the different

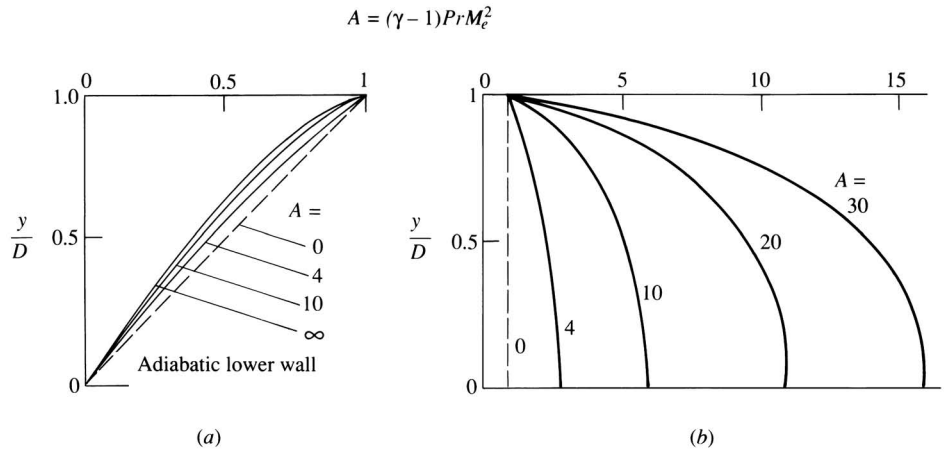


Figure 16.10 Velocity and temperature profiles for compressible Couette flow. Adiabatic lower wall. (*Data Source:* White, Reference 43.)

flow-field profiles for different values of the combined parameter $A = (\gamma - 1)\,\mathrm{Pr}\,M_e^2$. In particular, examining Figure 16.9 for the equal temperature, cold wall case, we note that:

1. From Figure 16.9*a*, the velocity profiles are not greatly affected by compressibility. The profile labeled $A = 0$ is the familiar linear incompressible case, and that labeled $A = 30$ corresponds to M_e approximately 10. Clearly, the velocity profile (in terms of u/u_e versus y/D) does not change greatly over such a large range of Mach number.
2. In contrast, from Figure 16.9*b*, there are huge temperature changes in the flow; these are due exclusively to viscous dissipation, which is a major effect at high Mach numbers. For example, for $A = 30$ ($M_e \approx 10$), the temperature in the middle of the flow is almost five times the wall temperature. Contrast this with the very small temperature increase calculated in Example 16.1 for an incompressible flow. This is why, on the scale in Figure 16.9*b*, the incompressible case ($A = 0$) is seen as essentially a vertical line.

For the adiabatic wall case shown in Figure 16.10, we note the following:

1. From Figure 16.10*a*, the velocity profiles show a pronounced curvature due to compressibility.
2. From Figure 16.10*b*, the temperature increases are larger than for the cold wall case. Note that, for $A = 30$ ($M_e \approx 10$), the maximum temperature is over 15 times that of the upper wall. Also, note the results, familiar from our discussion in Section 16.3, that the temperature is the largest at the adiabatic wall; that is, T_{aw} is the maximum temperature. As expected, Figure 16.10*b* shows that T_{aw} increases markedly as M_e increases.

In summary, in a general comparison between the incompressible flow discussed in Section 16.3 and the compressible flow discussed here, there is no tremendous *qualitative* change; that is, there is no discontinuous change in the flow-field behavior in going from subsonic to supersonic flow as is the case for an inviscid flow, such as discussed in Part 3. Qualitatively, a supersonic viscous flow is similar to a subsonic viscous flow. On the other hand, there are tremendous *quantitative* differences, especially in regard to the large temperature changes that occur due to massive viscous dissipation in a high-speed compressible viscous flow. The physical reason for this difference in viscous versus inviscid flow is as follows. In an inviscid flow, information is propagated via the mechanism of pressure waves traveling throughout the flow. This mechanism changes radically when the flow goes from subsonic to supersonic. In contrast, for a viscous flow, information is propagated by the diffusive transport mechanisms of μ and k (a molecular phenomenon), and these mechanisms are not basically changed when the flow goes from subsonic to supersonic. These statements hold in general for any viscous flow, not just for the Couette flow case treated here.

16.4.4 Some Analytical Considerations

For air temperatures up to 1000 K, the specific heats are essentially constant, thus justifying the assumption of a calorically perfect gas for this range. Moreover, the temperature variations of μ and k over this range are virtually identical. As a result, the Prandtl number, $\mu c_p/k$, is essentially *constant* up to temperatures on the order of 1000 K. This is shown in Figure 16.11, obtained from Schetz (Reference 53). Note that Pr $\approx$ 0.71 for air; this is the value that was used in Example 16.1.

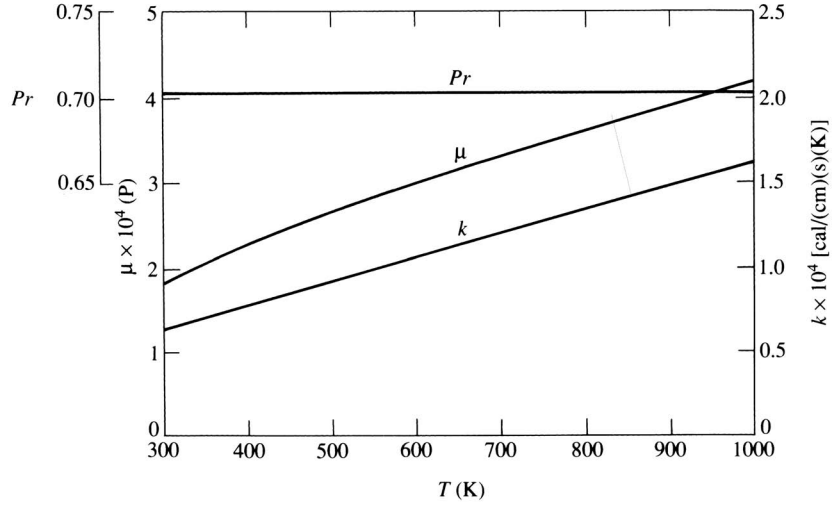


Figure 16.11 Variation of viscosity coefficient, thermal conductivity, and Prandtl number for air as a function of temperature. (*Data Source:* From Schetz, Reference 53.)

Question: How high a Mach number can exist before we would expect to encounter temperatures in the flow above 1000 K? *Answer:* An approximate answer is to calculate that Mach number at which the total temperature is 1000 K. Assuming a static temperature $T = 288$ K, from Equation (8.40),

$$M = \sqrt{\frac{2}{\gamma - 1}\left(\frac{T_0}{T} - 1\right)} = \sqrt{\frac{2}{0.4}\left(\frac{1000}{288} - 1\right)} = 3.5$$

Hence, for most aeronautical applications involving flight at a Mach number of 3.5 or less, the temperature within the viscous portions of the flow field will not exceed 1000 K. A Mach number of 3.5 or less encompasses virtually all operational aircraft today, with the exception of a few hypersonic test vehicles.

In light of the above, many viscous flow solutions are carried out making the justifiable assumption of a *constant Prandtl number*. For the case of compressible Couette flow, the assumption of Pr = constant allows the following analysis. Consider the energy equation, Equation (16.3), repeated below:

$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial y}\left(\mu u\frac{\partial u}{\partial y}\right) = 0 \tag{16.3}$$

Since $T = h/c_p$ and $\text{Pr} = \mu c_p/k$, Equation (16.3) can be written as

$$\frac{\partial}{\partial y}\left(\frac{\mu}{\text{Pr}}\frac{\partial h}{\partial y}\right) + \frac{\partial}{\partial y}\left(\mu u\frac{\partial u}{\partial y}\right) = 0 \tag{16.75}$$

or

$$\frac{\partial}{\partial y}\left[\mu\left(\frac{1}{\text{Pr}}\frac{\partial h}{\partial y} + u\frac{\partial u}{\partial y}\right)\right] = 0 \tag{16.76}$$

Integrating Equation (16.76) with respect to y, we have

$$\frac{1}{\text{Pr}}\frac{\partial h}{\partial y} + u\frac{\partial u}{\partial y} = \frac{a}{\mu} \tag{16.77}$$

Since $\tau = \mu(\partial u/\partial y)$, we have $\mu = \tau(\partial u/\partial y)^{-1}$. Also, recalling from Equation (16.61) that τ is constant, we can write the right-hand side of Equation (16.77) as

$$\frac{a}{\mu} = \frac{a}{\tau}\frac{\partial u}{\partial y} = b\frac{\partial u}{\partial y}$$

where b is a constant. With this, Equation (16.77) becomes

$$\frac{1}{\text{Pr}}\frac{\partial h}{\partial y} + \frac{\partial(u^2/2)}{\partial y} - b\frac{\partial u}{\partial y} = 0 \tag{16.78}$$

Integrating Equation (16.78) with respect to y, remembering that Pr = constant, we have

$$\frac{h}{\text{Pr}} + \frac{u^2}{2} - bu = c \tag{16.79}$$

where c is another constant of integration. Expressions for b and c can be obtained by evaluating Equation (16.79) at $y = 0$ and $y = D$. At $y = 0$, $h = h_w$ and $u = 0$; hence,

$$c = \frac{h_w}{\text{Pr}}$$

At $y = D$, $h = h_e$ and $u = u_e$; hence,

$$b = \frac{1}{u_e}\left(\frac{h_e - h_w}{\text{Pr}}\right) + \frac{u_e}{2}$$

Inserting b and c into Equation (16.79) and simplifying, we obtain

$$h + \text{Pr}\frac{u^2}{2} = h_w + \frac{u}{u_e}(h_e - h_w) + \frac{\text{Pr}}{2}(uu_e) \tag{16.80}$$

Assume the lower wall is adiabatic; that is, $(\partial h/\partial y)_w = 0$. Differentiating Equation (16.80) with respect to y, we have

$$\frac{\partial h}{\partial y} = -\text{Pr}\,u\frac{\partial u}{\partial y} + \left(\frac{h_e + h_w}{u_e}\right)\frac{\partial u}{\partial y} + \frac{u_e\,\text{Pr}}{2}\frac{\partial u}{\partial y}$$

or

$$\frac{\partial h}{\partial y} = \left(-u\,\text{Pr} + \frac{h_e - h_w}{u_e} + \frac{u_e\,\text{Pr}}{2}\right)\frac{\partial u}{\partial y} \tag{16.81}$$

Recall that the condition for an adiabatic wall is that $(\partial h/\partial y)_w = 0$. Applying Equation (16.81) at $y = 0$ for an adiabatic wall, where $u = 0$ and by definition $h_w = h_{aw}$, we have

$$\left(\frac{\partial h}{\partial y}\right)_w = \left(\frac{h_e - h_{aw}}{u_e} + \frac{u_e\,\text{Pr}}{2}\right)\left(\frac{\partial u}{\partial y}\right)_w = 0$$

Since $(\partial u/\partial y)_w$ is finite, then

$$\frac{h_e - h_{aw}}{u_e} + \frac{u_e\,\text{Pr}}{2} = 0$$

or

$$\boxed{h_{aw} = h_e + \text{Pr}\frac{u_e^2}{2}} \tag{16.82}$$

This is identical to Equation (16.39) obtained for incompressible flow. Hence, we have just proven that the recovery factor for compressible Couette flow, assuming constant Prandtl number, is also

$$r = \text{Pr} \tag{16.83}$$

Since the recovery factors for the incompressible and compressible cases are the same (as long as Pr = constant), what can we say about Reynolds analogy? Does Equation (16.59) hold for the compressible case? Let us examine this question. Return to Equation (16.3), repeated below:

$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial y}\left(\mu u\frac{\partial u}{\partial y}\right) = 0 \tag{16.3}$$

Recalling that, from the definitions,

$$\dot{q} = k\frac{\partial T}{\partial y} \tag{16.84}$$

and

$$\tau = \mu\frac{\partial u}{\partial y} \tag{16.85}$$

then Equation (16.3) can be written as

$$\frac{\partial \dot{q}}{\partial y} + \frac{\partial(\tau u)}{\partial y} = 0 \tag{16.86}$$

Integrating Equation (16.86) with respect to y, we have

$$\dot{q} + \tau u = a \tag{16.87}$$

where a is a constant of integration. Evaluating Equation (16.87) at $y = 0$, where $u = 0$ and $\dot{q} = q_w$, we find that

$$a = \dot{q}_w$$

Hence, Equation (16.87) is

$$\dot{q} + \tau u = \dot{q}_w \tag{16.88}$$

Inserting Equations (16.84) and (16.85) into (16.88), we have

$$\dot{q}_w = k\frac{\partial T}{\partial y} + \mu u\frac{\partial u}{\partial y} \tag{16.89}$$

or

$$\frac{\dot{q}_w}{\mu} = \frac{k}{\mu}\frac{\partial T}{\partial y} + u\frac{\partial u}{\partial y} \tag{16.90}$$

Recall that the shear stress is constant throughout the flow; hence,

$$\tau = \mu\frac{\partial u}{\partial y} = \tau_w$$

or

$$\mu = \frac{\tau_w}{\partial u/\partial y} \tag{16.91}$$

Also,

$$\frac{k}{\mu} = \frac{c_p}{\text{Pr}} \tag{16.92}$$

Inserting Equation (16.91) into the left-hand side of Equation (16.90), and Equation (16.92) into the right-hand side of Equation (16.90), we have

$$\frac{\dot{q}_w}{\tau_w}\frac{\partial u}{\partial y} = \frac{c_p}{\text{Pr}}\frac{\partial T}{\partial y} + \frac{\partial(u^2/2)}{\partial y} \tag{16.93}$$

Integrate Equation (16.93) between the two plates, keeping in mind that $\dot{q}_w$, τ_w, c_p, and Pr are all fixed values:

$$\frac{\dot{q}_w}{\tau_w}\int_0^D \frac{\partial u}{\partial y}dy = \frac{c_p}{\text{Pr}}\int_0^D \frac{\partial T}{\partial y}dy + \int_0^D \frac{\partial(u^2/2)}{\partial y}dy$$

or
$$\frac{\dot{q}_w}{\tau_w}\int_0^{u_e} du = \frac{c_p}{\text{Pr}}\int_{T_w}^{T_e} dT + \int_0^{u_e} d\left(\frac{u^2}{2}\right)$$

which yields

$$\frac{\dot{q}_w}{\tau_w}u_e = \frac{c_p}{\text{Pr}}(T_e - T_w) + \frac{u_e^2}{2} \tag{16.94}$$

Rearranging Equation (16.94), and recalling that $h = c_p T$, we have

$$\dot{q}_w = \frac{\tau_w}{u_e\,\text{Pr}}\left(h_e - h_w + \text{Pr}\,\frac{u_e^2}{2}\right) \tag{16.95}$$

Inserting Equation (16.82) into (16.95), we have

$$\dot{q}_w = \frac{\tau_w}{u_e\,\text{Pr}}(h_{aw} - h_w) \tag{16.96}$$

The skin-friction coefficient and Stanton number are defined by Equations (16.51) and (16.55), respectively. Thus, their ratio is

$$\frac{C_H}{c_f} = \frac{\dot{q}_w/[\rho_e u_e(h_{aw} - h_w)]}{\tau_w/\left(\frac{1}{2}\rho_e u_e^2\right)} = \frac{\dot{q}_w}{\tau_w}\left[\frac{u_e}{2(h_{aw} - h_w)}\right] \tag{16.97}$$

Inserting Equation (16.96) into (16.97), we have

$$\frac{C_H}{c_f} = \frac{(h_{aw} - h_w)}{u_e\,\text{Pr}}\left[\frac{u_e}{2(h_{aw} - h_w)}\right]$$

or
$$\boxed{\frac{C_H}{c_f} = \frac{1}{2}\text{Pr}^{-1}} \tag{16.98}$$

Equation (16.98) is *Reynolds analogy*—a relation between heat transfer and skin friction coefficients. Moreover, it is precisely the same result as obtained in Equation (16.59) for incompressible flow. Hence, for a constant Prandtl number, we have shown that Reynolds analogy is precisely the same form for incompressible and compressible flow.

EXAMPLE 16.2

Consider the geometry given in Figure 16.2. The two plates are separated by a distance of 0.01 in (the same as in Example 16.1). The temperature of the two plates is equal, at a value of 288 K (standard sea level temperature). The air pressure is constant throughout the flow and equal to 1 atm. The upper plate is moving at Mach 3. The shear stress at the lower wall is 72 N/m^2. (This is about 1.5 lb/ft^2—a much larger value than that associated with the low-speed case treated in Example 16.1.) Calculate the heat transfer to either plate. (Since the shear stress is constant throughout the flow, and the plates are at equal temperature, the heat transfer to the upper and lower plates is the same.)

■ Solution

The velocity of the upper plate is

$$u_e = M_e a_e = M_e\sqrt{\gamma RT_e} = 3\sqrt{(1.4)(288)(287)} = 1020 \text{ m/s}$$

The air density at both plates is (noting that 1 atm $= 1.01 \times 10^5$ N/m^2)

$$\rho_e = \frac{p_e}{RT_e} = \frac{1.01 \times 10^5}{(287)(288)} = 1.22 \text{ kg/m}^3$$

Hence, the skin-friction coefficient is

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho_e u_e^2} = \frac{72}{(0.5)(1.22)(1020)^2} = 1.13 \times 10^{-4}$$

From Reynolds analogy, Equation (16.92), we have

$$C_H = \frac{c_f}{2\,\text{Pr}} = \frac{1.13 \times 10^{-4}}{2(0.71)} = 8 \times 10^{-5}$$

The adiabatic wall enthalpy, from Equation (16.82), is

$$h_{aw} = h_e + \text{Pr}\,\frac{u_e^2}{2} = c_p T_e + \text{Pr}\,\frac{u_e^2}{2}$$

For air, $c_p = 1004.5$ J/kg · K. Thus,

$$h_{aw} = (1004.5)(288) + (0.71)\frac{(1020)^2}{2} = 6.59 \times 10^5 \text{ J/kg}$$

[*Note:* This gives $T_{aw} = h_{aw}/c_p = (6.59 \times 10^5)/1004.5 = 656$ K. In the adiabatic case, the wall would be quite warm.] Hence, from the definition of the Stanton number [Equation (16.55)], and noting that $h_w = c_p T_w = (1004.6)(288) = 2.89 \times 10^5$ J/kg,

$$\dot{q}_w = \rho_e u_e (h_{aw} - h_w) C_H = (1.22)(1020)[(6.59 - 2.89) \times 10^5](8 \times 10^{-5})$$

$$= \boxed{3.68 \times 10^4 \text{ W/m}^2}$$

16.5 SUMMARY

The parallel flow discussed in this chapter illustrates features common to many more complex viscous flows, with the added advantage of lending itself to a relatively straightforward solution. The purpose of this discussion has been to introduce many of the basic concepts of viscous flows in a fashion unencumbered by fluid dynamic complexities. In particular, we have studied Couette flow and found the following.

1. The driving force is the shear stress between the moving wall and the fluid. Shear stress is constant across the flow for both incompressible and compressible cases.

2. For incompressible Couette flow,

$$u = u_e \left(\frac{y}{D} \right) \tag{16.6}$$

$$\tau = \mu \left(\frac{u_e}{D} \right) \tag{16.9}$$

3. The heat transfer depends on the wall temperatures and the amount of viscous dissipation. For an adiabatic wall, the wall enthalpy is

$$h_{aw} = h_e + r \frac{u_e^2}{2} \tag{16.46a}$$

For incompressible and compressible Couette flow with a constant Prandtl number, the recovery factor is

$$r = \text{Pr}$$

and Reynolds analogy holds in both cases;

$$\frac{C_H}{c_f} = \frac{1}{2} \text{Pr}^{-1} \tag{16.59}$$

CHAPTER 17

Introduction to Boundary Layers

A very satisfactory explanation of the physical process in the boundary layer between a fluid and a solid body could be obtained by the hypothesis of an adhesion of the fluid to the walls, that is, by the hypothesis of a zero relative velocity between fluid and wall. If the viscosity was very small and the fluid path along the wall not too long, the fluid velocity ought to resume its normal value at a very short distance from the wall. In the thin transition layer however, the sharp changes of velocity, even with small coefficient of friction, produce marked results.

Ludwig Prandtl, 1904

PREVIEW BOX

The revelations in this chapter are simply amazing. Although the influence of friction is present at every point throughout every flow, on a practical basis it is usually of no consequence *except* in a thin region adjacent to the surface of a body immersed in the flow, or in the boundary region between two flows of widely different velocities. For the former, the thin region is called a *boundary layer,* and for the latter it is called a *shear layer*. After centuries of trying to solve flows taking into account friction, most to no avail, the introduction of the boundary-layer concept revolutionized fluid dynamics and made the analysis of flows with friction tractable. It is amazing that what goes on inside the thin boundary layer is the physical mechanism for skin friction and aerodynamic heating to the surface, and for flow separation from the surface. It is truly amazing that to calculate skin friction and aerodynamic heating at the surface, we have only to account for friction and thermal conduction within the thin boundary layer, and *not* in the large region of flow outside the boundary layer.

This chapter is all about boundary layers. Read on, and be prepared to be amazed.

17.1 INTRODUCTION

The above quotation is taken from an historic paper given by Ludwig Prandtl at the third Congress of Mathematicians at Heidelberg, Germany, in 1904. In this paper, the concept of the boundary layer was first introduced—a concept which eventually revolutionized the analysis of viscous flows in the twentieth century and which allowed the practical calculation of drag and flow separation over aerodynamic bodies. Before Prandtl's 1904 paper, the Navier-Stokes equations discussed in Chapter 15 were well known, but fluid dynamicists were frustrated in their attempts to solve these equations for practical engineering problems. After 1904, the picture changed completely. Using Prandtl's concept of a boundary layer adjacent to an aerodynamic surface, the Navier-Stokes equations can be reduced to a more tractable form called the *boundary-layer equations*. In turn, these boundary-layer equations can be solved to obtain the distributions of shear stress and aerodynamic heat transfer to the surface. Prandtl's boundary-layer concept was an advancement in the science of fluid mechanics of the caliber of a Nobel prize, although he never received that honor. The purpose of this chapter is to present the general concept of the boundary layer and to give a few representative samples of its application. Our purpose here is to provide only an introduction to boundary-layer theory; consult Reference 42 for a rigorous and thorough discussion of boundary-layer analysis and applications. Also, a general introduction to boundary layers was given in Section 1.11. If you have not read that section, now would be a good time to pause and return to Section 1.11 before progressing further.

What is a boundary layer? We have used this term in several places in our previous chapters, first introducing the idea in Section 1.10 and illustrating the concept in Figure 1.42. The boundary layer is the thin region of flow adjacent to a surface, where the flow is retarded by the influence of friction between a solid surface and the fluid. For example, a photograph of the flow over a supersonic body is shown in Figure 17.1, where the boundary layer (along with shock and expansion waves and the wake) is made visible by a special optical technique called a *shadowgraph* (see References 25 and 26 for discussions of the shadowgraph method). Note how thin the boundary layer is in comparison with the size of the body; however, although the boundary layer occupies geometrically only a small portion of the flow field, its influence on the drag and heat transfer to the body is immense—in Prandtl's own words as quoted above, it produces "marked results."

The purpose of the remaining chapters is to examine these "marked results." The road map for the present chapter is given in Figure 17.2. In the next section, we discuss some fundamental properties of boundary layers. This is followed by a development of the boundary-layer equations, which are the continuity, momentum, and energy equations written in a special form applicable to the flow in the thin viscous region adjacent to a surface. The boundary layer equations are partial differential equations that apply *inside* the boundary layer.

Finally, we note that this chapter represents the second of the three options discussed in Section 15.7 for the solution of the viscous flow equations, namely,

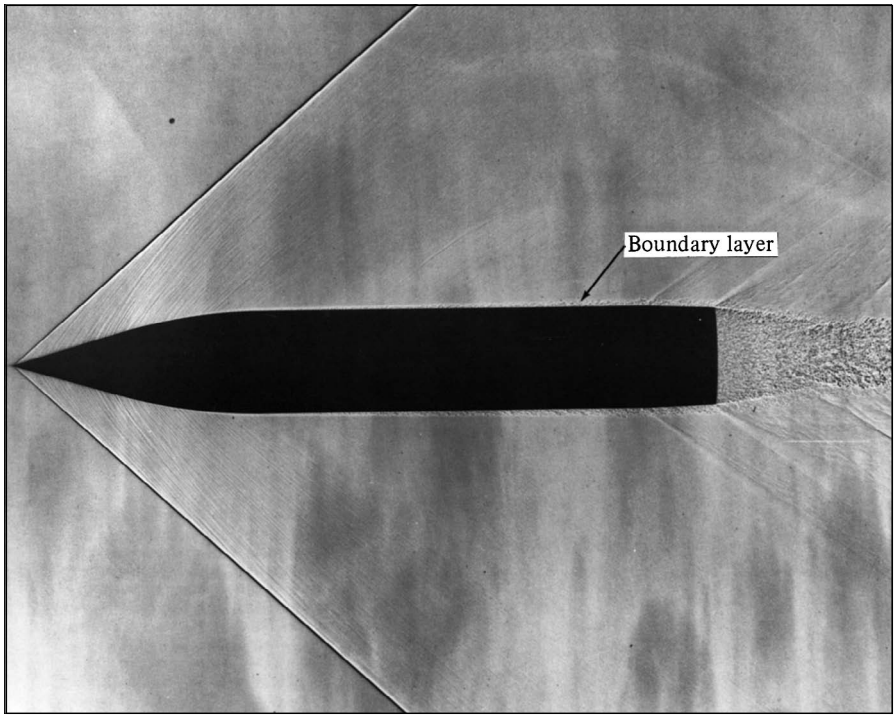


Figure 17.1 The boundary layer on an aerodynamic body. (*Courtesy of the U.S. Army Ballistics Laboratory, Aberdeen, Maryland.*)

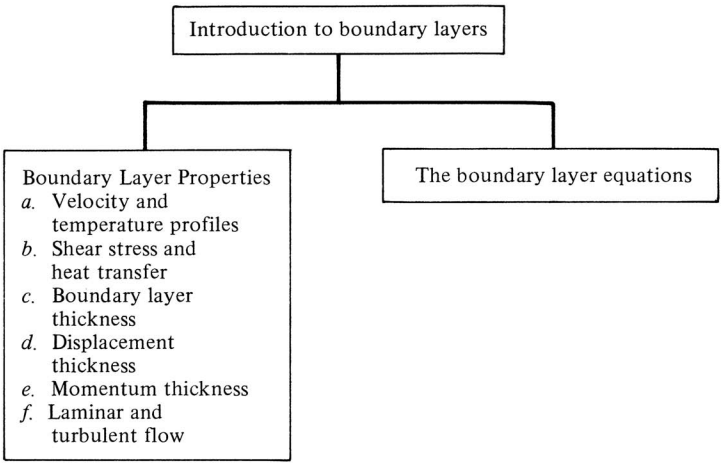


Figure 17.2 Road map for Chapter 17.

the simplification of the Navier-Stokes equations by neglecting certain terms that are smaller than other terms. This is an approximation, not a precise condition as in the case of Couette and Poiseuille flows in Chapter 16. In this chapter, we will see that the Navier-Stokes equations, when applied to the thin viscous boundary layer adjacent to a surface, can be reduced to simpler forms, albeit approximate, which lend themselves to simpler solutions. These simpler forms of the equations are called the boundary-layer equations—they are the subject of the present chapter.

17.2 BOUNDARY-LAYER PROPERTIES

Consider the viscous flow over a flat plate as sketched in Figure 17.3. The viscous effects are contained within a thin layer adjacent to the surface; the thickness is exaggerated in Figure 17.3 for clarity. Immediately at the surface, the flow velocity is zero; this is the "no-slip" condition discussed in Section 15.2. In addition, the temperature of the fluid immediately at the surface is equal to the temperature of the surface; this is called the *wall temperature* T_w, as shown in Figure 17.3. Above the surface, the flow velocity increases in the y direction until, for all practical purposes, it equals the freestream velocity. This will occur at a height above the wall equal to δ, as shown in Figure 17.3. More precisely, δ is defined as that distance above the wall where $u = 0.99u_e$; here, u_e is the velocity at the outer edge of the boundary layer. In Figure 17.3, which illustrates the flow over a flat plate, the velocity at the edge of the boundary layer will be V_∞; that is, $u_e = V_\infty$. For a body of general shape, u_e is the velocity obtained from an inviscid flow solution evaluated at the body surface (or at the "effective body" surface, as discussed later). The quantity δ is called the *velocity boundary-layer thickness*. At any given x station, the variation of u between $y = 0$ and $y = \delta$, that is, $u = u(y)$, is defined as the *velocity profile* within the boundary layer, as sketched in Figure 17.3. This profile is different for different x stations. Similarly, the flow temperature will change above the wall, ranging from $T = T_w$ at $y = 0$ to $T = 0.99T_e$ at $y = \delta_T$. Here, δ_T is defined as the *thermal boundary-layer thickness*. At any given x station, the variation of T between $y = 0$ and $y = \delta_T$, that is, $T = T(y)$, is called the *temperature profile* within the boundary layer, as sketched in Figure 17.3. (In the above, T_e is the temperature at the edge of the

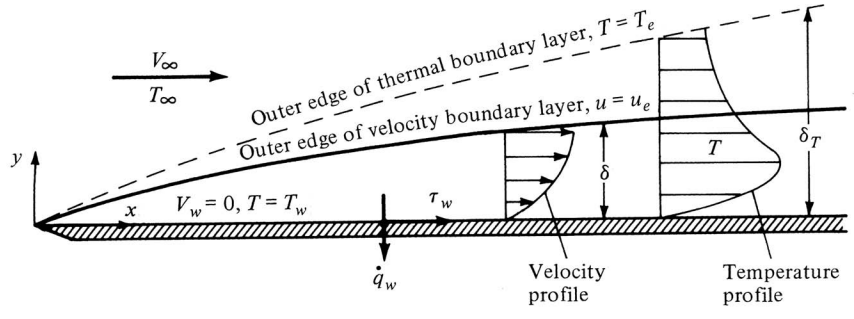


Figure 17.3 Boundary-layer properties.

thermal boundary layer. For the flow over a flat plate, as sketched in Figure 17.3, $T_e = T_\infty$. For a general body, T_e is obtained from an inviscid flow solution evaluated at the body surface, or at the "effective body" surface, to be discussed later.) Hence, two boundary layers can be defined: a velocity boundary layer with thickness δ and a temperature boundary layer with thickness δ_T. In general, $\delta_T \neq \delta$. The relative thicknesses depend on the Prandtl number: it can be shown that if $\Pr = 1$, then $\delta = \delta_T$; if $\Pr > 1$, then $\delta_T < \delta$; if $\Pr < 1$, then $\delta_T > \delta$. For air at standard conditions, $\Pr = 0.71$; hence, the thermal boundary layer is thicker than the velocity boundary layer, as shown in Figure 17.3. Note that both boundary-layer thicknesses increase with distance from the leading edge; that is, $\delta = \delta(x)$ and $\delta_T = \delta_T(x)$.

The consequence of the velocity gradient at the wall is the generation of shear stress at the wall,

$$\tau_w = \mu \left(\frac{\partial u}{\partial y} \right)_w \tag{17.1}$$

where $(\partial u/\partial y)_w$ is the velocity gradient evaluated at $y = 0$ (i.e., at the wall). Similarly, the temperature gradient at the wall generates heat transfer at the wall,

$$\dot{q}_w = -k \left(\frac{\partial T}{\partial y} \right)_w \tag{17.2}$$

where $(\partial T/\partial y)_w$ is the temperature gradient evaluated at $y = 0$ (i.e., at the wall). In general, both τ_w and $\dot{q}_w$ are functions of distance from the leading edge; that is, $\tau_w = \tau_w(x)$ and $\dot{q}_w = \dot{q}_w(x)$. One of the central purposes of boundary-layer theory is to compute τ_w and $\dot{q}_w$.

A frequently used boundary-layer property is the *displacement thickness* δ^*, defined as

$$\boxed{\delta^* \equiv \int_0^{y_1} \left(1 - \frac{\rho u}{\rho_e u_e} \right) dy \quad \delta \leq y_1 \to \infty} \tag{17.3}$$

The displacement thickness has two physical interpretations:

1. δ^* is an index proportional to the "missing mass flow" due to the presence of the boundary layer. Let us explain. Consider point y_1 above the boundary layer, as shown in Figure 17.4. Consider also the mass flow (per unit depth perpendicular to the page) across the vertical line connecting $y = 0$ and $y = y_1$. Then

$$A = \text{actual mass flow between 0 and } y_1 = \int_0^{y_1} \rho u \, dy$$

$$B = \begin{array}{l}\text{hypothetical mass flow} \\ \text{between 0 and } y_1 \textit{ if} \text{ boundary} \\ \text{layer were not present}\end{array} = \int_0^{y_1} \rho_e u_e \, dy$$

$$B - A = \begin{array}{l}\text{decrement in mass flow due to} \\ \text{presence of boundary layer, that is,} \\ \text{missing mass flow}\end{array} = \int_0^{y_1} (\rho_e u_e - \rho u) \, dy \tag{17.4}$$

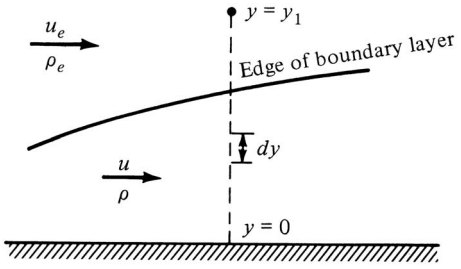


Figure 17.4 Construction for the discussion of displacement thickness.

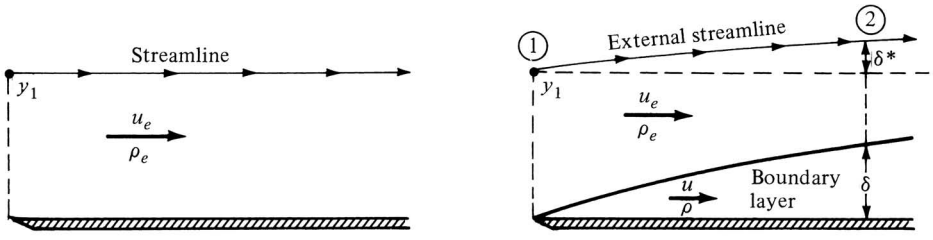


Figure 17.5 Displacement thickness is the distance by which an external flow streamline is displaced by the presence of the boundary layer.

Express this missing mass flow as the product of $\rho_e u_e$ and a height δ^*; that is,

$$\text{Missing mass flow} = \rho_e u_e \, \delta^* \tag{17.5}$$

Equating Equations (17.4) and (17.5), we have

$$\rho_e u_e \delta^* = \int_0^{y_1} (\rho_e u_e - \rho u) \, dy$$

or

$$\delta^* = \int_0^{y_1} \left(1 - \frac{\rho u}{\rho_e u_e} \right) dy \tag{17.6}$$

Equation (17.6) is identical to the definition of δ^* given in Equation (17.3). Hence, clearly δ^* is a height proportional to the missing mass flow. If this missing mass flow was crammed into a streamtube where the flow properties were constant at ρ_e and u_e, then Equation (17.5) says that δ^* is the height of this hypothetical streamtube.

2. The second physical interpretation of δ^* is more practical than the one discussed above. Consider the flow over a flat surface as sketched in Figure 17.5. At the left is a picture of the hypothetical inviscid flow over the surface; a streamline through point y_1 is straight and parallel to the surface.

The actual viscous flow is shown at the right of Figure 17.5; here, the retarded flow inside the boundary layer acts as a partial obstruction to the freestream flow. As a result, the streamline external to the boundary layer passing through point y_1 is deflected upward through a distance δ^*. We now prove that this δ^* is precisely the displacement thickness defined by Equation (17.3). At station 1 in Figure 17.5, the mass flow (per unit depth perpendicular to the page) between the surface and the external streamline is

$$\dot{m} = \int_0^{y_1} \rho_e u_e \, dy \tag{17.7}$$

At station 2, the mass flow between the surface and the external streamline is

$$\dot{m} = \int_0^{y_1} \rho u \, dy + \rho_e u_e \, \delta^* \tag{17.8}$$

Since the surface and the external streamline form the boundaries of a streamtube, the mass flows across stations 1 and 2 are equal. Hence, equating Equations (17.7) and (17.8), we have

$$\int_0^{y_1} \rho_e u_e \, dy = \int_0^{y_1} \rho u \, dy + \rho_e u_e \, \delta^*$$

or

$$\delta^* = \int_0^{y_1} \left(1 - \frac{\rho u}{\rho_e u_e}\right) dy \tag{17.9}$$

Hence, the height by which the streamline in Figure 17.5 is displaced upward by the presence of the boundary layer, namely, δ^*, is given by Equation (17.9). However, Equation (17.9) is precisely the definition of the displacement thickness given by Equation (17.3). Thus, the displacement thickness, first defined by Equation (17.3), is physically the distance through which the external inviscid flow is displaced by the presence of the boundary layer.

This second interpretation of δ^* gives rise to the concept of an *effective body*. Consider the aerodynamic shape sketched in Figure 17.6. The actual contour of

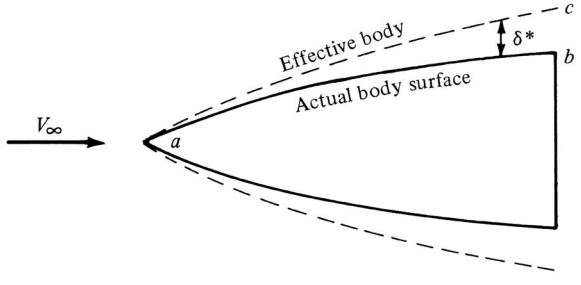


Figure 17.6 The "effective body," equal to the actual body shape plus the displacement thickness distribution.

the body is given by curve ab. However, due to the displacement effect of the boundary layer, the shape of the body effectively seen by the freestream is not given by curve ab; rather, the freestream sees an effective body given by curve ac. In order to obtain the conditions ρ_e, u_e, T_e, etc., at the outer edge of the boundary layer on the actual body ab, an inviscid flow solution should be carried out for the effective body, and ρ_e, u_e, T_e, etc., are obtained from this inviscid solution evaluated along curve ac.

Note that in order to solve for δ^* from Equation (17.3), we need the profiles of u and ρ from a solution of the boundary-layer flow. In turn, to solve the boundary-layer flow, we need ρ_e, u_e, T_e, etc. However, ρ_e, u_e, T_e, etc., depend on δ^*. This leads to an iterative solution. To calculate accurately the boundary-layer properties as well as the pressure distribution over the surface of the body in Figure 17.6, we proceed as follows:

1. Carry out an inviscid solution for the given body shape ab. Evaluate ρ_e, u_e, T_e, etc., along curve ab.
2. Using these values of ρ_e, u_e, T_e, etc., solve the boundary-layer equations (discussed in Sections 17.3 to 17.6) for $u = u(y)$, $\rho = \rho(y)$, etc., at various stations along the body.
3. Obtain δ^* at these stations from Equation (17.3). This will not be an accurate δ^* because ρ_e, u_e, T_e, etc., were evaluated on curve ab, not the proper effective body. Using this intermediate δ^*, calculate an effective body, given by a curve ac' (not shown in Figure 17.6).
4. Carry out an inviscid solution for the flow over the intermediate effective body ac', and evaluate new values of ρ_e, u_e, T_e, etc., along ac'.
5. Repeat steps 2 to 4 above until the solution at one iteration essentially does not deviate from the solution at the previous iteration. At this stage, a converged solution will be obtained, and the final results will pertain to the flow over the proper effective body ac shown in Figure 17.6.

In some cases, the boundary layers are so thin that the effective body can be ignored, and a boundary-layer solution can proceed directly from ρ_e, u_e, T_e, etc., obtained from an inviscid solution evaluated on the actual body surface (ab in Figure 17.6). However, for highly accurate solutions, and for cases where the boundary-layer thickness is relatively large (such as for hypersonic flow as discussed in Chapter 14), the iterative procedure described above should be carried out. Also, we note parenthetically that δ^* is usually smaller than δ; typically, $\delta^* \approx 0.3\,\delta$.

Another boundary-layer property of importance is the *momentum thickness* θ, defined as

$$\theta \equiv \int_0^{y_1} \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e}\right) dy \qquad \delta \leq y_1 \to \infty \tag{17.10}$$

To understand the physical interpretation of θ, return again to Figure 17.4. Consider the mass flow across a segment dy, given by $dm = \rho u\, dy$. Then

$$A = \text{momentum flow across } dy = dm\, u = \rho u^2\, dy$$

If this same elemental mass flow were associated with the freestream, where the velocity is u_e, then

$$B = \begin{cases} \text{momentum flow at freestream} \\ \text{velocity associated with mass } dm = dm\, u_e = (\rho u\, dy) u_e \end{cases}$$

Hence,

$$B - A = \begin{cases} \text{decrement in momentum flow} \\ \text{(missing momentum flow) associated} = \rho u(u_e - u)\, dy \\ \text{with mass } dm \end{cases} \tag{17.11}$$

The *total* decrement in momentum flow across the vertical line from $y = 0$ to $y = y_1$ in Figure 17.4 is the integral of Equation (17.11),

$$\left.\begin{array}{l} \text{Total decrement in momentum} \\ \text{flow, or missing momentum flow} \end{array}\right\} = \int_0^{y_1} \rho u(u_e - u)\, dy \tag{17.12}$$

Assume that the missing momentum flow is the product of $\rho_e u_e^2$ and a height θ. Then,

$$\text{Missing momentum flow} = \rho_e u_e^2 \theta \tag{17.13}$$

Equating Equations (17.12) and (17.13), we obtain

$$\rho_e u_e^2 \theta = \int_0^{y_1} \rho u(u_e - u)\, dy$$

$$\theta = \int_0^{y_1} \frac{\rho u}{\rho_e u_e} \left(1 - \frac{u}{u_e}\right) dy \tag{17.14}$$

Equation (17.14) is precisely the definition of the momentum thickness given by Equation (17.10). Therefore, θ is an index that is proportional to the decrement in momentum flow due to the presence of the boundary layer. It is the height of a hypothetical streamtube which is carrying the missing momentum flow at freestream conditions.

Note that $\theta = \theta(x)$. In more detailed discussions of boundary-layer theory, it can be shown that θ evaluated at a given station $x = x_1$ is proportional to the integrated friction drag coefficient from the leading edge to x_1; that is,

$$\theta(x_1) \propto \frac{1}{x_1} \int_0^{x_1} c_f\, dx = C_f$$

where c_f is the local skin-friction coefficient defined in Section 1.5 and C_f is the total skin-friction drag coefficient for the length of surface from $x = 0$ to $x = x_1$. Hence, the concept of momentum thickness is useful in the prediction of drag coefficient.

All the boundary-layer properties discussed above are general concepts; they apply to compressible as well as incompressible flows, and to turbulent as well

as laminar flows. The differences between turbulent and laminar flows were introduced in Section 15.2. Here, we extend that discussion by noting that the increased momentum and energy exchange that occur within a turbulent flow cause a turbulent boundary layer to be thicker than a laminar boundary layer. That is, for the same edge conditions, ρ_e, u_e, T_e, etc., we have $\delta_{\text{turbulent}} > \delta_{\text{laminar}}$ and $(\delta_T)_{\text{turbulent}} > (\delta_T)_{\text{laminar}}$. When a boundary layer changes from laminar to turbulent flow, as sketched in Figure 15.8, the boundary-layer thickness markedly increases. Similarly, δ^* and θ are larger for turbulent flows.

17.3 THE BOUNDARY-LAYER EQUATIONS

For the remainder of this chapter, we consider two-dimensional, steady flow. The nondimensionalized form of the x-momentum equation (one of the Navier-Stokes equations) was developed in Section 15.6 and was given by Equation (15.29):

$$\rho' u' \frac{\partial u'}{\partial x'} + \rho' v' \frac{\partial u'}{\partial y'} = -\frac{1}{\gamma M_\infty^2} \frac{\partial p'}{\partial x'} + \frac{1}{\text{Re}_\infty} \frac{\partial}{\partial y'} \left[\mu' \left(\frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'} \right) \right] \quad (15.29)$$

Let us now reduce Equation (15.29) to an approximate form which holds reasonably well within a boundary layer.

Consider the boundary layer along a flat plate of length c as sketched in Figure 17.7. The basic assumption of boundary-layer theory is that a boundary layer is very thin in comparison with the scale of the body; that is,

$$\boxed{\delta \ll c} \quad (17.15)$$

Consider the continuity equation for a steady, two-dimensional flow,

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \quad (17.16)$$

In terms of the nondimensional variables defined in Section 15.6, Equation (17.16) becomes

$$\frac{\partial(\rho' u')}{\partial x'} + \frac{\partial(\rho' v')}{\partial y'} = 0 \quad (17.17)$$

Because u' varies from 0 at the wall to 1 at the edge of the boundary layer, let us say that u' is of the *order of magnitude* equal to 1, symbolized by $O(1)$. Similarly,

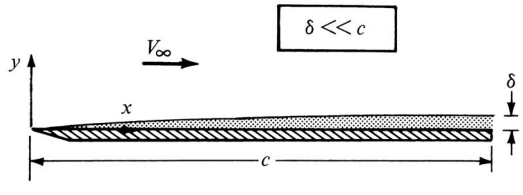


Figure 17.7 The basic assumption of boundary-layer theory: A boundary layer is very thin in comparison with the scale of the body.

$\rho' = O(1)$. Also, since x varies from 0 to c, $x' = O(1)$. However, since y varies from 0 to δ, where $\delta \ll c$, then y' is of the *smaller* order of magnitude, denoted by $y' = O(\delta/c)$. Without loss of generality, we can assume that c is a unit length. Therefore, $y' = O(\delta)$. Putting these orders of magnitude in Equation (17.17), we have

$$\frac{[O(1)][O(1)]}{O(1)} + \frac{[O(1)][v']}{O(\delta)} = 0 \tag{17.18}$$

Hence, from Equation (17.18), clearly v' must be of an order of magnitude equal to δ; that is, $v' = O(\delta)$. Now examine the order of magnitude of the terms in Equation (15.29). We have

$$\rho' u' \frac{\partial u'}{\partial x'} = O(1) \qquad \rho' v' \frac{\partial u'}{\partial y'} = O(1) \qquad \frac{\partial p'}{\partial x'} = O(1)$$

$$\frac{\partial}{\partial y'}\left(\mu' \frac{\partial v'}{\partial x'}\right) = O(1) \qquad \frac{\partial}{\partial y'}\left(\mu' \frac{\partial u'}{\partial y'}\right) = O\left(\frac{1}{\delta^2}\right)$$

Hence, the order-of-magnitude equation for Equation (15.29) can be written as

$$O(1) + O(1) = -\frac{1}{\gamma M_\infty^2} O(1) + \frac{1}{\mathrm{Re}_\infty}\left[O(1) + O\left(\frac{1}{\delta^2}\right)\right] \tag{17.19}$$

Let us now introduce another assumption of boundary-layer theory, namely, the *Reynolds number is large,* indeed, large enough such that

$$\boxed{\frac{1}{\mathrm{Re}_\infty} = O(\delta^2)} \tag{17.20}$$

Then, Equation (17.19) becomes

$$O(1) + O(1) = -\frac{1}{\gamma M_\infty^2} O(1) + O(\delta^2)\left[O(1) + O\left(\frac{1}{\delta^2}\right)\right] \tag{17.21}$$

In Equation (17.21), there is one term with an order of magnitude that is much smaller than the rest, namely, the product $O(\delta^2)[O(1)] = O(\delta^2)$. This term corresponds to $(1/\mathrm{Re}_\infty)\partial/\partial y'(\mu' \partial v'/\partial x')$ in Equation (15.29). Hence, *neglect* this term in comparison to the remaining terms in Equation (15.29). We obtain

$$\rho' u' \frac{\partial u'}{\partial x'} + \rho' v' \frac{\partial u'}{\partial y'} = -\frac{1}{\gamma M_\infty^2} \frac{\partial p'}{\partial x'} + \frac{1}{\mathrm{Re}_\infty} \frac{\partial}{\partial y'}\left(\mu' \frac{\partial u'}{\partial y'}\right) \tag{17.22}$$

In terms of dimensional variables, Equation (17.22) is

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial y}\left(\mu \frac{\partial u}{\partial y}\right) \tag{17.23}$$

Equation (17.23) is the approximate x-momentum equation which holds for flow in a thin boundary layer at a high Reynolds number.